

Lie Groups and Lie Algebras

兹白大姨の数理研究

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Motivation: Lie Groups and Lie Algebras in Physics

Road map

Symmetry means invariance under a transformation. Starting from this physical idea, this opening chapter explains why transformations should be bijections, why bijections naturally form groups, and why physicists need finite groups, discrete groups, Lie groups, Lie algebras, and their representations. The examples include mirror symmetry, Lorentz covariance of the continuity equation, permutations of identical particles, the Monster group, the Standard Model, crystal lattices, the Rubik's cube, matrix representations, particle-hole symmetry, and time reversal. The last section records the course map, expected background, and organisational options appearing in the handwritten lecture.

1.1 Symmetry as invariance under a transformation

Definition 1.1: Symmetry

A *symmetry* of an object or a physical law is a transformation under which the relevant object or law is invariant.

The words “relevant” and “invariant” depend on the problem. A geometric object may be required to occupy exactly the same set of points after a transformation. A physical equation may instead retain the same form in new coordinates, and a quantum state may be unchanged only up to an unobservable phase.

Example 1.1: A mirror-symmetric object

The front elevation of an idealised house is unchanged by reflection in its vertical central

axis.



The transformation moves individual points, but it sends the entire house to the same geometric figure.

Example 1.2: Lorentz covariance of the continuity equation

Let $j^\mu(x)$ be a four-current satisfying

$$\sum_{\mu=0}^3 \partial_\mu j^\mu = 0, \quad \text{or simply} \quad \partial_\mu j^\mu = 0.$$

Under a Lorentz transformation $x'^\mu = L^\mu{}_\nu x^\nu$, a contravariant four-vector and a derivative transform as

$$j'^\mu(x') = L^\mu{}_\nu j^\nu(x), \quad \partial'_\mu = (L^{-1})^\rho{}_\mu \partial_\rho.$$

The transformed divergence is

$$\begin{aligned} \partial'_\mu j'^\mu &= (L^{-1})^\rho{}_\mu \partial_\rho (L^\mu{}_\nu j^\nu) \\ &= (L^{-1})^\rho{}_\mu L^\mu{}_\nu \partial_\rho j^\nu \\ &= \delta^\rho{}_\nu \partial_\rho j^\nu \\ &= \partial_\nu j^\nu = 0. \end{aligned}$$

The Lorentz matrix is constant, so the derivative does not act on it. The middle two matrices multiply to the identity. Hence the continuity equation has the same form in every Lorentz frame. This is invariance of a law rather than invariance of one geometric picture.

Example 1.3: Permutations of identical particles

Consider a system of N particles and a permutation $p \in S_N$. On an ordered list of one-particle data it acts by

$$p \cdot (x_1, \dots, x_N) = (x_{p^{-1}(1)}, \dots, x_{p^{-1}(N)}).$$

If the particles are identical, relabelling them cannot change any physical prediction. In quantum mechanics the permutation operator $P(p)$ acts on the tensor-product Hilbert space. Bosonic states satisfy

$$P(p)\psi = \psi,$$

whereas fermionic states satisfy

$$P(p)\psi = \text{sgn}(p)\psi.$$

Thus permutation symmetry is built into the physical meaning of “identical”.

1.2 What exactly is a transformation?

The most general set-theoretic starting point is a map

$$f : S \longrightarrow S,$$

from a set of objects to itself. Depending on the example, S might be

- a set of geometric objects in Euclidean space;
- a set of four-vector fields $j^\mu(x)$;
- the set of ordered lists describing N particles.

Maps already possess three pieces of the algebraic structure suggested in the handwritten notes.

1. Two maps can be composed:

$$f_1 \circ f_2 : S \longrightarrow S, \quad (f_1 \circ f_2)(x) = f_1(f_2(x)).$$

Composition is closed because the output of f_2 is again an element of S .

2. Composition is associative:

$$(f_1 \circ f_2) \circ f_3 = f_1 \circ (f_2 \circ f_3).$$

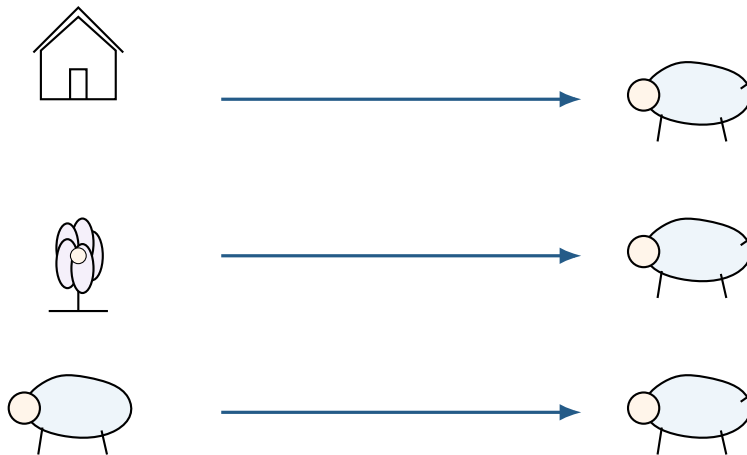
Evaluating either side at x gives $f_1(f_2(f_3(x)))$.

3. There is an identity map

$$\text{id}_S : S \longrightarrow S, \quad \text{id}_S(x) = x,$$

and $f \circ \text{id}_S = \text{id}_S \circ f = f$.

An arbitrary map need not be reversible. The following diagram reproduces the deliberately “bad” actions in the lecture: several distinct objects are sent to the same sheep.



The first two arrows destroy information: from the final sheep one cannot reconstruct whether the initial object was a house, a flower, or a sheep.

图 1.1: Non-bijective maps are not suitable reversible symmetries.

The diagram fails the intuitive requirement of a symmetry because the map is not bijective and therefore has no inverse. We are led to require

$$f : S \longrightarrow S \quad \text{to be bijective.}$$

Every bijection has an inverse bijection, and the inverse undoes the transformation exactly.

Proposition 1.1: Bijections form a group

For every set S , the set

$$\text{Sym}(S) = \{f : S \rightarrow S \mid f \text{ is bijective}\}$$

is a group under composition.

证明. The composition of two bijections is a bijection, so the operation is closed. Function composition is associative. The identity map is bijective and is an identity element. Finally, every bijection f has an inverse bijection f^{-1} , with

$$f \circ f^{-1} = f^{-1} \circ f = \text{id}_S.$$

These are precisely the group axioms. ■

1.3 From transformations to the abstract group concept

The group concept deliberately forgets the internal nature of the objects in S and retains only the transformations and their composition.

Definition 1.2: Group

A *group* is a set G together with a binary operation

$$\circ : G \times G \longrightarrow G, \quad (a, b) \longmapsto a \circ b,$$

such that

1. **associativity:** $a \circ (b \circ c) = (a \circ b) \circ c$ for all $a, b, c \in G$;
2. **identity:** there is $e \in G$ such that $e \circ a = a \circ e = a$ for every $a \in G$;
3. **inverses:** for every $a \in G$, there is $a^{-1} \in G$ such that $a \circ a^{-1} = a^{-1} \circ a = e$.

Closure is encoded by the fact that the operation has codomain G .

The notes place this definition in a short hierarchy:

structure	axioms
magma (called “groupoid” in the lecture)	a set with a closed binary operation
semigroup	a magma with an associative operation
monoid	a semigroup with an identity element
group	a monoid in which every element has an inverse

Modern terminology usually reserves *groupoid* for a category in which every arrow is invertible; *magma* is the unambiguous name for a bare set with a binary operation.

The group axioms are sparse, but their consequences are rich. This is why the same theory applies to permutations, matrices, rotations, lattice symmetries, internal particle symmetries, and many other settings.

1.4 The Monster: how rich can a finite group be?

The lecture pauses at this point to emphasise that an object satisfying only the short group definition can nevertheless have astonishing structure. Its example is the *Monster group*, also called the Fischer–Griess Monster or the Friendly Giant. It is the largest of the 26 sporadic finite simple groups.

Its order is

$$\begin{aligned} |\mathbb{M}| &= 808017424794512875886459904961710757005754368000000000 \\ &= 2^{46} 3^{20} 5^9 7^6 11^2 13^3 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \\ &\quad \cdot 31 \cdot 41 \cdot 47 \cdot 59 \cdot 71. \end{aligned}$$

This is roughly 8.08×10^{53} . The article reproduced in the notes compares it, evocatively rather than as a precise physical count, to the number of elementary particles in Jupiter.

1.4.1 Several complementary descriptions

The Monster can be approached through several mathematical structures.

1. It is the unique simple group of its order and the largest sporadic simple group.
2. Griess constructed a real algebra of dimension 196884. It is commutative but not associative, and its automorphism group is the Monster. This algebra splits as

$$\mathbb{R}\mathbf{1} \oplus V_{196883},$$

where V_{196883} is the smallest nontrivial irreducible representation highlighted in the lecture.

3. It is the automorphism group of the Monster vertex operator algebra. This is often regarded as the most conceptual construction.
4. The Monster acts by diagram automorphisms of an infinite-dimensional generalised Kac–Moody algebra. This object is called the Monster Lie algebra.

These descriptions illustrate an important theme of the course: an abstract group becomes accessible by allowing it to act on a vector space, algebra, geometry, or physical state space.

1.4.2 The first numerical glimpse of monstrous moonshine

Let

$$q = e^{2\pi i \tau}.$$

The elliptic modular j -function has Fourier expansion

$$j(\tau) = q^{-1} + 744 + 196884q + 21493760q^2 + \cdots.$$

The striking observations are

$$196884 = 196883 + 1$$

and

$$21493760 = 21296876 + 196883 + 1,$$

where the numbers on the right are dimensions of irreducible Monster representations. This suggested a graded representation

$$V = \bigoplus_{n \in \mathbb{Z}} V_n$$

whose graded dimensions reproduce the coefficients of the modular function. The McKay–Thompson series refine the dimension to traces

$$T_g(\tau) = \sum_n \text{tr}(g|_{V_n}) q^n.$$

The eventual construction of the Monster vertex operator algebra and the proof of the moonshine conjectures connected finite group theory, modular forms, vertex algebras, and ideas from string theory.

Why the original article pages are retained

The handwritten lecture embeds a two-page historical article by Richard E. Borcherds, with underlining and annotations. Those pages contain dense typeset material—the exact Monster order, Griess algebra, modular j -function, McKay–Thompson series, vertex algebra, Monster Lie algebra, and references. To preserve every printed and handwritten detail, the two source pages are reproduced below. The surrounding prose gives a beginner-friendly mathematical guide to what is on them.

the concept of a group is extremely rich !

↓ see:

WHAT IS . . .

The Monster?

Richard E. Borcherds

When I was a graduate student, my supervisor John Conway would bring into the department his one-year-old son, who was soon known as the baby monster. A more serious answer to the title question is that the monster is the largest of the (known!) sporadic simple groups. Its name comes from its size: The number of elements is

8080, 17424, 79451, 28758, 86459, 90496,

17107, 57005, 75436, 80000, 00000

$= 2^{46} \cdot 3^{20} \cdot 5^9 \cdot 7^6 \cdot 11^2 \cdot 13^3 \cdot 17 \cdot$

19 \cdot 23 \cdot 29 \cdot 31 \cdot 41 \cdot 47 \cdot 59 \cdot 71,

exactly about equal to the number of elementary particles in the planet Jupiter.

The monster was originally predicted to exist by B. Fischer and by R. L. Griess in the early 1970s. Griess constructed it a few years later in an extraordinary tour de force as the group of linear transformations on a vector space of dimension 196883 that preserve a certain commutative but nonassociative bilinear product, now called the Griess product.

Our knowledge of the structure and representations of the monster is now pretty good. The 194 irreducible complex representations were worked out by Fischer, D. Livingstone, and M. P. Thorne (before the monster was even shown to exist). These take up eight large pages in the atlas [A] of finite groups, which is the best single source of information about the monster (and other finite simple groups). The subgroup structure is mostly known; in particular there is an almost complete list of the maximal subgroups, and the main gaps in our knowledge concern embeddings of very small simple groups in the monster. If anyone

wishes to multiply elements of the monster explicitly, R. A. Wilson can supply two matrices that generate the monster. But there is a catch: Each matrix takes up about five gigabytes of storage, and to quote from Wilson's atlas page: "[S]tandard generators have now been made as 196882×196882 matrices over $\text{GF}(2)$ They have been multiplied together, using most of the computing resources of Lehrstuhl D für Mathematik, RWTH Aachen, for about 45 hours...." (The difficulty of multiplying two elements of the monster is caused not so much by its huge size as by the lack of "small" representations; for example, the symmetric group S_{50} is quite a lot bigger than the monster, but it only takes a few minutes to multiply two elements by hand.) Finally the modular representations of the monster for large primes were worked out by G. Hiss and K. Lux; the ones for small primes still seem to be out of reach at the moment.

In the late 1970s John McKay decided to switch from finite group theory to Galois theory. One function that turns up in Galois theory is the elliptic modular function

$$j(\tau) = q^{-1} + 744 + 196884q + 21493760q^2 + \cdots \\ = \sum c(n)q^n$$

($q = e^{2\pi i\tau}$), which is essentially the simplest non-constant function invariant under the action $\tau \mapsto (a\tau + b)/(c\tau + d)$ of $SL_2(\mathbb{Z})$ on the upper half plane $\{\tau \mid \Im(\tau) > 0\}$. He noticed that the coefficient 196884 of q^1 was almost equal to the degree 196883 of the smallest complex representation of the monster (up to a small experimental error). The term "moonshine" roughly means weird relations between sporadic groups and modular functions (and anything else) similar to this. It was clear to many people that this was just a meaningless coincidence; after all, if you have enough large integers from various areas of mathematics, then a few are going to be close just by chance, and John McKay was told that his observation was about as useful as looking at tea leaves. John Thompson took McKay's observation further and pointed out that the next few coefficients of the

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¹The announcement of the classification of the finite simple groups about twenty years ago was a little over-enthusiastic, but a recent 1,300-page preprint by M. Aschbacher and S. D. Smith should finally complete it.

图 1.2: The first historical page embedded in the handwritten lecture: Richard E. Borcherds, "What is the Monster?" The lecture's underlining marks the order, the 196883-dimensional representation, the Griess product, and the status of the finite-simple-group classification.

elliptic modular function were also simple linear combinations of dimensions of irreducible representations of the monster; for example, $21493760 = 21296876 + 196883 + 1$. He suggested that there should be a natural infinite-dimensional graded representation $V = \sum_{n \in \mathbb{Z}} V_n$ of the monster such that the dimension of V_n is the coefficient $c(n)$ of q^n in $j(\tau)$, at least for $n \neq 0$. (The constant term of $j(\tau)$ is arbitrary, since adding a constant to j still produces a function invariant under $SL_2(\mathbb{Z})$ and is set equal to 744 mainly for historical reasons.) Conway and Norton [C-N] followed up Thompson's suggestion of looking at the McKay-Thompson series $T_g(\tau) = \sum_n \text{Trace}(g|V_n)q^n$, whose coefficients are given by the traces of elements g of the monster on the representations V_n , and found by calculating the first few terms that these functions all seemed to be *Hauptmoduls* of genus 0. (A *Hauptmodul* is a function similar to j but invariant under some group other than $SL_2(\mathbb{Z})$.) A. O. L. Atkin, P. Fong, and S. D. Smith showed by computer calculation that there was indeed an infinite-dimensional graded representation of the monster whose McKay-Thompson series were the *Hauptmoduls* found by Conway and Norton, and soon afterwards I. B. Frenkel, J. Lepowsky, and A. Meurman explicitly constructed this representation using vertex operators.

If a group acts on a vector space it is natural to ask if it preserves any algebraic structure, such as a bilinear form or product. The monster module constructed by Frenkel-Lepowsky-Meurman has a vertex algebra structure invariant under the action of the monster. Unfortunately there is no easy way to explain what vertex algebras are; see [K] for the best introduction to them. Roughly speaking, vertex algebras can be thought of (at least in characteristic 0) as commutative rings with derivation where the ring multiplication is not quite defined everywhere; this is analogous to rational maps in algebraic geometry, which are also not quite defined everywhere. A more concrete but less intuitive definition of a vertex algebra is that it consists of a space with a countable number of bilinear products satisfying certain rather complicated identities. In the case of the monster vertex algebra $V = \bigoplus V_n$, this gives bilinear maps from $V_i \times V_j$ to V_k for all integers i, j, k , and the special case of the map from $V_2 \times V_2$ to V_2 is (essentially) the Griess product. So the Griess algebra is a sort of section of the monster vertex algebra.

Following an idea of Frenkel, one can use the monster vertex algebra and the Goddard-Thorn "no-ghost theorem" from string theory to construct the *monster Lie algebra*. This is a $\mathbb{Z}^2$ -graded Lie algebra whose piece of degree $(m, n) \in \mathbb{Z}^2$ has dimension $c(mn)$ whenever $(m, n) \neq (0, 0)$. The monster should be thought of as a group of "diagram automorphisms" of this Lie algebra, in the same way

that the symmetric group S_3 is a group of diagram automorphisms of the Lie algebra D_4 . The monster Lie algebra has a denominator formula, similar to the Weyl denominator formula for finite-dimensional Lie algebras and the Macdonald-Kac identities for affine Lie algebras, which looks like

$$j(\sigma) - j(\tau) = p^{-1} \prod_{\substack{m>0 \\ n \in \mathbb{Z}}} (1 - p^m q^n)^{c(mn)}$$

where $p = e^{2\pi i \sigma}$, $q = e^{2\pi i \tau}$. This formula was discovered independently in the 1980s by several people, including M. Koike, S. P. Norton, and D. Zagier. There are similar identities with $j(\tau)$ replaced by the McKay-Thompson series of any element of the monster, and C. J. Cummins and T. Gannon showed that any function satisfying such identities is a *Hauptmodul*. So this provides some sort of explanation of Conway and Norton's observation that the McKay-Thompson series are all *Hauptmoduls*.

So the question "What is the monster?" now has several reasonable answers:

1. It is the largest sporadic simple group or alternatively the unique simple group of its order.
2. It is the automorphism group of the Griess algebra.
3. It is the automorphism group of the monster vertex algebra. (This is probably the best answer.)
4. It is a group of diagram automorphisms of the monster Lie algebra.

Unfortunately none of these definitions is completely satisfactory. At the moment all constructions of the algebraic structures above seem artificial; they are constructed as sums of two or more apparently unrelated spaces, and it takes a lot of effort to define the algebraic structure on the sum of these spaces and to check that the monster acts on the resulting structure. It is still an open problem to find a really simple and natural construction of the monster vertex algebra.

References

- [A] J. H. CONWAY, R. T. CURTIS, S. P. NORTON, R. A. PARKER, and R. A. WILSON, *Atlas of Finite Groups*, Clarendon Press, Oxford, 1985; see also the online Atlas of Finite Group Representations at <http://www.mat.bham.ac.uk/atlas/v2.0/>.
- [C-N] J. H. CONWAY and S. P. NORTON, Monstrous moonshine, *Bull. London. Math. Soc.* **11** (1979), 308–39.
- [K] V. KAC, *Vertex Algebras for Beginners*, second edition, University Lecture Series, vol. 10, American Mathematical Society, Providence, RI, 1998.

The "WHAT IS..." column carries short (one- or two-page), nontechnical articles aimed at graduate students. Each article focuses on a single mathematical object, rather than a whole theory. The *Notices* welcomes feedback and suggestions for topics for future columns. Messages may be sent to notices-what@ams.org.

—Allyn Jackson

图 1.3: The continuation retained from the source lecture. It discusses monstrous moonshine, vertex algebras, the Monster Lie algebra, and four answers to the question "What is the Monster?"

1.5 Why study group theory?

The motivational list in the lecture ranges from the largest sporadic group to familiar laboratory symmetries. Each item points to a later topic.

1.5.1 Finite groups and classification

The Monster motivates the study of finite groups: their examples, properties, subgroups, quotients, representations, and classification. Even when a group is too large to list, its structure can be exposed through smaller subgroups and linear actions.

There are also groups of very different kinds:

$$\mathbb{M}, \quad \text{Sym}(S), \quad SU(N), \quad \dots$$

The first is a particular enormous finite group, the second is the full group of bijections of an arbitrary set, and the third is a continuous matrix group.

1.5.2 The Standard Model and continuous groups

The internal gauge group of the Standard Model is locally described by

$$U(1) \times SU(2) \times SU(3).$$

The three factors encode, respectively, hypercharge, weak isospin, and colour. These are continuous groups: their elements vary smoothly with real parameters. Understanding their local structure leads from *Lie groups* to their infinitesimal *Lie algebras*.

continuous internal symmetries

$$U(1) \quad SU(2) \quad SU(3)$$

one generator three generators eight generators
before symmetry breaking weak sector colour sector

1.5.3 Crystals and discrete groups

Crystallography asks which transformations preserve a lattice in a given dimension. Translations, rotations, reflections, and glide operations form discrete groups because only separated parameter values preserve the lattice.

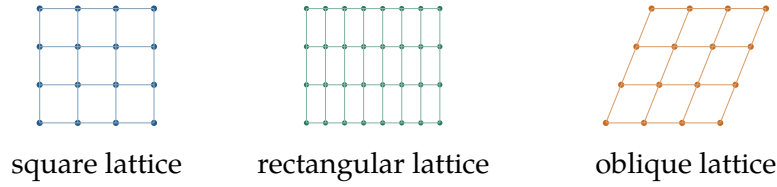


图 1.4: Three of the lattice sketches motivating discrete symmetry groups in crystallography.

1.5.4 The Rubik's cube

Legal moves of a Rubik's cube are permutations of its movable pieces, and sequences of moves form a finite non-Abelian group. The order matters:

$$ab \neq ba$$

for typical face turns a and b . Subgroups describe restricted families of moves, and conjugation

$$aba^{-1}$$

transports a move to a differently oriented position. This physical puzzle therefore gives concrete meaning to noncommutativity, subgroups, and conjugate transformations.

1.6 Representations make abstract groups calculable

Abstract elements need not initially be numbers or matrices. A *representation* replaces each abstract transformation by a linear operator in such a way that composition becomes operator multiplication. This converts group questions into calculations in linear algebra.

The handwritten notes illustrate this by assigning matrices

$$a \hat{=} \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \quad b \hat{=} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

If the representation is D , the rule to preserve is

$$D(a \circ b) = D(a)D(b).$$

In this example,

$$\begin{aligned} D(a)D(b) &= \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}. \end{aligned}$$

The matrix on the last line is the representative of the composite transformation $a \circ b$. Later chapters make this idea precise and study equivalent, reducible, unitary, direct-sum, and tensor-product representations.

1.7 Transformations of quantum systems

The state space of a quantum system supplies many physically important group actions.

Example 1.4: Particle–hole transformation

In fermionic Fock space, a particle–hole transformation exchanges a creation operator with an annihilation operator:

$$c_\alpha \longleftrightarrow c_\alpha^\dagger.$$

Because

$$\{c_\alpha, c_\beta^\dagger\} = \delta_{\alpha\beta}, \quad \{c_\alpha, c_\beta\} = 0,$$

a valid transformation must preserve the canonical anticommutation relations. On a single mode, the number operator changes as

$$n_\alpha = c_\alpha^\dagger c_\alpha \longmapsto c_\alpha c_\alpha^\dagger = 1 - n_\alpha.$$

Thus an occupied state and an empty state are interchanged.

Example 1.5: Time reversal

Time reversal sends

$$t \longmapsto -t.$$

In classical mechanics it also reverses momenta and angular momenta. In quantum mechanics it is represented by an antiunitary operator rather than an ordinary unitary operator. This is an early warning that the mathematical implementation of a symmetry can contain more structure than the visible transformation of coordinates.

These examples motivate representation theory and the systematic study of transformations of quantum systems. The main focus of the lectures is on

Lie groups, and therefore their Lie algebras.

1.8 Map of topics and physical applications

The source lecture organises the motivation in two directions:

1. physical problems identify the mathematics that is necessary;
2. mathematical concepts are repeatedly tested on examples from physics.

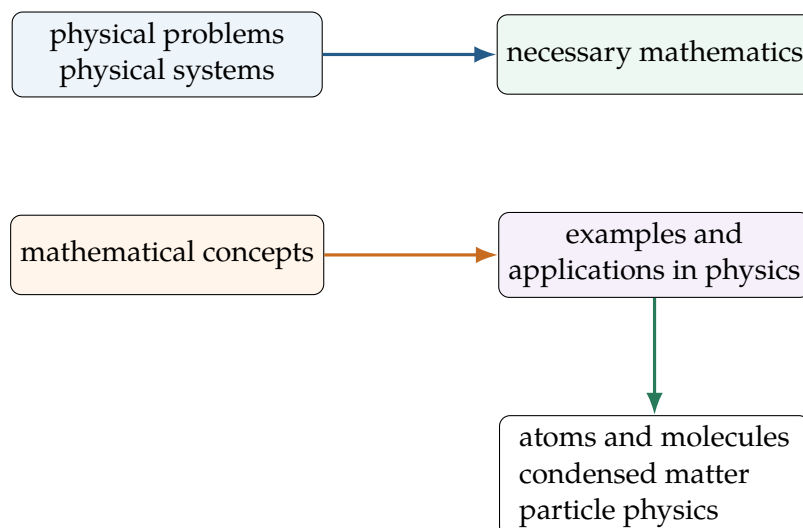


图 1.5: The two-way course philosophy reproduced from the introductory diagram.

The examples on the preceding pages also preview the syllabus:

motivation	topic developed later
Monster group	finite groups
different kinds of groups	examples, properties, and classification
Standard Model gauge group	continuous groups, Lie groups, Lie algebras
crystal lattices	discrete groups
Rubik's cube	basic group concepts
matrix calculations	representation theory
particle-hole and time reversal	transformations of quantum systems

1.8.1 Background checklist from the lecture

The introductory sheet lists the mathematical and physical background that students may bring to the course. The red circles in the source are unfilled checkboxes, so the list records possibilities rather than a claim that every item is mandatory.

Mathematics	Physics
☐ Linear algebra	☐ Theoretical Physics I
☐ Calculus	☐ Theoretical Physics II
☐ Group theory	☐ Theoretical Physics III
☐ Differential geometry	☐ Condensed-matter theory
	☐ Field theory
	☐ Special relativity and four-vector notation
	☐ Second quantisation

The short handwritten phrase “SRT, four notation” has been expanded here as special

relativity and four-vector notation, matching the Lorentz-index calculation at the start of the lecture.

1.9 Course organisation recorded in the source

The last page is an administrative questionnaire. No checkbox is marked, so all alternatives are retained:

Item	Options written in the lecture
Type of examination	☐ oral if the class size $n \leq 10$; ☐ written if $n \geq 10$
Script or printed notes	☐ yes ☐ no
Exercise class and problem solving	☐ at home ☐ in the instructor's presence ☐ no problem-solving session ☐ delegate the problem-solving session
Presentation speed	fast ☐ ☐ slow

A practical reading path

Readers without all the listed background can still use these notes. The early chapters develop abstract group theory from the definitions. Linear algebra is introduced again when matrix groups and representations appear, and the physical sections spell out the needed notation before applying it. The central strategy is always the same: identify transformations, work out how they compose, and then study what remains invariant.

What to retain from this chapter

1. Symmetry is invariance under a transformation. It can mean invariance of a geometric object, covariance of a physical equation, or invariance of physical predictions.
2. Maps $S \rightarrow S$ compose associatively and have an identity map. Requiring transformations to be bijective supplies inverses, so the bijections of S form a group.
3. An abstract group retains the composition law while forgetting what the transformed objects originally were.
4. The Monster demonstrates how much structure can follow from the simple group axioms. Its actions connect finite groups to nonassociative algebras, modular functions, vertex algebras, and Lie algebras.
5. Finite groups, crystallographic groups, and the Rubik's-cube group motivate discrete group theory. The Standard Model group $U(1) \times SU(2) \times SU(3)$ motivates Lie groups and Lie algebras.
6. A representation turns abstract composition into matrix or operator multiplication:

$$D(ab) = D(a)D(b).$$

7. Particle-hole symmetry and time reversal illustrate why quantum mechanics needs a careful theory of group actions on state spaces.

Some Groups and Basic Properties

Road map

This chapter begins with the order of a group and three fundamental examples. It then develops Cayley tables, cancellation, translations, Abelian and cyclic groups, the complete classification of groups of order four, subgroups and the subgroup test, all subgroups of $\mathbb{Z}$, presentations by generators and relations, dihedral groups, and the quaternion group. Every computation and diagram in the handwritten lecture is reproduced below and supplemented with the intermediate steps needed by a first-time reader.

2.1 Order and first examples

We first recall the structure that is assumed throughout the lecture.

Definition 2.1: Group

A *group* is a set G equipped with a binary operation

$$G \times G \longrightarrow G, \quad (a, b) \longmapsto ab,$$

such that the following four axioms hold.

1. **Closure:** $ab \in G$ whenever $a, b \in G$.
2. **Associativity:** $(ab)c = a(bc)$ for all $a, b, c \in G$.
3. **Identity:** there is an element $e \in G$ such that $ea = ae = a$ for every $a \in G$.
4. **Inverses:** for each $a \in G$ there is an element $a^{-1} \in G$ such that $aa^{-1} = a^{-1}a = e$.

Definition 2.2: Order of a group

The *order* of G , denoted by $|G|$, is the number of elements in G . A finite group has $|G| \in \mathbb{N}$; for an infinite group one often writes $|G| = \infty$.

Example 2.1: The unique group of order one

If $|G| = 1$, then $G = \{e\}$. There is only one possible operation:

$$ee = e.$$

The identity is its own inverse, and

$$(ee)e = e = e(ee),$$

so associativity also holds. Thus this is a group, and it is the only group of order one.

Example 2.2: The additive group of integers

For $|G| = \infty$, a basic example is

$$G = (\mathbb{Z}, +).$$

For $a, b, c \in \mathbb{Z}$,

$$a + (b + c) = (a + b) + c, \quad a + 0 = 0 + a = a, \quad a + (-a) = (-a) + a = 0.$$

Thus 0 is the identity and $-a$ is the inverse of a . This is an infinite discrete group.

Example 2.3: The circle group $U(1)$

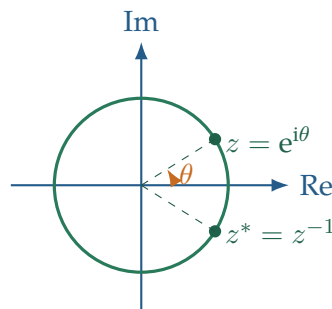
Another infinite group is

$$U(1) = \{z \in \mathbb{C} : |z| = 1\} = \{e^{i\theta} : \theta \in \mathbb{R}\},$$

with ordinary complex multiplication. Closure follows from $|zw| = |z||w| = 1$. Complex multiplication is associative, the identity is 1, and

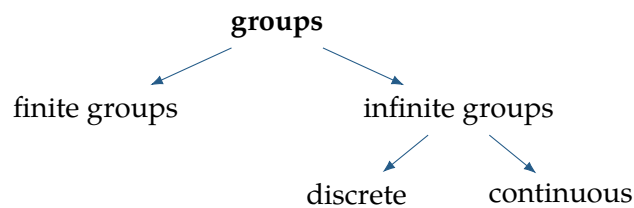
$$z^{-1} = \bar{z} = z^*$$

because $z\bar{z} = |z|^2 = 1$.



The inverse is therefore the reflection of z across the real axis, exactly as in the handwritten diagram.

The first broad classification in the notes is:



For example, finite groups are automatically discrete; $(\mathbb{Z}, +)$ is infinite and discrete; $U(1)$ is infinite and continuous.

2.2 Cayley tables, cancellation, and translations

2.2.1 Cayley tables

For a small finite group, the binary operation can be recorded completely in an operation table.

Definition 2.3: Cayley table

The *Cayley table* of a finite group explicitly specifies the map

$$G \times G \longrightarrow G, \quad (x, y) \longmapsto xy.$$

The entry in row x and column y is xy . Our convention is therefore “row element times column element.”

Example 2.4: The group of order two

Let $G = \{e, a\}$. The identity determines the first row and column:

$$\begin{array}{c|cc} \cdot & e & a \\ \hline e & e & a \\ a & a & ? \end{array}$$

We must have $a^2 = e$. Indeed, if $a^2 = a$, then

$$a^2 = a = ae,$$

and cancellation gives $a = e$, contradicting $|G| = 2$. Consequently

$$\begin{array}{c|cc} \cdot & e & a \\ \hline e & e & a \\ a & a & e \end{array} \quad \text{and} \quad G \cong \mathbb{Z}_2.$$

For $n \in \mathbb{N}$, the notation

$$a^n = \underbrace{a \cdot a \cdots a}_{n \text{ factors}}$$

will be used. We also set $a^0 = e$ and $a^{-n} = (a^{-1})^n$.

2.2.2 The cancellation rule

Lemma 2.1: Left and right cancellation

For $a, x, y \in G$,

$$ax = ay \implies x = y, \quad xa = ya \implies x = y.$$

More generally, each of the equations $ax = b$ and $xa = b$ has a unique solution:

$$ax = b \implies x = a^{-1}b, \quad xa = b \implies x = ba^{-1}.$$

证明. If $ax = ay$, multiply both sides on the left by a^{-1} :

$$a^{-1}(ax) = a^{-1}(ay),$$

$$(a^{-1}a)x = (a^{-1}a)y,$$

$$ex = ey,$$

$$x = y.$$

If $xa = ya$, multiply on the right by a^{-1} . The two displayed solutions are obtained in exactly the same way:

$$ax = b \Rightarrow a^{-1}ax = a^{-1}b \Rightarrow x = a^{-1}b,$$

$$xa = b \Rightarrow xaa^{-1} = ba^{-1} \Rightarrow x = ba^{-1}.$$

Uniqueness follows from cancellation. ■

Definition 2.4: Left and right translations

For fixed $a \in G$, define

$$\ell_a : G \longrightarrow G, \quad x \longmapsto ax, \quad r_a : G \longrightarrow G, \quad x \longmapsto xa.$$

These maps are called the *left translation* and *right translation* by a .

Proposition 2.1: Translations are bijective

For every $a \in G$, both ℓ_a and r_a are bijections.

证明. We prove the claim for ℓ_a .

Surjectivity. Pick $x \in G$ arbitrarily. Set

$$y = a^{-1}x.$$

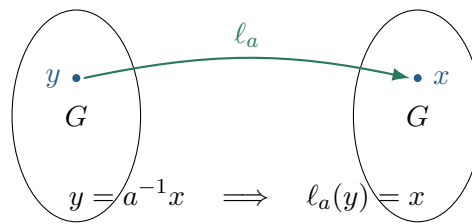
Then

$$\ell_a(y) = ay = a(a^{-1}x) = (aa^{-1})x = ex = x.$$

Thus every x has a preimage.

Injectivity. Suppose $\ell_a(x) = \ell_a(y)$. Then $ax = ay$, so left cancellation gives $x = y$. Therefore ℓ_a is one-to-one.

The proof for r_a is the same, using $y = xa^{-1}$ for surjectivity and right cancellation for injectivity. ■



The Sudoku rule for Cayley tables

The x -th row of a Cayley table lists the values of the bijection ℓ_x , while the y -th column lists the values of the bijection r_y . Hence every group element occurs exactly once in every row and exactly once in every column. This is the “Sudoku rule” used repeatedly in the handwritten notes.

Example 2.5: Translations in $\mathbb{Z}$ and $\mathbb{Z}_2$

In the additive group $\mathbb{Z}$,

$$\ell_k(m) = k + m = m + k = r_k(m).$$

Thus left and right translations coincide because addition is commutative. They shift the integer line by k :

$$\dots, -2, -1, 0, 1, 2, \dots \xrightarrow{\ell_k} \dots, k-2, k-1, k, k+1, k+2, \dots$$

In $G = \mathbb{Z}_2 = \{e, a\}$, the nontrivial translation interchanges the two elements:

$$\ell_a(e) = a, \quad \ell_a(a) = e.$$

For every group, $\ell_e = r_e = \text{id}_G$, because $ex = xe = x$.

2.3 Abelian and cyclic groups

Definition 2.5: Abelian group

A group G is *Abelian* if

$$ab = ba \quad \text{for all } a, b \in G.$$

The additive notation $a + b$ is often used for an Abelian group.

2.3.1 The unique group of order three

Proposition 2.2: There is essentially one group of order three

Every group of order three is cyclic and, after renaming its elements, is $\mathbb{Z}_3$.

证明. Write $G = \{e, a, b\}$. The identity first gives

$\cdot$	e	a	b
e	e	a	b
a	a	$?$	$?$
b	b	$?$	$?$

The entry a^2 cannot be a , by cancellation. It also cannot be e : if $a^2 = e$, the Sudoku rule forces $ab = b$, but $eb = b$, and right cancellation would imply $a = e$. Therefore $a^2 = b$. The row of a must then contain the remaining element e in its last position, so $ab = e$. The Sudoku rule now determines the final row:

$\cdot$	e	a	b
e	e	a	b
a	a	b	e
b	b	e	a

Since $b = a^2$ and $a^3 = e$, this is exactly the power table

$$G = \{e, a, a^2\} = \langle a \rangle.$$

Associativity is transparent from the power rule $(a^i a^j) a^k = a^{i+j+k} = a^i (a^j a^k)$, with exponents read modulo 3. ■

Although a table with three elements has $3^3 = 27$ individual associativity equalities

$$(xy)z = x(yz), \quad x, y, z \in G,$$

the power description proves all of them at once.

Definition 2.6: Cyclic group and generator

A group G is *cyclic* if there exists $a \in G$ such that

$$G = \langle a \rangle := \{a^k : k \in \mathbb{Z}\}.$$

The element a is then called a *generator* of G .

For all integers k, ℓ, m , the usual power rules hold:

$$\begin{aligned} (a^k a^\ell) a^m &= a^{k+\ell} a^m = a^{(k+\ell)+m} = a^{k+(\ell+m)} = a^k (a^\ell a^m), \\ a^0 a^k &= a^{0+k} = a^k, \\ a^{-k} a^k &= a^{-k+k} = a^0 = e, \\ a^k a^\ell &= a^{k+\ell} = a^{\ell+k} = a^\ell a^k. \end{aligned}$$

The last equality proves the next statement.

Proposition 2.3: Every cyclic group is Abelian

If $G = \langle a \rangle$, then G is Abelian.

证明. Given $x, y \in G$, choose $k, \ell \in \mathbb{Z}$ with $x = a^k$ and $y = a^\ell$. Then

$$xy = a^{k+\ell} = a^{\ell+k} = yx.$$

■

Example 2.6: $\mathbb{Z}$ and its generators

The additive group $\mathbb{Z}$ is cyclic:

$$\mathbb{Z} = \langle 1 \rangle.$$

In additive notation, the expression 1^n from multiplicative notation means a repeated sum:

$$\begin{cases} n = \underbrace{1 + \cdots + 1}_{n \text{ summands}}, & n \geq 1, \\ 0 = 1^0, & n = 0, \\ n = \underbrace{(-1) + \cdots + (-1)}_{|n| \text{ summands}}, & n < 0. \end{cases}$$

The element -1 is another generator. Indeed, for $n \geq 2$,

$$n = (-1)^{-n} = \underbrace{1 + \cdots + 1}_{n \text{ summands}}.$$

Thus a cyclic group need not have a unique generator.

Example 2.7: Two generators of $\mathbb{Z}_3$

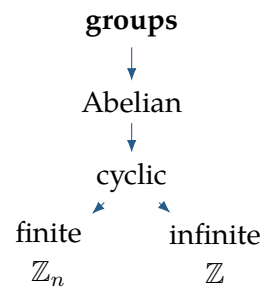
Let

$$G = \{a, a^2, a^3 = e\}.$$

Then a is a generator. The element a^2 is also a generator because

$$(a^2)^1 = a^2, \quad (a^2)^2 = a^4 = a, \quad (a^2)^3 = a^6 = e.$$

The hierarchy sketched in the notes is the following. The examples at the bottom are the cyclic groups; they are not intended to say that every Abelian group is cyclic.



2.4 Groups of order four

2.4.1 The cyclic possibility

Let $G = \{e, a, b, c\}$. One possible group is

$$\mathbb{Z}_4 = \{e, a, a^2 = b, a^3 = c\}, \quad a^4 = e,$$

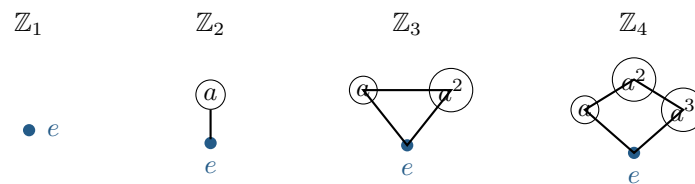
whose table is

$\cdot$	e	a	b	a^3
e	e	a	b	a^3
a	a	b	a^3	e
b	b	a^3	e	a
a^3	a^3	e	a	b

Equivalently, replacing the label a^3 by c ,

$\cdot$	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b	b	c	e	a
c	c	e	a	b

The cycle diagrams in the original notes give a graphical representation of the first four cyclic groups. An edge means “multiply by the displayed generator”; the filled vertex is the identity.



The diagrams immediately show that a^2 generates $\mathbb{Z}_3$ but does not generate $\mathbb{Z}_4$: in $\mathbb{Z}_4$, repeatedly multiplying by a^2 visits only e and a^2 .

2.4.2 Completing all order-four Cayley tables

The handwritten notes complete the table by a case split. We reproduce that argument explicitly. The identity row and column are fixed:

$\cdot$	e	a	b	c
e	e	a	b	c
a	a	$\square$	$\square$	$\square$
b	b	$\square$	$\square$	$\square$
c	c	$\square$	$\square$	$\square$

The entry a^2 cannot equal a .

Case A: $a^2 = b$. The Sudoku rule forces

$$G_A : \begin{array}{c|cccc} \cdot & e & a & b & c \\ \hline e & e & a & b & c \\ a & a & b & c & e \\ b & b & c & e & a \\ c & c & e & a & b \end{array} .$$

Here a generates the whole group, so $G_A \cong \mathbb{Z}_4$.

Case B: $a^2 = e$, but $c^2 = a$. The second row and second column are first fixed by $a^2 = e$. One possible completion is

$$G_B : \begin{array}{c|cccc} \cdot & e & a & b & c \\ \hline e & e & a & b & c \\ a & a & e & c & b \\ b & b & c & a & e \\ c & c & b & e & a \end{array} .$$

Now c is a generator:

$$G_B = \{e, c, c^2 = a, c^3 = b\} \cong \mathbb{Z}_4.$$

Thus G_A and G_B are the same group after a relabelling of elements.

Case C: $a^2 = b^2 = c^2 = e$. The other possible completion is

$$K_4 : \begin{array}{c|cccc} \cdot & e & a & b & c \\ \hline e & e & a & b & c \\ a & a & e & c & b \\ b & b & c & e & a \\ c & c & b & a & e \end{array} .$$

This is the *Klein four group*.

Theorem 2.1: Classification of groups of order four

Up to a renaming that preserves multiplication, there are exactly two groups of order four:

$$\mathbb{Z}_4 = \langle a \mid a^4 = e \rangle, \quad K_4 = \{e, a, b, c\}, \quad a^2 = b^2 = c^2 = e.$$

证明. The preceding case split exhausts the possible values of a^2 and the remaining Sudoku completions. Cases A and B are cyclic and become identical after relabelling. Case C is not cyclic because every nonidentity element has order 2. Therefore it cannot be the same group as $\mathbb{Z}_4$. ■

What does “the same group” mean?

The handwritten notes ask how to define “essentially equal” or “isomorphic.” The formal definition appears later, but the idea is already visible here. A bijection $\varphi : G \rightarrow H$ is an isomorphism when

$$\varphi(xy) = \varphi(x)\varphi(y).$$

It changes labels but preserves the multiplication structure. For example, the isomorphism $G_A \rightarrow G_B$ can send $e \mapsto e$, $a \mapsto c$, $b = a^2 \mapsto a = c^2$, and $c = a^3 \mapsto b = c^3$.

2.4.3 Several models of K_4

The table of K_4 is symmetric across its main diagonal, so K_4 is Abelian. Associativity does hold; for example, it follows immediately from any of the concrete models below.

Example 2.8: Odd units modulo 8

The set

$$\{1, 3, 5, 7\}$$

is a group under multiplication modulo 8. Every element is its own inverse:

$$1^2 \equiv 3^2 \equiv 5^2 \equiv 7^2 \equiv 1 \pmod{8}.$$

Its multiplication table has exactly the pattern of K_4 . Hence

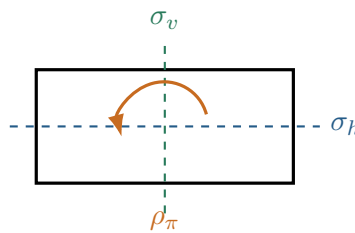
$$K_4 \cong \{1, 3, 5, 7\} \text{ under multiplication modulo 8.}$$

Example 2.9: The symmetry group of a rectangle

The symmetry group of a geometric object is the group of all transformations that leave the object invariant. A non-square rectangle has four such symmetries:

$$e, \quad \sigma_v \text{ (vertical reflection),} \quad \sigma_h \text{ (horizontal reflection),} \quad \rho_\pi \text{ (rotation by } 180^\circ\text{).}$$

Each nonidentity symmetry squares to e , and the product of any two different nonidentity symmetries is the third.



Thus the symmetry group of a rectangle is isomorphic to K_4 .

One may also write

$$K_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

The four pairs $(0, 0)$, $(1, 0)$, $(0, 1)$, $(1, 1)$, added componentwise modulo 2, again have precisely the same table.

2.5 Subgroups and the order of an element

The notes first point out that the highlighted 2×2 corner $\{e, a\}$ of the K_4 table is closed:

$\cdot$	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

Because the inherited operation keeps $\{e, a\}$ inside itself, and the remaining group axioms are inherited, $\{e, a\}$ is itself a group.

Definition 2.7: Subgroup

Let G be a group and $H \subseteq G$. We call H a *subgroup* of G , written $H \leq G$, if the operation of G , restricted to

$$H \times H \longrightarrow H,$$

makes H into a group.

The groups $\{e\}$ and G are always subgroups of G ; they are called the *trivial subgroups*. Associativity never needs to be re-proved for a subset: if $a, b, c \in H$, then $(ab)c = a(bc)$ because the same equality already holds in G . What must be checked is that the required products, identity, and inverses stay in H .

Lemma 2.2: One-step subgroup test

Let H be a nonempty subset of a group G . Then

$$H \leq G \iff ab^{-1} \in H \text{ for all } a, b \in H.$$

证明. *Necessity.* If $H \leq G$, then $b^{-1} \in H$ and closure gives $ab^{-1} \in H$.

Sufficiency. Assume $H \neq \emptyset$ and that $ab^{-1} \in H$ for every $a, b \in H$.

1. Choose $a \in H$. Taking the pair (a, a) gives

$$aa^{-1} = e \in H.$$

2. For any $a \in H$, apply the condition to e, a :

$$ea^{-1} = a^{-1} \in H.$$

3. If $a, b \in H$, then $b^{-1} \in H$ by the preceding step. Applying the condition to a, b^{-1} gives

$$a(b^{-1})^{-1} = ab \in H.$$

Thus H contains e , contains inverses, and is closed under products. Associativity is inherited from G , so $H \leq G$. ■

Proposition 2.4: The cyclic subgroup generated by an element

For every $a \in G$,

$$\langle a \rangle := \{a^k : k \in \mathbb{Z}\}$$

is a subgroup of G .

证明. The set is nonempty because $a^0 = e \in \langle a \rangle$. For $a^m, a^n \in \langle a \rangle$,

$$a^m(a^n)^{-1} = a^m a^{-n} = a^{m-n} \in \langle a \rangle.$$

The subgroup test completes the proof. ■

Definition 2.8: Order of an element

The *order* of $a \in G$ is

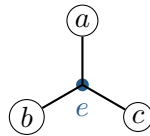
$$|a| := |\langle a \rangle|.$$

Equivalently, if some positive power of a is e , then $|a|$ is the smallest positive integer n with $a^n = e$. If no positive power equals e , then a has infinite order.

In K_4 , every nonidentity element has order 2:

$$\langle a \rangle = \{e, a\}, \quad \langle b \rangle = \{e, b\}, \quad \langle c \rangle = \{e, c\}.$$

The cycle graph drawn in the notes is therefore a three-armed star:



2.6 All subgroups of $\mathbb{Z}$

Definition 2.9: The subgroup $m\mathbb{Z}$

For $m \in \mathbb{N}$, define

$$m\mathbb{Z} := \{mk : k \in \mathbb{Z}\},$$

the set of all integer multiples of m . We also use $0\mathbb{Z} = \{0\}$.

The set $m\mathbb{Z}$ is a subgroup of $(\mathbb{Z}, +)$. Indeed, it is nonempty, and if $mk, m\ell \in m\mathbb{Z}$, then the additive form of xy^{-1} is a difference:

$$mk - (m\ell) = mk + m(-\ell) = m(k - \ell) \in m\mathbb{Z}.$$

The subgroup test applies.

Theorem 2.2: Subgroups of the integers

If $H \leq (\mathbb{Z}, +)$, then there is a unique $m \in \mathbb{N} \cup \{0\}$ such that

$$H = m\mathbb{Z}.$$

证明. If $H = \{0\}$, take $m = 0$. Now assume $H \neq \{0\}$. If $h \in H$ is nonzero, then either $h > 0$ or $-h > 0$, and $-h \in H$. Hence H contains a positive integer. By the well-ordering principle, there is a smallest positive integer $m \in H$.

Every integer multiple of m lies in H , so

$$m\mathbb{Z} \subseteq H.$$

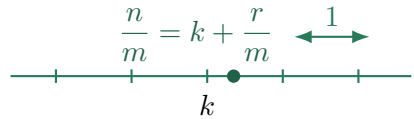
For the reverse inclusion, take any $n \in H$. By Euclidean division there are integers k, r such that

$$n = km + r, \quad 0 \leq r < m.$$

Equivalently,

$$\frac{n}{m} = k + \frac{r}{m}, \quad 0 \leq \frac{r}{m} < 1.$$

This is the decomposition illustrated by the number-line sketch:



Because $n, m \in H$, closure under integer multiples and subtraction gives

$$r = n - km \in H.$$

If $r > 0$, then r would be a positive element of H smaller than m , contradicting the choice of m . Thus $r = 0$, so $n = km \in m\mathbb{Z}$. Therefore $H \subseteq m\mathbb{Z}$, and hence $H = m\mathbb{Z}$.

For uniqueness, if $m\mathbb{Z} = n\mathbb{Z}$ with $m, n > 0$, then each is the smallest positive element of the same set, so $m = n$. The case $m = 0$ is also unique because $0\mathbb{Z} = \{0\}$. ■

2.7 Presentations: generators and relations

Definition 2.10: Presentation of a group

A presentation

$$G = \langle S \mid R \rangle$$

consists of a set S of *generators* and a set R of *relations* among those generators. Start with all finite words whose letters are the generators and their formal inverses, and then identify words using the relations in R .

Example 2.10: A presentation of K_4

The handwritten presentation is

$$K_4 = \langle a, b \mid a^2 = b^2 = (ab)^2 = e \rangle.$$

The relations imply $a^{-1} = a$ and $b^{-1} = b$. Moreover,

$$\begin{aligned} (ab)^2 = e &\implies abab = e, \\ &\implies bab = a^{-1} = a \quad (\text{multiply on the left by } a^{-1} = a), \\ &\implies ab = b^{-1}a = ba \quad (\text{multiply on the left by } b^{-1} = b). \end{aligned}$$

Thus a and b commute. Any word can now be rearranged into the form $a^i b^j$, and the squares can be deleted, leaving $i, j \in \{0, 1\}$. Consequently every word represents one of

$$e, \quad a, \quad b, \quad ab.$$

For example,

$$ba = ab, \quad a^2 = e, \quad b^2 a = a, \quad a^{-1} b^{-1} = ab.$$

These are precisely the four elements of K_4 .

Associativity does not have to be added as another relation: multiplication of words is concatenation, and concatenation is associative. Formally, a presentation constructs a quotient of a free group, so the group axioms are automatically satisfied.

Example 2.11: A presentation of $\mathbb{Z}_n$

$$\mathbb{Z}_n = \langle a \mid a^n = e \rangle = \{e, a, a^2, \dots, a^{n-1}\}.$$

Any higher or negative power is reduced by changing the exponent modulo n . For instance $a^{n+2} = a^2$ and $a^{-1} = a^{n-1}$.

What presentations do and do not make easy

- One group may have many different presentations.
- A presentation can define a group very compactly.
- Important properties, even the order of the group, may not be obvious from a presentation.
- Every group has a presentation: in a deliberately redundant one, use every group element as a generator and the entire Cayley table as the relations.

Definition 2.11: Free group

For a set $S = \{a, b, \dots\}$, the *free group on S* is

$$F(S) = \langle S \mid \emptyset \rangle.$$

Its elements are reduced words in the letters s and s^{-1} ($s \in S$), with only adjacent inverse pairs ss^{-1} and $s^{-1}s$ cancelled.

If S contains at least two generators, the free group is non-Abelian by construction: the words ab and ba are distinct because no relation permits one to be changed into the other.

2.8 The dihedral groups

2.8.1 The group D_3

Example 2.12: A six-element presentation

Consider

$$G = \langle a, b \mid a^3 = e, b^2 = e, bab^{-1} = a^2 \rangle.$$

Since $b^{-1} = b$ and $a^2 = a^{-1}$, the third relation is equivalently

$$ba = a^2b.$$

This rule moves every b to the right of every power of a . After using $a^3 = e$ and $b^2 = e$, every word reduces to one of

$$G = \{e, a, a^2, b, ab, a^2b\}.$$

The Cayley table is

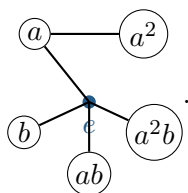
$\cdot$	e	a	a^2	b	ab	a^2b
e	e	a	a^2	b	ab	a^2b
a	a	a^2	e	ab	a^2b	b
a^2	a^2	e	a	a^2b	b	ab
b	b	a^2b	ab	e	a^2	a
ab	ab	b	a^2b	a	e	a^2
a^2b	a^2b	ab	b	a^2	a	e

The table is not symmetric; for example,

$$ab \neq ba = a^2b.$$

Therefore G is non-Abelian.

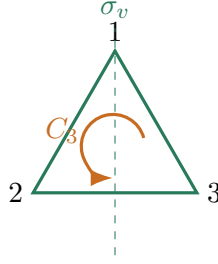
The cycle graph in the notes makes all cyclic subgroups visible. The rotation subgroup $\{e, a, a^2\}$ is a triangle, while b, ab, a^2b each generate a two-element subgroup.



Definition 2.12: Dihedral group

The *dihedral group* D_n is the symmetry group of a regular polygon with n sides: the group of all transformations that leave the polygon invariant. The integer n is called the degree.

For a regular triangle, let C_3 be counterclockwise rotation through $2\pi/3$, and let σ_v be reflection in the vertical axis.



As permutations of the three vertices,

$$C_3 = (123), \quad \sigma_v = (23).$$

Because $\sigma_v^{-1} = \sigma_v$,

$$\sigma_v C_3 \sigma_v^{-1} = (23)(123)(23) = (132) = C_3^2 = C_3^{-1}.$$

Equivalently,

$$\sigma_v C_3 = C_3^2 \sigma_v.$$

Thus

$$D_3 = \langle C_3, \sigma_v \mid C_3^3 = \sigma_v^2 = e, \sigma_v C_3 \sigma_v^{-1} = C_3^2 \rangle.$$

Identifying $a = C_3$ and $b = \sigma_v$ recovers the six-element group and the Cayley table above.

The smallest non-Abelian group

D_3 is the smallest non-Abelian group. Groups of orders 1, 2, 3 and 4 are Abelian by the classifications already discussed; a group of prime order 5 is cyclic (proved later using Lagrange's theorem). At order 6, D_3 provides the first non-Abelian example.

2.8.2 The general group D_n

Let C_n be rotation by $2\pi/n$, and let σ be a fixed reflection. Then

$$D_n = \langle C_n, \sigma \mid C_n^n = e, \sigma^2 = e, \sigma C_n \sigma^{-1} = C_n^{-1} \rangle.$$

The relation says that reflecting, rotating, and reflecting back reverses the direction of rotation. Every word can be reduced by moving σ to the right:

$$\sigma C_n = C_n^{-1} \sigma.$$

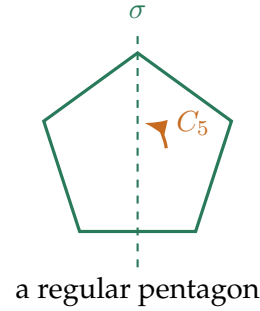
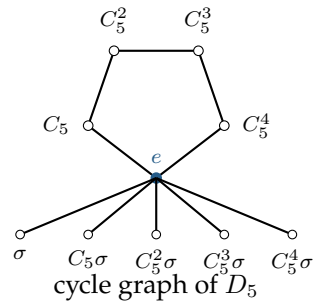
After also reducing powers modulo n , the elements are

$$D_n = \{e, C_n, C_n^2, \dots, C_n^{n-1}, \sigma, C_n \sigma, C_n^2 \sigma, \dots, C_n^{n-1} \sigma\}.$$

There are n rotations and n reflections, so

$$|D_n| = 2n.$$

For $n = 5$, the two sketches from the notes may be drawn together:



The small-order list recorded in the notes is:

$ G $	2	3	4	5	6	7
groups listed	$\mathbb{Z}_2$	$\mathbb{Z}_3$	$\mathbb{Z}_4, K_4$	$\mathbb{Z}_5$	$\mathbb{Z}_6, D_3$	$\mathbb{Z}_7$

For orders 2, 3, 5, 7, the cyclic group is the only possibility. At order 4 there are two groups; at order 6 the list already contains both an Abelian and a non-Abelian group. Starting at order 8, further kinds of groups appear.

2.9 The quaternion group Q_8

Example 2.13: Presentation and multiplication rules

The quaternion group has the presentation

$$Q_8 = \langle a, b \mid a^4 = e, b^2 = a^2, bab^{-1} = a^{-1} \rangle.$$

An equivalent and more familiar notation is

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\},$$

with

$$i^2 = j^2 = k^2 = ijk = -1, \quad (-1)^2 = 1.$$

Here 1 is the identity and -1 is a central element of order 2. The six imaginary units occur in inverse pairs:

$$-i = (-1)i = i^{-1}, \quad -j = j^{-1}, \quad -k = k^{-1}.$$

The fundamental products are

$$ij = k, \quad jk = i, \quad ki = j,$$

whereas reversing the order changes the sign:

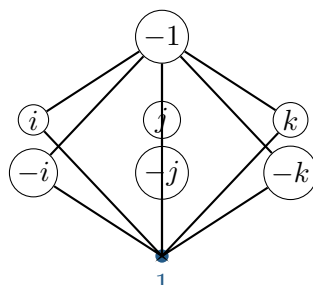
$$ji = -k, \quad kj = -i, \quad ik = -j.$$

In particular,

$$ij = k \neq -k = ji,$$

so Q_8 is non-Abelian and $|Q_8| = 8$.

The original cycle graph consists of three cyclic subgroups of order 4 sharing the vertices 1 and -1 :



each diamond is a four-element cyclic subgroup

The three diamonds are $\langle i \rangle$, $\langle j \rangle$, and $\langle k \rangle$. Their common top and bottom vertices are the common elements -1 and 1 .

The quaternion algebra is not a field

The symbols $1, i, j, k$ span the real quaternion algebra

$$\mathbb{H} = \{a + ib + jc + kd : a, b, c, d \in \mathbb{R}\}.$$

Although every nonzero quaternion has a multiplicative inverse, multiplication is not commutative; for example $ij \neq ji$. Under the standard definition, a field must have commutative multiplication, so $\mathbb{H}$ is a noncommutative division algebra rather than a field.

The lecture closes with the scale of finite-group classification: groups of order at most 2000 have been tabulated, and the note records

49,910,529,484

different isomorphism classes in that range. Even though the first few orders have very short lists, the number of possible group structures grows dramatically.

What to retain from this chapter

1. Cancellation makes every left and right translation a bijection, so every Cayley-table row and column is a permutation of the group.
2. Cyclic groups are Abelian, but an Abelian group need not be cyclic; K_4 is the first example in these notes.
3. The only groups of order four are $\mathbb{Z}_4$ and K_4 .
4. A nonempty subset $H \subseteq G$ is a subgroup exactly when $ab^{-1} \in H$ for all $a, b \in H$.
5. Every subgroup of $(\mathbb{Z}, +)$ is $m\mathbb{Z}$.
6. Presentations turn generators and relations into groups. They give compact descriptions of K_4 , D_n , and Q_8 .
7. D_3 is the first non-Abelian group, while Q_8 illustrates a genuinely new group structure at order eight.

The Symmetric Group and Cayley's Theorem

Road map

This chapter introduces the symmetric group of a finite set and carefully distinguishes it from the symmetry group of a geometric object. It then turns the left translations of an arbitrary finite group into a permutation group. The resulting faithful action proves Cayley's theorem: every finite group of order n is isomorphic to a subgroup of S_n .

3.1 The symmetric group of a finite set

Definition 3.1: Symmetric group

Let X be a finite set. The *symmetric group on X* is

$$S_X = \{f : X \longrightarrow X : f \text{ is bijective}\}.$$

Its operation is composition:

$$(f \circ g)(x) = f(g(x)).$$

Thus g acts first and f acts second.

Let us verify the group axioms, including the facts that are often left implicit.

- If $f, g : X \rightarrow X$ are bijections, then $f \circ g$ is a bijection: its inverse is $g^{-1} \circ f^{-1}$. Hence S_X is closed.
- Composition of maps is associative:

$$((f \circ g) \circ h)(x) = f(g(h(x))) = (f \circ (g \circ h))(x)$$

for every $x \in X$.

- The identity map $\text{id}_X(x) = x$ is the identity element.
- Every $f \in S_X$ has its inverse bijection $f^{-1} \in S_X$.

Consequently S_X is indeed a group.

Symmetric group versus symmetry group

The terms are related but are not synonymous.

- The *symmetric group* S_X contains all permutations of the set X .
- A *symmetry group* contains transformations that leave a chosen geometric or other structured object invariant.

For example, the symmetry group of a regular polygon is a dihedral group, whereas S_X forgets geometry and permits every bijection of its underlying set.

If X has n elements, choose a labelling

$$X = \{x_1, \dots, x_n\}$$

and identify x_i with i . Relabelling gives an isomorphism

$$S_X \cong S_n,$$

where

$$S_n := S_{\{1, 2, \dots, n\}}.$$

The group S_n is also called the *permutation group on n letters*. A permutation may be written as a bijection

$$f : \{1, \dots, n\} \longrightarrow \{1, \dots, n\}.$$

Proposition 3.1: Order of S_n

$$|S_n| = n!.$$

证明. There are n choices for $f(1)$. Once that value is chosen, bijectivity leaves $n - 1$ choices for $f(2)$, then $n - 2$ choices for $f(3)$, and so on, ending with one choice for $f(n)$. By the multiplication principle,

$$|S_n| = n(n - 1)(n - 2) \cdots 2 \cdot 1 = n!.$$

■

Example 3.1: S_3 and the triangle group

S_3 has $3! = 6$ elements. It has the same multiplication structure as the symmetry group D_3 of an equilateral triangle:

$$S_3 \cong D_3.$$

Indeed, every permutation of the three vertices is realized by exactly one rigid symmetry of the triangle. Under this correspondence, the 3-cycles are rotations and the transpositions are reflections. Hence their Cayley tables agree after the elements are relabelled.

3.2 Left translations as permutations

Let

$$G = \{a_1, \dots, a_n\}, \quad |G| = n,$$

be a finite group. For each $a \in G$, recall the left translation

$$\ell_a : G \longrightarrow G, \quad \ell_a(x) = ax.$$

The cancellation argument from Chapter 2 shows that every ℓ_a is bijective. Thus each left translation is an element of the symmetric group S_G .

Definition 3.2: The left regular permutation group

Define

$$L_G := \{\ell_a : a \in G\}.$$

This is the set of all left translations of G .

Proposition 3.2

L_G is a group under composition.

证明. We check every group axiom.

Closure. Let $a, b \in G$. For every $x \in G$,

$$\begin{aligned} (\ell_a \circ \ell_b)(x) &= \ell_a(\ell_b(x)) \\ &= \ell_a(bx) \\ &= a(bx) \\ &= (ab)x \\ &= \ell_{ab}(x). \end{aligned}$$

Two maps are equal when they have the same value at every input, so

$$\boxed{\ell_a \circ \ell_b = \ell_{ab}}.$$

Since $ab \in G$, the composite belongs to L_G .

Identity. For every $x \in G$,

$$\ell_e(x) = ex = x.$$

Hence $\ell_e = \text{id}_G$, and

$$\ell_e \circ \ell_a = \ell_a = \ell_a \circ \ell_e.$$

Inverses. Using the composition rule,

$$\ell_{a^{-1}} \circ \ell_a = \ell_{a^{-1}a} = \ell_e, \quad \ell_a \circ \ell_{a^{-1}} = \ell_{aa^{-1}} = \ell_e.$$

Therefore

$$(\ell_a)^{-1} = \ell_{a^{-1}}.$$

Associativity. Composition of functions is associative:

$$\ell_a \circ (\ell_b \circ \ell_c) = (\ell_a \circ \ell_b) \circ \ell_c.$$

Thus all group axioms hold. ■

3.3 The group G and the group L_G

Consider the map

$$\Phi : G \longrightarrow L_G, \quad a \longmapsto \ell_a.$$

Proposition 3.3

Φ is a bijection and preserves multiplication:

$$\Phi(ab) = \Phi(a) \circ \Phi(b).$$

证明. *Preservation of multiplication.* The composition calculation above gives

$$\Phi(a) \circ \Phi(b) = \ell_a \circ \ell_b = \ell_{ab} = \Phi(ab).$$

Surjectivity. By the definition of L_G , every one of its elements has the form $\ell_a = \Phi(a)$ for some $a \in G$. Thus Φ is onto.

Injectivity. Suppose $\Phi(a) = \Phi(b)$, so $\ell_a = \ell_b$. Equal maps agree at e , and therefore

$$a = \ell_a(e) = \ell_b(e) = b.$$

Equivalently, equality at every $x \in G$ gives $ax = bx$, and right cancellation gives $a = b$. Thus Φ is one-to-one. ■

Definition 3.3: Group isomorphism

A map $\varphi : G \rightarrow H$ is a *group isomorphism* if it is bijective and

$$\varphi(ab) = \varphi(a)\varphi(b) \quad \text{for all } a, b \in G.$$

If such a map exists, we write $G \cong H$.

The proposition proves

$$G \cong L_G.$$

It also explains in a literal way why the two groups have the same Cayley table structure. The element a corresponds to ℓ_a , and

$$ab = c \iff \ell_a \circ \ell_b = \ell_c.$$

Thus every entry of the Cayley table of G becomes the corresponding entry of the Cayley table of L_G .

Since Φ is bijective, there is a one-to-one correspondence

$$\begin{array}{ccc} G & \longleftrightarrow & L_G \\ a & \longleftrightarrow & \ell_a \end{array}$$

and consequently

$$|G| = |L_G| = n.$$

Example 3.2: The regular copy of $\mathbb{Z}_3$ inside S_3

Let $G = \{e, a, a^2\}$ with $a^3 = e$. In cycle notation for permutations of the set $\{e, a, a^2\}$,

$$\ell_e = \text{id}, \quad \ell_a = (e \ a \ a^2), \quad \ell_{a^2} = (e \ a^2 \ a).$$

Hence

$$L_G = \{\text{id}, (e \ a \ a^2), (e \ a^2 \ a)\}.$$

It is a three-element subgroup of the six-element group $S_G \cong S_3$, and it is isomorphic to $\mathbb{Z}_3$.

3.4 Cayley's theorem

Because every left translation is a permutation of the underlying set G , we have

$$L_G \subseteq S_G.$$

The group operations agree: both are composition of maps. Therefore the subgroup test, or the group verification above, gives

$$L_G \leq S_G.$$

If $|G| = n$, then a labelling of the elements of G gives

$$S_G \cong S_n.$$

Theorem 3.1: Cayley's theorem

Every finite group G of order n is isomorphic to a subgroup of S_n . More precisely,

$$G \cong L_G \leq S_G \cong S_n.$$

证明. The map

$$\Phi : G \longrightarrow S_G, \quad a \longmapsto \ell_a,$$

preserves multiplication because $\ell_a \circ \ell_b = \ell_{ab}$. It is injective because $\ell_a = \ell_b$ implies

$$a = \ell_a(e) = \ell_b(e) = b.$$

Its image is exactly L_G , which is a subgroup of S_G . Thus Φ is an isomorphism from G onto L_G . Finally, after labelling the n elements of G by $1, \dots, n$, the group S_G becomes S_n . ■

The theorem also records the size comparison

$$|G| = |L_G| = n, \quad |S_G| = n!.$$

When $G \neq \{e\}$, the subgroup L_G is nontrivial. For $n \geq 3$, most permutations of the underlying set are not left translations: L_G has only n elements, while S_G has $n!$ elements.

The theorem beyond finite groups

The same proof works for an arbitrary group G , finite or infinite:

$$G \cong L_G \leq S_G,$$

where S_G now means the group of all bijections $G \rightarrow G$. No counting by $n!$ is then intended. The handwritten note phrases this as the infinite, discrete version of Cayley's theorem; algebraically, no topology is needed for the proof.

What to retain from this chapter

1. S_X is the group of all bijections $X \rightarrow X$, with composition. If $|X| = n$, then $S_X \cong S_n$ and $|S_n| = n!$.
2. Left translations satisfy $\ell_a \circ \ell_b = \ell_{ab}$, $\ell_e = \text{id}_G$, and $(\ell_a)^{-1} = \ell_{a^{-1}}$.
3. The correspondence $a \leftrightarrow \ell_a$ is bijective and preserves multiplication, so $G \cong L_G$.
4. Since $L_G \leq S_G \cong S_n$, every group of order n occurs as a subgroup of S_n .

Cosets and Lagrange's Theorem

Road map

Let H be a subgroup of a group G . Multiplying every element of H by a fixed element of G produces a left or right *coset*. We shall prove carefully when two cosets are equal, show that cosets partition G , and count the pieces to obtain Lagrange's theorem. The chapter ends with complete left- and right-coset calculations in $D_3 \cong S_3$, including the example in which the two kinds of coset do not agree.

4.1 Left and right cosets

Definition 4.1: Cosets

Let G be a group, let $H \leq G$, and let $a \in G$. The *left coset* of H represented by a is

$$aH := \{ax : x \in H\},$$

whereas the *right coset* represented by a is

$$Ha := \{xa : x \in H\}.$$

The representative a need not belong to H . The notation simply means that we multiply every element of H by a , on the indicated side. In terms of the translations introduced in Chapter 2,

$$aH = \ell_a(H), \quad Ha = r_a(H),$$

where $\ell_a(x) = ax$ and $r_a(x) = xa$.

If G is Abelian, then $ax = xa$ for every $x \in H$, and hence

$$aH = Ha.$$

For example, in the additive Abelian group $\mathbb{Z}$, if $H = m\mathbb{Z}$, then

$$k + m\mathbb{Z} = m\mathbb{Z} + k = \{k + mn : n \in \mathbb{Z}\}.$$

In a non-Abelian group, left and right cosets may be different. We will see an explicit example in D_3 at the end of this chapter.

Proposition 4.1: Every coset has the same size as the subgroup

For every $a \in G$, left translation restricts to a bijection

$$\ell_a|_H : H \longrightarrow aH, \quad x \longmapsto ax.$$

Similarly, $r_a|_H : H \rightarrow Ha$ is a bijection. Consequently,

$$|aH| = |H| = |Ha|.$$

This statement is valid for finite and infinite sets; for finite sets it is an equality of ordinary integers.

证明. The map $x \mapsto ax$ lands in aH by the definition of aH , and it is onto because every element of aH has the form ax with $x \in H$. It is one-to-one: if $ax = ay$, the cancellation law gives $x = y$. Equivalently, its inverse is the restriction of $\ell_{a^{-1}}$, since

$$a^{-1}(ax) = x.$$

The proof for right translation is identical, with multiplication on the right. ■

4.2 When are two cosets equal?

Although a coset is written using a representative, the representative is not unique. The following criterion is the basic computational tool.

Lemma 4.1: Equality criterion for cosets

For $a, b \in G$,

$$aH = bH \iff a^{-1}b \in H \iff b^{-1}a \in H$$

and

$$Ha = Hb \iff ab^{-1} \in H \iff ba^{-1} \in H.$$

证明. We first prove the criterion for left cosets.

Forward implication. Suppose $aH = bH$. Since $e \in H$, the element $b = be$ belongs to $bH = aH$. Thus $b = ax$ for some $x \in H$, and multiplication on the left by a^{-1} gives

$$a^{-1}b = x \in H.$$

This is the shortest proof. To reproduce the calculation in the handwritten notes, choose $x, y \in H$ such that $ax = by$. Then

$$a = byx^{-1}, \quad b^{-1}a = yx^{-1} \in H.$$

Since a subgroup is closed under inverses,

$$a^{-1}b = (b^{-1}a)^{-1} \in H$$

as well.

Reverse implication. Suppose $a^{-1}b \in H$. Write $x = a^{-1}b$, so $x \in H$ and $b = ax$. If $y \in H$, then $xy \in H$, and therefore

$$by = a(xy) \in aH.$$

This proves $bH \subseteq aH$. Moreover,

$$b^{-1}a = (a^{-1}b)^{-1} \in H,$$

so the same argument with a and b interchanged gives $aH \subseteq bH$. Hence $aH = bH$.

The criterion for right cosets is obtained by the same argument on the other side. For example, $Ha = Hb$ implies $a \in Hb$, so $a = xb$ for some $x \in H$, and hence $ab^{-1} = x \in H$. Conversely, $ab^{-1} = x \in H$ means $a = xb$, from which $Ha \subseteq Hb$; using the inverse $ba^{-1} \in H$ gives the reverse inclusion. ■

A useful memory aid

For left cosets aH and bH , invert the element written on the *left*: test $a^{-1}b$. For right cosets Ha and Hb , invert the element written on the *right*: test ab^{-1} .

Example 4.1: Congruence classes in $\mathbb{Z}$

Take $H = m\mathbb{Z} \leq \mathbb{Z}$, using additive notation. The inverse of k is $-k$, so the left-coset criterion becomes

$$k + m\mathbb{Z} = \ell + m\mathbb{Z} \iff -k + \ell \in m\mathbb{Z} \iff m \mid (\ell - k).$$

Thus the cosets of $m\mathbb{Z}$ are precisely the familiar congruence classes modulo m .

4.3 Congruence modulo a subgroup

Definition 4.2: Left and right congruence

For a fixed subgroup $H \leq G$, define *left congruence modulo H* by

$$a \sim_L b \iff aH = bH \iff a^{-1}b \in H.$$

Define *right congruence modulo H* by

$$a \sim_R b \iff Ha = Hb \iff ab^{-1} \in H.$$

Proposition 4.2

Both $\sim_L$ and $\sim_R$ are equivalence relations on G .

证明. We give every step for $\sim_L$.

1. *Reflexivity.* For every $a \in G$,

$$a^{-1}a = e \in H,$$

so $a \sim_L a$.

2. *Symmetry.* If $a \sim_L b$, then $a^{-1}b \in H$. Since H contains inverses,

$$b^{-1}a = (a^{-1}b)^{-1} \in H.$$

Therefore $b \sim_L a$.

3. *Transitivity.* If $a \sim_L b$ and $b \sim_L c$, then $a^{-1}b \in H$ and $b^{-1}c \in H$. Closure of H gives

$$(a^{-1}b)(b^{-1}c) = a^{-1}(bb^{-1})c = a^{-1}c \in H.$$

Hence $a \sim_L c$.

For $\sim_R$, replace the tests above by $ab^{-1} \in H$. For instance, transitivity follows because

$$(ab^{-1})(bc^{-1}) = ac^{-1} \in H.$$

■

The equivalence class of a under $\sim_L$ is exactly aH :

$$[a]_{\sim_L} = \{b \in G : a^{-1}b \in H\} = \{ax : x \in H\} = aH.$$

Similarly, $[a]_{\sim_R} = Ha$.

Theorem 4.1: Cosets partition the group

The distinct left cosets of H are pairwise disjoint and their union is all of G . The same is true of the distinct right cosets.

证明. Every $g \in G$ lies in the left coset gH , because $g = ge$ and $e \in H$. Thus the union of all left cosets is G .

It remains to show that two left cosets cannot overlap without being equal. Suppose

$$x \in aH \cap bH.$$

Then $x = ah_1 = bh_2$ for some $h_1, h_2 \in H$. Rearranging,

$$a^{-1}b = h_1h_2^{-1} \in H.$$

The coset-equality criterion therefore gives $aH = bH$. Hence two left cosets are either equal or disjoint. The right-coset proof is analogous. ■

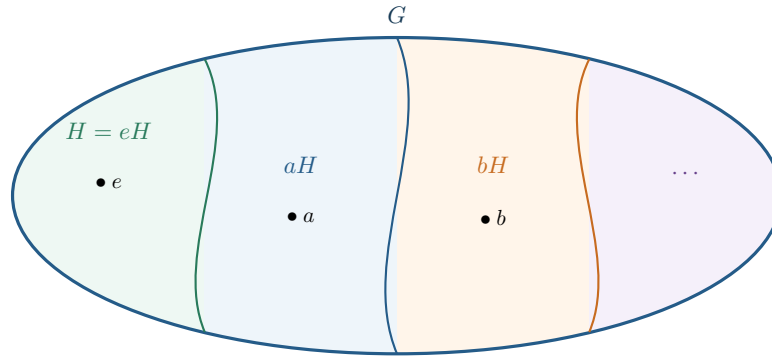


图 4.1: The subgroup H and its distinct left cosets form a partition of G . Each part has the same cardinality as H .

4.4 Coset spaces and index

Definition 4.3: Sets of cosets

The set of all left cosets of H in G is

$$G/H := \{aH : a \in G\}.$$

The set of all right cosets is

$$H \backslash G := \{Ha : a \in G\}.$$

A quotient set is not yet a quotient group

At this point, G/H means only a *set* of left cosets. A multiplication of cosets is well defined only under an additional condition—normality of H —which will be studied later. Also note the notation: the slash points away from the subgroup, so G/H contains left cosets and $H \backslash G$ contains right cosets.

Because $\mathbb{Z}$ is Abelian, left and right cosets of $m\mathbb{Z}$ coincide:

$$\mathbb{Z}/m\mathbb{Z} = m\mathbb{Z} \backslash \mathbb{Z}.$$

For example,

$$\mathbb{Z}/3\mathbb{Z} = \{3\mathbb{Z}, 1 + 3\mathbb{Z}, 2 + 3\mathbb{Z}\}.$$

Definition 4.4: Index

The *index* of H in G , written

$$[G : H] \quad \text{or equivalently} \quad |G : H|,$$

is the number of distinct left cosets of H in G .

For a finite group, the number of left cosets equals the number of right cosets. This will follow immediately from the counting formula below. There is also a direct bijection

$$G/H \longrightarrow H \backslash G, \quad aH \longmapsto Ha^{-1}.$$

Indeed, if $aH = bH$, then $a^{-1}b \in H$, and after taking inverses, $b^{-1}a \in H$. In particular $a^{-1}b \in H$, so the right-coset criterion applied to the representatives a^{-1} and b^{-1} gives $Ha^{-1} = Hb^{-1}$. The same formula applied twice gives the inverse map.

4.5 Lagrange's theorem

Theorem 4.2: Lagrange's theorem

Let G be a finite group and $H \leq G$. Then

$$|G| = [G : H] |H|$$

and therefore

$$[G : H] = \frac{|G|}{|H|}.$$

In particular, the order of every subgroup divides the order of the group.

证明. Let the distinct left cosets be

$$a_1H, a_2H, \dots, a_rH, \quad r = [G : H].$$

By the partition theorem,

$$G = a_1H \sqcup a_2H \sqcup \dots \sqcup a_rH,$$

where $\sqcup$ indicates a disjoint union. Every coset has $|H|$ elements because $h \mapsto a_ih$ is a bijection $H \rightarrow a_iH$. Counting the elements in the disjoint pieces gives

$$\begin{aligned} |G| &= |a_1H| + |a_2H| + \dots + |a_rH| \\ &= \underbrace{|H| + |H| + \dots + |H|}_{r \text{ terms}} \\ &= r|H| = [G : H]|H|. \end{aligned}$$

Dividing by $|H| \neq 0$ yields the formula for the index. ■

Proposition 4.3: Orders of elements divide the group order

If G is finite and $a \in G$, then

$$\text{ord}(a) \mid |G|.$$

证明. The cyclic subgroup generated by a is $\langle a \rangle \leq G$, and its order is precisely $\text{ord}(a)$. Applying Lagrange's theorem to $\langle a \rangle$ gives

$$|G| = [G : \langle a \rangle] |\langle a \rangle| = [G : \langle a \rangle] \text{ord}(a).$$
■

Proposition 4.4: Groups of prime order

If $|G| = p$ is prime, then the only subgroups of G are $\{e\}$ and G . Moreover, every $a \neq e$ generates G , so

$$G \cong \mathbb{Z}_p.$$

证明. For any subgroup $H \leq G$, Lagrange's theorem says that $|H|$ divides p . The only positive divisors of a prime are 1 and p . If $|H| = 1$, then $H = \{e\}$; if $|H| = p = |G|$, then $H = G$.

Now choose $a \neq e$. The subgroup $\langle a \rangle$ contains both e and a , so it is not $\{e\}$. It must therefore be all of G . Thus G is cyclic of order p , and hence isomorphic to $\mathbb{Z}_p$. ■

4.6 Complete coset calculations in D_3

Recall the presentation

$$D_3 = \langle a, b \mid a^3 = e, b^2 = e, ba = a^2b \rangle = \{e, a, a^2, b, ab, a^2b\}.$$

Geometrically, a is a rotation through 120° , while b is a reflection. The relation $ba = a^2b = a^{-1}b$ expresses the fact that performing a rotation and a reflection in the opposite order reverses the rotation.

The complete Cayley table from the handwritten notes is

$\cdot$	e	a	a^2	b	ab	a^2b
e	e	a	a^2	b	ab	a^2b
a	a	a^2	e	ab	a^2b	b
a^2	a^2	e	a	a^2b	b	ab
b	b	a^2b	ab	e	a^2	a
ab	ab	b	a^2b	a	e	a^2
a^2b	a^2b	ab	b	a^2	a	e

4.6.1 The subgroup $H_1 = \langle a \rangle$

Let

$$H_1 = \{e, a, a^2\} = \langle a \rangle \cong \mathbb{Z}_3.$$

Lagrange's theorem gives

$$[D_3 : H_1] = \frac{|D_3|}{|H_1|} = \frac{6}{3} = 2.$$

Thus there must be exactly two left cosets. One is H_1 ; choosing $b \notin H_1$ gives the other:

$$\begin{aligned} bH_1 &= \{be, ba, ba^2\} \\ &= \{b, a^2b, ab\}. \end{aligned}$$

Here $ba = a^2b$, while

$$ba^2 = (ba)a = (a^2b)a = a^2(ba) = a^2(a^2b) = a^4b = ab.$$

Therefore

$$D_3/H_1 = \{H_1, bH_1\}.$$

For the right cosets,

$$\begin{aligned} H_1b &= \{eb, ab, a^2b\} \\ &= \{b, ab, a^2b\}. \end{aligned}$$

As unordered sets,

$$bH_1 = \{b, a^2b, ab\} = \{b, ab, a^2b\} = H_1b.$$

Consequently,

$$D_3/H_1 = \{H_1, bH_1\} = \{H_1, H_1b\} = H_1 \backslash D_3.$$

In this example every left coset of H_1 is also a right coset.

4.6.2 The subgroup $H_2 = \langle b \rangle$

Now take

$$H_2 = \{e, b\} = \langle b \rangle \cong \mathbb{Z}_2.$$

Its index is

$$[D_3 : H_2] = \frac{|D_3|}{|H_2|} = \frac{6}{2} = 3.$$

The three left cosets are

$$\begin{aligned} H_2 &= \{e, b\}, \\ aH_2 &= \{a, ab\}, \\ a^2H_2 &= \{a^2, a^2b\}. \end{aligned}$$

They are disjoint, each contains two elements, and together they contain all six elements of D_3 . Thus

$$D_3/H_2 = \{H_2, aH_2, a^2H_2\}.$$

The three right cosets are different:

$$\begin{aligned} H_2 &= \{e, b\}, \\ H_2a &= \{a, ba\} = \{a, a^2b\}, \\ H_2a^2 &= \{a^2, ba^2\} = \{a^2, ab\}. \end{aligned}$$

The last equality uses the calculation $ba^2 = ab$ displayed above. Hence

$$H_2 \backslash D_3 = \{H_2, H_2a, H_2a^2\} \neq \{H_2, aH_2, a^2H_2\} = D_3/H_2.$$

For example,

$$aH_2 = \{a, ab\} \quad \text{whereas} \quad H_2a = \{a, a^2b\}.$$

This is the promised concrete demonstration that left and right cosets need not coincide in a non-Abelian group.

What to retain from this chapter

1. Left and right cosets are translates: $aH = \ell_a(H)$ and $Ha = r_a(H)$. Translation is a bijection, so every coset has the same size as H .

2. The equality tests are $aH = bH \iff a^{-1}b \in H$ and $Ha = Hb \iff ab^{-1} \in H$.
3. Left and right congruence modulo H are equivalence relations. Their equivalence classes are the corresponding cosets, which partition G .
4. If G is finite, then $|G| = [G : H]|H|$. Hence subgroup orders and element orders divide $|G|$.
5. A group of prime order has no nontrivial proper subgroup and is cyclic.
6. In D_3 , the cosets of $H_1 = \langle a \rangle$ agree on the left and right, whereas those of $H_2 = \langle b \rangle$ do not.

Conjugation

Road map

Conjugation compares elements that differ only by a change of viewpoint inside the group. We first prove that conjugacy is an equivalence relation and compute all conjugacy classes of $S_3 \cong D_3$. We then introduce the centralizer and construct a bijection between the cosets of a centralizer and a conjugacy class. This gives the formula

$$|\text{Cen}_G(a)| |\text{Cl}_G(a)| = |G|.$$

Finally, we study the center $Z(G)$ and derive the class equation. All tables and partition diagrams in the handwritten lecture are reproduced and explained below.

5.1 Conjugation and conjugate elements

Definition 5.1: Conjugation by an element

Let G be a group and let $g \in G$. The map

$$\gamma_g : G \longrightarrow G, \quad a \longmapsto gag^{-1},$$

is called *conjugation by g* .

The element g acts on a from the left and g^{-1} acts from the right. Parentheses may be omitted because multiplication is associative. Conjugation is a bijection: its inverse is conjugation by g^{-1} , since

$$\gamma_{g^{-1}}(\gamma_g(a)) = g^{-1}(gag^{-1})g = a.$$

It also preserves multiplication:

$$\begin{aligned} \gamma_g(ab) &= gabg^{-1} \\ &= ga(g^{-1}g)bg^{-1} \\ &= (gag^{-1})(gbg^{-1}) = \gamma_g(a)\gamma_g(b). \end{aligned}$$

Thus conjugation is an automorphism of G , called an *inner automorphism*.

Definition 5.2: Conjugate elements

Elements $a, b \in G$ are *conjugate*, written

$$a \sim b,$$

if there exists $g \in G$ such that

$$b = gag^{-1}.$$

Proposition 5.1

Conjugacy is an equivalence relation on every group.

证明. We verify the three defining properties.

1. *Reflexivity*. Since $e^{-1} = e$,

$$a = eae^{-1},$$

so $a \sim a$.

2. *Symmetry*. If $a \sim b$, then $b = gag^{-1}$ for some $g \in G$. Multiplying by g^{-1} on the left and by g on the right gives

$$a = g^{-1}bg.$$

This is conjugation of b by g^{-1} , because $(g^{-1})^{-1} = g$. Hence $b \sim a$.

3. *Transitivity*. Suppose $a \sim b$ and $b \sim c$. Choose $g, g' \in G$ such that

$$b = gag^{-1}, \quad c = g'bg'^{-1}.$$

Substitution yields

$$\begin{aligned} c &= g'(gag^{-1})g'^{-1} \\ &= (g'g)a(g^{-1}g'^{-1}) \\ &= (g'g)a(g'g)^{-1}. \end{aligned}$$

Therefore $a \sim c$. ■

Definition 5.3: Conjugacy class

The *conjugacy class* of $a \in G$ is

$$\text{Cl}_G(a) := \{b \in G : b \sim a\} = \{gag^{-1} : g \in G\}.$$

When the ambient group is clear, we write simply $\text{Cl}(a)$.

Because conjugacy is an equivalence relation, the distinct conjugacy classes form a partition of G . In particular, every element belongs to exactly one conjugacy class.

Proposition 5.2: Basic invariants under conjugacy

If $b = gag^{-1}$, then a and b have the same order. More precisely, for every integer n ,

$$b^n = (gag^{-1})^n = ga^n g^{-1}.$$

证明. For $n \geq 1$, all adjacent factors $g^{-1}g$ cancel:

$$(gag^{-1})^n = ga \underbrace{(g^{-1}g)}_e a \cdots \underbrace{(g^{-1}g)}_e ag^{-1} = ga^n g^{-1}.$$

The formula for $n = 0$ is immediate, and negative powers follow by taking inverses. Consequently, $b^n = e$ if and only if $ga^n g^{-1} = e$, which is equivalent to $a^n = e$. Their orders are therefore equal. ■

5.2 Conjugacy classes in $S_3 \cong D_3$

Use the presentation

$$S_3 \cong D_3 = \langle a, b \mid a^3 = b^2 = e, ba = a^2b \rangle = \{e, a, a^2, b, ab, a^2b\}.$$

The complete Cayley table in the notes is

$\cdot$	e	a	a^2	b	ab	a^2b
e	e	a	a^2	b	ab	a^2b
a	a	a^2	e	ab	a^2b	b
a^2	a^2	e	a	a^2b	b	ab
b	b	a^2b	ab	e	a^2	a
ab	ab	b	a^2b	a	e	a^2
a^2b	a^2b	ab	b	a^2	a	e

The identity is fixed by every conjugation:

$$geg^{-1} = gg^{-1} = e,$$

so

$$\text{Cl}(e) = \{e\}.$$

Conjugating a by b gives

$$bab^{-1} = bab = (ba)b = (a^2b)b = a^2.$$

Since conjugate elements have the same order, a cannot be conjugate to e or to a reflection, all of which have order 2. Hence

$$\text{Cl}(a) = \text{Cl}(a^2) = \{a, a^2\}.$$

For the reflection b ,

$$\begin{aligned} aba^{-1} &= aba^2 = a(ba^2) = a(ab) = a^2b, \\ a^2ba^{-2} &= a^2ba = a^2(ba) = a^2(a^2b) = a^4b = ab. \end{aligned}$$

Therefore

$$\text{Cl}(b) = \text{Cl}(ab) = \text{Cl}(a^2b) = \{b, ab, a^2b\}.$$

The three distinct conjugacy classes are thus

$$\{e\}, \quad \{a, a^2\}, \quad \{b, ab, a^2b\}.$$

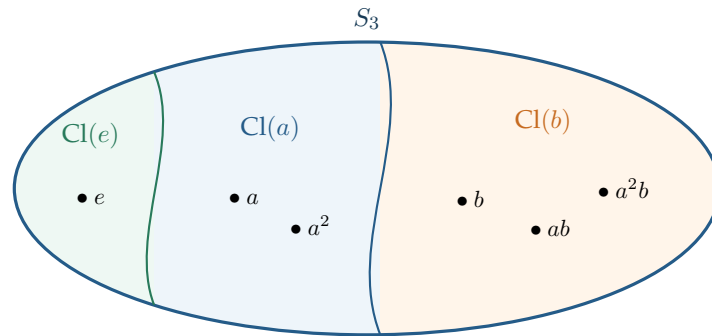


图 5.1: The partition of S_3 into its three conjugacy classes, of sizes 1, 2, and 3.

If G is Abelian, then

$$gag^{-1} = agg^{-1} = a$$

for every $a, g \in G$. Therefore every conjugacy class is a singleton:

$$\text{Cl}(a) = \{a\}.$$

The handwritten notes illustrate this with the Klein four-group

$$K_4 = \{e, a, b, c\},$$

whose four conjugacy classes are the four one-element sets.

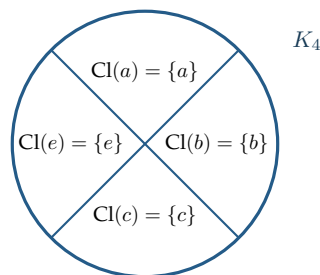


图 5.2: In the Abelian group K_4 , every conjugacy class is a singleton.

5.3 Centralizers

Definition 5.4: Centralizer of an element

For $a \in G$, the *centralizer* of a in G is

$$\text{Cen}_G(a) := \{b \in G : ab = ba\}.$$

It is the set of all elements of G that commute with a .

The defining condition has an equivalent conjugation form:

$$b \in \text{Cen}_G(a) \iff ba = ab \iff bab^{-1} = a.$$

Thus the centralizer is the set of group elements whose conjugation fixes a .

Proposition 5.3

For every $a \in G$, the centralizer $\text{Cen}_G(a)$ is a subgroup of G .

证明. The set is nonempty because $ea = a = ae$, so $e \in \text{Cen}_G(a)$. We use the one-step subgroup test. Let $x, y \in \text{Cen}_G(a)$. Then $ax = xa$ and $ay = ya$. The latter equality also implies $ay^{-1} = y^{-1}a$: from $ay = ya$, multiply by y^{-1} on the left and right to obtain $y^{-1}a = ay^{-1}$. Consequently,

$$\begin{aligned} a(xy^{-1}) &= (ax)y^{-1} \\ &= (xa)y^{-1} \\ &= x(ay^{-1}) \\ &= x(y^{-1}a) \\ &= (xy^{-1})a. \end{aligned}$$

Hence $xy^{-1} \in \text{Cen}_G(a)$, and the subgroup test gives $\text{Cen}_G(a) \leq G$. ■

If G is Abelian, every element commutes with a , so

$$\text{Cen}_G(a) = G \quad \text{for every } a \in G.$$

5.3.1 Centralizers in S_3

For $S_3 \cong D_3$, the centralizers recorded in the notes are

x	$\text{Cen}_{S_3}(x)$	$ \text{Cen}_{S_3}(x) $
e	S_3	6
a, a^2	$\{e, a, a^2\}$	3
b	$\{e, b\}$	2
ab	$\{e, ab\}$	2
a^2b	$\{e, a^2b\}$	2

For instance, an element x belongs to $\text{Cen}(a)$ exactly when the entry ax in the row of a agrees with the entry xa in the column of a . Reading those two lines of the Cayley table gives the comparison

x	ax	xa	$x \in \text{Cen}(a)?$
e	a	a	yes
a	a^2	a^2	yes
a^2	e	e	yes
b	ab	a^2b	no
ab	a^2b	b	no
a^2b	b	ab	no

In particular, the highlighted comparison in the handwritten table is

$$a(ab) = a^2b \neq (ab)a = b,$$

so $ab \notin \text{Cen}(a)$.

5.4 The size of a conjugacy class

The S_3 calculation suggests the identity

$$|\text{Cen}_G(a)| |\text{Cl}_G(a)| = |G|.$$

To prove it, we will show that the elements of $\text{Cl}_G(a)$ correspond one-to-one with the left cosets of $\text{Cen}_G(a)$.

Theorem 5.1: Centralizer–conjugacy-class formula

Let G be a group and $a \in G$. There is a bijection

$$G/\text{Cen}_G(a) \longrightarrow \text{Cl}_G(a), \quad g\text{Cen}_G(a) \longmapsto gag^{-1}.$$

If G is finite, then

$$|\text{Cl}_G(a)| = [G : \text{Cen}_G(a)]$$

and, equivalently,

$$|\text{Cen}_G(a)| |\text{Cl}_G(a)| = |G|.$$

证明. Write $C = \text{Cen}_G(a)$ and define

$$\Phi : G/C \longrightarrow \text{Cl}_G(a), \quad \Phi(gC) = gag^{-1}.$$

We verify all three requirements for a bijection.

Well-definedness. The same left coset may have different representatives. Suppose $gC = g'C$. The left-coset criterion gives

$$g^{-1}g' \in C.$$

Put $c = g^{-1}g'$, so $c \in C$ and $g' = gc$. Because c commutes with a , we have $cac^{-1} = a$. Therefore

$$\begin{aligned} g'ag'^{-1} &= (gc)a(gc)^{-1} \\ &= gcac^{-1}g^{-1} \\ &= gag^{-1}. \end{aligned}$$

Thus $\Phi(gC)$ does not depend on the chosen representative.

Surjectivity. If $b \in \text{Cl}_G(a)$, then by definition $b = gag^{-1}$ for some $g \in G$. Hence $b = \Phi(gC)$.

Injectivity. Suppose

$$\Phi(gC) = \Phi(g'C).$$

Then $gag^{-1} = g'ag'^{-1}$. Multiplying on the left by g^{-1} and on the right by g' yields

$$a(g^{-1}g') = (g^{-1}g')a.$$

Thus $g^{-1}g' \in C$, and the left-coset criterion gives $gC = g'C$. The map is therefore one-to-one.

We have proved that Φ is a bijection. When G is finite, Lagrange's theorem now gives

$$|\text{Cl}_G(a)| = |G/C| = [G : C] = \frac{|G|}{|C|}.$$

Multiplying by $|C| = |\text{Cen}_G(a)|$ produces the second formula. ■

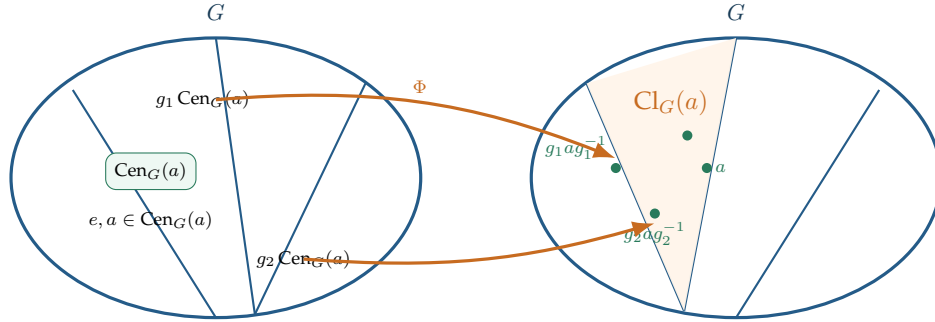


图 5.3: Each left coset of $\text{Cen}_G(a)$ determines exactly one conjugate of a . Different cosets determine different conjugates, which is precisely the bijection used in the theorem.

Two divisors of the group order

For a finite group, both $|\text{Cen}_G(a)|$ and $|\text{Cl}_G(a)|$ divide $|G|$: the first because the centralizer is a subgroup, and the second because it is the index of that subgroup.

Example 5.1: Checking the formula in S_3

Since $|S_3| = 6$,

representative x	$ \text{Cl}(x) $	$ \text{Cen}(x) $	$ \text{Cl}(x) \text{Cen}(x) $
e	1	6	$1 \cdot 6 = 6$
a	2	3	$2 \cdot 3 = 6$
b	3	2	$3 \cdot 2 = 6$

The notes explicitly record the first two lines:

$$|\text{Cl}(e)| = 1 = [S_3 : \text{Cen}(e)] = \frac{6}{6},$$

and

$$|\text{Cl}(a)| = 2 = [S_3 : \text{Cen}(a)] = \frac{6}{3}.$$

Proposition 5.4: Prime-order groups are Abelian

If G is finite and $|G| = p$ is prime, then G is Abelian.

证明. For every $a \in G$, the centralizer–class formula gives

$$|\text{Cl}(a)| |\text{Cen}(a)| = |G| = p.$$

If $a \neq e$, then the two distinct elements e and a both commute with a . Therefore

$$|\text{Cen}(a)| \geq 2,$$

so $|\text{Cen}(a)| \neq 1$. Since p is prime, the product formula forces

$$|\text{Cen}(a)| = p, \quad |\text{Cl}(a)| = 1.$$

The same singleton statement is obvious for $a = e$. Hence, for every $a, b \in G$,

$$bab^{-1} = a.$$

Multiplication on the right by b gives $ba = ab$. Thus every pair of elements commutes and G is Abelian. ■

There is also the Lagrange-theorem proof recorded in the notes. If $a \neq e$, then the cyclic subgroup

$$\langle a \rangle = \{a^k : k \in \mathbb{Z}\}$$

contains at least e and a , so $|\langle a \rangle| \geq 2$. Its order divides the prime p , hence $|\langle a \rangle| = p$ and $\langle a \rangle = G$. Therefore every group of prime order is in fact cyclic, which also implies that it is Abelian.

5.5 The center of a group

Definition 5.5: Center

The *center* of G is

$$Z(G) := \{z \in G : az = za \text{ for every } a \in G\}.$$

Equivalently,

$$Z(G) = \bigcap_{a \in G} \text{Cen}_G(a).$$

The centralizer $\text{Cen}_G(a)$ contains the elements that commute with one fixed element a ; the center contains the elements that commute with *every* element of G . Consequently,

$$Z(G) \leq \text{Cen}_G(a) \leq G \quad \text{for every } a \in G.$$

The inclusions need not be proper.

Proposition 5.5: Elementary properties of the center

For every group G :

1. $Z(G) \leq G$;
2. $Z(G)$ is Abelian;
3. G is Abelian if and only if $Z(G) = G$;
4. $z \in Z(G)$ if and only if $\text{Cl}_G(z) = \{z\}$.

证明. First, $e \in Z(G)$. Let $x, y \in Z(G)$. For every $a \in G$,

$$\begin{aligned} a(xy^{-1}) &= (ax)y^{-1} \\ &= (xa)y^{-1} \\ &= x(ay^{-1}) \\ &= x(y^{-1}a) \\ &= (xy^{-1})a. \end{aligned}$$

Thus $xy^{-1} \in Z(G)$, and the subgroup test proves $Z(G) \leq G$.

If $x, y \in Z(G)$, then x commutes with every element of G , in particular with y . Hence $xy = yx$, so $Z(G)$ is Abelian.

If G is Abelian, every element commutes with every other element and therefore $Z(G) = G$. Conversely, if $Z(G) = G$, then every element of G commutes with every other element, so G is Abelian.

Finally, $z \in Z(G)$ means

$$gzg^{-1} = z \quad \text{for every } g \in G,$$

which is exactly the assertion that the conjugacy class of z contains only z . ■

A group is called *centerless* if

$$Z(G) = \{e\}.$$

For example, S_3 is centerless. Indeed, its three conjugacy classes are $\{e\}$, $\{a, a^2\}$, and $\{b, ab, a^2b\}$; only e lies in a singleton conjugacy class.

5.6 The class equation

Let G be finite, and choose one representative a_i from each conjugacy class. Since the conjugacy classes partition G ,

$$|G| = \sum_i |\text{Cl}_G(a_i)|.$$

The centralizer–class formula turns each summand into an index:

$$|\text{Cl}_G(a_i)| = [G : \text{Cen}_G(a_i)].$$

If $a_i \in Z(G)$, then its conjugacy class is the singleton $\{a_i\}$. Thus the central elements contribute exactly $|Z(G)|$ to the sum. Separating them from all nonsingleton conjugacy classes yields the principal counting formula.

Theorem 5.2: Class equation

Let G be a finite group. Choose one representative a_i from each conjugacy class not contained in $Z(G)$. Then

$$|G| = |Z(G)| + \sum_{a_i \notin Z(G)} [G : \text{Cen}_G(a_i)].$$

Equivalently,

$$|G| = |Z(G)| + \sum_{a_i \notin Z(G)} |\text{Cl}_G(a_i)|.$$

证明. The distinct conjugacy classes form a disjoint partition of G , so their sizes add to $|G|$. A conjugacy class has one element precisely when its representative lies in $Z(G)$. There are $|Z(G)|$ such classes, each contributing 1. Every remaining class has a representative $a_i \notin Z(G)$ and has size $[G : \text{Cen}_G(a_i)]$. Adding these two kinds of contribution gives the displayed formula. ■

Example 5.2: The class equation of S_3

For S_3 ,

$$Z(S_3) = \{e\},$$

and representatives of the two noncentral conjugacy classes may be taken to be a and b . Therefore

$$\begin{aligned} |S_3| &= |Z(S_3)| + [S_3 : \text{Cen}(a)] + [S_3 : \text{Cen}(b)] \\ &= 1 + \frac{6}{3} + \frac{6}{2} \\ &= 1 + 2 + 3 = 6. \end{aligned}$$

Conjugacy class versus centralizer

The two sets associated with a answer complementary questions:

$$\text{Cl}_G(a) = \{bab^{-1} : b \in G\}$$

lists all elements obtained by moving a through conjugation, whereas

$$\text{Cen}_G(a) = \{b \in G : bab^{-1} = a\}$$

lists the group elements that leave a fixed. The larger the centralizer, the smaller the conjugacy class, exactly as expressed by

$$|\text{Cen}_G(a)| |\text{Cl}_G(a)| = |G|.$$

What to retain from this chapter

1. Conjugation is $\gamma_g(a) = gag^{-1}$, and conjugacy is an equivalence relation. Its equivalence classes partition G .
2. In $S_3 \cong D_3$, the classes are $\{e\}$, $\{a, a^2\}$, and $\{b, ab, a^2b\}$.
3. The centralizer $\text{Cen}_G(a) = \{b \in G : ab = ba\}$ is a subgroup.
4. Cosets of $\text{Cen}_G(a)$ correspond bijectively to conjugates of a , so

$$|\text{Cl}_G(a)| = [G : \text{Cen}_G(a)].$$

5. The center is $Z(G) = \bigcap_{a \in G} \text{Cen}_G(a)$, and central elements are precisely those whose conjugacy classes are singletons.
6. For a finite group,

$$|G| = |Z(G)| + \sum_{a_i \notin Z(G)} [G : \text{Cen}_G(a_i)].$$

Homomorphisms

Road map

A homomorphism is a map that preserves multiplication and hence preserves the algebraic structure of a group. We develop its basic properties, images, preimages, kernel, injectivity criterion, and composition. We then distinguish endomorphisms, isomorphisms, and automorphisms, and prove again—now from the homomorphism viewpoint—that conjugation is an automorphism. Detailed examples include multiplication and reduction modulo m on $\mathbb{Z}$, left translations and the left regular representation, the sign map on S_n , the alternating group A_n , and the complete classification of cyclic groups.

6.1 Definition and first properties

Homomorphisms play the same structural role for groups that linear maps play for vector spaces: they are the maps compatible with the operations.

Definition 6.1: Group homomorphism

Let G and G' be groups. A map

$$\varphi : G \longrightarrow G'$$

is a *group homomorphism* if

$$\varphi(ab) = \varphi(a)\varphi(b) \quad \text{for all } a, b \in G.$$

The two products in this equation belong to different groups: ab is computed in G , while $\varphi(a)\varphi(b)$ is computed in G' . If the operations are denoted explicitly by $\circ$ on G and $\circ'$ on G' , the same condition is

$$\varphi(a \circ b) = \varphi(a) \circ' \varphi(b).$$

As in the handwritten notes, we usually use a deliberately strict notation and let context distinguish the two products.

Theorem 6.1: Basic properties of homomorphisms

Let $\varphi : G \rightarrow G'$ be a homomorphism, with identity elements $e \in G$ and $e' \in G'$. Then:

1. $\varphi(e) = e'$;

2. $\varphi(a^{-1}) = \varphi(a)^{-1}$ for every $a \in G$;
3. $\varphi(a^n) = \varphi(a)^n$ for every $a \in G$ and $n \in \mathbb{Z}$.

证明. Because $e = e^2$,

$$\varphi(e) = \varphi(e^2) = \varphi(e)\varphi(e).$$

Multiplying on the left by $\varphi(e)^{-1}$ gives $\varphi(e) = e'$.

Next,

$$e' = \varphi(e) = \varphi(aa^{-1}) = \varphi(a)\varphi(a^{-1}).$$

Thus $\varphi(a^{-1})$ is the inverse of $\varphi(a)$, so

$$\varphi(a^{-1}) = \varphi(a)^{-1}.$$

For $n > 0$, repeated use of the homomorphism property gives

$$\varphi(a^n) = \varphi(\underbrace{a \cdots a}_{n \text{ factors}}) = \underbrace{\varphi(a) \cdots \varphi(a)}_{n \text{ factors}} = \varphi(a)^n.$$

The case $n = 0$ is the identity calculation, and the result for $n < 0$ follows from the inverse calculation. ■

A consequence for element orders

If a has finite order n , then

$$\varphi(a)^n = \varphi(a^n) = \varphi(e) = e'.$$

Consequently, the order of $\varphi(a)$ divides the order of a . The order may decrease when φ is not injective.

6.2 Images and preimages of subgroups

Definition 6.2: Image and preimage

For a subset $H \subseteq G$, its *image* under φ is

$$\varphi(H) := \{\varphi(h) : h \in H\} \subseteq G'.$$

For a subset $H' \subseteq G'$, its *preimage* is

$$\varphi^{-1}(H') := \{a \in G : \varphi(a) \in H'\} \subseteq G.$$

The notation $\varphi^{-1}(H')$ denotes a preimage of a set and does not require φ to possess an inverse function.

Proposition 6.1: Subgroups pass through homomorphisms

Let $\varphi : G \rightarrow G'$ be a homomorphism.

1. If $H \leq G$, then $\varphi(H) \leq G'$.

2. If $H' \leq G'$, then $\varphi^{-1}(H') \leq G$.

证明. Suppose $H \leq G$. Since $e \in H$,

$$e' = \varphi(e) \in \varphi(H),$$

so the image is nonempty. Let $a', b' \in \varphi(H)$. There are $a, b \in H$ with $a' = \varphi(a)$ and $b' = \varphi(b)$. Then

$$\begin{aligned} a'(b')^{-1} &= \varphi(a)\varphi(b)^{-1} \\ &= \varphi(a)\varphi(b^{-1}) \\ &= \varphi(ab^{-1}). \end{aligned}$$

Because $H \leq G$, the element ab^{-1} belongs to H . Hence $a'(b')^{-1} \in \varphi(H)$, and the subgroup test proves $\varphi(H) \leq G'$.

Now suppose $H' \leq G'$. Since $\varphi(e) = e' \in H'$, the preimage is nonempty. Let $a, b \in \varphi^{-1}(H')$. Then $\varphi(a), \varphi(b) \in H'$, so

$$\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} \in H'.$$

Thus $ab^{-1} \in \varphi^{-1}(H')$, and the subgroup test gives $\varphi^{-1}(H') \leq G$. ■

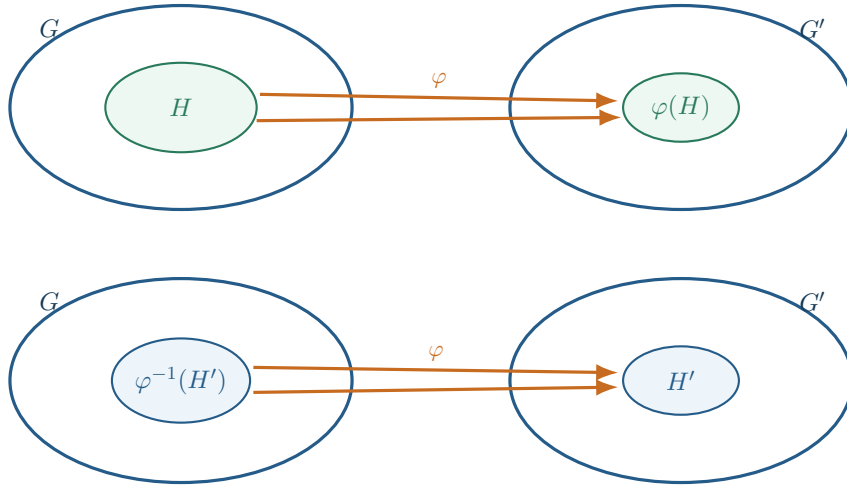


图 6.1: Top: the image $\varphi(H)$ of a subgroup $H \leq G$. Bottom: the full preimage $\varphi^{-1}(H')$ of a subgroup $H' \leq G'$. Both are subgroups.

6.3 Image, kernel, and injectivity

Definition 6.3: Image and kernel of a homomorphism

The *image* and *kernel* of $\varphi : G \rightarrow G'$ are

$$\text{im } \varphi := \{\varphi(a) : a \in G\} = \varphi(G)$$

and

$$\text{ker } \varphi := \{a \in G : \varphi(a) = e'\} = \varphi^{-1}(\{e'\}).$$

The previous proposition immediately gives

$$\text{im } \varphi \leq G', \quad \ker \varphi \leq G.$$

Indeed, these are the special cases $H = G$ and $H' = \{e'\}$.

Theorem 6.2: Kernel criterion for injectivity

A homomorphism $\varphi : G \rightarrow G'$ is injective if and only if

$$\ker \varphi = \{e\}.$$

证明. Suppose first that φ is injective. If $a \in \ker \varphi$, then

$$\varphi(a) = e' = \varphi(e).$$

Injectivity gives $a = e$. Thus $\ker \varphi = \{e\}$.

Conversely, suppose $\ker \varphi = \{e\}$, and assume $\varphi(a) = \varphi(b)$. Then

$$\begin{aligned} \varphi(a^{-1}b) &= \varphi(a)^{-1}\varphi(b) \\ &= \varphi(a)^{-1}\varphi(a) = e'. \end{aligned}$$

Therefore $a^{-1}b \in \ker \varphi = \{e\}$, so $a^{-1}b = e$, and hence $a = b$. Thus φ is injective. ■

Proposition 6.2: Composition

If

$$\varphi_1 : G \rightarrow G' \quad \text{and} \quad \varphi_2 : G' \rightarrow G''$$

are homomorphisms, then

$$\varphi_2 \circ \varphi_1 : G \rightarrow G''$$

is a homomorphism.

证明. For $a, b \in G$,

$$\begin{aligned} (\varphi_2 \circ \varphi_1)(ab) &= \varphi_2(\varphi_1(ab)) \\ &= \varphi_2(\varphi_1(a)\varphi_1(b)) \\ &= \varphi_2(\varphi_1(a))\varphi_2(\varphi_1(b)) \\ &= (\varphi_2 \circ \varphi_1)(a)(\varphi_2 \circ \varphi_1)(b). \end{aligned}$$

■

6.4 Endomorphisms, isomorphisms, and automorphisms

Definition 6.4: Three kinds of homomorphism

Let $\varphi : G \rightarrow G'$ be a homomorphism.

- It is an *endomorphism* if $G' = G$.
- It is an *isomorphism* if it is bijective.

- It is an *automorphism* if it is a bijective endomorphism, that is, if $G' = G$ and φ is bijective.

If an isomorphism $G \rightarrow G'$ exists, the groups are called *isomorphic*, and we write

$$G \cong G'.$$

Isomorphic groups have the same group-theoretic structure with different names for the elements. For finite groups, an isomorphism is a bijection, so

$$|G| = |G'|.$$

After corresponding elements are placed in the same order, their Cayley tables are identical.

Proposition 6.3

The inverse of an isomorphism is an isomorphism.

证明. Let $\varphi : G \rightarrow G'$ be an isomorphism. Its inverse function exists because φ is bijective. Take $x', y' \in G'$, and write $x' = \varphi(x)$, $y' = \varphi(y)$. Then

$$\begin{aligned} \varphi^{-1}(x'y') &= \varphi^{-1}(\varphi(x)\varphi(y)) \\ &= \varphi^{-1}(\varphi(xy)) \\ &= xy = \varphi^{-1}(x')\varphi^{-1}(y'). \end{aligned}$$

Thus φ^{-1} is a homomorphism; it is automatically bijective. ■

Definition 6.5: Automorphism group

The set of all automorphisms of G is

$$\text{Aut}(G) := \{\varphi : G \rightarrow G : \varphi \text{ is a bijective homomorphism}\}.$$

With composition of functions, $\text{Aut}(G)$ is called the *automorphism group* of G .

Proposition 6.4

$\text{Aut}(G)$ is a subgroup of the symmetric group S_G .

证明. Every automorphism is a bijection $G \rightarrow G$, so $\text{Aut}(G) \subseteq S_G$. The identity map is an automorphism. Composition of automorphisms is a homomorphism by the composition proposition and is bijective because it is a composition of bijections. The inverse of an automorphism is an automorphism by the preceding proposition. Hence $\text{Aut}(G) \leq S_G$. ■

6.5 Conjugation as an automorphism

Theorem 6.3: Conjugation is an automorphism

Let $g \in G$. The conjugation map

$$\gamma_g : G \longrightarrow G, \quad a \longmapsto gag^{-1},$$

is an automorphism. Moreover, if $H \leq G$, then

$$\gamma_g(H) = gHg^{-1} \leq G \quad \text{and} \quad H \cong gHg^{-1}.$$

证明. For $a, b \in G$,

$$\begin{aligned} \gamma_g(ab) &= gabg^{-1} \\ &= ga(g^{-1}g)bg^{-1} \\ &= (gag^{-1})(gbg^{-1}) \\ &= \gamma_g(a)\gamma_g(b). \end{aligned}$$

Thus γ_g is a homomorphism.

It is surjective: given $a' \in G$, set $a = g^{-1}a'g$. Then

$$\gamma_g(a) = g(g^{-1}a'g)g^{-1} = a'.$$

It is injective: if $\gamma_g(a) = \gamma_g(b)$, then

$$gag^{-1} = gbg^{-1}.$$

Multiplication by g^{-1} on the left and by g on the right gives $a = b$. Therefore $\gamma_g \in \text{Aut}(G)$.

If $H \leq G$, the image theorem gives $\gamma_g(H) \leq G$. The restriction

$$\gamma_g|_H : H \longrightarrow \gamma_g(H)$$

is a homomorphism, is one-to-one because γ_g is, and is onto by the definition of the image. It is an isomorphism, so $H \cong \gamma_g(H) = gHg^{-1}$. ■

The subgroup gHg^{-1} is called the *conjugate subgroup of H by g* . Conjugation can therefore produce differently situated subgroups of G , but every conjugate subgroup has exactly the same abstract group structure as H .

6.6 Homomorphisms involving the integers

6.6.1 Multiplication by an integer

Fix $m \in \mathbb{Z}$, and define

$$\varphi_m : \mathbb{Z} \longrightarrow \mathbb{Z}, \quad k \longmapsto mk.$$

Both groups are additive. For $k_1, k_2 \in \mathbb{Z}$,

$$\varphi_m(k_1 + k_2) = m(k_1 + k_2) = mk_1 + mk_2 = \varphi_m(k_1) + \varphi_m(k_2),$$

so φ_m is a homomorphism.

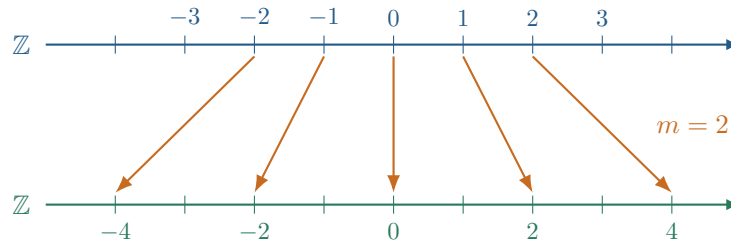


图 6.2: The homomorphism $\varphi_2(k) = 2k$. Its image contains exactly the even integers.

The complete behaviour is:

m	$\ker \varphi_m$	$\text{im } \varphi_m$	type
0	$\mathbb{Z}$	$\{0\}$	neither injective nor surjective
1	$\{0\}$	$\mathbb{Z}$	automorphism
-1	$\{0\}$	$\mathbb{Z}$	automorphism
$ m \geq 2$	$\{0\}$	$m\mathbb{Z}$	injective, not surjective

In particular, for the nontrivial nonsurjective cases,

$$\text{im } \varphi_m = m\mathbb{Z} < \mathbb{Z}.$$

6.6.2 Reduction modulo m

For $m \geq 1$, define

$$\pi_m : \mathbb{Z} \longrightarrow \mathbb{Z}/m\mathbb{Z}, \quad k \longmapsto k + m\mathbb{Z}.$$

Then

$$\begin{aligned} \pi_m(k_1 + k_2) &= (k_1 + k_2) + m\mathbb{Z} \\ &= (k_1 + m\mathbb{Z}) + (k_2 + m\mathbb{Z}) \\ &= \pi_m(k_1) + \pi_m(k_2). \end{aligned}$$

Thus π_m is a homomorphism. It is surjective because every coset has a representative $k \in \mathbb{Z}$. Its kernel is

$$\begin{aligned} \ker \pi_m &= \{k \in \mathbb{Z} : k + m\mathbb{Z} = m\mathbb{Z}\} \\ &= m\mathbb{Z}. \end{aligned}$$

For $m > 1$, it is not injective because every multiple of m maps to the identity coset.

6.7 Translations and the left regular representation

For $a \in G$, left translation is the bijection

$$\ell_a : G \rightarrow G, \quad x \longmapsto ax.$$

It is important not to confuse a bijection of the underlying set with a group homomorphism. Indeed,

$$\ell_a(xy) = axy,$$

whereas

$$\ell_a(x)\ell_a(y) = (ax)(ay) = axay.$$

If ℓ_a is a homomorphism, it must preserve the identity; hence

$$a = \ell_a(e) = e.$$

Conversely, $\ell_e = \text{id}_G$ is a homomorphism. Therefore

$$\ell_a : G \rightarrow G \text{ is a homomorphism if and only if } a = e.$$

There is nevertheless a homomorphism built from all left translations:

$$L : G \longrightarrow S_G, \quad a \longmapsto \ell_a.$$

Since

$$\ell_{ab}(x) = (ab)x = a(bx) = \ell_a(\ell_b(x)),$$

we have

$$\ell_{ab} = \ell_a \circ \ell_b \quad \text{and hence} \quad L(ab) = L(a)L(b).$$

Thus L is a homomorphism. It is injective because

$$\ell_a = \ell_b \implies a = \ell_a(e) = \ell_b(e) = b.$$

Its image is the left regular group

$$L_G = \{\ell_a : a \in G\} \leq S_G.$$

Therefore

$$L : G \longrightarrow L_G$$

is an isomorphism, and

$$G \cong L_G.$$

This is the homomorphism formulation of Cayley's theorem.

6.8 The sign homomorphism

Write a permutation $\sigma \in S_n$ in two-line notation as

$$\sigma = \begin{pmatrix} 1 & 2 & \cdots & n \\ \sigma(1) & \sigma(2) & \cdots & \sigma(n) \end{pmatrix}.$$

Definition 6.6: Sign of a permutation

The *sign* of $\sigma \in S_n$ is

$$\text{sgn}(\sigma) := \prod_{1 \leq i < j \leq n} \frac{\sigma(j) - \sigma(i)}{j - i}.$$

Equivalently,

$$\operatorname{sgn}(\sigma) = (-1)^{N(\sigma)},$$

where $N(\sigma)$ is the number of inversions, that is, pairs (i, j) with $i < j$ but $\sigma(i) > \sigma(j)$.

The numerator contains the same absolute differences as the denominator, only permuted and possibly sign-reversed. Hence the quotient is always $+1$ or -1 . A permutation is called *even* when its sign is $+1$, and *odd* when its sign is -1 .

Example 6.1

Let

$$\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}.$$

The pairs $(1, 3)$ and $(2, 3)$ are inversions, while $(1, 2)$ is not. Thus $N(\sigma) = 2$ and

$$\operatorname{sgn}(\sigma) = (-1)^2 = 1.$$

The product formula gives the same calculation:

$$\begin{aligned} \operatorname{sgn}(\sigma) &= \frac{\sigma(2) - \sigma(1)}{2 - 1} \frac{\sigma(3) - \sigma(1)}{3 - 1} \frac{\sigma(3) - \sigma(2)}{3 - 2} \\ &= \frac{3 - 2}{1} \frac{1 - 2}{2} \frac{1 - 3}{1} \\ &= 1 \cdot \left(-\frac{1}{2}\right) \cdot (-2) = 1. \end{aligned}$$

For a transposition

$$\tau = \begin{pmatrix} 1 & \cdots & i & \cdots & j & \cdots & n \\ 1 & \cdots & j & \cdots & i & \cdots & n \end{pmatrix},$$

the parity of the number of inversions is odd, so

$$\operatorname{sgn}(\tau) = -1.$$

Theorem 6.4: The sign is a homomorphism

The map

$$\operatorname{sgn} : S_n \longrightarrow \{+1, -1\}$$

is a homomorphism:

$$\operatorname{sgn}(\sigma \circ \rho) = \operatorname{sgn}(\sigma) \operatorname{sgn}(\rho).$$

证明. Insert the intermediate differences $\rho(j) - \rho(i)$:

$$\begin{aligned} \operatorname{sgn}(\sigma \circ \rho) &= \prod_{i < j} \frac{\sigma(\rho(j)) - \sigma(\rho(i))}{j - i} \\ &= \left(\prod_{i < j} \frac{\sigma(\rho(j)) - \sigma(\rho(i))}{\rho(j) - \rho(i)} \right) \left(\prod_{i < j} \frac{\rho(j) - \rho(i)}{j - i} \right). \end{aligned}$$

The second parenthesis is $\text{sgn}(\rho)$. In the first, the unordered pairs

$$\{\rho(i), \rho(j)\}, \quad i < j,$$

run exactly once through all unordered pairs of distinct elements of $\{1, \dots, n\}$. Reversing the order of a pair changes the signs of both numerator and denominator and therefore leaves their ratio unchanged. We may consequently reorder the factors with their smaller index first:

$$\prod_{i < j} \frac{\sigma(\rho(j)) - \sigma(\rho(i))}{\rho(j) - \rho(i)} = \prod_{p < q} \frac{\sigma(q) - \sigma(p)}{q - p} = \text{sgn}(\sigma).$$

Multiplying the two parentheses proves the result. ■

6.9 The alternating group

Definition 6.7: Alternating group

The *alternating group of degree n* is

$$A_n := \ker(\text{sgn}) = \{\sigma \in S_n : \text{sgn}(\sigma) = +1\}.$$

Thus A_n is the group of all even permutations.

Because a kernel is a subgroup,

$$A_n \leq S_n.$$

For $n \geq 2$, the sign map is onto: the identity has sign $+1$, while any transposition τ has sign -1 . Its two fibers are

$$A_n \quad \text{and} \quad \tau A_n.$$

Indeed, if σ is odd, then

$$\text{sgn}(\tau^{-1}\sigma) = \text{sgn}(\tau)^{-1} \text{sgn}(\sigma) = (-1)^{-1}(-1) = 1,$$

so $\tau^{-1}\sigma \in A_n$ and $\sigma \in \tau A_n$. Consequently,

$$[S_n : A_n] = 2.$$

Lagrange's theorem yields

$$|A_n| = \frac{|S_n|}{2} = \frac{n!}{2} \quad (n \geq 2).$$

6.10 Classification of cyclic groups

Theorem 6.5: Classification of cyclic groups

Let $G = \langle g \rangle$ be cyclic.

1. If G is infinite, then $G \cong \mathbb{Z}$.
2. If $|G| = m < \infty$, then $G \cong \mathbb{Z}/m\mathbb{Z} \cong \mathbb{Z}_m$.

证明. *Infinite case.* Define

$$\varphi : \mathbb{Z} \longrightarrow G, \quad k \longmapsto g^k.$$

Using additive notation on $\mathbb{Z}$,

$$\varphi(k + \ell) = g^{k+\ell} = g^k g^\ell = \varphi(k)\varphi(\ell),$$

so φ is a homomorphism. It is onto because every element of $\langle g \rangle$ is a power of g . It is one-to-one because an infinite cyclic group has no repeated powers: if $g^k = g^\ell$ with $k \neq \ell$, then $g^{k-\ell} = e$, and the powers of g would repeat with a finite positive period, making G finite. Thus φ is an isomorphism and $G \cong \mathbb{Z}$.

Finite case. Suppose $|G| = m$. Then

$$G = \{e, g, g^2, \dots, g^{m-1}\},$$

and m is the smallest positive integer for which $g^m = e$. Define

$$\psi : \mathbb{Z}/m\mathbb{Z} \longrightarrow G, \quad k + m\mathbb{Z} \longmapsto g^k.$$

We first prove that ψ is well defined. If

$$k + m\mathbb{Z} = \ell + m\mathbb{Z},$$

then $k - \ell \in m\mathbb{Z}$, so $k - \ell = qm$ for some $q \in \mathbb{Z}$. Therefore

$$g^{k-\ell} = g^{qm} = (g^m)^q = e,$$

and hence $g^k = g^\ell$.

Next,

$$\begin{aligned} \psi((k + m\mathbb{Z}) + (\ell + m\mathbb{Z})) &= \psi(k + \ell + m\mathbb{Z}) \\ &= g^{k+\ell} \\ &= g^k g^\ell \\ &= \psi(k + m\mathbb{Z})\psi(\ell + m\mathbb{Z}), \end{aligned}$$

so ψ is a homomorphism. It is onto because every element of G is a power g^k . Finally, if

$$\psi(k + m\mathbb{Z}) = \psi(\ell + m\mathbb{Z}),$$

then $g^{k-\ell} = e$. Since g has order m , this means $m \mid (k - \ell)$, and therefore

$$k + m\mathbb{Z} = \ell + m\mathbb{Z}.$$

Thus ψ is injective as well as surjective. It is an isomorphism, and

$$G \cong \mathbb{Z}/m\mathbb{Z} \cong \mathbb{Z}_m.$$

■

What to retain from this chapter

1. A homomorphism satisfies $\varphi(ab) = \varphi(a)\varphi(b)$; it preserves identities, inverses, and integer powers.
2. Images of subgroups and preimages of subgroups are subgroups. In particular, $\text{im } \varphi \leq G'$ and $\ker \varphi \leq G$.
3. A homomorphism is injective exactly when its kernel is $\{e\}$, and a composition of homomorphisms is a homomorphism.
4. Isomorphisms are bijective homomorphisms; automorphisms are isomorphisms $G \rightarrow G$, and $\text{Aut}(G) \leq S_G$.
5. Conjugation is an automorphism, and $H \cong gHg^{-1}$ for every subgroup $H \leq G$.
6. The sign map is a homomorphism $S_n \rightarrow \{\pm 1\}$, with kernel A_n , so $[S_n : A_n] = 2$ and $|A_n| = n!/2$ for $n \geq 2$.
7. Every infinite cyclic group is isomorphic to $\mathbb{Z}$, and every cyclic group of order m is isomorphic to $\mathbb{Z}_m$.

Normal Subgroups and Factor Groups

Road map

The set G/H of left cosets exists for every subgroup $H \leq G$, but coset multiplication

$$(aH)(bH) := (ab)H$$

is independent of representatives only for special subgroups. These are the normal subgroups. We prove the equivalence between normality, equality of left and right cosets, and invariance under conjugation; build the factor group G/N ; and show that kernels are normal. We then study canonical projections, centers, simple groups, the classification picture for finite simple groups, cyclic simple groups, A_4 , and the quotient S_n/A_n .

7.1 Why coset multiplication needs a condition

Recall that for a group G and a subgroup $H \leq G$,

$$G/H = \{aH : a \in G\}$$

is the set of all left cosets. A coset aH is generally not a subgroup: it contains the identity only when $a \in H$, in which case $aH = H$.

A natural attempt to multiply cosets is

$$aH \circ bH := (ab)H.$$

If this operation is well defined, its group axioms would follow from those of G :

- the identity would be $H = eH$, because

$$aH \circ H = (ae)H = aH \quad \text{and} \quad H \circ aH = (ea)H = aH;$$

- the inverse of aH would be $a^{-1}H$, because

$$aH \circ a^{-1}H = (aa^{-1})H = H;$$

- associativity would follow from

$$\begin{aligned} aH \circ (bH \circ cH) &= aH \circ (bc)H = (a(bc))H, \\ (aH \circ bH) \circ cH &= (ab)H \circ cH = ((ab)c)H, \end{aligned}$$

and $a(bc) = (ab)c$.

The real issue is *well-definedness*. A coset has many representatives. If

$$aH = a'H, \quad bH = b'H,$$

must it follow that

$$(ab)H = (a'b')H?$$

For an arbitrary subgroup, the answer can be no.

7.2 Normal subgroups

Definition 7.1: Normal subgroup

A subgroup $N \leq G$ is *normal* in G , written

$$N \trianglelefteq G,$$

if

$$aN = Na \quad \text{for every } a \in G.$$

Thus every left coset of N is the corresponding right coset.

Theorem 7.1: Equivalent forms of normality

Let $N \leq G$. The following are equivalent:

1. $N \trianglelefteq G$;
2. $gN = Ng$ for every $g \in G$;
3. $gNg^{-1} = N$ for every $g \in G$;
4. $gNg^{-1} \subseteq N$ for every $g \in G$.

证明. The first two statements are the same by definition. If $gN = Ng$, multiply both sets on the right by g^{-1} :

$$gNg^{-1} = Ng g^{-1} = N.$$

Conversely, if $gNg^{-1} = N$, multiply on the right by g to obtain $gN = Ng$. Thus (2) and (3) are equivalent.

Clearly (3) implies (4). Suppose (4) holds. Applying it to g^{-1} gives

$$g^{-1}Ng \subseteq N.$$

Conjugating this inclusion by g gives

$$N \subseteq gNg^{-1}.$$

Together with $gNg^{-1} \subseteq N$, this yields equality. Hence (4) implies (3). ■

Geometric meaning

Conjugation by g sends the subgroup N to the conjugate subgroup gNg^{-1} . Normality means that this operation never produces a new subgroup:

$$gNg^{-1} = N \quad \text{for all } g \in G.$$

The subgroup is invariant under all inner automorphisms.

7.3 Normality is exactly what makes the product well defined

Theorem 7.2: Well-defined coset multiplication

Let $N \leq G$. The rule

$$(aN)(bN) := (ab)N$$

defines a well-defined binary operation on G/N if and only if $N \trianglelefteq G$.

证明. Suppose first that $N \trianglelefteq G$. Let

$$aN = a'N, \quad bN = b'N.$$

By the coset-equality criterion, there are $n_1, n_2 \in N$ such that

$$a' = an_1, \quad b' = bn_2.$$

Then

$$\begin{aligned} a'b' &= (an_1)(bn_2) \\ &= ab(b^{-1}n_1b)n_2. \end{aligned}$$

Normality gives $b^{-1}n_1b \in N$, and closure of N gives

$$(b^{-1}n_1b)n_2 \in N.$$

Therefore $a'b' \in abN$, which is equivalent to

$$(a'b')N = (ab)N.$$

The product is independent of the chosen representatives.

Conversely, suppose the product is well defined. Fix $n \in N$ and $b \in G$. Since $nN = N = eN$, well-definedness gives

$$(nN)(bN) = (eN)(bN).$$

Therefore

$$nbN = bN,$$

so the left-coset criterion gives

$$b^{-1}nb \in N.$$

Since n was arbitrary,

$$b^{-1}Nb \subseteq N \quad \text{for every } b \in G.$$

The equivalent-normality theorem now shows that $N \trianglelefteq G$. ■

This calculation is the detailed version of the chain recorded in the handwritten notes. Starting from arbitrary representatives and applying the coset equality lemma eventually forces

$$Hb \subseteq bH \quad \text{and} \quad bH \subseteq Hb,$$

and hence $bH = Hb$ for every $b \in G$.

7.4 Factor groups

Definition 7.2: Factor group or quotient group

Let $N \trianglelefteq G$. The set

$$G/N = \{aN : a \in G\},$$

equipped with

$$aN \cdot bN := (ab)N,$$

is called the *factor group* or *quotient group* of G by N .

Theorem 7.3

If $N \trianglelefteq G$, then G/N is a group. Its identity is N , and

$$(aN)^{-1} = a^{-1}N.$$

证明. The preceding theorem proves that multiplication is well defined. Closure is automatic because $(ab)N \in G/N$. The calculations at the beginning of the chapter verify associativity, the identity N , and the inverse $a^{-1}N$. Thus all group axioms hold. ■

If G is finite, then

$$|G/N| = [G : N] = \frac{|G|}{|N|}.$$

Here $|G/N|$ counts cosets, not the number of elements inside one coset.

7.5 Kernels and images

Theorem 7.4: Kernels are normal

Let $\varphi : G \rightarrow G'$ be a homomorphism. Then

$$\ker \varphi \trianglelefteq G.$$

If $N \trianglelefteq G$, then

$$\varphi(N) \trianglelefteq \varphi(G) \leq G'.$$

In particular, if φ is surjective, then $\varphi(N) \trianglelefteq G'$.

证明. We already know that $\ker \varphi \leq G$. Let $a \in G$ and $x \in \ker \varphi$. Then

$$\begin{aligned} \varphi(axa^{-1}) &= \varphi(a)\varphi(x)\varphi(a)^{-1} \\ &= \varphi(a)e'\varphi(a)^{-1} = e'. \end{aligned}$$

Hence $axa^{-1} \in \ker \varphi$, and therefore

$$a(\ker \varphi)a^{-1} \subseteq \ker \varphi.$$

The normality criterion gives $\ker \varphi \trianglelefteq G$.

Now suppose $N \trianglelefteq G$. The image theorem gives $\varphi(N) \leq \varphi(G)$. Take $a' \in \varphi(G)$ and $x' \in \varphi(N)$. Choose $a \in G$ and $x \in N$ with

$$a' = \varphi(a), \quad x' = \varphi(x).$$

Then

$$\begin{aligned} a'x'(a')^{-1} &= \varphi(a)\varphi(x)\varphi(a)^{-1} \\ &= \varphi(axa^{-1}). \end{aligned}$$

Since $N \trianglelefteq G$, we have $axa^{-1} \in N$, so the last element belongs to $\varphi(N)$. Therefore

$$a'\varphi(N)(a')^{-1} \subseteq \varphi(N) \quad (a' \in \varphi(G)),$$

and $\varphi(N) \trianglelefteq \varphi(G)$. ■

Proposition 7.1: Subgroups of Abelian groups

If G is Abelian, every subgroup $N \leq G$ is normal.

证明. For $a \in G$ and $n \in N$,

$$ana^{-1} = naa^{-1} = n \in N.$$

Thus $aNa^{-1} = N$ for every a , and $N \trianglelefteq G$. ■

For example,

$$m\mathbb{Z} \trianglelefteq \mathbb{Z}, \quad \mathbb{Z}/m\mathbb{Z} \cong \mathbb{Z}_m.$$

7.6 The canonical projection

Let $N \trianglelefteq G$. Define

$$\pi : G \longrightarrow G/N, \quad a \longmapsto aN.$$

This is the *canonical projection* or *quotient map*. It is a homomorphism because

$$\begin{aligned}\pi(a)\pi(b) &= (aN)(bN) \\ &= (ab)N \\ &= \pi(ab).\end{aligned}$$

It is surjective by the definition of G/N . Its kernel is

$$\begin{aligned}\ker \pi &= \{a \in G : \pi(a) = N\} \\ &= \{a \in G : aN = N\} \\ &= N.\end{aligned}$$

This is a consistency check on the theory: every normal subgroup occurs as the kernel of its canonical projection.

Proposition 7.2: The center is normal

For every group G ,

$$Z(G) \trianglelefteq G.$$

Moreover, $Z(G)$ is Abelian.

证明. If $z \in Z(G)$ and $a \in G$, then $az = za$, so

$$aza^{-1} = zaa^{-1} = z.$$

Thus $aZ(G)a^{-1} = Z(G)$, proving normality. If $x, y \in Z(G)$, then x commutes with every element of G , in particular with y ; hence $xy = yx$. Therefore $Z(G)$ is Abelian. ■

7.7 Simple groups

Definition 7.3: Simple group

A nontrivial group G is called *simple* if its only normal subgroups are

$$\{e\} \quad \text{and} \quad G.$$

Simple groups are the indivisible building blocks of finite group theory: they have no nontrivial proper normal subgroup from which a smaller quotient can be formed.

7.7.1 The classification picture for finite simple groups

The handwritten lecture presents the classification of finite simple groups as a major milestone in mathematics. Its main branches are shown in Figure 7.1.

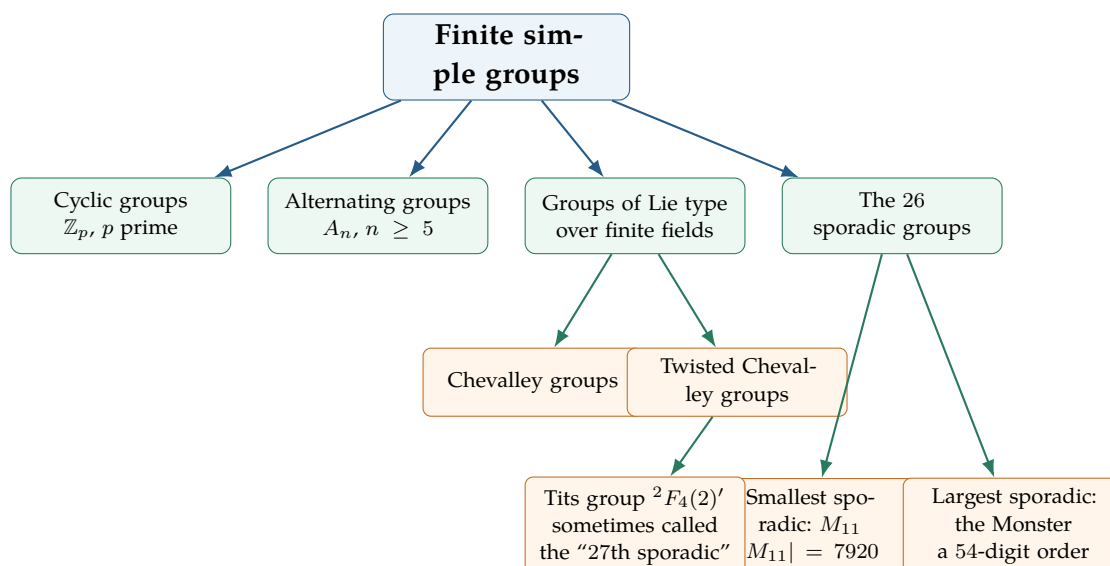


图 7.1: The classification tree recorded in the notes: three infinite families and 26 sporadic exceptions. The Tits group belongs to the Lie-type side but is historically nicknamed the “27th sporadic group.”

The details written around the original diagram include the following.

- The cyclic simple groups form the series $\mathbb{Z}_p$ for primes p .
- The alternating groups A_n are simple for $n \geq 5$.
- The groups of Lie type include Chevalley groups and twisted Chevalley groups constructed over finite fields.
- The smallest sporadic simple group is the Mathieu group M_{11} , with

$$|M_{11}| = 2^4 \cdot 3^2 \cdot 5 \cdot 11 = 7920.$$

- The largest sporadic group is the Monster, also called the Fischer–Griess Monster or the “Friendly Giant.” Its order is

$$|\mathbb{M}| = 808017424794512875886459904961710757005754368 \\ 000000000,$$

a 54-digit integer. The notes describe it through rotations in a 196883-dimensional space; more precisely, 196883 is the dimension of its smallest nontrivial complex irreducible representation.

- The Tits group ${}^2F_4(2)'$ has order

$$17\,971\,200 \approx 1.8 \times 10^7.$$

It is a finite simple group of Lie type, although it is often informally called the “27th sporadic group.”

- The handwritten note summarizes the classification proof as a collaboration of hundreds of authors over roughly 1955–2004, spread over thousands of pages.

For a field $\mathbb{F}$, the characteristic mentioned in the notes is

$$\text{char } \mathbb{F} := \min\{p \in \mathbb{N} : \underbrace{1 + \cdots + 1}_{p \text{ terms}} = 0\},$$

when such a positive integer exists; otherwise the characteristic is 0. Equivalently, for positive characteristic,

$$\text{char } \mathbb{F} = |\langle 1 \rangle_+|,$$

the order of the cyclic subgroup generated by 1 in the *additive* group of the field.

7.7.2 Exactly which cyclic groups are simple?

Theorem 7.5

The finite cyclic group $\mathbb{Z}_n$ is simple if and only if n is prime.

证明. If n is prime, Lagrange's theorem shows that a subgroup of $\mathbb{Z}_n$ has order 1 or n . Hence the only subgroups, and therefore the only normal subgroups, are $\{0\}$ and $\mathbb{Z}_n$.

Conversely, suppose n is composite. Choose a proper divisor k with

$$1 < k < n, \quad n = kr.$$

Let a be a generator of $\mathbb{Z}_n$ in multiplicative notation and put $b = a^k$. Then

$$b^r = (a^k)^r = a^{kr} = a^n = e.$$

The subgroup

$$\langle b \rangle = \{e, b, b^2, \dots, b^{r-1}\}$$

is nontrivial and proper. Since $\mathbb{Z}_n$ is Abelian, every subgroup is normal, so

$$\{e\} < \langle b \rangle < \mathbb{Z}_n \quad \text{and} \quad \langle b \rangle \trianglelefteq \mathbb{Z}_n.$$

Therefore $\mathbb{Z}_n$ is not simple. ■

Example 7.1: A_4 is not simple

Inside A_4 , consider the Klein four-subgroup

$$V_4 = \{e, (12)(34), (13)(24), (14)(23)\}.$$

This is the subgroup displayed in the handwritten notes in one-line notation:

$$\{(1234), (2143), (3412), (4321)\}.$$

For every $\sigma \in A_4$,

$$\sigma(ij)(k\ell)\sigma^{-1} = (\sigma(i) \sigma(j))(\sigma(k) \sigma(\ell)).$$

Thus conjugation permutes the three double transpositions and fixes e . Hence

$$\sigma V_4 \sigma^{-1} = V_4 \quad (\sigma \in A_4),$$

so $V_4 \trianglelefteq A_4$. Because

$$\{e\} < V_4 < A_4,$$

the group A_4 is not simple. This contrasts with the theorem that A_n is simple for every $n \geq 5$.

7.8 The quotient S_n/A_n

The sign homomorphism satisfies

$$A_n = \ker(\text{sgn}),$$

so the kernel theorem gives

$$A_n \trianglelefteq S_n.$$

For $n \geq 2$, let τ be any transposition. The two cosets are

$$S_n/A_n = \{A_n, \tau A_n\}.$$

Their multiplication is

$\cdot$	A_n	τA_n
A_n	A_n	τA_n
τA_n	τA_n	A_n

because $\tau^2 = e$. This is the Cayley table of $\mathbb{Z}_2$, and hence

$$S_n/A_n \cong \mathbb{Z}_2.$$

A quotient group is not usually a subgroup

The elements of S_n/A_n are the two sets A_n and τA_n ; they are not individual permutations. Although the quotient is abstractly isomorphic to the two-element subgroup $\{e, \tau\}$, that chosen copy need not be normal in S_n .

For example, in S_3 , let

$$\tau = (12), \quad \sigma = (123), \quad H = \{e, \tau\}.$$

Then

$$\sigma\tau = (13), \quad \tau\sigma = (23),$$

so

$$\sigma H = \{\sigma, (13)\} \neq \{\sigma, (23)\} = H\sigma.$$

Therefore H is not normal in S_3 , even though $S_3/A_3 \cong H \cong \mathbb{Z}_2$. Isomorphism does not identify a quotient group with a particular subset of the original group.

What to retain from this chapter

1. A subgroup $N \leq G$ is normal exactly when $gN = Ng$, equivalently $gNg^{-1} = N$, for every $g \in G$.
2. Coset multiplication $(aN)(bN) = (ab)N$ is well defined exactly when $N \trianglelefteq G$; it then makes G/N a group.
3. Kernels are normal, and the image of a normal subgroup is normal in the image of the homomorphism.
4. The canonical projection $G \rightarrow G/N, a \mapsto aN$, is surjective and has kernel N .
5. Every subgroup of an Abelian group is normal, and $Z(G) \trianglelefteq G$.
6. A simple group has no nontrivial proper normal subgroup. The cyclic group $\mathbb{Z}_n$ is simple exactly when n is prime, while A_4 is not simple because $V_4 \trianglelefteq A_4$.
7. $A_n \trianglelefteq S_n$ and $S_n/A_n \cong \mathbb{Z}_2$, but a quotient group is not in general a subgroup of its parent group.

The Symplectic Group

Road map

The symplectic group is the linear symmetry group underlying classical Hamiltonian mechanics. We shall first write Hamilton's equations as one matrix equation. A change of phase-space coordinates is canonical precisely when its Jacobian preserves the corresponding skew-symmetric matrix. We then prove the canonical form and uniqueness of nondegenerate skew-symmetric matrices, establish the reciprocal pairing of symplectic eigenvalues and the identity $\det A = 1$, reformulate canonicity by Poisson brackets, and derive all canonical transformations of the form $q' = f(q)$.

8.1 Hamiltonian mechanics in matrix form

8.1.1 Phase space

Consider a mechanical system with n degrees of freedom. Its generalized coordinates and conjugate momenta are

$$q = (q_1, \dots, q_n)^T, \quad p = (p_1, \dots, p_n)^T.$$

A physical state is represented by the phase-space point

$$x = (q, p)^T = (q_1, \dots, q_n, p_1, \dots, p_n)^T \in \mathbb{R}^{2n}.$$

As time varies, the system traces a phase-space trajectory

$$t \mapsto x(t) = (q(t), p(t))^T = (x_1(t), \dots, x_{2n}(t))^T.$$

The Hamiltonian

$$H = H(q, p) = H(x)$$

is the energy function that determines the motion. For example, a particle of mass m moving in three-dimensional Euclidean space in an external potential V has

$$H(q, p) = \sum_{i=1}^3 \frac{p_i^2}{2m} + V(q_1, q_2, q_3).$$

Hamilton's canonical equations are

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}, \quad i = 1, \dots, n. \quad (8.1)$$

8.1.2 The standard symplectic matrix

Let I_n denote the $n \times n$ identity matrix and define

$$J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}. \quad (8.2)$$

This $2n \times 2n$ matrix satisfies

$$J^T = -J, \quad J^2 = -I_{2n}, \quad J^{-1} = -J, \quad \det J = 1.$$

Writing

$$\nabla_x H = \begin{pmatrix} \nabla_q H \\ \nabla_p H \end{pmatrix},$$

we obtain

$$J \nabla_x H = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix} \begin{pmatrix} \nabla_q H \\ \nabla_p H \end{pmatrix} = \begin{pmatrix} \nabla_p H \\ -\nabla_q H \end{pmatrix}.$$

Component by component,

$$[J \nabla H]_i = \begin{cases} \frac{\partial H}{\partial p_i}, & 1 \leq i \leq n, \\ -\frac{\partial H}{\partial q_{i-n}}, & n+1 \leq i \leq 2n. \end{cases}$$

Consequently, the $2n$ scalar equations (8.1) are exactly the single vector equation

$$\boxed{\dot{x} = J \nabla_x H(x)}. \quad (8.3)$$

8.2 Coordinate representations and canonical transformations

8.2.1 The same physical state in two coordinate systems

It is useful to separate a physical state from the coordinates used to describe it. Let M be the phase-space manifold and let

$$X : M \supset U \longrightarrow X(U) \subset \mathbb{R}^{2n}, \quad Y : M \supset U \longrightarrow Y(U) \subset \mathbb{R}^{2n}$$

be two coordinate charts. If $m = m(t) \in M$, define

$$x = X(m), \quad y = Y(m).$$

The coordinate change is

$$y = f(x) = Y \circ X^{-1}(x), \quad x = f^{-1}(y) = X \circ Y^{-1}(y).$$

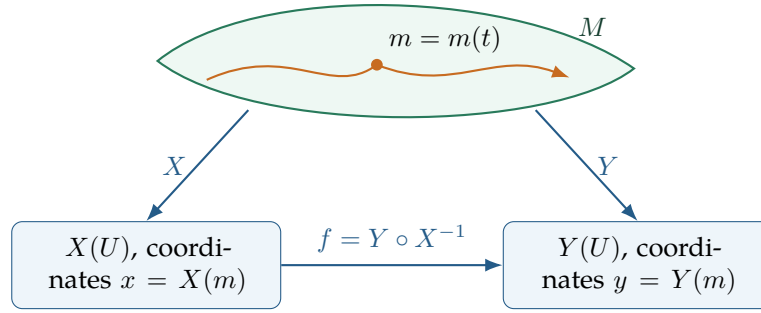


图 8.1: One phase-space point and two coordinate representations. The orange curve represents the same physical trajectory in M , independently of the chosen chart.

The Hamiltonian is intrinsically a scalar function $H : M \rightarrow \mathbb{R}$. Its two coordinate expressions are

$$H_X(x) = H(X^{-1}(x)), \quad H_Y(y) = H(Y^{-1}(y)).$$

Because $y = f(x)$ implies $Y^{-1}(y) = X^{-1}(x)$, we have

$$H_X(x) = H_Y(f(x)) = H_Y(y). \quad (8.4)$$

Thus the numerical energy is unchanged; only its formula in coordinates may look different.

8.2.2 Deriving the Jacobian condition

Let $f : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ be a smooth invertible coordinate transformation and write

$$y = f(x) =: y(x).$$

Its Jacobian is

$$S(x) := Df(x) = \frac{\partial f}{\partial x}(x), \quad S_{ij}(x) = \frac{\partial y_i}{\partial x_j}(x).$$

Since f is a local diffeomorphism, $S(x) \in GL(2n, \mathbb{R})$. At $y = f(x)$, the inverse Jacobian is

$$Df^{-1}(y) = S(x)^{-1}, \quad \frac{\partial x_i}{\partial y_j} = [S(x)^{-1}]_{ij}.$$

Along a trajectory,

$$\dot{y}(t) = S(x(t))\dot{x}(t). \quad (8.5)$$

Moreover, differentiating (8.4) gives

$$\frac{\partial H_X}{\partial x_i} = \sum_{j=1}^{2n} \frac{\partial H_Y}{\partial y_j} \frac{\partial y_j}{\partial x_i}, \quad \text{so} \quad \nabla_x H_X = S(x)^T \nabla_y H_Y.$$

Substituting this and $\dot{x} = J\nabla_x H_X$ into (8.5) yields

$$\dot{y} = S(x)JS(x)^T \nabla_y H_Y. \quad (8.6)$$

Therefore the transformed equation has the same standard Hamiltonian form $\dot{y} = J\nabla_y H_Y$ exactly when

$$\boxed{S(x)JS(x)^T = J \quad \text{for every } x.} \quad (8.7)$$

Definition 8.1: Canonical transformation

A smooth diffeomorphism f of phase space is called *canonical*, or a *symplectomorphism*, when its Jacobian satisfies (8.7) at every point. Such a transformation preserves the form of Hamilton's equations.

Why one also sees $S^T JS = J$

Our matrix $S_{ij} = \partial y_i / \partial x_j$ pushes coordinate velocities forward, and the direct calculation above produces $SJS^T = J$. If instead one describes the pullback of the bilinear symplectic form, the standard condition is $S^T JS = J$. For an invertible matrix and the standard J , the two conditions are equivalent: $S^T JS = J$ implies

$$S^{-1} = -JS^T J,$$

and hence $SJS^T = J$; the converse follows in the same way. The apparent transpose difference is therefore a matter of which direction of the coordinate change is being represented, not a difference in the geometry.

8.3 Definition and elementary group properties

More generally, let Γ be a fixed real, nondegenerate, skew-symmetric $2n \times 2n$ matrix:

$$\Gamma^T = -\Gamma, \quad \det \Gamma \neq 0.$$

Definition 8.2: Symplectic group associated with Γ

$$\mathrm{Sp}(\Gamma) := \{A \in GL(2n, \mathbb{R}) : A^T \Gamma A = \Gamma\}.$$

For the standard matrix J in (8.2), one writes

$$\mathrm{Sp}(2n, \mathbb{R}) = \mathrm{Sp}(J).$$

For the standard matrix $J^{-1} = -J$, the two equations

$$A^T J A = J \quad \text{and} \quad A J A^T = J$$

are equivalent. In particular $A \in \mathrm{Sp}(2n, \mathbb{R})$ if and only if $A^T \in \mathrm{Sp}(2n, \mathbb{R})$.

Proposition 8.1

$\mathrm{Sp}(\Gamma)$ is a group under matrix multiplication.

证明. Associativity is inherited from matrix multiplication, and $I_{2n}^T \Gamma I_{2n} = \Gamma$, so the identity belongs to the set. If $A_1, A_2 \in \mathrm{Sp}(\Gamma)$, then

$$(A_1 A_2)^T \Gamma (A_1 A_2) = A_2^T (A_1^T \Gamma A_1) A_2 = A_2^T \Gamma A_2 = \Gamma,$$

which proves closure. Finally, from $A^T \Gamma A = \Gamma$, multiply on the left by A^{-T} and on the right by

A^{-1} :

$$(A^{-1})^T \Gamma A^{-1} = \Gamma.$$

Thus $A^{-1} \in \text{Sp}(\Gamma)$. ■

8.4 Canonical form of a nondegenerate skew matrix

Lemma 8.1: Orthogonal block form

Let Γ be a real nondegenerate skew-symmetric matrix. Then its dimension is even, say $2n$, and there is an orthogonal matrix $O \in O(2n)$ such that

$$O^T \Gamma O = \begin{pmatrix} 0 & \Lambda \\ -\Lambda & 0 \end{pmatrix}, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_n), \quad \lambda_j > 0. \quad (8.8)$$

Equivalently, after interleaving the basis vectors, the matrix is the block diagonal matrix

$$\text{diag} \left(\begin{pmatrix} 0 & \lambda_1 \\ -\lambda_1 & 0 \end{pmatrix}, \dots, \begin{pmatrix} 0 & \lambda_n \\ -\lambda_n & 0 \end{pmatrix} \right).$$

证明. First observe that

$$(i\Gamma)^\dagger = -i\Gamma^T = i\Gamma.$$

Thus $i\Gamma$ is Hermitian, so its eigenvalues are real. Consequently, the eigenvalues of Γ are purely imaginary. Since Γ has real entries, nonreal eigenvalues occur in conjugate pairs, and since $\det \Gamma \neq 0$, zero is not an eigenvalue. We may therefore list the eigenvalues as

$$i\lambda_1, -i\lambda_1, \dots, i\lambda_n, -i\lambda_n, \quad \lambda_j > 0.$$

This already proves that the dimension is even.

Choose unit eigenvectors $u_j \in \mathbb{C}^{2n}$ such that

$$\Gamma u_j = i\lambda_j u_j.$$

Complex conjugation gives

$$\Gamma \overline{u_j} = -i\lambda_j \overline{u_j}.$$

The spectral theorem permits the vectors

$$u_1, \overline{u_1}, \dots, u_n, \overline{u_n}$$

to be chosen as a unitary basis of $\mathbb{C}^{2n}$. Define the real vectors

$$a_j = \frac{u_j + \overline{u_j}}{\sqrt{2}}, \quad b_j = \frac{u_j - \overline{u_j}}{\sqrt{2}i}.$$

They are real because $u_j + \overline{u_j} = 2 \text{Re } u_j$ and $(u_j - \overline{u_j})/i = 2 \text{Im } u_j$. Unitarity gives

$$a_j^T a_k = \delta_{jk}, \quad b_j^T b_k = \delta_{jk}, \quad a_j^T b_k = 0.$$

Hence

$$O = (a_1, \dots, a_n, b_1, \dots, b_n)$$

is a real orthogonal matrix.

It remains to compute the action of Γ . From the eigenvector equations,

$$\begin{aligned}\Gamma a_j &= \frac{i\lambda_j u_j - i\lambda_j \bar{u}_j}{\sqrt{2}} = -\lambda_j b_j, \\ \Gamma b_j &= \frac{i\lambda_j u_j + i\lambda_j \bar{u}_j}{\sqrt{2}i} = \lambda_j a_j.\end{aligned}$$

Therefore

$$\begin{aligned}a_i^T \Gamma a_j &= 0, & a_i^T \Gamma b_j &= \lambda_j \delta_{ij}, \\ b_i^T \Gamma a_j &= -\lambda_j \delta_{ij}, & b_i^T \Gamma b_j &= 0.\end{aligned}$$

These four identities give precisely (8.8). ■

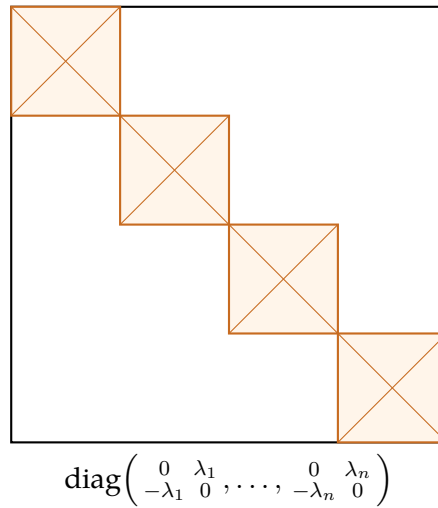


图 8.2: The 2×2 skew blocks in the interleaved real basis $(a_1, b_1, \dots, a_n, b_n)$.

8.5 All nondegenerate symplectic forms give isomorphic groups

Lemma 8.2: Uniqueness up to congruence

Let Γ and Γ' be real nondegenerate skew-symmetric $2n \times 2n$ matrices. Then there is $Q \in GL(2n, \mathbb{R})$ such that

$$\Gamma = Q^T \Gamma' Q.$$

Moreover,

$$\text{Sp}(\Gamma) \cong \text{Sp}(\Gamma').$$

证明. By the orthogonal block-form lemma above, there are orthogonal matrices O, O' for which

$$D = O^T \Gamma O, \quad D' = O'^T \Gamma' O'$$

are block diagonal with nonzero blocks

$$\begin{pmatrix} 0 & \lambda_j \\ -\lambda_j & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & \lambda'_j \\ -\lambda'_j & 0 \end{pmatrix}.$$

Choose a real invertible block diagonal matrix

$$R = \text{diag}(R_1, \dots, R_n)$$

so that $R_j^T D'_j R_j = D_j$. For instance,

$$R_j = \begin{pmatrix} \sqrt{\lambda_j/\lambda'_j} & 0 \\ 0 & \sqrt{\lambda_j/\lambda'_j} \end{pmatrix}$$

works when the λ_j, λ'_j have been chosen positive. Hence

$$D = R^T D' R.$$

With

$$Q = O' R O^T$$

we obtain

$$Q^T \Gamma' Q = O R^T O'^T \Gamma' O' R O^T = O D O^T = \Gamma.$$

Now let $A \in \text{Sp}(\Gamma)$. Since $\Gamma = Q^T \Gamma' Q$,

$$Q^T \Gamma' Q = A^T Q^T \Gamma' Q A.$$

Multiplying by Q^{-T} on the left and Q^{-1} on the right gives

$$\Gamma' = (Q A Q^{-1})^T \Gamma' (Q A Q^{-1}).$$

Thus $Q A Q^{-1} \in \text{Sp}(\Gamma')$. Conjugation

$$\Phi_Q : A \mapsto Q A Q^{-1}$$

is an automorphism of $GL(2n, \mathbb{R})$, and its inverse is conjugation by Q^{-1} . Its restriction is therefore the required group isomorphism. ■

In particular, every group defined by a real nondegenerate skew-symmetric form is isomorphic to the standard group

$$\text{Sp}(2n, \mathbb{R}) = \text{Sp} \left(\begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix} \right).$$

8.6 Eigenvalues of a symplectic matrix

Theorem 8.1: Reciprocal and conjugate pairing

Let $A \in \text{Sp}(\Gamma)$, where $\Gamma^T = -\Gamma$ and $\det \Gamma \neq 0$. If $\lambda_0 \in \mathbb{C}$ is an eigenvalue of A of algebraic multiplicity k , then

$$\overline{\lambda_0} \quad \text{and} \quad \lambda_0^{-1}$$

are also eigenvalues of algebraic multiplicity k . Consequently, $(\overline{\lambda_0})^{-1}$ has the same multiplicity as well.

证明. The symplectic identity gives

$$A^T \Gamma A = \Gamma \implies A^{-T} = \Gamma A \Gamma^{-1}.$$

Let

$$\chi_A(\lambda) = \det(A - \lambda I_{2n})$$

be the characteristic polynomial in the convention used in the handwritten derivation. Similar matrices and transposed matrices have the same determinant, so

$$\begin{aligned} \chi_A(\lambda) &= \det(\Gamma(A - \lambda I)\Gamma^{-1}) \\ &= \det(A^{-T} - \lambda I) \\ &= \det(A^{-1} - \lambda I). \end{aligned}$$

Next,

$$\begin{aligned} \det(A^{-1} - \lambda I) &= \det(A^{-1}) \det(I - \lambda A) \\ &= \det(A)^{-1} \lambda^{2n} \det(\lambda^{-1} I - A) \\ &= \det(A)^{-1} \lambda^{2n} \det(A - \lambda^{-1} I), \end{aligned}$$

because the dimension $2n$ is even. Taking determinants in $A^T \Gamma A = \Gamma$ gives

$$(\det A)^2 \det \Gamma = \det \Gamma,$$

so at this stage $\det A = \pm 1$. Hence

$$\chi_A(\lambda) = \pm \lambda^{2n} \chi_A(\lambda^{-1}). \quad (8.9)$$

Suppose

$$\chi_A(\lambda) = (\lambda - \lambda_0)^k r(\lambda), \quad r(\lambda_0) \neq 0.$$

Since A is invertible, $\lambda_0 \neq 0$. Substituting this factorization into (8.9) gives

$$\chi_A(\lambda) = \pm \lambda^{2n-k} (1 - \lambda \lambda_0)^k r(\lambda^{-1}).$$

Thus $\lambda = \lambda_0^{-1}$ is a zero of multiplicity at least k . Applying the same argument to λ_0^{-1} shows that the multiplicity cannot exceed k , so it is exactly k .

Finally, A is real, and therefore

$$\overline{\chi_A(\lambda)} = \chi_A(\bar{\lambda}).$$

Complex roots of a real polynomial occur with the same multiplicity as their complex conjugates. Hence $\bar{\lambda}_0$ is also a k -fold eigenvalue. ■

The four possible partners

Away from the real axis and the unit circle, one eigenvalue produces the quadruple

$$\lambda, \quad \bar{\lambda}, \quad \lambda^{-1}, \quad \bar{\lambda}^{-1}.$$

Some of these can coincide. For example, when $|\lambda| = 1$, $\lambda^{-1} = \bar{\lambda}$.

Proposition 8.2: A symplectic matrix has determinant 1

If $A \in \text{Sp}(2n, \mathbb{R})$, then

$$\det A = 1.$$

Thus $\text{Sp}(2n, \mathbb{R}) \subset \text{SL}(2n, \mathbb{R})$.

证明. For a fully rigorous argument, including the self-reciprocal eigenvalues $\lambda = \pm 1$, use the Pfaffian. For every $2n \times 2n$ skew-symmetric matrix B , the Pfaffian is a polynomial $\text{pf}(B)$ satisfying

$$\text{pf}(B)^2 = \det B, \quad \text{pf}(P^T B P) = \det(P) \text{pf}(B).$$

The second identity follows directly by applying P to every argument of the alternating 2-form represented by B and comparing the coefficient of the top-degree volume form. Apply it to $A^T J A = J$:

$$\text{pf}(J) = \text{pf}(A^T J A) = \det(A) \text{pf}(J).$$

Because J is nondegenerate, $\text{pf}(J)^2 = \det J = 1$, so $\text{pf}(J) \neq 0$. Cancellation gives $\det A = 1$.

The reciprocal-pair picture provides the same intuition: whenever $\lambda \neq \pm 1$, the factors λ and λ^{-1} multiply to 1. The Pfaffian argument also handles the fixed reciprocal points $\lambda = \pm 1$ and does not require A to be diagonalizable. ■

8.7 Examples of canonical transformations

Example 8.1: The identity

For $y = x$, the Jacobian is $S = I_{2n}$, and therefore

$$SJS^T = I_{2n}JI_{2n} = J.$$

The identity transformation is canonical.

Example 8.2: Exchange positions and momenta

Let

$$q' = p, \quad p' = -q.$$

Then $y = Jx$, so $S = J$ is constant. Since $J^T = -J$ and $J^2 = -I$,

$$SJS^T = JJ(-J) = J.$$

Hence the transformation is canonical.

Example 8.3: Why $q' = f(q)$, $p' = g(q)$ cannot work

Suppose both new coordinates and new momenta depend only on the old coordinates. With

$$F = \frac{\partial f}{\partial q}, \quad G = \frac{\partial g}{\partial q},$$

the Jacobian is

$$S = \begin{pmatrix} F & 0 \\ G & 0 \end{pmatrix}.$$

Its last n columns vanish, so S is singular; directly,

$$SJS^T = 0 \neq J.$$

Such a map discards all dependence on the old momenta and cannot be an invertible canonical transformation.

8.7.1 The first restrictions on a point transformation

Now take

$$q' = f(q), \quad p' = g(q, p).$$

Put

$$F = \frac{\partial f}{\partial q}, \quad G = \frac{\partial g}{\partial q}, \quad P = \frac{\partial g}{\partial p}.$$

Then

$$S = \begin{pmatrix} F & 0 \\ G & P \end{pmatrix}, \quad SJS^T = \begin{pmatrix} 0 & FP^T \\ -PF^T & GP^T - PG^T \end{pmatrix}.$$

The condition $SJS^T = J$ is therefore equivalent to

$$FP^T = I_n, \quad GP^T - PG^T = 0. \quad (8.10)$$

The first equation gives

$$P = F^{-T}.$$

Since F depends only on q , integration with respect to p shows that p' must be affine in p :

$$p' = F(q)^{-T}p + h(q) \quad (8.11)$$

for some vector-valued function h . The second equation in (8.10) supplies the remaining integrability condition, which we derive more transparently by Poisson brackets below.

8.8 Poisson brackets

Definition 8.3: Poisson bracket

For smooth phase-space functions $A = A(q, p)$ and $B = B(q, p)$, define

$$\{A, B\} = \sum_{i=1}^n \left(\frac{\partial A}{\partial q_i} \frac{\partial B}{\partial p_i} - \frac{\partial B}{\partial q_i} \frac{\partial A}{\partial p_i} \right) \quad (8.12)$$

$$= (\nabla A)^T J \nabla B. \quad (8.13)$$

For $y = f(x)$, one has

$$\begin{aligned} \{y_i, y_j\} &= (\nabla_x y_i)^T J \nabla_x y_j \\ &= \sum_{k, \ell=1}^{2n} \frac{\partial y_i}{\partial x_k} J_{k\ell} \frac{\partial y_j}{\partial x_\ell} \\ &= (SJS^T)_{ij}. \end{aligned}$$

We have therefore proved the following convenient criterion.

Proposition 8.3: Canonical Poisson relations

A smooth invertible transformation $y = f(x)$ is canonical if and only if

$$\boxed{\{y_i, y_j\} = J_{ij} \quad (1 \leq i, j \leq 2n).}$$

Writing $y = (q', p')$, this is equivalent to the four families

$$\begin{aligned} \{q'_i, q'_j\} &= 0, & \{q'_i, p'_j\} &= \delta_{ij}, \\ \{p'_i, q'_j\} &= -\delta_{ij}, & \{p'_i, p'_j\} &= 0. \end{aligned}$$

8.9 All canonical transformations with $q' = f(q)$

Assume $q' = f(q)$, with $F = \partial q' / \partial q$ invertible. Because q' has no p -dependence,

$$\{q'_i, q'_j\} = 0$$

automatically. The mixed relation gives

$$\delta_{ij} = \{q'_i, p'_j\} = \sum_{k=1}^n \frac{\partial q'_i}{\partial q_k} \frac{\partial p'_j}{\partial p_k}.$$

In matrix form,

$$F \left(\frac{\partial p'}{\partial p} \right)^T = I_n,$$

and hence

$$\frac{\partial p'}{\partial p} = F^{-T}.$$

This recovers

$$p' = A(q)p + h(q), \quad A(q) = F(q)^{-T} = \left(\frac{\partial q}{\partial q'} \right)^T.$$

It remains to impose $\{p'_i, p'_j\} = 0$. Since

$$p'_i = \sum_m A_{im}(q) p_m + h_i(q),$$

we have

$$\frac{\partial p'_i}{\partial q_k} = \sum_m \frac{\partial A_{im}}{\partial q_k} p_m + \frac{\partial h_i}{\partial q_k}, \quad \frac{\partial p'_i}{\partial p_k} = A_{ik}.$$

Therefore

$$\begin{aligned} \{p'_i, p'_j\} &= \sum_{k,m} \left(\frac{\partial A_{im}}{\partial q_k} A_{jk} - \frac{\partial A_{jm}}{\partial q_k} A_{ik} \right) p_m \\ &\quad + \sum_k \left(\frac{\partial h_i}{\partial q_k} A_{jk} - \frac{\partial h_j}{\partial q_k} A_{ik} \right). \end{aligned}$$

The coefficients of p_m cancel. To see the cancellation rather than merely assert it, use

$$A_{jk} = \frac{\partial q_k}{\partial q'_j}$$

and the chain rule:

$$\begin{aligned} \sum_k \frac{\partial A_{im}}{\partial q_k} A_{jk} &= \sum_k \frac{\partial}{\partial q_k} \left(\frac{\partial q_m}{\partial q'_i} \right) \frac{\partial q_k}{\partial q'_j} \\ &= \frac{\partial}{\partial q'_j} \left(\frac{\partial q_m}{\partial q'_i} \right) = \frac{\partial^2 q_m}{\partial q'_j \partial q'_i}. \end{aligned}$$

Interchanging i and j gives the same result because mixed partial derivatives commute. Thus

only the h -terms remain:

$$\begin{aligned} 0 = \{p'_i, p'_j\} &= \sum_k \left(\frac{\partial h_i}{\partial q_k} \frac{\partial q_k}{\partial q'_j} - \frac{\partial h_j}{\partial q_k} \frac{\partial q_k}{\partial q'_i} \right) \\ &= \frac{\partial h_i}{\partial q'_j} - \frac{\partial h_j}{\partial q'_i}. \end{aligned}$$

This is precisely the local integrability condition for the one-form $\sum_i h_i dq'_i$. On a simply connected coordinate neighbourhood, there is a scalar function S such that

$$h_i = \frac{\partial S}{\partial q'_i}, \quad h = \nabla_{q'} S.$$

If the same scalar is expressed as a function of q , the chain rule gives

$$h_i = \sum_k \frac{\partial q_k}{\partial q'_i} \frac{\partial S}{\partial q_k} = [A(q) \nabla_q S(q)]_i.$$

Combining the results gives the complete local answer.

Theorem 8.2: Canonical point transformations

Let f be a local diffeomorphism of configuration space. The most general local canonical transformation whose new coordinates depend only on the old coordinates is

$$\begin{aligned} q' &= f(q), \\ p' &= \left(\frac{\partial q}{\partial q'} \right)^T [p + \nabla_q S(q)], \end{aligned}$$

where S is an arbitrary smooth scalar function. Equivalently,

$$p' = \left(\frac{\partial q'}{\partial q} \right)^{-T} [p + \nabla_q S(q)].$$

Local versus global statement

The equation $\partial h_i / \partial q'_j = \partial h_j / \partial q'_i$ guarantees a local potential S . A single global S exists under an additional topological hypothesis, for example on a simply connected domain. The handwritten computation is local, as are ordinary coordinate charts.

8.10 Reducing a general skew form to the standard form

Suppose the equations of motion are written with an arbitrary constant real nondegenerate skew-symmetric matrix Γ :

$$\dot{x} = \Gamma \nabla_x H(x). \quad (8.14)$$

By the uniqueness-up-to-congruence lemma, there is an invertible constant matrix Q such that

$$\Gamma = Q^T J Q.$$

Define new coordinates

$$y = Q^{-T} x, \quad \text{so that} \quad x = Q^T y.$$

If $H_Y(y) = H_X(Q^T y)$, then

$$\frac{\partial H_X}{\partial x_i} = \sum_j \frac{\partial H_Y}{\partial y_j} \frac{\partial y_j}{\partial x_i} = \sum_j (Q^{-T})_{ji} \frac{\partial H_Y}{\partial y_j},$$

and hence

$$\nabla_x H_X = Q^{-1} \nabla_y H_Y.$$

Using $\dot{y} = Q^{-T} \dot{x}$ in (8.14), we find

$$\begin{aligned} \dot{y} &= Q^{-T} \Gamma \nabla_x H_X \\ &= Q^{-T} (Q^T J Q) Q^{-1} \nabla_y H_Y \\ &= J \nabla_y H_Y. \end{aligned}$$

Thus every constant nondegenerate skew form can be brought to the standard Hamiltonian form by a linear change of variables.

8.11 Equations of motion as Poisson-bracket equations

For the coordinate function x_i , its gradient is the i -th standard basis vector e_i . Therefore

$$\{x_i, H\} = e_i^T J \nabla H = [J \nabla H]_i.$$

Hamilton's equation $\dot{x} = J \nabla H$ may consequently be written as

$$\boxed{\dot{x}_i = \{x_i, H\}, \quad \text{or symbolically} \quad \dot{x} = \{x, H\}.} \quad (8.15)$$

For $i = 1, \dots, n$, this reads

$$\dot{q}_i = \{q_i, H\} = \frac{\partial H}{\partial p_i},$$

whereas for the momentum coordinate,

$$\dot{p}_i = \{p_i, H\} = -\frac{\partial H}{\partial q_i}.$$

Thus the bracket notation contains both families of Hamilton equations without losing any component information.

8.12 Geometric discussion and symmetry groups

Let $y = f(x)$ be canonical. If $x(t)$ solves

$$\dot{x} = J\nabla_x H_X(x),$$

then $y(t) = f(x(t))$ solves

$$\dot{y} = J\nabla_y H_Y(y).$$

The physical trajectory has not changed: it has merely been represented in different coordinates. One may therefore choose whichever representation makes the Hamiltonian or the equations easiest to analyze.

The invariant structure can be described in three equivalent languages:

1. the form of Hamilton's equations;
2. all Poisson brackets;
3. the antisymmetric bilinear symplectic form

$$\Omega(u, v) = u^T J v.$$

With the tangent-vector convention, preservation of Ω means

$$\Omega(Su, Sv) = u^T S^T J S v = u^T J v,$$

or $S^T J S = J$.

Definition 8.4: The full symplectomorphism group

The group of smooth canonical transformations of $\mathbb{R}^{2n}$ is

$$\text{Symp}(\mathbb{R}^{2n}, J) = \left\{ f : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n} : \begin{array}{l} f \text{ is a smooth diffeomorphism,} \\ Df(x)^T J Df(x) = J \text{ for every } x \end{array} \right\}.$$

This is an extremely large, infinite-dimensional Lie group. Its linear canonical subgroup is

$$\text{Sp}(2n, \mathbb{R}) = \{S \in GL(2n, \mathbb{R}) : S^T J S = J\} \subset \text{Symp}(\mathbb{R}^{2n}, J),$$

where the associated transformation is simply $f(x) = Sx$ and $Df(x) = S$ is constant.

Representation symmetry is not automatically a conservation law

A canonical coordinate transformation preserves the mathematical structure used to represent the dynamics. This alone does not say that a particular Hamiltonian is unchanged, so it does not by itself give a conserved quantity. A conservation law requires an actual symmetry of the Hamiltonian (together with the appropriate continuous one-parameter structure). Canonical symmetry in classical mechanics plays a role analogous to unitary changes of representation in quantum mechanics.

What to retain from this chapter

1. Hamilton's component equations combine into $\dot{x} = J\nabla H$, with $J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$.
2. A phase-space diffeomorphism is canonical precisely when its pointwise Jacobian is symplectic; equivalently, it preserves all canonical Poisson brackets.
3. The linear symplectic group is $\text{Sp}(2n, \mathbb{R}) = \{A : A^T J A = J\}$, and every one of its matrices has determinant 1.
4. Every real nondegenerate skew-symmetric matrix has even size and can be reduced to 2×2 skew blocks by an orthogonal basis. All resulting symplectic groups are isomorphic.
5. The spectrum of a real symplectic matrix is closed under complex conjugation and reciprocation, including algebraic multiplicity.
6. For $q' = f(q)$, every local canonical lift to phase space has

$$p' = \left(\frac{\partial q}{\partial q'} \right)^T [p + \nabla_q S(q)].$$

7. Canonical changes preserve the representation-independent Hamiltonian structure, but a change of representation is not by itself a physical conservation law.

Direct Products of Groups

Road map

This chapter begins with the First Isomorphism Theorem, which explains why a kernel naturally produces a quotient. We then construct the external direct product $G_1 \times G_2$, analyze its presentations and canonical factor subgroups, and prove the internal direct-product criterion. The final sections determine exactly when a product of finite cyclic groups is cyclic and record how subgroups, quotients, conjugacy classes, centralizers, and centers behave under products.

9.1 From a homomorphism to a quotient

Let

$$\varphi : G \longrightarrow G'$$

be a group homomorphism. Its kernel is a normal subgroup of G :

$$\ker \varphi \trianglelefteq G.$$

The quotient $G/\ker \varphi$ is therefore defined. Recall also that if $H \trianglelefteq G$, then

$$(aH)^{-1} = a^{-1}H,$$

because

$$(aH)(a^{-1}H) = aa^{-1}H = H,$$

the identity element of G/H .

The natural question is whether the quotient by the kernel recovers the part of G' actually reached by φ . The answer is yes.

Theorem 9.1: First Isomorphism Theorem

Let $\varphi : G \rightarrow G'$ be a group homomorphism. Define

$$\overline{\varphi} : G/\ker \varphi \longrightarrow \operatorname{im} \varphi, \quad \overline{\varphi}(a \ker \varphi) = \varphi(a).$$

Then $\bar{\varphi}$ is an isomorphism. Hence

$$G/\ker \varphi \cong \text{im } \varphi.$$

证明. We check every required point.

Homomorphism. For $a, b \in G$,

$$\begin{aligned} \bar{\varphi}((a \ker \varphi)(b \ker \varphi)) &= \bar{\varphi}(ab \ker \varphi) \\ &= \varphi(ab) \\ &= \varphi(a)\varphi(b) \\ &= \bar{\varphi}(a \ker \varphi)\bar{\varphi}(b \ker \varphi). \end{aligned}$$

Well-definedness and injectivity. For $a, b \in G$,

$$\begin{aligned} a \ker \varphi = b \ker \varphi &\iff a^{-1}b \in \ker \varphi \\ &\iff \varphi(a^{-1}b) = e_{G'} \\ &\iff \varphi(a)^{-1}\varphi(b) = e_{G'} \\ &\iff \varphi(a) = \varphi(b). \end{aligned}$$

The forward implication shows that the displayed definition does not depend on the representative of the coset. The reverse implication shows that two cosets with the same image are equal, so $\bar{\varphi}$ is injective.

Surjectivity onto the image. If $a' \in \text{im } \varphi$, then $a' = \varphi(a)$ for some $a \in G$, and

$$\bar{\varphi}(a \ker \varphi) = a'.$$

Thus $\bar{\varphi}$ is surjective onto $\text{im } \varphi$, completing the proof. ■

9.2 The external direct product

Definition 9.1: Direct product of two groups

Let G_1 and G_2 be groups. Their Cartesian product is

$$G_1 \times G_2 = \{(a_1, a_2) : a_1 \in G_1, a_2 \in G_2\}.$$

Define multiplication component by component:

$$\begin{aligned} \mu : (G_1 \times G_2) \times (G_1 \times G_2) &\longrightarrow G_1 \times G_2, \\ ((a_1, a_2), (b_1, b_2)) &\longmapsto (a_1 b_1, a_2 b_2). \end{aligned}$$

The resulting group is called the *external direct product* of G_1 and G_2 .

Proposition 9.1

The componentwise multiplication above makes $G_1 \times G_2$ a group. Its identity and inverse are

$$e_{G_1 \times G_2} = (e_1, e_2), \quad (a_1, a_2)^{-1} = (a_1^{-1}, a_2^{-1}).$$

证明. For three elements,

$$\begin{aligned} ((a_1, a_2)(b_1, b_2))(c_1, c_2) &= (a_1 b_1, a_2 b_2)(c_1, c_2) \\ &= ((a_1 b_1)c_1, (a_2 b_2)c_2) \\ &= (a_1(b_1 c_1), a_2(b_2 c_2)) \\ &= (a_1, a_2)((b_1, b_2)(c_1, c_2)), \end{aligned}$$

so associativity follows from associativity in each factor. Also,

$$(e_1, e_2)(a_1, a_2) = (a_1, a_2) = (a_1, a_2)(e_1, e_2),$$

and

$$(a_1, a_2)(a_1^{-1}, a_2^{-1}) = (e_1, e_2) = (a_1^{-1}, a_2^{-1})(a_1, a_2).$$

■

9.2.1 Immediate properties

1. If both groups are finite, then

$$|G_1 \times G_2| = |G_1| |G_2|.$$

This is the ordinary counting rule for a Cartesian product.

2. The product $G_1 \times G_2$ is Abelian if and only if both G_1 and G_2 are Abelian. Indeed,

$$(a_1, a_2)(b_1, b_2) = (b_1, b_2)(a_1, a_2)$$

for all pairs exactly when $a_1 b_1 = b_1 a_1$ and $a_2 b_2 = b_2 a_2$ for all components.

3. The sets $G_1 \times G_2$ and $G_2 \times G_1$ are generally not literally equal, but the swap map

$$(a_1, a_2) \mapsto (a_2, a_1)$$

is an isomorphism. Hence $G_1 \times G_2 \cong G_2 \times G_1$.

4. Direct products are associative up to the natural isomorphism

$$((a_1, a_2), a_3) \mapsto (a_1, (a_2, a_3)).$$

Thus one normally writes $G_1 \times G_2 \times G_3$ without parentheses.

Example 9.1: $\mathbb{Z} \times \mathbb{Z}$

In additive notation,

$$\mathbb{Z} \times \mathbb{Z} = \{(m_1, m_2) : m_1, m_2 \in \mathbb{Z}\},$$

with

$$(m_1, m_2) + (n_1, n_2) = (m_1 + n_1, m_2 + n_2).$$

This is the integer lattice in the plane under vector addition.

Example 9.2: $\mathbb{Z}_2 \times \mathbb{Z}_2$ and the Klein four group

Write the two copies of $\mathbb{Z}_2$ multiplicatively as

$$\mathbb{Z}_2^{(1)} = \{e_1, a_1\}, \quad \mathbb{Z}_2^{(2)} = \{e_2, a_2\},$$

with $a_1^2 = e_1$ and $a_2^2 = e_2$. The product has four elements:

$$(e_1, e_2), \quad (a_1, e_2), \quad (e_1, a_2), \quad (a_1, a_2).$$

Its Cayley table is

$\cdot$	(e_1, e_2)	(a_1, e_2)	(e_1, a_2)	(a_1, a_2)
(e_1, e_2)	(e_1, e_2)	(a_1, e_2)	(e_1, a_2)	(a_1, a_2)
(a_1, e_2)	(a_1, e_2)	(e_1, e_2)	(a_1, a_2)	(e_1, a_2)
(e_1, a_2)	(e_1, a_2)	(a_1, a_2)	(e_1, e_2)	(a_1, e_2)
(a_1, a_2)	(a_1, a_2)	(e_1, a_2)	(a_1, e_2)	(e_1, e_2)

Every nonidentity element has order 2. Under

$$(e_1, e_2) \mapsto e, \quad (a_1, e_2) \mapsto a, \quad (e_1, a_2) \mapsto b, \quad (a_1, a_2) \mapsto c,$$

this table becomes

$\cdot$	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

the Cayley table of the Klein four group $K_4 = V_4$. Therefore

$$\boxed{\mathbb{Z}_2 \times \mathbb{Z}_2 \cong K_4.}$$

Example 9.3: $\mathbb{Z}_3 \times \mathbb{Z}_2$

Let

$$G_1 = \langle a : a^3 = e_1 \rangle \cong \mathbb{Z}_3, \quad G_2 = \langle b : b^2 = e_2 \rangle \cong \mathbb{Z}_2.$$

Using the abbreviations

$$a = (a, e_2), \quad b = (e_1, b),$$

the product is

$$G_1 \times G_2 = \{e, a, a^2, b, ab, a^2b\} = \langle a, b : a^3 = e, b^2 = e, ab = ba \rangle.$$

The relation $ab = ba$ is necessary because the two canonical factors commute. The element

a^2b generates the entire product:

$$\begin{array}{lll} (a^2b)^1 = a^2b, & (a^2b)^2 = a, & (a^2b)^3 = b, \\ (a^2b)^4 = a^2, & (a^2b)^5 = ab, & (a^2b)^6 = e. \end{array}$$

Hence

$$\mathbb{Z}_3 \times \mathbb{Z}_2 \cong \mathbb{Z}_6 \cong \mathbb{Z}_2 \times \mathbb{Z}_3.$$

9.3 Presentations of direct products

Suppose

$$G_1 = \langle S_1 : R_1 \rangle, \quad G_2 = \langle S_2 : R_2 \rangle,$$

where the generating sets S_1 and S_2 are disjoint. In the direct product, every generator from the first factor commutes with every generator from the second. Thus

$$G_1 \times G_2 = \langle S_1 \cup S_2 : R_1 \cup R_2 \cup \{st = ts : s \in S_1, t \in S_2\} \rangle.$$

The original relations R_1 and R_2 control each factor, while the additional cross-relations force the two factors to commute.

9.4 The canonical factor subgroups

Let e_1, e_2 be the identity elements of G_1, G_2 . Define

$$G'_1 = G_1 \times \{e_2\} = \{(a_1, e_2) : a_1 \in G_1\},$$

$$G'_2 = \{e_1\} \times G_2 = \{(e_1, a_2) : a_2 \in G_2\}.$$

The maps

$$G_1 \longrightarrow G'_1, \quad a_1 \longmapsto (a_1, e_2),$$

and

$$G_2 \longrightarrow G'_2, \quad a_2 \longmapsto (e_1, a_2),$$

are isomorphisms. Hence $G'_1 \cong G_1$ and $G'_2 \cong G_2$, and both are subgroups of $G_1 \times G_2$.

Theorem 9.2: Canonical factors are normal

The canonical factor subgroups satisfy

$$G'_1 \trianglelefteq G_1 \times G_2, \quad G'_2 \trianglelefteq G_1 \times G_2,$$

and

$$(G_1 \times G_2)/G'_1 \cong G_2, \quad (G_1 \times G_2)/G'_2 \cong G_1.$$

证明. Consider the projection

$$\pi_2 : G_1 \times G_2 \longrightarrow G_2, \quad \pi_2(a_1, a_2) = a_2.$$

It is a homomorphism because

$$\pi_2((a_1, a_2)(b_1, b_2)) = \pi_2(a_1b_1, a_2b_2) = a_2b_2 = \pi_2(a_1, a_2)\pi_2(b_1, b_2).$$

Its kernel is

$$\ker \pi_2 = \{(a_1, e_2) : a_1 \in G_1\} = G'_1,$$

so $G'_1 \trianglelefteq G_1 \times G_2$. Since π_2 is surjective, the First Isomorphism Theorem gives

$$(G_1 \times G_2)/G'_1 \cong G_2.$$

Interchanging the roles of the two factors proves the other two statements. ■

It is instructive to see the quotient cosets directly. For $(a_1, a_2) \in G_1 \times G_2$,

$$\begin{aligned} (a_1, a_2)G'_1 &= \{(a_1, a_2)(g_1, e_2) : g_1 \in G_1\} \\ &= \{(a_1g_1, a_2) : g_1 \in G_1\} \\ &= G_1 \times \{a_2\}. \end{aligned}$$

This coset depends only on a_2 , not on a_1 . The map

$$(a_1, a_2)G'_1 \mapsto a_2$$

is therefore well defined. It is injective because equality of the two cosets is equivalent to equality of their second components, it is surjective because every $a_2 \in G_2$ occurs, and it is a homomorphism because coset multiplication multiplies the second components.

9.5 Recognizing an internal direct product

Inside $G_1 \times G_2$, the canonical copies have four conspicuous properties:

1. $G'_1 \cap G'_2 = \{(e_1, e_2)\}$;
2. $G'_1 G'_2 = G_1 \times G_2$;
3. every element of G'_1 commutes with every element of G'_2 ;
4. G'_1 and G'_2 are normal in $G_1 \times G_2$.

Indeed,

$$(a_1, e_2)(e_1, a_2) = (a_1, a_2) = (e_1, a_2)(a_1, e_2).$$

The next theorem shows that these properties characterize direct-product decompositions inside an arbitrary group.

Theorem 9.3: Internal direct-product criterion

Let $H, K \leq G$. Suppose

$$H \cap K = \{e\}, \quad HK = G, \quad H \trianglelefteq G, \quad K \trianglelefteq G.$$

Then every $h \in H$ commutes with every $k \in K$, and

$$\kappa : H \times K \longrightarrow G, \quad \kappa(h, k) = hk,$$

is an isomorphism. Thus

$$G \cong H \times K.$$

证明. We first prove the elementwise commutation that is needed for κ to be a homomorphism. Let $h \in H$ and $k \in K$. Normality of H gives

$$kh^{-1}k^{-1} \in H,$$

and hence

$$(hk)(kh)^{-1} = hkh^{-1}k^{-1} \in H.$$

Normality of K similarly gives

$$hkh^{-1} \in K,$$

so

$$(hk)(kh)^{-1} = hkh^{-1}k^{-1} \in K.$$

Therefore

$$(hk)(kh)^{-1} \in H \cap K = \{e\}.$$

It follows that $hk = kh$.

Now take $(h_1, k_1), (h_2, k_2) \in H \times K$. Using the commutation just proved,

$$\begin{aligned} \kappa((h_1, k_1)(h_2, k_2)) &= \kappa(h_1h_2, k_1k_2) \\ &= h_1h_2k_1k_2 \\ &= h_1k_1h_2k_2 \\ &= \kappa(h_1, k_1)\kappa(h_2, k_2). \end{aligned}$$

Thus κ is a homomorphism.

The equality $HK = G$ says that every $g \in G$ has the form $g = hk$, so κ is surjective. Finally, if $(h, k) \in \ker \kappa$, then $hk = e$, and therefore

$$k = h^{-1} \in H.$$

Since also $k \in K$, we have $k \in H \cap K = \{e\}$, so $k = e$ and $h = e$. Hence

$$\ker \kappa = \{(e, e)\},$$

and κ is injective. ■

Proposition 9.2: Equivalent commuting version

In the internal direct-product criterion, the two normality assumptions may be replaced by

$$hk = kh \quad (h \in H, k \in K).$$

Thus either of the following sets of assumptions is sufficient:

$$H \cap K = \{e\}, \quad HK = G, \quad H, K \trianglelefteq G,$$

or

$$H \cap K = \{e\}, \quad HK = G, \quad [H, K] = \{e\}.$$

证明. Only the normality implication needs proof. Assume $HK = G$ and that elements of H commute with elements of K . Write an arbitrary $g \in G$ as $g = h_0 k_0$. Then

$$\begin{aligned} gHg^{-1} &= h_0 k_0 H k_0^{-1} h_0^{-1} \\ &= h_0 H h_0^{-1} \\ &= H, \end{aligned}$$

because k_0 commutes with every element of H , and a subgroup is invariant under conjugation by its own elements. Hence $H \trianglelefteq G$. The same argument gives $K \trianglelefteq G$. ■

Example 9.4: An internal decomposition of K_4

Using the table above, let

$$H = \{e, a\} = \langle a \rangle, \quad K = \{e, b\} = \langle b \rangle.$$

The Klein group is Abelian, so $H, K \trianglelefteq K_4$. Moreover,

$$H \cap K = \{e\}, \quad HK = \{e, a, b, ab\} = \{e, a, b, c\} = K_4.$$

The internal direct-product criterion gives

$$K_4 \cong H \times K \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

9.6 Products of finite cyclic groups

Let

$$\mathbb{Z}_{m_1} = \langle g_1 \rangle, \quad \mathbb{Z}_{m_2} = \langle g_2 \rangle,$$

where $g_1^{m_1} = e_1$ and $g_2^{m_2} = e_2$. Then

$$|\mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2}| = m_1 m_2.$$

For any positive integer k ,

$$(g_1, g_2)^k = (e_1, e_2)$$

if and only if $m_1 \mid k$ and $m_2 \mid k$. Therefore

$$\text{ord}(g_1, g_2) = \text{lcm}(m_1, m_2). \tag{9.1}$$

More generally,

$$\text{ord}(x_1, x_2) = \text{lcm}(\text{ord } x_1, \text{ord } x_2).$$

Theorem 9.4: When a product of two cyclic groups is cyclic

$$\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn} \iff \gcd(m, n) = 1.$$

证明. If $\gcd(m, n) = 1$, then

$$\text{lcm}(m, n) = mn.$$

By (9.1), the element (g_1, g_2) has order mn , which equals the order of the entire product. It therefore generates the product, and the product is cyclic of order mn .

Conversely, if the product is cyclic, it contains an element of order mn . Every element (x_1, x_2) has order

$$\text{lcm}(\text{ord } x_1, \text{ord } x_2) \leq \text{lcm}(m, n).$$

Thus $mn \leq \text{lcm}(m, n)$. Since

$$\text{lcm}(m, n) = \frac{mn}{\gcd(m, n)} \leq mn,$$

equality must hold and $\gcd(m, n) = 1$. ■

In additive quotient notation, the explicit isomorphism in the coprime case is

$$\Phi : \mathbb{Z}/(mn)\mathbb{Z} \longrightarrow \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}, \quad [r]_{mn} \mapsto ([r]_m, [r]_n).$$

Its kernel consists of integers divisible by both m and n . When $\gcd(m, n) = 1$, this means divisibility by mn , so the induced map is injective; the finite groups on the two sides have the same order, so it is also surjective. This is the group-theoretic form of the Chinese Remainder Theorem.

Proposition 9.3: Several cyclic factors

If $m_1, \dots, m_r$ are pairwise coprime, then

$$\mathbb{Z}_{m_1} \times \cdots \times \mathbb{Z}_{m_r} \cong \mathbb{Z}_{m_1 \cdots m_r}.$$

证明. If g_i generates $\mathbb{Z}_{m_i}$, then

$$\text{ord}(g_1, \dots, g_r) = \text{lcm}(m_1, \dots, m_r).$$

Pairwise coprimality makes this least common multiple equal to the product $m_1 \cdots m_r$, which is also the order of the whole direct product. ■

Example 9.5

Because 3 and 4 are coprime,

$$\mathbb{Z}_{12} \cong \mathbb{Z}_3 \times \mathbb{Z}_4.$$

On the other hand,

$$\mathbb{Z}_4 \not\cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

The group $\mathbb{Z}_4$ contains an element of order 4, whereas every nonidentity element of $\mathbb{Z}_2 \times \mathbb{Z}_2$ has order 2.

9.7 Further properties of direct products

9.7.1 Subgroups, normal subgroups, and quotients

If $H_1 \leq G_1$ and $H_2 \leq G_2$, then

$$H_1 \times H_2 \leq G_1 \times G_2.$$

Closure and inverses are checked componentwise. If, in addition,

$$H_1 \trianglelefteq G_1, \quad H_2 \trianglelefteq G_2,$$

then

$$H_1 \times H_2 \trianglelefteq G_1 \times G_2,$$

because

$$\begin{aligned} (g_1, g_2)(h_1, h_2)(g_1, g_2)^{-1} \\ = (g_1 h_1 g_1^{-1}, g_2 h_2 g_2^{-1}) \in H_1 \times H_2. \end{aligned}$$

Proposition 9.4: A quotient of a product

If $H_i \trianglelefteq G_i$ for $i = 1, 2$, then

$$\frac{G_1 \times G_2}{H_1 \times H_2} \cong \frac{G_1}{H_1} \times \frac{G_2}{H_2}.$$

证明. Define

$$\Psi : G_1 \times G_2 \longrightarrow G_1/H_1 \times G_2/H_2, \quad \Psi(g_1, g_2) = (g_1 H_1, g_2 H_2).$$

It is a surjective homomorphism, and

$$\begin{aligned} \ker \Psi &= \{(g_1, g_2) : g_1 \in H_1, g_2 \in H_2\} \\ &= H_1 \times H_2. \end{aligned}$$

The result follows from the First Isomorphism Theorem. ■

Not every subgroup is a product subgroup

For any group G , the diagonal

$$\Delta_G = \{(a, a) : a \in G\}$$

is a subgroup of $G \times G$, because

$$(a, a)(b, b) = (ab, ab), \quad (a, a)^{-1} = (a^{-1}, a^{-1}).$$

Except in trivial cases, it is not of the form $H \times H$: a product subgroup permits independent choices in the two coordinates, whereas the diagonal requires them to be equal.

9.7.2 Conjugacy classes and centralizers

Conjugation in a direct product is componentwise:

$$\begin{aligned} (g_1, g_2)(a_1, a_2)(g_1, g_2)^{-1} \\ = (g_1 a_1 g_1^{-1}, g_2 a_2 g_2^{-1}). \end{aligned}$$

It follows that each conjugacy class is a Cartesian product:

$$\text{Cl}_{G_1 \times G_2}(a_1, a_2) = \text{Cl}_{G_1}(a_1) \times \text{Cl}_{G_2}(a_2).$$

Similarly, (g_1, g_2) commutes with (a_1, a_2) exactly when g_1 commutes with a_1 and g_2 commutes with a_2 . Hence

$$\text{Cen}_{G_1 \times G_2}(a_1, a_2) = \text{Cen}_{G_1}(a_1) \times \text{Cen}_{G_2}(a_2).$$

Intersecting the centralizers of all elements gives the center:

$$\boxed{Z(G_1 \times G_2) = Z(G_1) \times Z(G_2).}$$

9.7.3 Products of more than two groups

For groups $G_1, \dots, G_r$, define

$$G_1 \times \cdots \times G_r = \{(a_1, \dots, a_r) : a_i \in G_i \text{ for } i = 1, \dots, r\},$$

with componentwise multiplication

$$(a_1, \dots, a_r)(b_1, \dots, b_r) = (a_1 b_1, \dots, a_r b_r).$$

Its identity and inverses are also componentwise:

$$e = (e_1, \dots, e_r), \quad (a_1, \dots, a_r)^{-1} = (a_1^{-1}, \dots, a_r^{-1}).$$

What to retain from this chapter

1. The First Isomorphism Theorem states $G/\ker \varphi \cong \text{im } \varphi$.
2. The direct product $G_1 \times G_2$ uses componentwise multiplication, identity, and inverses; its order is $|G_1||G_2|$ when the factors are finite.
3. The canonical factor copies are normal, and quotienting by one recovers the other factor.
4. If $H, K \trianglelefteq G$, $H \cap K = \{e\}$, and $HK = G$, then the factors commute and $G \cong H \times K$. Elementwise commutation can replace the two normality assumptions.
5. The product $\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic of order mn exactly when $\gcd(m, n) = 1$.
6. Product subgroups and quotient groups behave componentwise, but subgroups such as the diagonal need not themselves be products.
7. Conjugacy classes, centralizers, and centers of direct products are the Cartesian products of the corresponding structures in the factors.

Matrix Groups

Road map

Matrix groups provide the central examples for Lie theory. We define the general and special linear groups, the orthogonal and unitary groups and their special subgroups, and briefly recall the symplectic group. Determinant homomorphisms identify the basic quotient groups. We then analyze scalar multiplication maps in detail, obtaining decompositions of $GL(n, \mathbb{R})$, $U(n)$, and $GL(n, \mathbb{C})$, with separate conclusions in odd and even real dimensions.

10.1 Fields, matrices, and characteristic

Throughout this chapter,

$$K = \mathbb{R} \quad \text{or} \quad K = \mathbb{C}.$$

Both fields have characteristic zero. In general, the *characteristic* of a field K , denoted $\text{char } K$, is the least positive integer m for which

$$\underbrace{1_K + \cdots + 1_K}_{m \text{ terms}} = 0_K,$$

if such an integer exists; otherwise the characteristic is 0. Thus no positive multiple of 1 vanishes in $\mathbb{R}$ or $\mathbb{C}$.

Let

$$\text{Mat}(n, K)$$

be the set of all $n \times n$ matrices with entries in K . Under matrix multiplication this set is not a group, because a singular matrix has no multiplicative inverse.

10.2 The general and special linear groups

Definition 10.1: General linear group

$$GL(n, K) = \{A \in \text{Mat}(n, K) : \det A \neq 0\}.$$

It is called the *general linear group of degree n over K* .

Proposition 10.1

$GL(n, K)$ is a group under matrix multiplication.

证明. Associativity is inherited from matrix multiplication. If $A, B \in GL(n, K)$, then

$$\det(AB) = \det A \det B \neq 0,$$

so $AB \in GL(n, K)$. The identity matrix has determinant 1, and therefore lies in the group. Finally,

$$\det A \neq 0 \iff A \text{ is invertible,}$$

and

$$\det(A^{-1}) = (\det A)^{-1} \neq 0.$$

Thus $A^{-1} \in GL(n, K)$. ■

Definition 10.2: Special linear group

$$SL(n, K) = \{A \in GL(n, K) : \det A = 1\}.$$

It is called the *special linear group of degree n over K* .

The word “special” records the additional determinant-one condition. The subgroup inclusions are

$$\begin{array}{ccc} SL(n, \mathbb{R}) & < & GL(n, \mathbb{R}), \\ \cap & & \cap \\ SL(n, \mathbb{C}) & < & GL(n, \mathbb{C}), \end{array}$$

where the vertical inclusions regard a real matrix as a complex matrix.

10.2.1 The determinant homomorphism

Let

$$K^\times = K \setminus \{0\}$$

be the multiplicative group of nonzero field elements.

Theorem 10.1

$$SL(n, K) \trianglelefteq GL(n, K)$$

and

$$GL(n, K)/SL(n, K) \cong K^\times.$$

证明. Consider

$$\det : GL(n, K) \longrightarrow K^\times.$$

The target is a group under multiplication, with identity 1 and inverse $\lambda^{-1} = 1/\lambda$. The determinant is a homomorphism:

$$\det(AB) = \det A \det B.$$

Its kernel is exactly

$$\ker(\det) = \{A \in GL(n, K) : \det A = 1\} = SL(n, K).$$

Therefore $SL(n, K)$ is normal.

The determinant map is surjective. Given $\lambda \in K^\times$, the diagonal matrix

$$D_\lambda = \text{diag}(1, \dots, 1, \lambda)$$

is invertible and satisfies $\det D_\lambda = \lambda$. Hence the First Isomorphism Theorem gives

$$GL(n, K)/SL(n, K) \cong K^\times.$$

■

10.2.2 The quotient isomorphism written directly

The induced map is

$$\tau : GL(n, K)/SL(n, K) \longrightarrow K^\times, \quad \tau(A SL(n, K)) = \det A.$$

It is a homomorphism because

$$\begin{aligned} \tau((A SL)(B SL)) &= \tau(AB SL) \\ &= \det(AB) \\ &= \det A \det B \\ &= \tau(A SL)\tau(B SL). \end{aligned}$$

It is well defined and injective because

$$\begin{aligned} \tau(A SL) = \tau(B SL) &\iff \det A = \det B \\ &\iff \det(A^{-1}B) = 1 \\ &\iff A^{-1}B \in SL(n, K) \\ &\iff A SL(n, K) = B SL(n, K). \end{aligned}$$

Finally, the matrices D_λ above show directly that τ is surjective.

10.3 Orthogonal, unitary, and symplectic groups

For a complex matrix A , write

$$A^\dagger = \overline{A}^T$$

for its conjugate transpose.

Definition 10.3: Orthogonal group

$$O(n) = \{A \in GL(n, \mathbb{R}) : AA^T = I_n\}.$$

The equation $AA^T = I$ is equivalent to $A^{-1} = A^T$, and also to $A^T A = I$. If $A, B \in O(n)$, then

$$(AB)(AB)^T = ABB^T A^T = AA^T = I,$$

and $A^{-1} = A^T \in O(n)$. Thus $O(n)$ is a group.

Definition 10.4: Unitary group

$$U(n) = \{A \in GL(n, \mathbb{C}) : AA^\dagger = I_n\}.$$

The same calculation, with T replaced by $\dagger$, proves that $U(n)$ is a group. A unitary matrix satisfies

$$A^{-1} = A^\dagger.$$

Definition 10.5: Symplectic group

Let Γ be a fixed real nondegenerate skew-symmetric $2n \times 2n$ matrix. Then

$$\mathrm{Sp}(2n, \mathbb{R}) = \{A \in GL(2n, \mathbb{R}) : A^T \Gamma A = \Gamma\}.$$

As proved in Chapter 8, this is a group, its isomorphism type is independent of the chosen nondegenerate skew form, and

$$\det A = 1 \quad (A \in \mathrm{Sp}(2n, \mathbb{R})).$$

The symplectic group is the structure-preserving linear group of classical Hamiltonian dynamics.

10.4 Special orthogonal and special unitary groups

10.4.1 The orthogonal determinant

If $A \in O(n)$, then

$$1 = \det(AA^T) = \det A \det A^T = (\det A)^2.$$

Thus $\det A = \pm 1$, and the determinant restricts to a homomorphism

$$\det : O(n) \longrightarrow \{+1, -1\} \cong \mathbb{Z}_2.$$

Definition 10.6: Special orthogonal group

$$SO(n) = \{A \in O(n) : \det A = 1\}.$$

The kernel of the restricted determinant is $SO(n)$. The map is surjective because

$$\mathrm{diag}(-1, 1, \dots, 1) \in O(n)$$

has determinant -1 . Therefore

$$SO(n) \trianglelefteq O(n), \quad \boxed{O(n)/SO(n) \cong \mathbb{Z}_2.}$$

10.4.2 The unitary determinant

If $A \in U(n)$, then

$$1 = \det(AA^\dagger) = \det A \det A^\dagger = \det A \overline{\det A} = |\det A|^2.$$

Hence $\det A$ lies on the unit circle

$$U(1) = \{e^{i\theta} : 0 \leq \theta < 2\pi\}.$$

Definition 10.7: Special unitary group

$$SU(n) = \{A \in U(n) : \det A = 1\}.$$

The determinant homomorphism

$$\det : U(n) \longrightarrow U(1)$$

is surjective: for every $e^{i\theta} \in U(1)$, the diagonal matrix

$$\text{diag}(1, \dots, 1, e^{i\theta})$$

is unitary and has that determinant. Its kernel is $SU(n)$. Thus

$$SU(n) \trianglelefteq U(n), \quad \boxed{U(n)/SU(n) \cong U(1)}.$$

10.5 Scalar matrices and real general linear groups

Let

$$\overline{\mathbb{R}^\times} = \{\lambda I_n : \lambda \in \mathbb{R}^\times\} \leq GL(n, \mathbb{R}).$$

This scalar subgroup is isomorphic to $\mathbb{R}^\times$ and is central, hence normal.

10.5.1 Odd dimension

Theorem 10.2: Odd-dimensional real decomposition

If n is odd, then

$$\boxed{GL(n, \mathbb{R}) \cong SL(n, \mathbb{R}) \times \mathbb{R}^\times}.$$

More precisely, the multiplication map

$$\kappa : SL(n, \mathbb{R}) \times \mathbb{R}^\times \longrightarrow GL(n, \mathbb{R}), \quad \kappa(A, \lambda) = \lambda A,$$

is an isomorphism.

证明. Scalar matrices commute with every matrix, so

$$\kappa((A, \lambda)(B, \mu)) = \lambda\mu AB = (\lambda A)(\mu B) = \kappa(A, \lambda)\kappa(B, \mu).$$

Thus κ is a homomorphism.

If $\kappa(A, \lambda) = I$, then $A = \lambda^{-1}I$. Since $A \in SL(n, \mathbb{R})$,

$$1 = \det A = \det(\lambda^{-1}I) = \lambda^{-n}.$$

For odd n , the real equation $\lambda^n = 1$ has the unique solution $\lambda = 1$. Hence $A = I$, and the kernel is trivial.

Finally, let $B \in GL(n, \mathbb{R})$. Because n is odd, the real number

$$\lambda = (\det B)^{1/n}$$

exists and is nonzero even when $\det B < 0$. Set

$$A = \lambda^{-1}B.$$

Then

$$\det A = \lambda^{-n} \det B = 1,$$

so $A \in SL(n, \mathbb{R})$, and $\kappa(A, \lambda) = B$. Thus κ is surjective and therefore an isomorphism. ■

This is also an internal direct product:

$$SL(n, \mathbb{R}) \cap \overline{\mathbb{R}^\times} = \{I\}, \quad SL(n, \mathbb{R}) \overline{\mathbb{R}^\times} = GL(n, \mathbb{R})$$

when n is odd.

10.5.2 Even dimension

For even n , the same map is not onto all of $GL(n, \mathbb{R})$, because

$$\det(\lambda A) = \lambda^n \det A = \lambda^n > 0$$

when $A \in SL(n, \mathbb{R})$. Define

$$GL_+(n, \mathbb{R}) = \{B \in GL(n, \mathbb{R}) : \det B > 0\}.$$

Theorem 10.3: Even-dimensional positive-determinant decomposition

If n is even, then

$$\frac{SL(n, \mathbb{R}) \times \mathbb{R}^\times}{\mathbb{Z}_2} \cong GL_+(n, \mathbb{R}),$$

where the embedded $\mathbb{Z}_2$ is

$$\{(I, 1), (-I, -1)\}.$$

证明. Use the homomorphism

$$\kappa(A, \lambda) = \lambda A.$$

If $\kappa(A, \lambda) = I$, then $A = \lambda^{-1}I$ and

$$1 = \det A = \lambda^{-n}.$$

For real $\lambda \neq 0$ and even n , the equation $\lambda^n = 1$ has the two solutions $\lambda = \pm 1$. Thus

$$\ker \kappa = \{(I, 1), (-I, -1)\} \cong \mathbb{Z}_2.$$

We already observed that $\text{im } \kappa \subset GL_+(n, \mathbb{R})$. Conversely, let $B \in GL_+(n, \mathbb{R})$ and put

$$d = \det B > 0, \quad \lambda = d^{1/n} > 0, \quad A = \lambda^{-1}B.$$

Then $\det A = d^{-1} \det B = 1$, and $\kappa(A, \lambda) = B$. Therefore $\text{im } \kappa = GL_+(n, \mathbb{R})$. The First Isomorphism Theorem now gives the stated quotient. ■

Why this differs from $GL(n, \mathbb{R})/SL(n, \mathbb{R})$

For every n ,

$$GL(n, \mathbb{R})/SL(n, \mathbb{R}) \cong \mathbb{R}^\times$$

via the determinant. The scalar multiplication map is a different homomorphism. In even dimension its image misses all negative-determinant matrices and its kernel contains the simultaneous sign change $(A, \lambda) \mapsto (-A, -\lambda)$.

10.6 The relation between $U(n)$ and $SU(n) \times U(1)$

Define

$$\kappa : SU(n) \times U(1) \longrightarrow U(n), \quad \kappa(S, z) = zS,$$

where z acts as the scalar matrix zI_n .

Proposition 10.2

The map κ is a surjective homomorphism and

$$\ker \kappa = \left\{ \left(e^{-2\pi i k/n} I_n, e^{2\pi i k/n} \right) : k = 0, \dots, n-1 \right\} \cong \mathbb{Z}_n.$$

Consequently,

$$\frac{SU(n) \times U(1)}{\mathbb{Z}_n} \cong U(n).$$

证明. Because a scalar matrix commutes with every S ,

$$\begin{aligned} \kappa((S_1, z_1)(S_2, z_2)) &= \kappa(S_1 S_2, z_1 z_2) \\ &= z_1 z_2 S_1 S_2 \\ &= (z_1 S_1)(z_2 S_2), \end{aligned}$$

so κ is a homomorphism.

If $(S, z) \in \ker \kappa$, then $zS = I$, so

$$S = z^{-1}I.$$

The condition $\det S = 1$ becomes

$$1 = \det(z^{-1}I) = z^{-n},$$

or $z^n = 1$. The n solutions in $U(1)$ are

$$z = e^{2\pi i k/n}, \quad k = 0, \dots, n-1.$$

This gives the displayed kernel. Under multiplication, adding the indices modulo n multiplies the corresponding pairs, so the kernel is isomorphic to the additive cyclic group $\mathbb{Z}_n$.

To prove surjectivity, take $U \in U(n)$. Since $|\det U| = 1$, write

$$\det U = e^{i\theta} \quad (0 \leq \theta < 2\pi).$$

Set

$$z = e^{i\theta/n}, \quad S = z^{-1}U.$$

Then S is unitary, because multiplication by a unit-modulus scalar preserves unitarity, and

$$\det S = z^{-n} \det U = e^{-i\theta} e^{i\theta} = 1.$$

Thus $S \in SU(n)$ and $\kappa(S, z) = U$. The First Isomorphism Theorem finishes the proof. ■

Two related but different quotient statements

$$U(n)/SU(n) \cong U(1)$$

comes from the determinant $U(n) \rightarrow U(1)$, whereas

$$(SU(n) \times U(1))/\mathbb{Z}_n \cong U(n)$$

comes from multiplying a special unitary matrix by a scalar phase. The two statements describe different homomorphisms.

10.7 The complex general linear group

The same construction works with all nonzero complex scalars:

$$\kappa : SL(n, \mathbb{C}) \times \mathbb{C}^\times \longrightarrow GL(n, \mathbb{C}), \quad \kappa(A, z) = zA.$$

Theorem 10.4: Scalar decomposition over $\mathbb{C}$

The map κ is surjective and has kernel

$$\left\{ \left(e^{-2\pi i k/n} I_n, e^{2\pi i k/n} \right) : k = 0, \dots, n-1 \right\} \cong \mathbb{Z}_n.$$

Therefore

$$\frac{SL(n, \mathbb{C}) \times \mathbb{C}^\times}{\mathbb{Z}_n} \cong GL(n, \mathbb{C}).$$

证明. The scalar matrices are central, so κ is a homomorphism. If $zA = I$, then $A = z^{-1}I$, and

$$1 = \det A = z^{-n}.$$

Thus $z^n = 1$, giving exactly the displayed kernel.

For surjectivity, take $B \in GL(n, \mathbb{C})$. Write its nonzero determinant in polar form:

$$\det B = r e^{i\theta}, \quad r > 0, \quad 0 \leq \theta < 2\pi.$$

Choose one n -th root

$$z = r^{1/n} e^{i\theta/n}$$

and set $A = z^{-1}B$. Then

$$\det A = z^{-n} \det B = 1,$$

so $A \in SL(n, \mathbb{C})$, and $\kappa(A, z) = B$. The result follows from the First Isomorphism Theorem. ■

For comparison, the determinant gives the simpler quotient

$$GL(n, \mathbb{C})/SL(n, \mathbb{C}) \cong \mathbb{C}^\times.$$

Again, this is not the same quotient construction as the scalar multiplication theorem.

What to retain from this chapter

1. $GL(n, K)$ consists of the invertible $n \times n$ matrices, and $SL(n, K)$ is its determinant-one normal subgroup.
2. The determinant gives $GL(n, K)/SL(n, K) \cong K^\times$.
3. $O(n)$ and $U(n)$ preserve the real and complex standard inner products; $SO(n)$ and $SU(n)$ impose determinant 1.
4. $O(n)/SO(n) \cong \mathbb{Z}_2$ and $U(n)/SU(n) \cong U(1)$.
5. For odd n , $GL(n, \mathbb{R}) \cong SL(n, \mathbb{R}) \times \mathbb{R}^\times$. For even n , scalar multiplication instead gives

$$(SL(n, \mathbb{R}) \times \mathbb{R}^\times)/\mathbb{Z}_2 \cong GL_+(n, \mathbb{R}).$$

6. Scalar multiplication also gives

$$(SU(n) \times U(1))/\mathbb{Z}_n \cong U(n), \quad (SL(n, \mathbb{C}) \times \mathbb{C}^\times)/\mathbb{Z}_n \cong GL(n, \mathbb{C}).$$

7. The finite kernels $\mathbb{Z}_n$ arise from the n -th roots of unity and record the nonuniqueness of extracting a scalar n -th root of the determinant.

Gauge Invariance

Road map

Gauge theory distinguishes physical fields from redundant potentials used to describe them. Maxwell theory is the paradigmatic example. We first review Minkowski notation and express Maxwell's equations by the antisymmetric field tensor. Differential forms explain locally why a four-potential exists. The potential displays gauge freedom, permits Lorenz gauge fixing, and leads to a compact action principle. We finish by deriving the field equations from that action and explaining how global and local symmetries enter Noether's first and second theorems.

11.1 The language of gauge theory

The following terms must be kept conceptually distinct.

Gauge freedom

A description contains redundant degrees of freedom: more than one mathematical field represents the same physical state.

Gauge transformation

A transformation moves from one representative to another without changing the physical observables.

Gauge invariance

Observable fields and the physical predictions are unchanged under gauge transformations.

Gauge fixing

One chooses a representative from every equivalence class by imposing an additional condition.

Gauge group

The set of all gauge transformations, with composition as its group operation.

Gauge field

A connection-like field introduced so that an interacting theory can respect a local symmetry.

Maxwell electromagnetism makes all of these ideas explicit and is most transparent in relativistic Lagrangian form.

11.2 Minkowski spacetime and index notation

Use spacetime coordinates

$$x^\mu = (ct, x^1, x^2, x^3) = (ct, x, y, z), \quad \mu = 0, 1, 2, 3.$$

The underlying spacetime is denoted M , and in standard inertial coordinates it is $\mathbb{R}^4$. With signature $(+, -, -, -)$, the metric is

$$\eta_{\mu\nu} = \eta^{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

It lowers and raises indices:

$$x_\mu = \eta_{\mu\nu} x^\nu, \quad x^\mu = \eta^{\mu\nu} x_\nu.$$

Therefore

$$x_\mu = (ct, -x, -y, -z).$$

From now on, a repeated upper–lower index is summed from 0 to 3; this is the Einstein summation convention.

The covariant derivative components are

$$\partial_\mu := \frac{\partial}{\partial x^\mu} = \left(\frac{1}{c} \frac{\partial}{\partial t}, \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right),$$

and

$$\partial^\mu = \eta^{\mu\nu} \partial_\nu.$$

Under a Lorentz transformation $x \mapsto x'$,

$$x'^\mu = \Lambda^\mu_\nu x^\nu.$$

The metric is the structure preserved by Lorentz transformations:

$$\Lambda^T \eta \Lambda = \eta.$$

11.3 Maxwell theory in tensor form

11.3.1 Charge and current density

Combine the charge density $\rho = \rho(t, \mathbf{x})$ and current density $\mathbf{j} = \mathbf{j}(t, \mathbf{x})$ into the four-current

$$j^\mu = (c\rho, \mathbf{j}).$$

This field is the source of the electromagnetic field.

11.3.2 The electromagnetic field tensor

The electric and magnetic fields are assembled into the antisymmetric tensor

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix}, \quad F^{\mu\nu} = -F^{\nu\mu}.$$

Lowering both indices gives

$$F_{\mu\nu} = \eta_{\mu\alpha}\eta_{\nu\beta}F^{\alpha\beta} = \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & -B_z & B_y \\ -E_y/c & B_z & 0 & -B_x \\ -E_z/c & -B_y & B_x & 0 \end{pmatrix}.$$

The time–space entries change sign when both indices are lowered, whereas the purely spatial entries do not.

11.3.3 The inhomogeneous equations

Gauss's law and the Ampère–Maxwell law combine into

$$\partial_\mu F^{\mu\nu} = \mu_0 j^\nu. \quad (11.1)$$

For $\nu = 0$, this is

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}.$$

For the three spatial values of ν , it is

$$\nabla \times \mathbf{B} - \varepsilon_0 \mu_0 \frac{\partial \mathbf{E}}{\partial t} = \mu_0 \mathbf{j}.$$

Here

$$c^{-2} = \varepsilon_0 \mu_0.$$

11.3.4 The homogeneous equations

The absence of magnetic charge and Faraday's law combine into the cyclic identity

$$\partial^\lambda F^{\mu\nu} + \partial^\mu F^{\nu\lambda} + \partial^\nu F^{\lambda\mu} = 0. \quad (11.2)$$

In three-vector notation these equations are

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0.$$

11.3.5 Charge conservation

Apply ∂_ν to (11.1):

$$\mu_0 \partial_\nu j^\nu = \partial_\nu \partial_\mu F^{\mu\nu}.$$

The derivatives commute, so $\partial_\nu \partial_\mu$ is symmetric under $\mu \leftrightarrow \nu$, while $F^{\mu\nu}$ is antisymmetric. Explicitly,

$$\begin{aligned}\partial_\nu \partial_\mu F^{\mu\nu} &= \frac{1}{2} \partial_\nu \partial_\mu (F^{\mu\nu} + F^{\nu\mu}) \\ &= 0.\end{aligned}$$

Therefore

$$\boxed{\partial_\mu j^\mu = 0.} \quad (11.3)$$

In ordinary notation,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0,$$

the continuity equation for electric charge.

11.4 Differential forms and the four-potential

11.4.1 The field strength as a two-form

Regard the field tensor as the differential two-form

$$F = \frac{1}{2} F_{\mu\nu} dx^\mu \wedge dx^\nu.$$

For an ordinary scalar function f ,

$$df = (\partial_\lambda f) dx^\lambda,$$

and the exterior derivative obeys the graded Leibniz rule

$$d(f\omega) = df \wedge \omega + f d\omega.$$

Because coordinate one-forms are exact,

$$d(dx^\mu) = 0.$$

Hence

$$\begin{aligned}dF &= \frac{1}{2} (\partial_\lambda F_{\mu\nu}) dx^\lambda \wedge dx^\mu \wedge dx^\nu \\ &= \frac{1}{3!} (\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu}) dx^\lambda \wedge dx^\mu \wedge dx^\nu.\end{aligned}$$

The second line follows by antisymmetrizing the coefficient, because the wedge product is completely antisymmetric. Thus the homogeneous Maxwell equations are exactly

$$\boxed{dF = 0.}$$

11.4.2 Poincaré's lemma

Poincaré's lemma states that on a contractible coordinate neighbourhood, every closed form is exact. Therefore $dF = 0$ implies locally that

$$F = dA$$

for a one-form

$$A = A_\mu dx^\mu.$$

Now

$$\begin{aligned} dA &= (\partial_\nu A_\mu) dx^\nu \wedge dx^\mu \\ &= \frac{1}{2}(\partial_\mu A_\nu - \partial_\nu A_\mu) dx^\mu \wedge dx^\nu. \end{aligned}$$

Consequently,

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu. \quad (11.4)$$

The vector field $A^\mu(x)$ is the electromagnetic four-potential.

Substituting (11.4) into the cyclic expression in (11.2) gives six second-derivative terms that cancel pairwise, since partial derivatives commute. Equivalently,

$$dF = d^2A = 0.$$

Thus the homogeneous Maxwell equations hold automatically once $F = dA$.

The three-dimensional analogy

For a conservative force,

$$\nabla \times \mathbf{F} = 0$$

implies locally that $\mathbf{F} = -\nabla V$. Writing the force in terms of the scalar potential automatically enforces the curl equation. The four-dimensional statement $F = dA$ plays the same role for electromagnetism.

11.5 The electromagnetic action

Let

$$S[A] = \int \mathcal{L} d^4x$$

with Lagrangian density

$$\mathcal{L} = -\frac{1}{4\mu_0} F^{\mu\nu} F_{\mu\nu} - j_\mu A^\mu. \quad (11.5)$$

It is local: at a spacetime point it depends only on $A^\mu(x)$, its first derivatives, and the prescribed source. The classical field equation follows from

$$\delta S = 0$$

for variations that vanish at the boundary.

The Euler–Lagrange equation for A^ν is

$$\frac{\partial \mathcal{L}}{\partial A^\nu} - \partial^\mu \left[\frac{\partial \mathcal{L}}{\partial (\partial^\mu A^\nu)} \right] = 0.$$

The source term gives

$$\frac{\partial \mathcal{L}}{\partial A^\nu} = -j_\nu.$$

To differentiate the kinetic term, first note

$$F_{\alpha\beta} = \partial_\alpha A_\beta - \partial_\beta A_\alpha.$$

Varying it gives

$$\delta F_{\alpha\beta} = \partial_\alpha \delta A_\beta - \partial_\beta \delta A_\alpha.$$

Therefore

$$\begin{aligned} \delta \left(-\frac{1}{4\mu_0} F^{\alpha\beta} F_{\alpha\beta} \right) &= -\frac{1}{2\mu_0} F^{\alpha\beta} \delta F_{\alpha\beta} \\ &= -\frac{1}{2\mu_0} F^{\alpha\beta} (\partial_\alpha \delta A_\beta - \partial_\beta \delta A_\alpha) \\ &= -\frac{1}{\mu_0} F^{\alpha\beta} \partial_\alpha \delta A_\beta. \end{aligned}$$

In the last step, interchange α and β in the second term and use $F^{\beta\alpha} = -F^{\alpha\beta}$. Thus

$$\frac{\partial \mathcal{L}}{\partial (\partial^\mu A^\nu)} = -\frac{1}{\mu_0} F_{\mu\nu}.$$

The Euler–Lagrange equation becomes

$$-j_\nu + \frac{1}{\mu_0} \partial^\mu F_{\mu\nu} = 0,$$

or, after raising the free index,

$$\partial_\mu F^{\mu\nu} = \mu_0 j^\nu.$$

This reproduces the inhomogeneous Maxwell equations.

The action formulation is valuable because symmetries and conserved quantities can be analyzed systematically, and it provides the natural starting point for canonical quantization.

11.6 Gauge transformations and gauge freedom

Definition 11.1: Electromagnetic gauge transformation

For any smooth real scalar function $\Lambda = \Lambda(x)$, define

$$A^\mu(x) \mapsto A'^\mu(x) = A^\mu(x) + \partial^\mu \Lambda(x).$$

The field strength is unchanged:

$$\begin{aligned} F'^{\mu\nu} &= \partial^\mu A'^\nu - \partial^\nu A'^\mu \\ &= \partial^\mu A^\nu - \partial^\nu A^\mu + \partial^\mu \partial^\nu \Lambda - \partial^\nu \partial^\mu \Lambda \\ &= F^{\mu\nu}. \end{aligned}$$

Thus the observable electric and magnetic fields are gauge invariant. The potential is not unique: all potentials related by such a transformation describe the same electromagnetic field.

In the source-free theory, the Lagrangian density depends only on F and is exactly invariant. With a prescribed source,

$$\begin{aligned} -j_\mu A'^\mu &= -j_\mu A^\mu - j_\mu \partial^\mu \Lambda \\ &= -j_\mu A^\mu - \partial^\mu (j_\mu \Lambda) + \Lambda \partial^\mu j_\mu. \end{aligned}$$

If the source is conserved, the change is a total derivative and therefore does not alter the action under the usual boundary conditions.

11.6.1 The gauge group

For standard Abelian Maxwell theory, the gauge parameters form the additive group

$$\mathcal{G} = C^\infty(M, \mathbb{R}) = \{\Lambda : M \rightarrow \mathbb{R} : \Lambda \text{ smooth}\},$$

with

$$(\Lambda_1 + \Lambda_2)(x) = \Lambda_1(x) + \Lambda_2(x).$$

Composing the transformations associated with Λ_1 and Λ_2 produces the transformation associated with $\Lambda_1 + \Lambda_2$. For this reason Maxwell theory is an Abelian gauge theory.

Why the potential is called a gauge field

When charged matter is included, promoting a constant phase symmetry of the matter field to a spacetime-dependent symmetry requires the electromagnetic potential and a covariant derivative. The transformation of A_μ compensates for derivatives of the local phase. In this sense the gauge field makes a local symmetry possible in the interacting theory.

11.7 Lorenz gauge and the wave equation

Definition 11.2: Lorenz gauge

The Lorenz gauge condition is

$$\partial_\mu A^\mu = 0.$$

Using (11.4),

$$\begin{aligned} \mu_0 j^\nu &= \partial_\mu F^{\mu\nu} \\ &= \partial_\mu (\partial^\mu A^\nu - \partial^\nu A^\mu) \\ &= \partial_\mu \partial^\mu A^\nu - \partial^\nu (\partial_\mu A^\mu). \end{aligned}$$

Define the d'Alembert operator

$$\square := \partial_\mu \partial^\mu = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2.$$

Under the Lorenz condition, the second term vanishes and Maxwell's equations reduce to four decoupled inhomogeneous wave equations:

$$\square A^\nu = \mu_0 j^\nu. \quad (11.6)$$

11.7.1 Why Lorenz gauge can be imposed

Start with an arbitrary potential A^μ and seek

$$A'^\mu = A^\mu + \partial^\mu \Lambda$$

satisfying $\partial_\mu A'^\mu = 0$. The condition is

$$\partial_\mu A^\mu + \partial_\mu \partial^\mu \Lambda = 0,$$

or

$$\square \Lambda = -\partial_\mu A^\mu. \quad (11.7)$$

This is an inhomogeneous wave equation for Λ , and locally it can be solved for suitable initial and boundary data. With such a solution, A'^μ represents the same $F^{\mu\nu}$ and satisfies Lorenz gauge.

Residual gauge freedom

Lorenz gauge does not always select a unique representative. If Λ_0 satisfies $\square \Lambda_0 = 0$, then

$$A^\mu \mapsto A^\mu + \partial^\mu \Lambda_0$$

preserves $\partial_\mu A^\mu = 0$. A complete gauge fixing may require additional boundary or initial conditions.

11.8 A nonrelativistic analogy

For a particle in a conservative force field,

$$m\ddot{\mathbf{r}} = \mathbf{F}(\mathbf{r}), \quad \nabla \times \mathbf{F} = 0.$$

Locally there is a scalar potential V with

$$\mathbf{F} = -\nabla V,$$

so the equation of motion is

$$m\ddot{\mathbf{r}} = -\nabla V.$$

The potential is not unique:

$$V \mapsto V + \Delta V, \quad \Delta V \in \mathbb{R} \text{ constant},$$

leaves F unchanged. The additive group $(\mathbb{R}, +)$ is the gauge group of this elementary example. A condition such as

$$V(r_0) = 0$$

fixes the additive freedom. This simple model separates the same four ideas: nonunique potential, gauge transformation, invariant force, and gauge fixing.

11.9 Noether's first theorem

Let fields $\psi_i(x)$ have a Lagrangian density

$$\mathcal{L} = \mathcal{L}(\psi_i, \partial_\mu \psi_i, x).$$

Consider an infinitesimal transformation

$$\psi_i(x) \mapsto \psi_i(x) + \delta\psi_i(x), \quad \delta\psi_i = \varepsilon F_i(\psi_1, \psi_2, \dots),$$

where ε is an infinitesimal constant parameter. The first variation is

$$\delta\mathcal{L} = \sum_i \frac{\partial\mathcal{L}}{\partial\psi_i} \delta\psi_i + \sum_{i,\mu} \frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi_i)} \delta(\partial_\mu\psi_i).$$

Because variation and differentiation commute,

$$\delta(\partial_\mu\psi_i) = \partial_\mu\delta\psi_i.$$

Apply the product rule:

$$\begin{aligned} \frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi_i)} \partial_\mu\delta\psi_i &= \partial_\mu \left[\frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi_i)} \delta\psi_i \right] \\ &\quad - \partial_\mu \left[\frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi_i)} \right] \delta\psi_i. \end{aligned}$$

Therefore

$$\begin{aligned} \delta\mathcal{L} &= \sum_i \left[\frac{\partial\mathcal{L}}{\partial\psi_i} - \partial_\mu \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi_i)} \right) \right] \delta\psi_i \\ &\quad + \partial_\mu \left[\sum_i \frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi_i)} \delta\psi_i \right]. \end{aligned} \tag{11.8}$$

On solutions of the Euler–Lagrange equations, the first line vanishes. If the transformation leaves the Lagrangian invariant, $\delta\mathcal{L} = 0$, then

$$\partial_\mu J^\mu = 0,$$

where

$$J^\mu = \sum_i \frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi_i)} \delta\psi_i \tag{11.9}$$

is the Noether current.

Theorem 11.1: Noether's first theorem, basic field form

Every differentiable one-parameter global symmetry of the action gives, on solutions of the field equations, a conserved current

$$\partial_\mu J^\mu = 0.$$

11.10 Local gauge symmetry and Noether's second theorem

11.10.1 Why a constant gauge parameter is trivial for A_μ

If $\delta\Lambda$ is constant, then

$$\delta A_\mu = \partial_\mu \delta\Lambda = 0.$$

Thus the “global” part of the gauge transformation acts trivially on the electromagnetic potential alone, and the current obtained from (11.9) is zero. Electric charge conservation is already implied by the Maxwell equation and, in a full matter theory, is associated with the global phase symmetry of the charged matter field.

11.10.2 Arbitrary local parameters produce an identity

For the source-free electromagnetic field, take

$$\delta A_\nu = \partial_\nu \Lambda(x)$$

with arbitrary $\Lambda(x)$. The action is invariant because $\delta F_{\mu\nu} = 0$. The Euler–Lagrange expression is

$$E^\nu := \frac{1}{\mu_0} \partial_\mu F^{\mu\nu}.$$

Gauge invariance of the action gives

$$\begin{aligned} 0 = \delta S &= \int E^\nu \partial_\nu \Lambda \, d^4x \\ &= - \int (\partial_\nu E^\nu) \Lambda \, d^4x, \end{aligned}$$

after discarding a boundary term. Since $\Lambda(x)$ is arbitrary,

$$\partial_\nu E^\nu \equiv 0.$$

Indeed,

$$\partial_\nu E^\nu = \frac{1}{\mu_0} \partial_\nu \partial_\mu F^{\mu\nu} \equiv 0$$

by the symmetric–antisymmetric contraction. This is not a new equation of motion; it is a differential identity among the field equations.

Theorem 11.2: Noether's second theorem, informal form

If an action is invariant under transformations depending on arbitrary functions of space-time, then its Euler–Lagrange equations satisfy corresponding differential identities. Local gauge symmetry therefore signals redundancy among the field equations rather than an independent ordinary conserved charge for every choice of the function.

With an external current, the Euler–Lagrange expression is

$$E^\nu = \frac{1}{\mu_0} \partial_\mu F^{\mu\nu} - j^\nu.$$

The same identity becomes

$$\partial_\nu E^\nu = -\partial_\nu j^\nu.$$

Thus gauge invariance of the sourced action requires the current conservation law $\partial_\nu j^\nu = 0$, consistently with (11.3).

What to retain from this chapter

1. Maxwell's equations are

$$\partial_\mu F^{\mu\nu} = \mu_0 j^\nu, \quad \partial^\lambda F^{\mu\nu} + \partial^\mu F^{\nu\lambda} + \partial^\nu F^{\lambda\mu} = 0.$$

2. The homogeneous equations say $dF = 0$; locally Poincaré's lemma gives $F = dA$, or $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.
3. The transformation $A^\mu \mapsto A^\mu + \partial^\mu \Lambda$ leaves $F^{\mu\nu}$ unchanged. This is the local Abelian gauge freedom.
4. Lorenz gauge $\partial_\mu A^\mu = 0$ reduces the sourced equations to $\square A^\nu = \mu_0 j^\nu$, and it can be reached by solving a wave equation for the gauge parameter.
5. Varying $\mathcal{L} = -F^{\mu\nu} F_{\mu\nu}/(4\mu_0) - j_\mu A^\mu$ reproduces the inhomogeneous Maxwell equations.
6. Noether's first theorem relates global one-parameter symmetries to conserved currents. Local gauge symmetry instead yields differential identities among Euler–Lagrange equations, the content of Noether's second theorem.
7. Charge conservation follows from $\partial_\nu \partial_\mu F^{\mu\nu} = 0$, and it is also the consistency condition for coupling a source to a gauge-invariant electromagnetic action.

Non-Abelian Gauge Groups and $SO(n)$ Yang–Mills Theory

Road map

This chapter develops $SO(n)$ gauge theory from first principles. We begin with n free real scalar fields and separate spacetime symmetry from internal symmetry. We then derive the Lie algebra $\mathfrak{so}(n)$, count its generators, compute all Noether currents, and explain the passage from a finite-dimensional global structure group to the infinite-dimensional group of spacetime-dependent gauge transformations. The gauge principle forces the introduction of a connection A_μ and the covariant derivative D_μ . Their transformation laws are derived rather than guessed. Finally, the commutator of covariant derivatives produces the non-Abelian field strength, whose invariant quadratic term is the Yang–Mills Lagrangian. The comparison table, the geometric fibre picture, and the elementary interaction vertex in the handwritten lecture are all reproduced below.

12.1 A free multiplet of real scalar fields

12.1.1 Fields, spacetime, and the action

Let M be four-dimensional Minkowski spacetime and let

$$\varphi_i : M \longrightarrow \mathbb{R}, \quad x \longmapsto \varphi_i(x), \quad i = 1, \dots, n,$$

be n real scalar fields. The index i labels the fields; it is an *internal* index and is completely passive under a change of spacetime coordinates. We use

$$x = x^\mu = (x^0, x^1, x^2, x^3) = (ct, x^1, x^2, x^3), \quad \mu = 0, 1, 2, 3,$$

and, unless units are being displayed explicitly, set $c = 1$. With the mostly-minus Minkowski metric $\eta = \text{diag}(1, -1, -1, -1)$, indices are raised by $\partial^\mu = \eta^{\mu\nu} \partial_\nu$.

The free action in the notes is

$$\begin{aligned} S[\varphi_1, \dots, \varphi_n] &= \int_M d^4x \sum_{i=1}^n \left[\frac{1}{2} \sum_{\mu=0}^3 \partial_\mu \varphi_i(x) \partial^\mu \varphi_i(x) - \frac{1}{2} m^2 \varphi_i(x)^2 \right] \\ &= \int_M d^4x \mathcal{L}(\Phi, \partial_\mu \Phi), \end{aligned} \quad (12.1)$$

where the field multiplet is the column vector

$$\Phi(x) = \begin{pmatrix} \varphi_1(x) \\ \varphi_2(x) \\ \vdots \\ \varphi_n(x) \end{pmatrix} \in \mathbb{R}^n.$$

In vector notation the Lagrangian density becomes

$$\mathcal{L}_{\text{mat}} = \frac{1}{2} \partial_\mu \Phi^T \partial^\mu \Phi - \frac{1}{2} m^2 \Phi^T \Phi. \quad (12.2)$$

It is quadratic in both the fields and their first derivatives. It therefore describes a free theory: the equations of motion will be linear and their solutions may be superposed.

12.1.2 Lorentz invariance and the wave equations

Under a Lorentz transformation,

$$x'^\mu = \Lambda^\mu_\nu x^\nu, \quad \Lambda^T \eta \Lambda = \eta.$$

Because every φ_i is a scalar, $\varphi'_i(x') = \varphi_i(x)$, whereas the derivatives transform as a covector:

$$\partial'_\mu = (\Lambda^{-1})^\nu_\mu \partial_\nu.$$

Consequently,

$$\partial'_\mu \varphi'_i(x') \partial'^\mu \varphi'_i(x') = \partial_\mu \varphi_i(x) \partial^\mu \varphi_i(x),$$

and $d^4x' = d^4x$ for a proper Lorentz transformation. Thus the action is Lorentz invariant. Noether's theorem associates the Lorentz rotations and boosts with conservation of relativistic angular momentum.

For each field, the Euler–Lagrange equation is

$$0 = \frac{\partial \mathcal{L}}{\partial \varphi_i} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi_i)}.$$

The two derivatives appearing here are

$$\frac{\partial \mathcal{L}}{\partial \varphi_i} = -m^2 \varphi_i, \quad \frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi_i)} = \partial^\mu \varphi_i.$$

Substitution gives

$$\begin{aligned} 0 &= -m^2\varphi_i - \partial_\mu\partial^\mu\varphi_i, \\ (\partial_\mu\partial^\mu + m^2)\varphi_i &= 0. \end{aligned}$$

Writing $\square = \partial_\mu\partial^\mu$, all n fields satisfy the Klein–Gordon equation

$$(\square + m^2)\varphi_i = 0, \quad i = 1, \dots, n. \quad (12.3)$$

For a plane-wave mode

$$\varphi_i(x) = c_i e^{-ip_\mu x^\mu},$$

one has $\partial_\mu\varphi_i = -ip_\mu\varphi_i$, so (12.3) becomes

$$(-p_\mu p^\mu + m^2)\varphi_i = 0.$$

A nonzero mode therefore obeys the mass-shell condition

$$p_\mu p^\mu = m^2, \quad E^2 = \mathbf{p}^2 + m^2.$$

The general real solution is a linear superposition of positive- and negative-frequency modes $e^{-iEt+i\mathbf{p}\cdot\mathbf{x}}$ and $e^{+iEt-i\mathbf{p}\cdot\mathbf{x}}$, with $E = \sqrt{\mathbf{p}^2 + m^2}$, combined so that the result is real.

12.2 The internal structure group $SO(n)$

12.2.1 The fundamental action on the multiplet

The theory has a symmetry unrelated to spacetime: one may rotate the n field components among themselves. For

$$R \in SO(n) = \{R \in M_n(\mathbb{R}) : R^T R = I, \det R = 1\},$$

the defining, or fundamental, representation is

$$D : SO(n) \longrightarrow GL(n, \mathbb{R}), \quad D(R) = R.$$

It is a representation because

$$D(R_1 R_2) = R_1 R_2 = D(R_1) D(R_2).$$

The action on fields is pointwise:

$$\Phi(x) \longmapsto D(R)\Phi(x) = R\Phi(x), \quad \varphi_i(x) \longmapsto \sum_{j=1}^n R_{ij}\varphi_j(x). \quad (12.4)$$

The spacetime point x is unchanged. This is why $SO(n)$ is called an *internal* structure group here.

For a *global* transformation, R is independent of x . Hence

$$\partial_\mu(R\Phi) = R\partial_\mu\Phi.$$

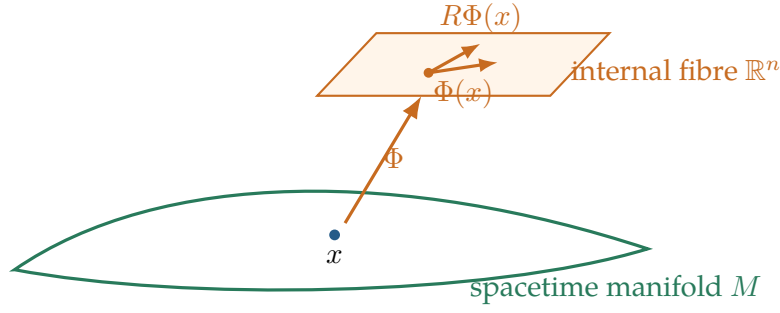


图 12.1: An internal rotation changes the vector in the fibre over x ; it does not move the base point $x \in M$.

Using $R^T R = I$,

$$\begin{aligned} (R\Phi)^T (R\Phi) &= \Phi^T R^T R \Phi = \Phi^T \Phi, \\ \partial_\mu (R\Phi)^T \partial^\mu (R\Phi) &= \partial_\mu \Phi^T R^T R \partial^\mu \Phi = \partial_\mu \Phi^T \partial^\mu \Phi. \end{aligned}$$

Both terms in (12.2) are invariant. Therefore the free action has global $SO(n)$ internal symmetry.

12.2.2 Infinitesimal rotations and the Lie algebra

Let $R(\varepsilon)$ be a differentiable curve in $SO(n)$ through the identity:

$$R(0) = I, \quad R(\varepsilon) = I + \varepsilon T + O(\varepsilon^2).$$

Orthogonality for every ε requires

$$\begin{aligned} I &= R(\varepsilon)^T R(\varepsilon) \\ &= (I + \varepsilon T^T + O(\varepsilon^2))(I + \varepsilon T + O(\varepsilon^2)) \\ &= I + \varepsilon(T^T + T) + O(\varepsilon^2). \end{aligned}$$

The coefficient of ε must vanish, so

$$T^T = -T.$$

Thus the generators of $SO(n)$ are real skew-symmetric matrices.

Definition 12.1: The Lie algebra of $SO(n)$

$$\mathfrak{so}(n) = \{T \in M_n(\mathbb{R}) : T^T = -T\}.$$

It is the tangent space at the identity of $SO(n)$, equipped with the matrix commutator $[T_1, T_2] = T_1 T_2 - T_2 T_1$.

Skew-symmetry fixes every diagonal entry to zero and fixes the lower triangle from the upper triangle:

$$T_{ii} = 0, \quad T_{ji} = -T_{ij}.$$

There is one independent entry for each pair $1 \leq i < j \leq n$. Therefore

$$\dim \mathfrak{so}(n) = \binom{n}{2} = \frac{1}{2}n(n-1). \quad (12.5)$$

Geometrically, each generator produces a rotation in one of the coordinate two-planes.

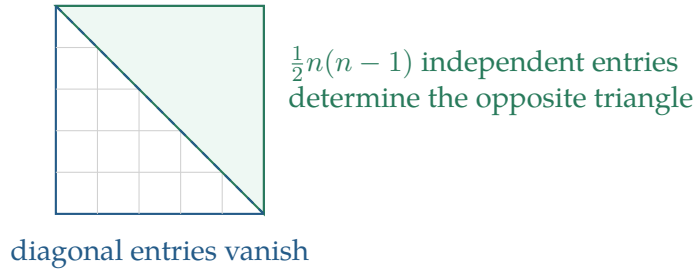


图 12.2: The independent data of a skew-symmetric matrix occupy one strict triangular half.

The Lie algebra is closed under real linear combinations. It is also closed under the commutator. Indeed, if $T_1^T = -T_1$ and $T_2^T = -T_2$, then

$$\begin{aligned}
 [T_1, T_2]^T &= (T_1 T_2 - T_2 T_1)^T \\
 &= T_2^T T_1^T - T_1^T T_2^T \\
 &= (-T_2)(-T_1) - (-T_1)(-T_2) \\
 &= T_2 T_1 - T_1 T_2 \\
 &= -[T_1, T_2].
 \end{aligned}$$

Hence $[T_1, T_2] \in \mathfrak{so}(n)$.

Choose a basis

$$T^1, \dots, T^d, \quad d = \frac{1}{2}n(n-1),$$

of $\mathfrak{so}(n)$. The infinitesimal rotation generated by T^a acts as

$$\begin{aligned}
 \Phi &\mapsto (I + \varepsilon T^a)\Phi + O(\varepsilon^2), \\
 \varphi_i(x) &\mapsto \varphi_i(x) + \varepsilon \sum_{j=1}^n T_{ij}^a \varphi_j(x) + O(\varepsilon^2), \\
 \delta_a \varphi_i(x) &= \varepsilon \sum_{j=1}^n T_{ij}^a \varphi_j(x).
 \end{aligned} \tag{12.6}$$

Thus there are exactly d linearly independent infinitesimal internal rotations.

12.3 Noether currents for the global internal symmetry

12.3.1 Direct check of infinitesimal invariance

For the variation (12.6),

$$\delta_a(\partial_\mu \varphi_i) = \partial_\mu(\delta_a \varphi_i) = \varepsilon \sum_j T_{ij}^a \partial_\mu \varphi_j,$$

because ε and T^a are constant. The first variation of the Lagrangian is

$$\begin{aligned}\delta_a \mathcal{L} &= \sum_i \left[\frac{\partial \mathcal{L}}{\partial \varphi_i} \delta_a \varphi_i + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi_i)} \delta_a (\partial_\mu \varphi_i) \right] \\ &= -\varepsilon m^2 \sum_{i,j} \varphi_i T_{ij}^a \varphi_j + \varepsilon \sum_{i,j} \partial^\mu \varphi_i T_{ij}^a \partial_\mu \varphi_j.\end{aligned}$$

Each quadratic contraction vanishes. For any real vector v ,

$$v^T T^a v = (v^T T^a v)^T = v^T (T^a)^T v = -v^T T^a v,$$

so $v^T T^a v = 0$. Apply this once to $v = \Phi$ and once, for each μ , to $v = \partial_\mu \Phi$. Therefore

$$\delta_a \mathcal{L} = 0.$$

12.3.2 The $\frac{1}{2}n(n-1)$ conserved currents

Noether's first theorem gives one conserved current for every generator. Omitting the common infinitesimal parameter ε , the current is

$$\begin{aligned}j^{\mu a} &= \sum_i \frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi_i)} \sum_j T_{ij}^a \varphi_j \\ &= \sum_{i,j} \partial^\mu \varphi_i T_{ij}^a \varphi_j = \partial^\mu \Phi^T T^a \Phi.\end{aligned}\tag{12.7}$$

On solutions of the field equations,

$$\partial_\mu j^{\mu a} = 0, \quad a = 1, \dots, \frac{1}{2}n(n-1).\tag{12.8}$$

This can also be checked directly:

$$\begin{aligned}\partial_\mu j^{\mu a} &= (\Box \Phi)^T T^a \Phi + \partial_\mu \Phi^T T^a \partial^\mu \Phi \\ &= (-m^2 \Phi)^T T^a \Phi + 0 = 0.\end{aligned}$$

The mathematical generators T^a are real and skew-symmetric. Physics often uses instead the Hermitian generators

$$H^a = iT^a, \quad (H^a)^\dagger = H^a.$$

Multiplying (12.7) by i gives the equally conserved current

$$\hat{j}^{\mu a} = \partial^\mu \Phi^T iT^a \Phi = \partial^\mu \Phi^T H^a \Phi, \quad \partial_\mu \hat{j}^{\mu a} = 0.\tag{12.9}$$

This is the convention in which the generators are represented by Hermitian observables. It amounts to working in the complexification of $\mathfrak{so}(n)$; no physical conservation law is changed.

12.4 From a global symmetry to a gauge symmetry

12.4.1 Structure group versus gauge group

We now promote the constant internal rotation to a spacetime-dependent one:

$$R \longrightarrow R(x).$$

The corresponding gauge group is

$$\mathcal{G} = C^\infty(M, SO(n)),$$

with pointwise multiplication $(R_1 R_2)(x) = R_1(x) R_2(x)$. The distinction made in the lecture is summarized in the following table.

Structure group $G = SO(n)$	Gauge group $\mathcal{G} = C^\infty(M, SO(n))$
A Lie group acting on the n -dimensional internal field space: $\Phi \mapsto R\Phi$.	The group of smooth functions from spacetime to G : $\Phi(x) \mapsto R(x)\Phi(x)$.
A finite number $d = \frac{1}{2}n(n-1)$ of independent generators T^a .	An independent infinitesimal coefficient may be chosen at every spacetime point: $T^a \mapsto \alpha^a(x)T^a$.
Finite-dimensional Lie group.	Infinite-dimensional Lie group.

表 12.1: Global internal rotations and local gauge transformations.

Demanding invariance under arbitrary $R(x)$ is much more restrictive than demanding invariance under a constant R . Starting with a free theory, this demand necessarily introduces additional fields and interactions. This construction is the *gauge principle*: it is a systematic recipe for building interacting field theories from a chosen internal structure group. In this sense, fundamental interactions arise from local symmetry requirements.

Quantum electrodynamics and the Standard Model are gauge theories. Their structure group is

$$U(1) \times SU(2) \times SU(3),$$

where the three factors describe, respectively, the electromagnetic, weak, and strong gauge sectors. Nothing in the construction is restricted to these examples: any suitable Lie group can be used. Non-Abelian groups such as $SU(2)$, $SU(3)$, and $SO(n)$ are particularly interesting because their gauge particles interact with one another.

A condensed-matter analogy

For a classical Heisenberg model on a lattice, one may write

$$H = \sum_{\langle x, x' \rangle} J_{xx'} \mathbf{S}_x \cdot \mathbf{S}_{x'}, \quad \mathbf{S}_x^2 = \text{constant}.$$

Replacing a normalized spin by components $\mathbf{S}_x = (\varphi_{1x}, \dots, \varphi_{nx})$ gives the constraint

$$\sum_i \varphi_{ix}^2 = 1.$$

In a continuum limit these become fields $\varphi_i(x)$. A description in which $\Phi(x)$ is replaced by $R(x)\Phi(x)$ without changing the physical configuration contains a local redundancy. This is a useful elementary picture of how gauge structure can arise naturally; the local transformation is a redundancy of description, not an operation producing a different physical state.

12.4.2 Why the ordinary derivative fails

The global transformation law (12.4) becomes

$$\Phi'(x) = R(x)\Phi(x), \quad R(x) \in SO(n).$$

The mass term remains invariant pointwise:

$$\Phi'^T \Phi' = \Phi^T R^T R \Phi = \Phi^T \Phi.$$

The kinetic term does not. The product rule gives

$$\partial_\mu \Phi' = (\partial_\mu R)\Phi + R \partial_\mu \Phi \neq R \partial_\mu \Phi \quad (12.10)$$

unless R is constant. The extra term $(\partial_\mu R)\Phi$ compares internal vectors using frames that vary from point to point. To remove it, we need a rule for parallel comparison: a connection.

12.5 The covariant derivative and the gauge field

12.5.1 Minimal substitution

Introduce a matrix-valued field $A_\mu(x)$, the *gauge field*, and a real coupling constant g . In the Hermitian-generator convention define

$$D_\mu = \partial_\mu - igA_\mu(x). \quad (12.11)$$

Replacing ∂_μ with D_μ is the *minimal substitution*. The locally coupled matter Lagrangian is

$$\mathcal{L}_{\text{matter}} = \frac{1}{2}(D_\mu \Phi)^T D^\mu \Phi - \frac{1}{2}m^2 \Phi^T \Phi. \quad (12.12)$$

We require the covariant derivative of a field to transform in the same way as the field itself:

$$D'_\mu \Phi' = R D_\mu \Phi. \quad (12.13)$$

Then

$$(D'_\mu \Phi')^T D'^\mu \Phi' = (R D_\mu \Phi)^T (R D^\mu \Phi) = (D_\mu \Phi)^T D^\mu \Phi,$$

so (12.12) is locally gauge invariant.

12.5.2 Derivation of the connection transformation law

We now determine A'_μ from (12.13). The right-hand side is

$$R D_\mu \Phi = R \partial_\mu \Phi - ig R A_\mu \Phi.$$

The left-hand side is

$$\begin{aligned} D'_\mu \Phi' &= (\partial_\mu - igA'_\mu)(R\Phi) \\ &= (\partial_\mu R)\Phi + R\partial_\mu \Phi - igA'_\mu R\Phi. \end{aligned}$$

After cancelling $R\partial_\mu \Phi$, equality for every Φ gives

$$-igRA_\mu = \partial_\mu R - igA'_\mu R.$$

Equivalently,

$$RA_\mu = \frac{i}{g}\partial_\mu R + A'_\mu R,$$

and multiplication on the right by R^{-1} yields

$$A'_\mu = RA_\mu R^{-1} - \frac{i}{g}(\partial_\mu R)R^{-1}. \quad (12.14)$$

Differentiate $RR^{-1} = I$:

$$0 = \partial_\mu(RR^{-1}) = (\partial_\mu R)R^{-1} + R\partial_\mu R^{-1}.$$

Thus $-(\partial_\mu R)R^{-1} = R\partial_\mu R^{-1}$, and the equivalent form recorded in the notes is

$$\boxed{A'_\mu = RA_\mu R^{-1} + \frac{i}{g}R\partial_\mu R^{-1}.} \quad (12.15)$$

The inhomogeneous derivative term is what distinguishes a connection from an ordinary matter field.

There is also a concise operator derivation. Equation (12.13) implies, for every field,

$$D'_\mu R\Phi = RD_\mu \Phi.$$

Therefore $D'_\mu R = RD_\mu$, and hence

$$\boxed{D'_\mu = RD_\mu R^{-1}.} \quad (12.16)$$

Expanding this identity reproduces (12.15).

12.5.3 Why the transformed field remains Lie-algebra valued

Let T^a be real skew-symmetric generators and put

$$H^a = iT^a.$$

The matrices H^a are Hermitian, and we expand the gauge field as

$$A_\mu(x) = \sum_{a=1}^d A_\mu^a(x)H^a, \quad d = \frac{1}{2}n(n-1). \quad (12.17)$$

The real functions $A_\mu^a(x)$ are the component gauge fields. This is the physics convention corresponding to the complexified Lie algebra. In the real skew convention one may instead define

$$B_\mu = -igA_\mu \in \mathfrak{so}(n), \quad D_\mu = \partial_\mu + B_\mu.$$

Then

$$B'_\mu = RB_\mu R^{-1} + R\partial_\mu R^{-1},$$

and both terms are manifestly real and skew-symmetric.

For completeness, this closure can be checked directly. First,

$$(RB_\mu R^{-1})^T = (R^{-1})^T B_\mu^T R^T = R(-B_\mu)R^{-1} = -RB_\mu R^{-1}.$$

Second, differentiating $R^T R = I$ gives

$$(\partial_\mu R^T)R + R^T \partial_\mu R = 0.$$

It follows that

$$(R\partial_\mu R^{-1})^T = -R\partial_\mu R^{-1}.$$

Thus $B'_\mu \in \mathfrak{so}(n)$, or equivalently A'_μ remains in the Hermitian-generator version of the same Lie algebra.

12.6 Interactions forced by the gauge principle

12.6.1 Expansion of the matter Lagrangian

Expanding the covariant derivatives in (12.12) gives

$$\begin{aligned} (D_\mu \Phi)^T D^\mu \Phi &= (\partial_\mu \Phi - igA_\mu \Phi)^T (\partial^\mu \Phi - igA^\mu \Phi) \\ &= \partial_\mu \Phi^T \partial^\mu \Phi - ig \Phi^T A_\mu^T \partial^\mu \Phi - ig \partial_\mu \Phi^T A^\mu \Phi - g^2 \Phi^T A_\mu^T A^\mu \Phi. \end{aligned}$$

Therefore

$$\mathcal{L}_{\text{matter}} = \mathcal{L}_{\text{free}} + \mathcal{L}_{\text{int}},$$

where

$$\begin{aligned} \mathcal{L}_{\text{free}} &= \frac{1}{2} \partial_\mu \Phi^T \partial^\mu \Phi - \frac{1}{2} m^2 \Phi^T \Phi, \\ \mathcal{L}_{\text{int}} &= \frac{1}{2} [-ig \Phi^T A_\mu^T \partial^\mu \Phi - ig \partial_\mu \Phi^T A^\mu \Phi - g^2 \Phi^T A_\mu^T A^\mu \Phi]. \end{aligned} \quad (12.18)$$

The first two terms are cubic in the fields and first order in g ; the last is quartic and second order in g . Thus the attempt to make the free matter theory locally invariant has automatically generated interactions.

After quantization, the cubic terms describe elementary processes in which a matter quantum changes its internal component while emitting or absorbing a gauge quantum. A schematic first-order vertex from the notes is:

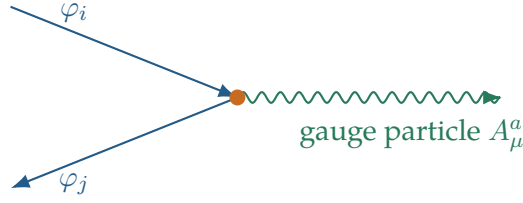


图 12.3: A schematic matter–matter–gauge-field interaction generated at first order in g .

12.7 Dynamics of the gauge field

12.7.1 Why the Maxwell-type first guess is incomplete

The field A_μ needs its own dynamical term. By analogy with electromagnetism, a first guess would be

$$f_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a, \quad -\frac{1}{4} \sum_a f_{\mu\nu}^a f^{\mu\nu a}.$$

This is Lorentz invariant, but for a non-Abelian group it is not gauge invariant. The transformation (12.15) contains both a conjugation and a derivative term, and differentiating it produces additional terms that do not cancel in the Abelian expression.

The correct replacement is to use covariant derivatives:

$$F_{\mu\nu} = D_\mu A_\nu - D_\nu A_\mu. \quad (12.19)$$

Expanding $D_\mu = \partial_\mu - igA_\mu$ yields

$$\begin{aligned} F_{\mu\nu} &= (\partial_\mu - igA_\mu)A_\nu - (\partial_\nu - igA_\nu)A_\mu \\ &= \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu]. \end{aligned} \quad (12.20)$$

The commutator term is the essential non-Abelian correction. It vanishes for $U(1)$, where all Lie-algebra elements commute, and Maxwell's field tensor is recovered.

12.7.2 Curvature as a commutator of covariant derivatives

The same tensor follows intrinsically from the commutator $[D_\mu, D_\nu]$. Acting on an arbitrary multiplet Φ ,

$$\begin{aligned} [D_\mu, D_\nu]\Phi &= [\partial_\mu - igA_\mu, \partial_\nu - igA_\nu]\Phi \\ &= -ig(\partial_\mu A_\nu)\Phi + ig(\partial_\nu A_\mu)\Phi - g^2[A_\mu, A_\nu]\Phi. \end{aligned}$$

The terms $A_\nu\partial_\mu\Phi$ and $A_\mu\partial_\nu\Phi$ cancel pairwise after the product rule is applied. Factoring out $-ig$ gives

$$[D_\mu, D_\nu]\Phi = -ig(\partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu])\Phi.$$

Hence

$$F_{\mu\nu} = \frac{i}{g}[D_\mu, D_\nu]. \quad (12.21)$$

This formula explains the geometric meaning of $F_{\mu\nu}$: it measures the failure of covariant derivatives in two spacetime directions to commute, so it is the curvature of the connection.

12.7.3 Gauge covariance of the field strength

Use the operator law (12.16):

$$\begin{aligned} F'_{\mu\nu} &= \frac{i}{g} [D'_\mu, D'_\nu] \\ &= \frac{i}{g} [RD_\mu R^{-1}, RD_\nu R^{-1}] \\ &= \frac{i}{g} R [D_\mu, D_\nu] R^{-1}. \end{aligned}$$

Therefore

$$\boxed{F'_{\mu\nu} = R F_{\mu\nu} R^{-1}.} \quad (12.22)$$

The field strength is not generally unchanged; it transforms *covariantly* by conjugation. This is sufficient to construct a gauge invariant scalar by taking a trace.

Because $\partial_\mu A_\nu$ lies in the same vector space as A_ν , and because a Lie algebra is closed under commutators,

$$F_{\mu\nu} \in \mathfrak{so}(n)$$

in the real convention, or in its Hermitian-generator realization used above. If

$$[H^a, H^b] = i f^{ab}{}_c H^c,$$

then

$$\begin{aligned} F_{\mu\nu} &= \sum_c F_{\mu\nu}^c H^c, \\ F_{\mu\nu}^c &= \partial_\mu A_\nu^c - \partial_\nu A_\mu^c + g f^{ab}{}_c A_\mu^a A_\nu^b. \end{aligned} \quad (12.23)$$

The final quadratic term is absent in an Abelian theory and is responsible for gauge-boson self-interactions.

12.8 The Yang–Mills Lagrangian

12.8.1 Trace form and proof of gauge invariance

The gauge-field Lagrangian proposed in the lecture is

$$\mathcal{L}_{\text{YM}} = -\frac{1}{2} \text{tr}(F_{\mu\nu} F^{\mu\nu}). \quad (12.24)$$

Under a gauge transformation,

$$\begin{aligned} \text{tr}(F'_{\mu\nu} F'^{\mu\nu}) &= \text{tr}(R F_{\mu\nu} R^{-1} R F^{\mu\nu} R^{-1}) \\ &= \text{tr}(R F_{\mu\nu} F^{\mu\nu} R^{-1}) \\ &= \text{tr}(F_{\mu\nu} F^{\mu\nu}), \end{aligned}$$

where cyclicity of the trace was used in the last line. Thus $\mathcal{L}_{\text{YM}}$ is gauge invariant.

Choose the generator normalization written in the notes,

$$\mathrm{tr}(H^a H^b) = \frac{1}{2} \delta^{ab}. \quad (12.25)$$

Then

$$\begin{aligned} \mathcal{L}_{\mathrm{YM}} &= -\frac{1}{2} \sum_{\mu, \nu} \sum_{a, b} F_{\mu\nu}^a F^{\mu\nu b} \mathrm{tr}(H^a H^b) \\ &= -\frac{1}{4} \sum_{\mu, \nu} \sum_a F_{\mu\nu}^a F^{\mu\nu a}. \end{aligned}$$

Since $F_{\nu\mu} = -F_{\mu\nu}$, summing over all ordered pairs (μ, ν) counts every independent pair twice. Equivalently,

$$\mathcal{L}_{\mathrm{YM}} = -\frac{1}{2} \sum_{\mu < \nu} \sum_a F_{\mu\nu}^a F^{\mu\nu a}. \quad (12.26)$$

The factor $1/4$ in the usual formula is the familiar free-field normalization together with this antisymmetric double counting.

12.8.2 Self-interactions of non-Abelian gauge particles

Substituting (12.20) into (12.24) shows three types of terms:

$$\mathcal{L}_{\mathrm{YM}} \sim (\partial A)^2 + g(\partial A)A^2 + g^2 A^4.$$

The first is the gauge-field kinetic term. The second and third are, respectively, cubic and quartic gauge-field interactions. They exist because $[A_\mu, A_\nu] \neq 0$. Thus, unlike the photon in pure Abelian electromagnetism, non-Abelian gauge particles carry the gauge charge and interact with one another.

12.9 The complete gauge-invariant theory

Combining the matter, interaction, and gauge-field pieces gives

$$\mathcal{L} = \mathcal{L}_{\mathrm{free}} + \mathcal{L}_{\mathrm{int}} + \mathcal{L}_{\mathrm{YM}}.$$

In compact covariant form,

$$\mathcal{L} = \frac{1}{2} (D_\mu \Phi)^T D^\mu \Phi - \frac{1}{2} m^2 \Phi^T \Phi - \frac{1}{2} \mathrm{tr}(F_{\mu\nu} F^{\mu\nu}). \quad (12.27)$$

Writing out every term gives the final expression displayed in the handwritten lecture:

$$\begin{aligned} \mathcal{L} &= \frac{1}{2} \partial_\mu \Phi^T \partial^\mu \Phi - \frac{1}{2} m^2 \Phi^T \Phi \\ &\quad + \frac{1}{2} [-ig \Phi^T A_\mu^T \partial^\mu \Phi - ig \partial_\mu \Phi^T A^\mu \Phi - g^2 \Phi^T A_\mu^T A^\mu \Phi] \\ &\quad - \frac{1}{2} \mathrm{tr}[(\partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu])(\partial^\mu A^\nu - \partial^\nu A^\mu - ig[A^\mu, A^\nu])]. \end{aligned} \quad (12.28)$$

Every term is Lorentz invariant. The three pieces are not separately arbitrary: their precise combination and their relative interaction coefficients are fixed by local $SO(n)$ gauge invariance.

What to retain from this chapter

1. The free multiplet $\Phi = (\varphi_1, \dots, \varphi_n)^T$ obeys $(\square + m^2)\Phi = 0$ and has global $SO(n)$ internal symmetry.
2. Infinitesimal $SO(n)$ generators are skew-symmetric matrices:

$$\mathfrak{so}(n) = \{T : T^T = -T\}, \quad \dim \mathfrak{so}(n) = \frac{1}{2}n(n-1).$$

3. Global internal symmetry gives one conserved Noether current $j^{\mu a} = \partial^\mu \Phi^T T^a \Phi$ for every generator.
4. Allowing $R = R(x)$ produces the infinite-dimensional gauge group $C^\infty(M, SO(n))$, but the ordinary derivative of Φ then fails to transform covariantly.
5. The covariant derivative and connection transform as

$$D_\mu = \partial_\mu - igA_\mu, \quad D'_\mu = RD_\mu R^{-1},$$

$$A'_\mu = RA_\mu R^{-1} + \frac{i}{g}R\partial_\mu R^{-1}.$$

6. The curvature is

$$F_{\mu\nu} = \frac{i}{g}[D_\mu, D_\nu] = \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu],$$

and transforms covariantly: $F'_{\mu\nu} = RF_{\mu\nu}R^{-1}$.

7. The trace removes the conjugation, so $\mathcal{L}_{\text{YM}} = -\frac{1}{2} \text{tr}(F_{\mu\nu}F^{\mu\nu})$ is gauge invariant.
8. Minimal coupling generates matter–gauge interactions, while the commutator in $F_{\mu\nu}$ generates cubic and quartic self-interactions of non-Abelian gauge fields.

Lie Groups: Topology, Connectedness, and Matrix Examples

Road map

Continuous groups contain infinitely many elements that can be arbitrarily close to one another, and their elements are locally described by real parameters. To make these statements precise, this chapter first reviews topological spaces, continuous maps, closure, interior, boundary, compactness, connectedness, simple connectedness, and homeomorphisms. It then introduces topological groups and proves that the identity component is a normal subgroup. Two central matrix examples are treated in detail:

$$O(n) \text{ is compact,} \quad O(n)^\circ = SO(n),$$

and

$$GL(n, \mathbb{R})^\circ = GL^+(n, \mathbb{R}) = \{A \in GL(n, \mathbb{R}) : \det A > 0\}.$$

Finally, we pass from topological groups to Lie groups and linear Lie groups, state the closed subgroup theorem and Ado's theorem in the form needed here, and construct classical, triangular, unipotent, and affine examples.

13.1 Why continuous groups require topology

Unlike a finite group, a continuous group has infinitely many elements and a notion of one element being close to another. For example, a rotation through an angle φ is close to a rotation through $\varphi + \Delta\varphi$ when $\Delta\varphi \rightarrow 0$. Its elements can be parameterized, at least locally, by real numbers. The set of group elements therefore carries extra structure:

- a topology, which makes closeness and continuity meaningful;
- frequently a manifold structure, which permits differentiation.

The main examples in these notes are *linear Lie groups*, realized by matrices. They are especially useful in physics and in quantum mechanics. Historically, some early physicists jokingly called the rapid entrance of group theory into quantum mechanics the *Gruppenpest*, or “group plague.” The matrix approach used here keeps the differential-geometric prerequisites as light as possible without hiding the topology that is actually needed.

Definition 13.1: Topological group

A group G , equipped with a topology, is a *topological group* if the maps

$$m : G \times G \longrightarrow G, \quad (g_1, g_2) \longmapsto g_1 g_2,$$

and

$$\iota : G \longrightarrow G, \quad g \longmapsto g^{-1},$$

are continuous. The domain $G \times G$ carries the product topology.

13.2 Topological spaces and continuous maps

13.2.1 Open sets and the axioms of a topology

Definition 13.2: Topological space

Let X be a set. A *topology* on X is a collection $\mathcal{T} \subseteq \mathcal{P}(X)$ of subsets, whose members are called *open sets*, such that:

1. $\emptyset, X \in \mathcal{T}$;
2. if $\{U_j\}_{j \in J} \subseteq \mathcal{T}$, where J may be infinite, then

$$\bigcup_{j \in J} U_j \in \mathcal{T};$$

3. if $U_1, \dots, U_r \in \mathcal{T}$, where r is finite, then

$$\bigcap_{j=1}^r U_j \in \mathcal{T}.$$

The pair $(X, \mathcal{T})$ is a *topological space*.

The asymmetry between arbitrary unions and only finite intersections is essential. For example,

$$\bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\},$$

which is not open in the usual topology of $\mathbb{R}$.

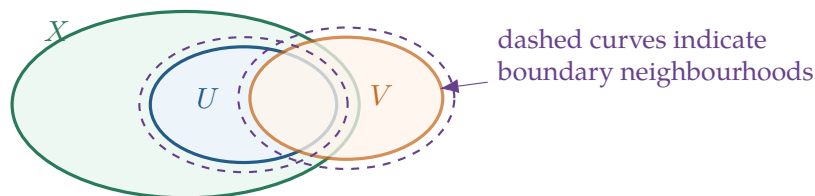


图 13.1: Open sets contain none of their boundary merely by definition; each of their points has a neighbourhood remaining inside the set.

Example 13.1: Metric topology

Let (X, d) be a metric space. For $x \in X$ and $\varepsilon > 0$, the open ball is

$$B_\varepsilon(x) = \{y \in X : d(x, y) < \varepsilon\}.$$

Declare a set open if it is a union of open balls. Arbitrary unions and finite intersections of such sets are again open, so this defines the metric topology. Euclidean spaces $\mathbb{R}^n$ and $\mathbb{C}^n$ always carry this standard topology unless stated otherwise.

Example 13.2: Two extreme topologies

On any set X ,

$$\mathcal{T}_{\text{disc}} = \mathcal{P}(X)$$

is the *discrete topology*: every subset is open. At the opposite extreme,

$$\mathcal{T}_{\text{ind}} = \{\emptyset, X\}$$

is the *indiscrete*, or trivial, topology.

13.2.2 Continuity as preservation of open preimages**Definition 13.3: Continuous map**

Let X and Y be topological spaces. A map $f : X \rightarrow Y$ is *continuous* if

$$U \subseteq Y \text{ open} \implies f^{-1}(U) \subseteq X \text{ open}.$$

The definition concerns inverse images, not images. In fact, every map $f : X \rightarrow Y$ induces on X the collection of inverse images

$$\{f^{-1}(U) : U \subseteq Y \text{ open}\};$$

requiring these sets to be open is exactly what continuity means.

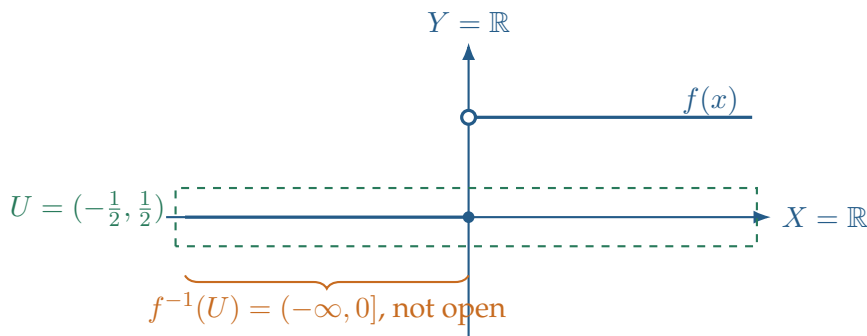


图 13.2: The step function $f(x) = 0$ for $x \leq 0$ and $f(x) = 1$ for $x > 0$ is not continuous: the open set $U = (-\frac{1}{2}, \frac{1}{2})$ has a non-open inverse image.

13.3 Closed sets, closure, interior, and boundary

Definition 13.4: Closed set

A subset $A \subseteq X$ is *closed* if its complement is open:

$$X \setminus A \in \mathcal{T}.$$

Definition 13.5: Closure

The *closure* $\bar{A}$ is the smallest closed subset of X containing A . Equivalently,

$$\bar{A} = \bigcap \{F \subseteq X : F \text{ closed and } A \subseteq F\}.$$

Writing every closed set as $X \setminus U$, this is the formula used in the lecture:

$$\bar{A} = \bigcap \{X \setminus U : U \in \mathcal{T}, A \subseteq X \setminus U\}.$$

Definition 13.6: Interior and boundary

The *interior* of A is the largest open set contained in A :

$$A^\circ = \bigcup \{U \in \mathcal{T} : U \subseteq A\}.$$

The *boundary* of A is

$$\partial A = \bar{A} \setminus A^\circ.$$

These definitions immediately imply the identities recorded in the notes:

$$A \text{ open} \implies A \cap \partial A = \emptyset,$$

$$A \text{ closed} \implies \partial A \subseteq A,$$

$$A^\circ \cap \partial A = \emptyset,$$

$$\bar{A} \cap \partial A = \partial A,$$

$$A^\circ \cup \partial A = \bar{A},$$

$$\bar{A} \cup \partial A = \bar{A}.$$

For $a < b$, the four intervals

$$[a, b], \quad [a, b), \quad (a, b], \quad (a, b)$$

all have the same closure, interior, and boundary in $\mathbb{R}$:

$$\bar{A} = [a, b], \quad A^\circ = (a, b), \quad \partial A = \{a, b\}.$$

13.4 Covers and compactness

Definition 13.7: Cover and open cover

A family $\{A_i\}_{i \in I}$ of subsets of X is a *cover* of X if

$$\bigcup_{i \in I} A_i = X.$$

If every A_i is open, it is an *open cover*.

Definition 13.8: Compact space

A topological space X is *compact* if every open cover $\{U_i\}_{i \in I}$ contains a finite subcover: there exist $i_1, \dots, i_r \in I$ such that

$$X = U_{i_1} \cup \dots \cup U_{i_r}.$$

Theorem 13.1: Heine–Borel theorem

A subset $X \subseteq \mathbb{R}^n$, with the subspace topology, is compact if and only if it is closed and bounded.

The lecture used this theorem without proof. Some immediate examples are:

- $\mathbb{R}_{>0} = \{x \in \mathbb{R} : x > 0\}$ is not compact; it is neither closed in $\mathbb{R}$ nor bounded.
- $[0, \pi] \cup [2, 3]$ is closed and bounded, hence compact. Since $2 < 3 < \pi$, this particular union actually equals $[0, \pi]$.
- The sphere

$$S^n = \{x \in \mathbb{R}^{n+1} : \|x\| = 1\}$$

is closed and bounded, hence compact.

13.5 Connectedness and homeomorphisms

13.5.1 Connected and simply connected spaces

Definition 13.9: Connected space

A topological space X is *connected* if it cannot be written as

$$X = X_1 \cup X_2$$

with X_1 and X_2 nonempty, disjoint, open subsets of X . Such a decomposition, if it exists, is a *separation*; then X is *disconnected*.

Definition 13.10: Loop and simple connectedness

A *loop* in X based at x_0 is a continuous map

$$\gamma : [0, 1] \longrightarrow X, \quad \gamma(0) = \gamma(1) = x_0.$$

A path-connected space is *simply connected* if every loop can be continuously contracted to a point. Explicitly, there must be a continuous map $H : [0, 1] \times [0, 1] \rightarrow X$ such that

$$H(0, t) = \gamma(t), \quad H(1, t) = x_0, \quad H(s, 0) = H(s, 1) = x_0.$$

The n -torus

$$T^n = S^1 \times \cdots \times S^1$$

is connected but not simply connected. Already on T^2 , a loop winding once around the central hole cannot shrink to a point without leaving the surface.

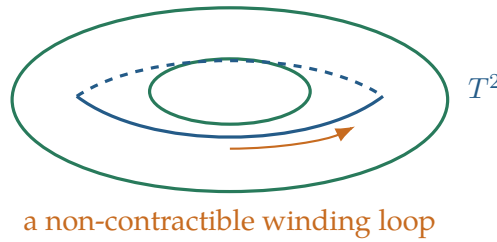


图 13.3: The torus is connected, but the displayed loop cannot be contracted to a point on the torus.

13.5.2 Homeomorphisms

Definition 13.11: Homeomorphism

Let X_1 and X_2 be topological spaces. A map $f : X_1 \rightarrow X_2$ is a *homeomorphism* if it is bijective, continuous, and its inverse $f^{-1} : X_2 \rightarrow X_1$ is continuous. If such a map exists, we write

$$X_1 \cong X_2 \quad \text{or} \quad X_1 \sim X_2$$

and say that the spaces are homeomorphic.

Being homeomorphic is an equivalence relation. The identity gives reflexivity, inverse maps give symmetry, and composition gives transitivity.

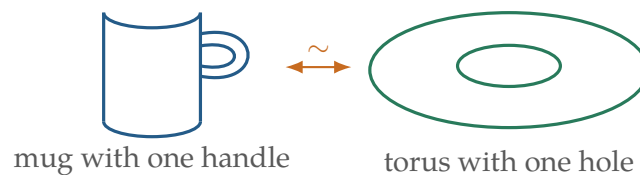


图 13.4: The familiar mug–torus correspondence: topology remembers the number of holes, not lengths or angles.

Similarly, two disjoint circles are homeomorphic as an abstract topological space to two linked circles. Linking is a property of a particular embedding in $\mathbb{R}^3$, not of the circles as a space by themselves. This illustrates why topology permits continuous bending and stretching but not cutting or gluing.

13.6 Topological groups and basic examples

13.6.1 Subgroups and standard examples

Proposition 13.1: Subgroups inherit a topological group structure

If G is a topological group and $H \leq G$ has the subspace topology, then H is a topological group.

Proof. The multiplication on H is the restriction

$$m|_{H \times H} : H \times H \rightarrow H,$$

and inversion is the restriction $\iota|_H : H \rightarrow H$. Restrictions of continuous maps are continuous. ■

Every abstract group becomes a topological group when given the discrete topology. Important nondiscrete examples include:

1. $(\mathbb{R}^n, +)$ with its standard topology.
2. The subgroup $(\mathbb{Z}^n, +) \leq (\mathbb{R}^n, +)$, whose subspace topology is discrete.
3. $(\mathbb{C}, +)$ and $(\mathbb{C}^\times, \cdot)$, where $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$.

4.

$$S^1 = \{z \in \mathbb{C} : |z| = 1\} \leq \mathbb{C}^\times.$$

5. $(\mathbb{R}^\times, \cdot)$ and its connected subgroup $(\mathbb{R}_{>0}, \cdot)$.

6.

$$GL(n, \mathbb{R}) = \{A \in M_n(\mathbb{R}) : \det A \neq 0\}.$$

Matrix multiplication is polynomial in the entries, and

$$A^{-1} = \frac{\text{adj}(A)}{\det A}$$

is continuous wherever $\det A \neq 0$.

7. The real matrix subgroups $SL(n, \mathbb{R})$, $O(n)$, $SO(n)$, and $Sp(2n, \mathbb{R})$.
8. Their complex analogues $GL(n, \mathbb{C})$, $SL(n, \mathbb{C})$, $U(n)$, and $SU(n)$.

13.6.2 Homomorphisms of topological groups

Definition 13.12: Topological-group homomorphism and isomorphism

For topological groups G and H , a map $f : G \rightarrow H$ is a *homomorphism of topological groups* if it is both a group homomorphism and continuous. It is an *isomorphism of topological groups* if it is a group isomorphism and both f and f^{-1} are continuous.

The examples in the lecture are as follows.

1. If $H \leq G$, the inclusion $H \hookrightarrow G$ is a continuous homomorphism.
2. The exponential map

$$\exp : (\mathbb{R}, +) \longrightarrow (S^1, \cdot), \quad t \longmapsto e^{it},$$

is a continuous surjective homomorphism with

$$\ker(\exp) = 2\pi\mathbb{Z},$$

a discrete topological subgroup of $\mathbb{R}$.

3.

$$\exp : (\mathbb{R}, +) \longrightarrow (\mathbb{R}_{>0}, \cdot), \quad t \longmapsto e^t,$$

is a topological-group isomorphism; its inverse is $\log$.

4. The subgroup

$$H = \left\{ \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} : t \in \mathbb{R} \right\} \leq GL(2, \mathbb{R})$$

is topologically isomorphic to $(\mathbb{R}, +)$ through

$$t \longmapsto \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix},$$

because multiplication adds the upper-right entries.

5. The determinant

$$\det : GL(n, \mathbb{R}) \longrightarrow \mathbb{R}^\times$$

is a continuous group homomorphism.

13.7 The identity component

Definition 13.13: Identity component

Let G be a topological group with identity e . The *path-connected component of the identity*, denoted G° , is

$$G^\circ = \left\{ g \in G : \begin{array}{l} \text{there is a continuous path } c : [0, 1] \rightarrow G \\ \text{with } c(0) = e \text{ and } c(1) = g \end{array} \right\}.$$

In the finite-dimensional Lie groups considered below, this agrees with the usual connected component of the identity.

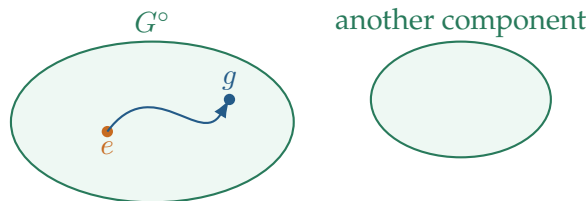


图 13.5: The identity component consists of the elements reachable from e by continuous paths.

Theorem 13.2: Normality of the identity component

For every topological group G , the path component G° is a normal subgroup of G .

Proof. We verify the subgroup properties and normality.

First take $h \in G^\circ$. Choose a path c with $c(0) = e$ and $c(1) = h$. Define

$$\bar{c}(t) = c(1-t)h^{-1}.$$

This is continuous because group multiplication and inversion are continuous. Its endpoints are

$$\bar{c}(0) = c(1)h^{-1} = e, \quad \bar{c}(1) = c(0)h^{-1} = h^{-1}.$$

Hence $h^{-1} \in G^\circ$.

Next let $h_1, h_2 \in G^\circ$, with paths c_1, c_2 from e to h_1, h_2 . Pointwise multiplication gives a continuous path

$$c(t) = c_1(t)c_2(t),$$

and

$$c(0) = ee = e, \quad c(1) = h_1h_2.$$

Thus $h_1h_2 \in G^\circ$, so $G^\circ \leq G$.

Finally, let $g \in G$ and $h \in G^\circ$. If c joins e to h , then

$$c^g(t) = gc(t)g^{-1}$$

is continuous and satisfies

$$c^g(0) = e, \quad c^g(1) = ghg^{-1}.$$

Therefore $ghg^{-1} \in G^\circ$, proving $G^\circ \trianglelefteq G$. ■

13.8 The topology of the orthogonal group

Theorem 13.3: Compactness and identity component of $O(n)$

$$O(n) \text{ is compact,} \quad O(n)^\circ = SO(n).$$

In particular, $SO(n) \trianglelefteq O(n)$.

Compactness

Give $M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$ the Frobenius norm

$$\|A\|_F^2 = \text{tr}(AA^T) = \sum_{i,j} A_{ij}^2.$$

If $A \in O(n)$, then $AA^T = I$, so

$$\|A\|_F^2 = \text{tr}(I) = n.$$

Hence $O(n)$ is bounded.

The map

$$F : M_n(\mathbb{R}) \longrightarrow M_n(\mathbb{R}), \quad F(A) = AA^T,$$

is continuous because its entries are polynomials in the entries of A . Moreover,

$$O(n) = F^{-1}(\{I\}).$$

The singleton $\{I\}$ is closed, and a continuous inverse image of a closed set is closed. Thus $O(n)$ is closed in $\mathbb{R}^{n^2}$. By the Heine–Borel theorem, it is compact.

$SO(n)$ lies in the identity component

Every real orthogonal matrix is orthogonally conjugate to a block diagonal matrix made of 2×2 rotation blocks

$$R(\alpha) = \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix}$$

and possible 1×1 blocks (1) or (-1) . When $\det A = 1$, the (-1) -blocks occur in pairs and may be regarded as $R(\pi)$. Consequently, for $A \in SO(n)$ there is an orthogonal P such that

$$P^{-1}AP = \text{diag}(R(\alpha_1), \dots, R(\alpha_r), 1, \dots, 1).$$

The path

$$c(t) = P \text{diag}(R(t\alpha_1), \dots, R(t\alpha_r), 1, \dots, 1)P^{-1}$$

lies in $SO(n)$ and satisfies $c(0) = I$, $c(1) = A$. Hence $SO(n)$ is path connected and

$$SO(n) \subseteq O(n)^\circ.$$

No determinant- -1 matrix lies in the identity component

Suppose that a path $c : [0, 1] \rightarrow O(n)$ joined I to a matrix B with $\det B = -1$. The continuous function

$$t \mapsto \det c(t)$$

would start at $+1$ and end at -1 . The intermediate value theorem would force $\det c(t_0) = 0$ for some t_0 , impossible for an orthogonal matrix. Thus

$$O(n)^\circ \subseteq SO(n).$$

Together with the opposite inclusion, this proves

$$O(n)^\circ = SO(n).$$

Normality also follows either from the identity-component theorem or from

$$SO(n) = \ker(\det : O(n) \rightarrow \{\pm 1\}).$$

13.9 The two components of $GL(n, \mathbb{R})$

Define

$$GL^+(n, \mathbb{R}) = \{A : \det A > 0\}, \quad GL^-(n, \mathbb{R}) = \{A : \det A < 0\}.$$

Theorem 13.4: Identity component of the real general linear group

$$GL(n, \mathbb{R})^\circ = GL^+(n, \mathbb{R}).$$

Thus $GL(n, \mathbb{R})$ has precisely two connected components, distinguished by the sign of the determinant.

Proof. Let $A \in GL^+(n, \mathbb{R})$. Gram–Schmidt, equivalently QR decomposition, produces a unique factorization

$$A = QR,$$

where $Q \in SO(n)$ and R is upper triangular with positive diagonal entries. Write

$$R = DN,$$

where

$$D = \text{diag}(\lambda_1, \dots, \lambda_n), \quad \lambda_i > 0,$$

and N is upper triangular with all diagonal entries equal to 1. Thus

$$A = QDN. \tag{13.1}$$

This is the elementary KDN , or Iwasawa-type, decomposition used in the lecture.

Each factor is path connected to I :

- $Q \in SO(n)$, and the preceding theorem supplies a path $Q(t)$ from I to Q .

-

$$D(t) = \text{diag}(1 - t + t\lambda_1, \dots, 1 - t + t\lambda_n)$$

has strictly positive diagonal entries for all $t \in [0, 1]$, so it joins I to D inside $GL^+(n, \mathbb{R})$.

- Write $N = I + U$, where U is strictly upper triangular. Then

$$N(t) = I + tU$$

remains upper triangular with diagonal entries 1, so it is invertible and joins I to N .

The product

$$c(t) = Q(t)D(t)N(t)$$

is a path in $GL^+(n, \mathbb{R})$ from I to A . Therefore

$$GL^+(n, \mathbb{R}) \subseteq GL(n, \mathbb{R})^\circ.$$

Conversely, no matrix in $GL^-(n, \mathbb{R})$ can be connected to I : the same intermediate-value argument would make the determinant vanish somewhere along the path. Hence

$$GL(n, \mathbb{R})^\circ = GL^+(n, \mathbb{R}).$$

Multiplication by $\text{diag}(-1, 1, \dots, 1)$ is a homeomorphism from $GL^+(n, \mathbb{R})$ to $GL^-(n, \mathbb{R})$, so the negative-determinant part is the other connected component. ■

13.10 Lie groups and linear Lie groups

13.10.1 General definition

Definition 13.14: Lie group

A finite-dimensional *Lie group* is a set G such that:

1. G is a group;
2. G is a finite-dimensional smooth manifold;
3. multiplication $G \times G \rightarrow G$ and inversion $G \rightarrow G$ are smooth.

Smooth maps are continuous, so every Lie group is automatically a topological group. Its dimension is its dimension as a manifold. The hierarchy emphasized in the notes is:

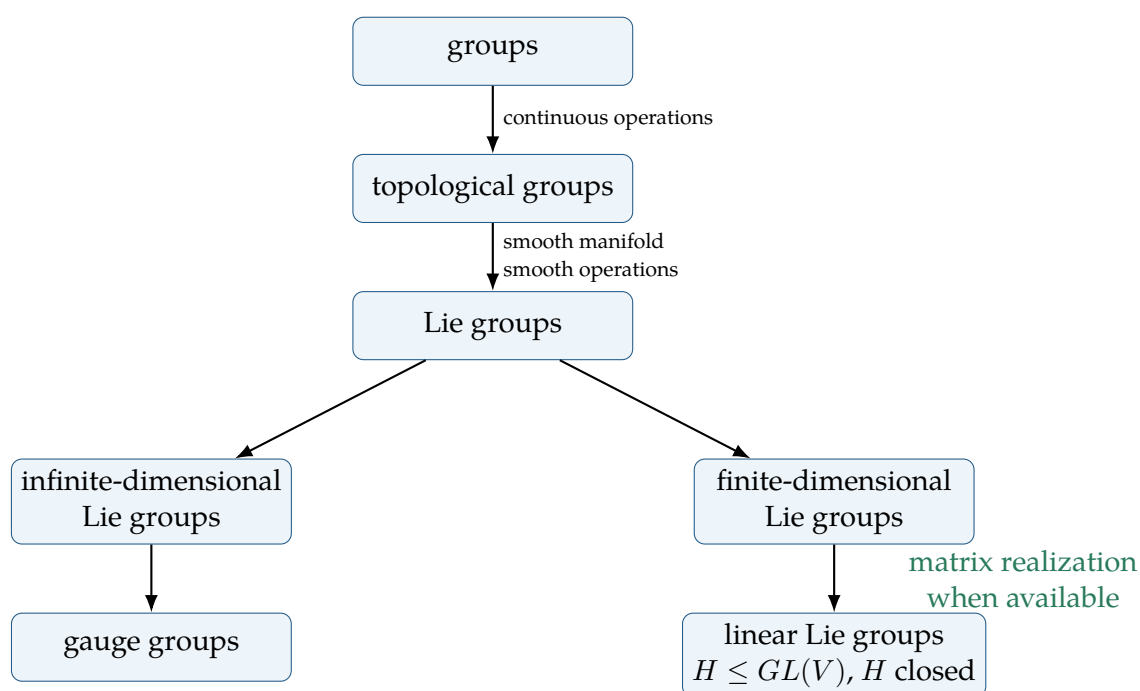


图 13.6: From abstract groups to the matrix Lie groups used in the present course.

Infinite-dimensional gauge groups such as $C^\infty(M, G)$ are important exceptions to the finite-dimensional framework and require additional functional analysis.

13.10.2 Linear Lie groups and the closed subgroup theorem

Definition 13.15: Linear Lie group

For $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, a *linear Lie group* is a closed subgroup of $GL(n, \mathbb{K})$ for some n .

Theorem 13.5: Closed subgroup theorem, matrix form

Every closed subgroup H of a finite-dimensional Lie group G is an embedded Lie subgroup. Conversely, an embedded subgroup carrying the subspace topology is closed precisely

when it is a closed embedded Lie subgroup.

In the abbreviated terminology of the handwritten lecture: for a subgroup $H \leq G$, being a matrix Lie subgroup is equivalent to being closed in G . If $H \trianglelefteq G$, then a closed normal subgroup is a normal Lie subgroup.

Theorem 13.6: Ado's theorem, Lie-algebra form

Every finite-dimensional Lie algebra over $\mathbb{R}$ or $\mathbb{C}$ has a faithful finite-dimensional matrix representation.

Ado's theorem explains why matrix methods capture every finite-dimensional Lie algebra. A useful precision is that it is a theorem about Lie *algebras*; a Lie group can still have a global topological obstruction to being faithfully represented by finite matrices. Thus linear Lie groups form the central, but not literally exhaustive, class of finite-dimensional Lie groups.

13.11 Examples of linear Lie groups

13.11.1 Classical groups as automorphism groups of forms

Let $\Phi \in GL(n, \mathbb{K})$ represent a nondegenerate bilinear form. Its automorphism group is

$$\text{Aut}(\Phi) = \{A \in GL(n, \mathbb{K}) : A^T \Phi A = \Phi\}. \quad (13.2)$$

For a sesquilinear form over $\mathbb{C}$, replace transpose by conjugate transpose:

$$\text{Aut}(\Phi) = \{A \in GL(n, \mathbb{C}) : A^\dagger \Phi A = \Phi\}. \quad (13.3)$$

Symmetric, skew-symmetric, and Hermitian choices of Φ produce the orthogonal, symplectic, and unitary families.

These groups are closed. To see this in the bilinear case, suppose $A_r \in \text{Aut}(\Phi)$ and $A_r \rightarrow A \in GL(n, \mathbb{K})$. Then continuity of matrix multiplication and transpose gives

$$A_r^T \Phi A_r = \Phi \implies A^T \Phi A = \lim_{r \rightarrow \infty} A_r^T \Phi A_r = \Phi.$$

Hence $A \in \text{Aut}(\Phi)$. The same argument, with $A^\dagger$, proves the sesquilinear case. Therefore these automorphism groups are linear Lie groups by the closed subgroup theorem.

If $G \leq GL(n, \mathbb{K})$ is a linear Lie group, then

$$H = \{A \in G : \det A = 1\} = (\det|_G)^{-1}(\{1\})$$

is closed because the determinant is continuous. Hence H is again a linear Lie group. This produces, among others, $SO(n)$ from $O(n)$ and $SU(n)$ from $U(n)$.

13.11.2 The Borel and unitriangular groups

The upper triangular, or standard Borel, group is

$$UT_n(\mathbb{K}) = \{A \in GL(n, \mathbb{K}) : A_{ij} = 0 \text{ whenever } i > j\}.$$

Each equation $A_{ij} = 0$ defines a closed condition, so $UT_n(\mathbb{K})$ is closed in $GL(n, \mathbb{K})$ and is therefore a linear Lie group. Schematically, its matrices have the shape

$$\begin{pmatrix} * & * & \cdots & * \\ 0 & * & \cdots & * \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & * \end{pmatrix}, \quad \text{all diagonal entries nonzero.}$$

The special unitriangular group is

$$SUT_n(\mathbb{K}) = \{A \in UT_n(\mathbb{K}) : A_{ii} = 1 \text{ for every } i\}.$$

Its elements are $I + N$ with N strictly upper triangular. Since $N^n = 0$, every such matrix is unipotent:

$$(A - I)^n = 0.$$

This is the *unipotent group of rank n* mentioned in the notes.

13.11.3 The affine group

The affine group is the closed matrix subgroup

$$\text{Aff}_n(\mathbb{K}) = \left\{ \begin{pmatrix} A & t \\ 0 & 1 \end{pmatrix} : A \in GL(n, \mathbb{K}), t \in \mathbb{K}^n \right\} \leq GL(n+1, \mathbb{K}).$$

Embed $x \in \mathbb{K}^n$ into homogeneous coordinates as $\begin{pmatrix} x \\ 1 \end{pmatrix}$. Then

$$\begin{pmatrix} A & t \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ 1 \end{pmatrix} = \begin{pmatrix} Ax + t \\ 1 \end{pmatrix}.$$

Thus $\text{Aff}_n(\mathbb{K})$ acts on $\mathbb{K}^n$ by affine transformations

$$x \mapsto Ax + t.$$

Multiplication displays the semidirect-product law:

$$(A, t)(B, s) = (AB, As + t),$$

so

$$\text{Aff}_n(\mathbb{K}) \cong GL(n, \mathbb{K}) \ltimes \mathbb{K}^n.$$

What to retain from this chapter

1. A topological group is a group whose multiplication and inversion are continuous.
2. A topology is closed under arbitrary unions and finite intersections. Continuity means that inverse images of open sets are open.
3. Closure, interior, and boundary satisfy

$$\partial A = \overline{A} \setminus A^\circ, \quad \overline{A} = A^\circ \cup \partial A.$$

4. Compactness means that every open cover has a finite subcover; in $\mathbb{R}^n$, compactness is equivalent to being closed and bounded.
5. The identity component G° is always a normal subgroup.

6.

$$O(n) \text{ is compact,} \quad O(n)^\circ = SO(n).$$

7.

$$GL(n, \mathbb{R})^\circ = GL^+(n, \mathbb{R}) = \{A : \det A > 0\}.$$

A *QDN* decomposition gives an explicit path to the identity.

8. A Lie group is simultaneously a smooth manifold and a group with smooth operations. A linear Lie group is a closed subgroup of $GL(n, \mathbb{R})$ or $GL(n, \mathbb{C})$.
9. The classical groups, determinant-one subgroups, and the triangular, unitriangular, and affine groups above are all linear Lie groups.

Lie Algebras and Matrix Lie Algebras

Road map

The local, infinitesimal structure of a Lie group is encoded by a Lie algebra. This chapter first defines Lie algebras abstractly, derives the component form of antisymmetry and the Jacobi identity, and explains how structure constants determine the bracket. It then develops subalgebras, homomorphisms, and the fundamental matrix example $\mathfrak{gl}(V)$. By differentiating the defining equations of matrix groups at the identity, we obtain

$$\mathfrak{so}(n), \quad \mathfrak{sp}(2n, \mathbb{R}), \quad \mathfrak{u}(n), \quad \mathfrak{su}(n), \quad \mathfrak{sl}(n, \mathbb{K}),$$

including their dimensions and useful bases. A table collects all group–algebra pairs from the lecture. The final sections introduce the derived algebra and derive the precise transformation rule of the structure constants under a basis change.

14.1 From a Lie group to its infinitesimal algebra

The basic correspondence has two directions:

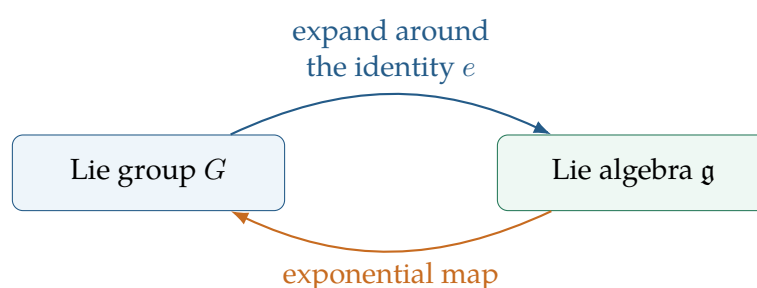


图 14.1: The local Lie correspondence. This chapter constructs and studies the algebra side; the exponential map and the global correspondence are developed in the following lecture.

For matrix groups, “expand around e ” means taking derivatives of smooth curves through the identity matrix. The result is a vector space with a bilinear product measuring noncommutativity. We first axiomatize that structure independently of any group.

14.2 Definition and immediate consequences

Definition 14.1: Lie algebra

Let V be a d -dimensional vector space over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. A *Lie bracket* on V is a map

$$[\cdot, \cdot] : V \times V \longrightarrow V, \quad (X, Y) \longmapsto [X, Y],$$

such that, for all $X, Y, Z \in V$ and $\alpha, \beta \in \mathbb{K}$:

1. **Bilinearity:**

$$[\alpha X + \beta Y, Z] = \alpha[X, Z] + \beta[Y, Z],$$

$$[X, \alpha Y + \beta Z] = \alpha[X, Y] + \beta[X, Z].$$

2. **Antisymmetry:**

$$[X, Y] = -[Y, X].$$

3. **Jacobi identity:**

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0.$$

The pair $\mathfrak{g} = (V, [\cdot, \cdot])$ is a real or complex *Lie algebra*, and $\dim \mathfrak{g} = \dim V = d$.

Several identities follow at once:

$$[X, X] = -[X, X] \implies [X, X] = 0,$$

because the fields $\mathbb{R}$ and $\mathbb{C}$ have characteristic zero. Bilinearity in the second argument follows from bilinearity in the first together with antisymmetry, and

$$[X, 0] = [X, Y - Y] = [X, Y] - [X, Y] = 0.$$

Definition 14.2: Abelian Lie algebra

A Lie algebra $\mathfrak{g}$ is *Abelian* if

$$[X, Y] = 0 \quad \text{for every } X, Y \in \mathfrak{g}.$$

14.3 Generators and structure constants

Let

$$\{T_1, \dots, T_d\}$$

be a basis of $\mathfrak{g}$. In the terminology of the lecture, the basis vectors T_i are *generators*. Each bracket of basis vectors has a unique expansion

$$[T_i, T_j] = \sum_{k=1}^d f^k_{ij} T_k. \quad (14.1)$$

The scalars $f^k_{ij} \in \mathbb{K}$ are the *structure constants* of $\mathfrak{g}$ with respect to this basis.

Antisymmetry gives

$$\sum_k f^k_{ij} T_k = [T_i, T_j] = -[T_j, T_i] = -\sum_k f^k_{ji} T_k.$$

Uniqueness of basis coefficients implies

$$f^k_{ij} = -f^k_{ji}. \quad (14.2)$$

Now impose the Jacobi identity on T_i, T_j, T_k :

$$\begin{aligned} 0 &= [T_i, [T_j, T_k]] + [T_j, [T_k, T_i]] + [T_k, [T_i, T_j]] \\ &= \sum_\ell f^\ell_{jk} [T_i, T_\ell] + \sum_\ell f^\ell_{ki} [T_j, T_\ell] + \sum_\ell f^\ell_{ij} [T_k, T_\ell] \\ &= \sum_{\ell, m} \left(f^\ell_{jk} f^m_{i\ell} + f^\ell_{ki} f^m_{j\ell} + f^\ell_{ij} f^m_{k\ell} \right) T_m. \end{aligned}$$

Therefore the component Jacobi constraints are

$$\boxed{\sum_{\ell=1}^d \left(f^\ell_{jk} f^m_{i\ell} + f^\ell_{ki} f^m_{j\ell} + f^\ell_{ij} f^m_{k\ell} \right) = 0} \quad (14.3)$$

for every i, j, k, m .

Conversely, suppose a d -dimensional vector space has a basis $\{T_i\}$, and scalars f^k_{ij} satisfy (14.2) and (14.3). For

$$X = \sum_i x_i T_i, \quad Y = \sum_j y_j T_j,$$

define

$$[X, Y] = \sum_{i,j,k} x_i y_j f^k_{ij} T_k. \quad (14.4)$$

Bilinearity is built into the formula, antisymmetry follows from (14.2), and the Jacobi identity follows from (14.3). Thus suitable structure constants completely determine a Lie algebra.

14.4 First examples

14.4.1 Every one-dimensional Lie algebra is Abelian

Let $\dim \mathfrak{g} = 1$ and choose a basis $\{T\}$. There is a scalar f with

$$[T, T] = fT.$$

But $[T, T] = 0$, so $f = 0$. If $X = xT$ and $Y = yT$, then

$$[X, Y] = xy[T, T] = 0.$$

Hence every Lie bracket on a one-dimensional vector space is Abelian.

14.4.2 $\mathbb{R}^3$ with the cross product

Let

$$\mathfrak{g} = (\mathbb{R}^3, \times), \quad [X, Y] = X \times Y.$$

The cross product is bilinear and antisymmetric. Its Jacobi identity can be checked with the vector triple-product identity

$$A \times (B \times C) = B(A \cdot C) - C(A \cdot B).$$

Indeed,

$$\begin{aligned} X \times (Y \times Z) + Y \times (Z \times X) + Z \times (X \times Y) \\ = Y(X \cdot Z) - Z(X \cdot Y) \\ + Z(Y \cdot X) - X(Y \cdot Z) \\ + X(Z \cdot Y) - Y(Z \cdot X) = 0. \end{aligned}$$

For the canonical basis e_1, e_2, e_3 ,

$$[e_i, e_j] = e_i \times e_j = \sum_k \varepsilon_{ijk} e_k.$$

Thus the structure constants are the Levi-Civita symbols:

$$f^k_{ij} = \varepsilon_{ijk}.$$

14.4.3 The general linear Lie algebra

Example 14.1: Matrix general linear algebra

$$\mathfrak{gl}(n, \mathbb{K}) = M_n(\mathbb{K})$$

becomes a Lie algebra under the commutator

$$[X, Y] = XY - YX.$$

Antisymmetry is immediate. For the Jacobi identity, expand:

$$\begin{aligned} [X, [Y, Z]] &= XYZ - XZY - YZX + ZYX, \\ [Y, [Z, X]] &= YZX - YXZ - ZXY + XZY, \\ [Z, [X, Y]] &= ZXY - ZYX - XYZ + YXZ. \end{aligned}$$

Every monomial cancels, so the sum is zero.

For an n -dimensional vector space V , define

$$\mathfrak{gl}(V) = \text{End}_{\mathbb{K}}(V)$$

with

$$[f_1, f_2] = f_1 \circ f_2 - f_2 \circ f_1.$$

This is the *general linear Lie algebra of degree n* . Its dimension is

$$\dim_{\mathbb{K}} \mathfrak{gl}(V) = n^2.$$

If $v_1, \dots, v_n$ is a basis of V , let E_{ij} be the matrix unit that sends v_j to v_i and annihilates the other basis vectors. Then $\{E_{ij}\}_{1 \leq i, j \leq n}$ is a basis of $\mathfrak{gl}(V)$, and

$$[E_{ij}, E_{kl}] = \delta_{jk} E_{il} - \delta_{li} E_{kj}. \quad (14.5)$$

14.5 Subalgebras and homomorphisms

Definition 14.3: Lie subalgebra

A Lie subalgebra $\mathfrak{h} \leq \mathfrak{g}$ is a linear subspace closed under the bracket:

$$X, Y \in \mathfrak{h} \implies [X, Y] \in \mathfrak{h}.$$

The whole algebra $\mathfrak{g}$ and the zero algebra $\{0\}$ are the trivial Lie subalgebras.

Definition 14.4: Lie-algebra homomorphism and isomorphism

For Lie algebras $\mathfrak{g}, \mathfrak{g}'$ over the same field, a map $\psi : \mathfrak{g} \rightarrow \mathfrak{g}'$ is a *homomorphism* if it is linear and

$$\psi([X, Y]) = [\psi(X), \psi(Y)] \quad \text{for all } X, Y \in \mathfrak{g}.$$

A bijective homomorphism is an *isomorphism*. If an isomorphism exists, we write

$$\mathfrak{g} \cong \mathfrak{g}'.$$

Example 14.2: Matrices of endomorphisms

Choose a basis $v_1, \dots, v_n$ of V . If

$$f(v_j) = \sum_i A_{ij} v_i,$$

let $\psi(f) = A$. Then

$$\psi : \mathfrak{gl}(V) \longrightarrow \mathfrak{gl}(n, \mathbb{K})$$

is linear and bijective. Since matrix multiplication represents composition,

$$\psi([f_1, f_2]) = \psi(f_1 \circ f_2 - f_2 \circ f_1) = \psi(f_1)\psi(f_2) - \psi(f_2)\psi(f_1),$$

so

$$\mathfrak{gl}(V) \cong \mathfrak{gl}(n, \mathbb{K}).$$

14.5.1 $(\mathbb{R}^3, \times) \cong \mathfrak{so}(3)$

Define

$$T_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad T_y = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad T_z = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Direct multiplication gives

$$[T_i, T_j] = \sum_k \varepsilon_{ijk} T_k.$$

Their span is

$$\text{span}_{\mathbb{R}}\{T_x, T_y, T_z\} = \left\{ \begin{pmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{pmatrix} : x_1, x_2, x_3 \in \mathbb{R} \right\} = \mathfrak{so}(3).$$

The map

$$\psi : \mathbb{R}^3 \longrightarrow \mathfrak{so}(3), \quad (x_1, x_2, x_3) \longmapsto x_1 T_x + x_2 T_y + x_3 T_z,$$

is linear and bijective. On basis elements,

$$\psi([e_i, e_j]) = \sum_k \varepsilon_{ijk} \psi(e_k) = \sum_k \varepsilon_{ijk} T_k = [T_i, T_j].$$

Bilinearity extends this equality to all X, Y , hence

$$(\mathbb{R}^3, \times) \cong \mathfrak{so}(3). \quad (14.6)$$

Theorem 14.1: Ado's theorem

Every finite-dimensional Lie algebra over $\mathbb{R}$ or $\mathbb{C}$ is isomorphic to a Lie subalgebra of $\mathfrak{gl}(n, \mathbb{K})$ for some n .

The lecture states Ado's theorem without proof. It justifies concentrating on matrix, or *linear*, Lie algebras when studying finite-dimensional Lie algebras.

14.6 Extracting a Lie algebra from a matrix Lie group

Let $G \leq GL(N, \mathbb{K})$ be a matrix Lie group. Consider all smooth curves

$$A : (-\epsilon, \epsilon) \longrightarrow G, \quad A(0) = I.$$

Their velocity vectors at the identity form the tangent space

$$T_I G = \{\dot{A}(0) : A(t) \in G, A(0) = I\}.$$

For matrix Lie groups, this tangent space is the associated Lie algebra $\mathfrak{g}$. Its bracket is the matrix commutator. In the next chapter we justify this construction and its connection with one-parameter subgroups; here we use it to derive examples directly.

14.6.1 The orthogonal and special orthogonal algebras

Let $A(t) \in O(n)$ and $A(0) = I$. Since

$$A(t)^T A(t) = I,$$

write

$$A(t) = I + tX + O(t^2), \quad X = \dot{A}(0).$$

Then

$$\begin{aligned} I &= (I + tX^T + O(t^2))(I + tX + O(t^2)) \\ &= I + t(X^T + X) + O(t^2). \end{aligned}$$

Therefore $X^T + X = 0$. This gives

$$\mathfrak{o}(n) = \mathfrak{so}(n) = \{X \in M_n(\mathbb{R}) : X^T = -X\}. \quad (14.7)$$

It is a vector subspace and is closed under commutators, as proved explicitly in the gauge-theory chapter. The independent entries lie above the diagonal, so

$$\dim \mathfrak{so}(n) = \frac{1}{2}n(n-1).$$

Every skew-symmetric matrix has zero trace. Thus the determinant-one condition defining $SO(n)$ imposes no further infinitesimal constraint, which explains why $O(n)$ and $SO(n)$ have the same Lie algebra. For $n = 3$, the basis T_x, T_y, T_z above makes the comparison with $(\mathbb{R}^3, \times)$ explicit.

14.6.2 Upper triangular matrices

The group

$$UT_n(\mathbb{K}) = \{A \in GL(n, \mathbb{K}) : A_{ij} = 0 \text{ for } i > j\}$$

has Lie algebra

$$\mathfrak{ut}_n(\mathbb{K}) = \{X \in M_n(\mathbb{K}) : X_{ij} = 0 \text{ for } i > j\}.$$

Products and commutators of upper triangular matrices remain upper triangular. A basis is

$$\{E_{ij} : 1 \leq i \leq j \leq n\},$$

and

$$\dim_{\mathbb{K}} \mathfrak{ut}_n(\mathbb{K}) = \frac{1}{2}n(n+1).$$

14.7 The symplectic Lie algebra

Recall

$$Sp(2n, \mathbb{R}) = \{A \in GL(2n, \mathbb{R}) : A^T J A = J\}, \quad J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}.$$

For a path $A(t) = I + tX + O(t^2)$ through the identity,

$$\begin{aligned} J &= A(t)^T J A(t) \\ &= (I + tX^T + O(t^2))J(I + tX + O(t^2)) \\ &= J + t(X^T J + JX) + O(t^2). \end{aligned}$$

Thus

$$\mathfrak{sp}(2n, \mathbb{R}) = \{X \in M_{2n}(\mathbb{R}) : X^T J + JX = 0\}. \quad (14.8)$$

Closure under the commutator can be verified exactly as in the notes. If $X_1^T J = -JX_1$ and $X_2^T J = -JX_2$, then

$$\begin{aligned} [X_1, X_2]^T J &= (X_2^T X_1^T - X_1^T X_2^T)J \\ &= X_2^T(-JX_1) - X_1^T(-JX_2) \\ &= JX_2X_1 - JX_1X_2 \\ &= -J[X_1, X_2]. \end{aligned}$$

Hence $[X_1, X_2] \in \mathfrak{sp}(2n, \mathbb{R})$.

14.7.1 Block form and dimension

Write

$$X = \begin{pmatrix} A & B \\ C & D \end{pmatrix}.$$

Then

$$\begin{aligned} X^T J + JX &= \begin{pmatrix} A^T & C^T \\ B^T & D^T \end{pmatrix} \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} + \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} \\ &= \begin{pmatrix} -C^T & A^T \\ -D^T & B^T \end{pmatrix} + \begin{pmatrix} C & D \\ -A & -B \end{pmatrix} \\ &= \begin{pmatrix} C - C^T & A^T + D \\ -D^T - A & B^T - B \end{pmatrix}. \end{aligned}$$

The condition is therefore equivalent to

$$B = B^T, \quad C = C^T, \quad D = -A^T.$$

The free block A contributes n^2 real parameters, and each symmetric block contributes $n(n+1)/2$. Hence

$$\dim \mathfrak{sp}(2n, \mathbb{R}) = n^2 + 2 \frac{n(n+1)}{2} = n(2n+1). \quad (14.9)$$

14.7.2 The rank-one case and a correction

For $n = 1$, the condition is simply $\text{tr } X = 0$, so

$$\mathfrak{sp}(2, \mathbb{R}) = \mathfrak{sl}(2, \mathbb{R}).$$

A basis used in the handwritten notes is

$$T_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad T_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad T_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Direct calculation gives

$$[T_1, T_2] = 2T_3, \quad [T_2, T_3] = 2T_1, \quad [T_3, T_1] = -2T_2. \quad (14.10)$$

Correction to the handwritten $\mathfrak{sp}(2, \mathbb{R})$ comparison

The notes write $[T_i, T_j] = 2 \sum_k \varepsilon_{ijk} T_k$, rescale $T'_i = \frac{1}{2} T_i$, and propose

$$T'_1 \mapsto T_x, \quad T'_2 \mapsto T_y, \quad T'_3 \mapsto T_z$$

as an isomorphism with $\mathfrak{so}(3)$. Equation (14.10) shows the missing sign:

$$[T'_3, T'_1] = -T'_2, \quad [T_z, T_x] = T_y.$$

Therefore the proposed map does not preserve the bracket. Without rescaling, the factor also fails:

$$\psi([T_1, T_2]) = 2T_z \neq [T_x, T_y] = T_z.$$

The correct real statement is

$$\boxed{\mathfrak{sp}(2, \mathbb{R}) \cong \mathfrak{sl}(2, \mathbb{R}) \not\cong \mathfrak{so}(3).}$$

The two three-dimensional real algebras are different real forms; after complexification one has

$$\mathfrak{sp}(2, \mathbb{C}) \cong \mathfrak{sl}(2, \mathbb{C}) \cong \mathfrak{so}(3, \mathbb{C}).$$

14.8 Unitary and special unitary Lie algebras

Let

$$SU(n) = \{A \in GL(n, \mathbb{C}) : A^\dagger A = I, \det A = 1\}.$$

For a path $A(t) = I + tX + O(t^2)$,

$$\begin{aligned} I &= A(t)^\dagger A(t) \\ &= (I + tX^\dagger + O(t^2))(I + tX + O(t^2)) \\ &= I + t(X^\dagger + X) + O(t^2). \end{aligned}$$

Thus $X^\dagger = -X$. In addition,

$$1 = \det A(t) = \det(I + tX + O(t^2)) = 1 + t \operatorname{tr} X + O(t^2),$$

so $\operatorname{tr} X = 0$. Therefore

$$\begin{aligned} \mathfrak{u}(n) &= \{X \in M_n(\mathbb{C}) : X^\dagger = -X\}, \\ \mathfrak{su}(n) &= \{X \in M_n(\mathbb{C}) : X^\dagger = -X, \operatorname{tr} X = 0\}. \end{aligned} \quad (14.11)$$

These are *real* Lie algebras. If $X^\dagger = -X$ and $z \in \mathbb{C}$, then

$$(zX)^\dagger = z^* X^\dagger = -z^* X.$$

This equals $-zX$ for every X only when $z = z^*$, that is, when $z \in \mathbb{R}$. Closure under real scalars, not arbitrary complex scalars, is the reason $\mathfrak{u}(n)$ and $\mathfrak{su}(n)$ are regarded here as real Lie algebras.

14.8.1 A basis and dimension for $\mathfrak{su}(2)$

The three matrices

$$T_1 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \quad T_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad T_3 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$$

are the matrices $i\sigma_1, i\sigma_2, i\sigma_3$, where the σ_i are the Pauli matrices. Each is anti-Hermitian and traceless. Every $X \in \mathfrak{su}(2)$ has the form

$$X = \begin{pmatrix} ia & b \\ -b^* & -ia \end{pmatrix}, \quad a \in \mathbb{R}, \quad b \in \mathbb{C}.$$

Thus $a, \operatorname{Re} b, \operatorname{Im} b$ are three real parameters, and

$$\dim_{\mathbb{R}} \mathfrak{su}(2) = 3.$$

14.8.2 Dimension of $\mathfrak{su}(n)$

An anti-Hermitian matrix is determined by:

- one arbitrary complex entry above the diagonal for each pair $i < j$, contributing $2 \cdot n(n-1)/2 = n(n-1)$ real parameters;
- n purely imaginary diagonal entries.

The trace-zero condition removes one real diagonal parameter. Therefore

$$\dim_{\mathbb{R}} \mathfrak{su}(n) = n(n-1) + (n-1) = n^2 - 1. \quad (14.12)$$

14.9 The special linear algebra

$$\mathfrak{sl}(n, \mathbb{C}) = \{X \in \mathfrak{gl}(n, \mathbb{C}) : \operatorname{tr} X = 0\}$$

is a complex Lie algebra. If X_i are traceless and $\lambda_i \in \mathbb{C}$, then $\sum_i \lambda_i X_i$ is traceless, and

$$\operatorname{tr}[X, Y] = \operatorname{tr}(XY) - \operatorname{tr}(YX) = 0.$$

A convenient complex basis consists of

$$E_{ij} \quad (i \neq j), \quad H_r = E_{rr} - E_{r+1, r+1} \quad (r = 1, \dots, n-1).$$

There are $n(n-1) + (n-1) = n^2 - 1$ basis elements, so

$$\dim_{\mathbb{C}} \mathfrak{sl}(n, \mathbb{C}) = n^2 - 1.$$

Viewed as a real Lie algebra, a real basis is obtained by adjoining iE_{ij} and iH_r , and

$$\dim_{\mathbb{R}} \mathfrak{sl}(n, \mathbb{C}) = 2n^2 - 2.$$

Similarly,

$$\mathfrak{sl}(n, \mathbb{R}) = \{X \in M_n(\mathbb{R}) : \operatorname{tr} X = 0\}, \quad \dim_{\mathbb{R}} \mathfrak{sl}(n, \mathbb{R}) = n^2 - 1.$$

14.10 Important groups and their Lie algebras

Group	Lie algebra	$\dim_{\mathbb{R}}$	$\dim_{\mathbb{C}}$	Matrix condition
$GL(n, \mathbb{R})$	$\mathfrak{gl}(n, \mathbb{R})$	n^2	–	$X_{ij} \in \mathbb{R}$
$GL(n, \mathbb{C})$	$\mathfrak{gl}(n, \mathbb{C})$	$2n^2$	n^2	$X_{ij} \in \mathbb{C}$
$SL(n, \mathbb{R})$	$\mathfrak{sl}(n, \mathbb{R})$	$n^2 - 1$	–	$X \in M_n(\mathbb{R}), \operatorname{tr} X = 0$
$SL(n, \mathbb{C})$	$\mathfrak{sl}(n, \mathbb{C})$	$2n^2 - 2$	$n^2 - 1$	$X \in M_n(\mathbb{C}), \operatorname{tr} X = 0$
$O(n)$	$\mathfrak{o}(n) = \mathfrak{so}(n)$	$\frac{1}{2}n(n-1)$	–	$X \in M_n(\mathbb{R}), X^T = -X$
$SO(n)$	$\mathfrak{so}(n)$	$\frac{1}{2}n(n-1)$	–	same tangent condition as $O(n)$
$U(n)$	$\mathfrak{u}(n)$	n^2	–	$X \in M_n(\mathbb{C}), X^\dagger = -X$
$SU(n)$	$\mathfrak{su}(n)$	$n^2 - 1$	–	$X \in \mathfrak{u}(n), \operatorname{tr} X = 0$
$Sp(2n, \mathbb{R})$	$\mathfrak{sp}(2n, \mathbb{R})$	$n(2n+1)$	–	$X \in M_{2n}(\mathbb{R}), X^T J = -JX$

表 14.1: The important matrix Lie groups and Lie algebras collected in the handwritten lecture. A dash means that the algebra is being treated as real, not as a complex vector space.

14.11 Brackets of subalgebras and the derived algebra

Definition 14.5: Bracket of two subalgebras

If $\mathfrak{h}_1, \mathfrak{h}_2 \leq \mathfrak{g}$, define

$$[\mathfrak{h}_1, \mathfrak{h}_2] = \operatorname{span}_{\mathbb{K}}\{[X_1, X_2] : X_1 \in \mathfrak{h}_1, X_2 \in \mathfrak{h}_2\}.$$

Definition 14.6: Derived Lie algebra

The *derived algebra*, or commutator algebra, is

$$\mathfrak{g}' = [\mathfrak{g}, \mathfrak{g}].$$

Proposition 14.1

The derived algebra $\mathfrak{g}'$ is a Lie subalgebra of $\mathfrak{g}$. In fact, it is an ideal.

Proof. It is a vector subspace by its definition as a span. If $X, Y \in \mathfrak{g}'$, then $X, Y \in \mathfrak{g}$, and the single commutator $[X, Y]$ belongs to the set whose span defines $\mathfrak{g}'$. Hence $[X, Y] \in \mathfrak{g}'$.

For the stronger ideal statement, take $Z \in \mathfrak{g}$ and a generator $[X, Y]$ of $\mathfrak{g}'$. Jacobi gives

$$[Z, [X, Y]] = [[Z, X], Y] + [X, [Z, Y]],$$

which is a sum of commutators and therefore lies in $\mathfrak{g}'$. Linearity extends this to every element of $\mathfrak{g}'$. ■

The notes also express an arbitrary element of the derived algebra in a basis:

$$X = \sum_{i,j} x_{ij} [X_i, X_j], \quad x_{ij} = -x_{ji}.$$

The antisymmetric coefficients suffice because

$$\sum_{i,j} x_{ij} [X_i, X_j] = \sum_{i,j} \frac{x_{ij} - x_{ji}}{2} [X_i, X_j];$$

the symmetric part makes no contribution.

Proposition 14.2: Characterization of Abelian algebras

$$\mathfrak{g} \text{ is Abelian} \iff \mathfrak{g}' = \{0\}.$$

Proof. If $\mathfrak{g}$ is Abelian, every commutator is zero, so its span is zero. Conversely, if $\mathfrak{g}' = \{0\}$, then every $[X, Y] \in \mathfrak{g}'$ is zero. ■

14.12 Structure constants, isomorphisms, and basis changes

14.12.1 Comparing two algebras in chosen bases

Let $\mathfrak{g}, \mathfrak{g}'$ have the same dimension, with bases $\{T_1, \dots, T_d\}$ and $\{T'_1, \dots, T'_d\}$, and structure constants

$$[T_i, T_j] = \sum_k f^k_{ij} T_k, \quad [T'_i, T'_j] = \sum_k f'^k_{ij} T'_k.$$

Define the linear bijection $\psi(T_i) = T'_i$. For

$$X = \sum_i x_i T_i, \quad Y = \sum_j y_j T_j,$$

we find

$$\begin{aligned} \psi([X, Y]) &= \sum_{i,j,k} x_i y_j f^k_{ij} T'_k, \\ [\psi(X), \psi(Y)] &= \sum_{i,j,k} x_i y_j f'^k_{ij} T'_k. \end{aligned}$$

These are equal for all coefficients x_i, y_j if and only if

$$f^k_{ij} = f'^k_{ij} \quad \text{for every } i, j, k.$$

Thus equality of structure constants in matched bases proves that the basis-matching map is a Lie-algebra isomorphism.

14.12.2 How the constants transform under a basis change

Now keep one Lie algebra $\mathfrak{g}$ but change the basis. Let

$$T'_i = \sum_k S_{ki} T_k, \quad S \in GL(d, \mathbb{K}).$$

The inverse relation is

$$T_m = \sum_n (S^{-1})_{nm} T'_n.$$

Then

$$\begin{aligned} [T'_i, T'_j] &= \left[\sum_k S_{ki} T_k, \sum_\ell S_{\ell j} T_\ell \right] \\ &= \sum_{k,\ell} S_{ki} S_{\ell j} [T_k, T_\ell] \\ &= \sum_{k,\ell,m} S_{ki} S_{\ell j} f^m_{k\ell} T_m \\ &= \sum_{k,\ell,m,n} S_{ki} S_{\ell j} f^m_{k\ell} (S^{-1})_{nm} T'_n. \end{aligned}$$

Therefore

$$\boxed{f^m_{ij} = \sum_{k,\ell,m} S_{ki} S_{\ell j} f^m_{k\ell} (S^{-1})_{nm}.} \quad (14.13)$$

There are two factors of S , one for each lower input index, and one factor of S^{-1} for the upper output index. In this precise sense the structure constants transform as the components of a tensor of type $(1, 2)$, or equivalently $(2, 1)$ if one lists the two input slots first. They are basis dependent, but the Lie bracket they describe is intrinsic.

What to retain from this chapter

1. A Lie algebra is a vector space with a bilinear, antisymmetric bracket satisfying the Jacobi identity.

2. In a basis,

$$[T_i, T_j] = f^k_{ij} T_k.$$

Antisymmetry and Jacobi become algebraic constraints on the f^k_{ij} , and conversely these constraints define a bracket.

3. The commutator makes $M_n(\mathbb{K})$ and $\text{End}(V)$ into Lie algebras. Ado's theorem says every finite-dimensional Lie algebra has a faithful matrix realization.

4. The cross-product algebra satisfies

$$(\mathbb{R}^3, \times) \cong \mathfrak{so}(3).$$

5. Differentiating the defining matrix equation at the identity gives the associated matrix Lie algebra:

$$\mathfrak{so}(n) = \{X : X^T = -X\},$$

$$\mathfrak{sp}(2n, \mathbb{R}) = \{X : X^T J + JX = 0\},$$

$$\mathfrak{su}(n) = \{X : X^\dagger = -X, \text{tr } X = 0\}.$$

- 6.

$$\dim \mathfrak{so}(n) = \frac{1}{2}n(n-1), \quad \dim \mathfrak{sp}(2n, \mathbb{R}) = n(2n+1), \quad \dim \mathfrak{su}(n) = n^2 - 1.$$

7. Over $\mathbb{R}$, $\mathfrak{sp}(2, \mathbb{R}) \cong \mathfrak{sl}(2, \mathbb{R})$, not $\mathfrak{so}(3)$; their complexifications are isomorphic.
8. The derived algebra $\mathfrak{g}' = [\mathfrak{g}, \mathfrak{g}]$ is an ideal, and $\mathfrak{g}$ is Abelian exactly when $\mathfrak{g}' = \{0\}$.
9. Structure constants change according to (14.13); the bracket itself does not depend on a basis.

The Lie Correspondence and the Exponential Map

Road map

The bridge from a matrix Lie algebra to its matrix Lie group is the exponential map. We first prove that the matrix-exponential series converges and establish its basic identities, including $\det(e^X) = e^{\text{tr } X}$. We then construct the local matrix logarithm, classify all one-parameter subgroups of $\text{GL}(n, \mathbb{K})$, and prove the Lie–Trotter product formula. After an application to quantum statistical mechanics, we define the Lie algebra of a closed matrix group and prove its vector-space, bracket, conjugation, and tangent-space properties. The final sections explain local and global surjectivity of the exponential map, products of exponentials in connected groups, and the relation between a group and its identity component. Here and throughout, $\mathbb{K}$ means either $\mathbb{R}$ or $\mathbb{C}$.

15.1 The matrix exponential

15.1.1 Definition and norm convergence

Definition 15.1: Matrix exponential

For $X \in \text{Mat}_n(\mathbb{K})$, define

$$\exp X = e^X := \sum_{k=0}^{\infty} \frac{X^k}{k!} = I + X + \frac{X^2}{2!} + \frac{X^3}{3!} + \cdots. \quad (15.1)$$

Thus the matrix exponential is initially a map

$$\exp : \text{Mat}_n(\mathbb{K}) \longrightarrow \text{Mat}_n(\mathbb{K}).$$

We shall prove that its values are automatically invertible, so its actual codomain may be taken to be $\text{GL}(n, \mathbb{K})$.

To make the infinite series rigorous, put on $\text{Mat}_n(\mathbb{K})$ the Frobenius, or Hilbert–Schmidt,

norm

$$\|X\|_F := \sqrt{\operatorname{tr}(X^\dagger X)} = \left(\sum_{i,j=1}^n |X_{ij}|^2 \right)^{1/2}.$$

For a real matrix $X^\dagger = X^T$. This is indeed a norm:

$$\|X + Y\|_F \leq \|X\|_F + \|Y\|_F, \quad \|\lambda X\|_F = |\lambda| \|X\|_F,$$

and $\|X\|_F \geq 0$, with equality only for $X = 0$.

The property needed for multiplying power series is submultiplicativity.

Lemma 15.1: Submultiplicativity of the Frobenius norm

For $X, Y \in \operatorname{Mat}_n(\mathbb{K})$,

$$\|XY\|_F \leq \|X\|_F \|Y\|_F. \quad (15.2)$$

证明. Let x_i be the i -th row of X , and let y_j be the j -th column of Y . With the standard Hermitian inner product on $\mathbb{K}^n$,

$$(XY)_{ij} = \sum_{k=1}^n X_{ik} Y_{kj} = \langle x_i^*, y_j \rangle.$$

The Cauchy–Schwarz inequality gives

$$|(XY)_{ij}|^2 \leq \|x_i\|_{\mathbb{K}^n}^2 \|y_j\|_{\mathbb{K}^n}^2.$$

Summing first over i, j and then separating the two finite sums,

$$\begin{aligned} \|XY\|_F^2 &= \sum_{i,j} |(XY)_{ij}|^2 \\ &\leq \sum_{i,j} \|x_i\|_{\mathbb{K}^n}^2 \|y_j\|_{\mathbb{K}^n}^2 \\ &= \left(\sum_i \|x_i\|_{\mathbb{K}^n}^2 \right) \left(\sum_j \|y_j\|_{\mathbb{K}^n}^2 \right) \\ &= \|X\|_F^2 \|Y\|_F^2. \end{aligned}$$

Taking square roots proves (15.2). ■

Theorem 15.1: The exponential series is well defined

For every $X \in \operatorname{Mat}_n(\mathbb{K})$, the series (15.1) converges absolutely in the Frobenius norm. Moreover,

$$\|e^X\|_F \leq e^{\|X\|_F}. \quad (15.3)$$

证明. Repeated use of (15.2) gives $\|X^k\|_F \leq \|X\|_F^k$. Hence

$$\sum_{k=0}^{\infty} \left\| \frac{X^k}{k!} \right\|_F \leq \sum_{k=0}^{\infty} \frac{\|X\|_F^k}{k!} = e^{\|X\|_F} < \infty.$$

The vector space of $n \times n$ matrices is finite dimensional and therefore complete. Absolute convergence consequently implies convergence. The same estimate, together with the triangle inequality, proves (15.3). ■

15.1.2 First algebraic properties

Proposition 15.1: Commuting exponential law

If $X, Y \in \text{Mat}_n(\mathbb{K})$ commute, then

$$e^{X+Y} = e^X e^Y = e^Y e^X. \quad (15.4)$$

In particular,

$$(e^X)^{-1} = e^{-X},$$

so $e^X \in \text{GL}(n, \mathbb{K})$ for every X .

证明. When $XY = YX$, the ordinary binomial formula remains valid:

$$(X + Y)^k = \sum_{r=0}^k \binom{k}{r} X^r Y^{k-r}.$$

Absolute convergence allows the terms to be regrouped. Therefore

$$\begin{aligned} e^{X+Y} &= \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{r=0}^k \binom{k}{r} X^r Y^{k-r} \\ &= \sum_{r,s \geq 0} \frac{X^r Y^s}{r! s!} = \left(\sum_{r=0}^{\infty} \frac{X^r}{r!} \right) \left(\sum_{s=0}^{\infty} \frac{Y^s}{s!} \right). \end{aligned}$$

This is $e^X e^Y$. Taking $Y = -X$, which certainly commutes with X , gives

$$e^{-X} e^X = e^0 = I = e^X e^{-X}.$$

■

Why commutativity matters

For arbitrary matrices it is generally false that $e^{X+Y} = e^X e^Y$. The missing information is measured, to first nontrivial order, by the commutator $[X, Y] = XY - YX$. The Lie–Trotter formula in Section 15.4 will recover e^{X+Y} without assuming that X and Y commute.

Proposition 15.2: Equivariance under similarity

For $A \in \text{GL}(n, \mathbb{K})$ and $X \in \text{Mat}_n(\mathbb{K})$,

$$e^{AXA^{-1}} = A e^X A^{-1}. \quad (15.5)$$

证明. Induction gives $(AXA^{-1})^k = AX^kA^{-1}$. Substituting this into the series,

$$e^{AXA^{-1}} = \sum_{k=0}^{\infty} \frac{AX^kA^{-1}}{k!} = A \left(\sum_{k=0}^{\infty} \frac{X^k}{k!} \right) A^{-1}.$$

■

Theorem 15.2: Determinant of an exponential

For every $X \in \text{Mat}_n(\mathbb{K})$,

$$\det(e^X) = e^{\text{tr} X}. \quad (15.6)$$

证明. First suppose that X is diagonalizable over $\mathbb{C}$:

$$X = SDS^{-1}, \quad D = \text{diag}(d_1, \dots, d_n).$$

Then (15.5) yields

$$\det(e^X) = \det(Se^D S^{-1}) = \det(e^D) = \prod_{j=1}^n e^{d_j} = e^{\sum_j d_j} = e^{\text{tr} X}.$$

The handwritten argument then observes that complex-diagonalizable matrices are dense and that both sides are continuous, which extends the equality to all matrices.

For completeness, one can avoid the density step by using complex Schur triangularization. Every complex matrix has the form $X = UTU^\dagger$, where U is unitary and T is upper triangular. The matrix e^T is upper triangular with diagonal entries $e^{T_{11}}, \dots, e^{T_{nn}}$; hence

$$\det(e^X) = \det(e^T) = \prod_j e^{T_{jj}} = e^{\text{tr} T} = e^{\text{tr} X}.$$

A real matrix may be regarded as a complex matrix, so the same calculation also proves the real case. ■

15.2 The local matrix logarithm

Definition 15.2: Logarithm near the identity

If $A \in \text{Mat}_n(\mathbb{K})$ satisfies $\|A - I\|_F < 1$, define

$$\log A := \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} (A - I)^k. \quad (15.7)$$

This is the matrix version of

$$\log(1 + a) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} a^k, \quad |a| < 1.$$

Proposition 15.3: Convergence and inverse identity

If $\|A - I\|_F < 1$, the series (15.7) converges absolutely and

$$e^{\log A} = A. \quad (15.8)$$

证明. Put $B = A - I$. Then

$$\sum_{k=1}^{\infty} \left\| \frac{(-1)^{k+1}}{k} B^k \right\|_F \leq \sum_{k=1}^{\infty} \frac{\|B\|_F^k}{k} < \infty.$$

Thus the logarithm converges in norm.

All terms involved are powers of the single matrix B , so they belong to the commutative subalgebra generated by B . Within that commutative algebra, absolutely convergent power series may be composed exactly as scalar power series. The scalar analytic identity $\exp(\log(1+z)) = 1+z$ for $|z| < 1$ therefore gives

$$\exp(\log(I+B)) = I+B = A.$$

This argument also explains why diagonalizing A is unnecessary. In particular, it remains valid for a non-diagonalizable matrix. ■

Correction to a tempting diagonalization argument

One may verify (15.8) first for diagonalizable matrices and then use density and continuity. However, for a non-unitary similarity $A = SDS^{-1}$, the inequality $\|D - I\|_F \leq \|A - I\|_F$ need not hold. The power-series proof above is both shorter and fully rigorous. Over $\mathbb{C}$, the logarithm used here is the local branch determined by the series; for scalar eigenvalues it agrees with the principal logarithm whenever that branch is defined.

The exponential also has a local inverse in the other direction. Indeed,

$$\|e^X - I\|_F \leq \sum_{k=1}^{\infty} \frac{\|X\|_F^k}{k!} = e^{\|X\|_F} - 1.$$

Consequently, if $\|X\|_F < \log 2$, then $\|e^X - I\|_F < 1$, so $\log(e^X)$ is defined. By composing the scalar series inside the commutative algebra generated by X ,

$$\log(e^X) = X \quad \text{for all sufficiently small } X. \quad (15.9)$$

Theorem 15.3: Local exponential chart

There are neighbourhoods $\mathcal{U}$ of 0 in $\text{Mat}_n(\mathbb{K})$ and $\mathcal{V}$ of I in $\text{GL}(n, \mathbb{K})$ for which

$$\exp : \mathcal{U} \longrightarrow \mathcal{V}$$

is a bijection with inverse $\log$. One may take $\mathcal{U}$ inside $\{X : \|X\|_F < \log 2\}$ and $\mathcal{V}$ inside $\{A : \|A - I\|_F < 1\}$.

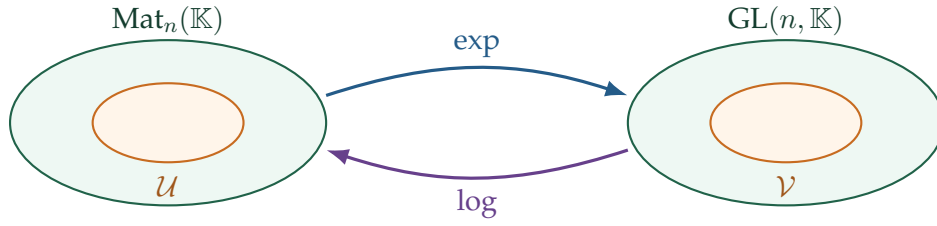


图 15.1: The exponential and logarithm are inverse maps near 0 and I . They are not generally inverse on the whole spaces.

15.3 One-parameter subgroups

Definition 15.3: One-parameter subgroup

Let G be a topological group. A *one-parameter subgroup* of G is a continuous group homomorphism

$$c : (\mathbb{R}, +) \longrightarrow G.$$

Equivalently,

$$c(t + t') = c(t)c(t'), \quad c(0) = e_G. \quad (15.10)$$

Its image $\{c(t) : t \in \mathbb{R}\}$ is an Abelian subgroup of G , because

$$c(t)c(t') = c(t + t') = c(t' + t) = c(t')c(t).$$

If $c(t)$ is close enough to the identity matrix, define

$$X(t) := \log c(t).$$

Since $c(t)$ and $c(t')$ commute, the local logarithm converts their product into a sum whenever all three matrices remain in the logarithm neighbourhood:

$$X(t + t') = \log(c(t)c(t')) = \log c(t) + \log c(t') = X(t) + X(t'). \quad (15.11)$$

Thus the local logarithm turns the multiplicative homomorphism law into an additive one.

Example 15.1: The subgroup generated by a matrix

For any $X \in \text{Mat}_n(\mathbb{K})$, set

$$c_X(t) := e^{tX}. \quad (15.12)$$

Because tX and $t'X$ commute,

$$c_X(t)c_X(t') = e^{tX}e^{t'X} = e^{(t+t')X} = c_X(t + t').$$

Moreover,

$$c_X(0) = I, \quad \dot{c}_X(t) = Xe^{tX} = e^{tX}X, \quad \dot{c}_X(0) = X.$$

Hence c_X is not only continuous but infinitely differentiable.

Theorem 15.4: Classification of one-parameter matrix subgroups

Every continuous one-parameter subgroup

$$c : \mathbb{R} \longrightarrow \mathrm{GL}(n, \mathbb{K})$$

has the form

$$c(t) = e^{tX} \quad (t \in \mathbb{R}) \quad (15.13)$$

for a unique $X \in \mathrm{Mat}_n(\mathbb{K})$. In particular, every continuous one-parameter subgroup is automatically smooth.

证明. Uniqueness. If $c(t) = e^{tX}$, differentiation at $t = 0$ gives

$$X = \dot{c}(0).$$

Thus at most one matrix X can work.

Existence near $t = 0$. Continuity and $c(0) = I$ allow us to choose $t_0 > 0$ so small that $\log c(t)$ is defined for $|t| \leq t_0$. Put

$$X := \frac{1}{t_0} \log c(t_0).$$

Then $c(t_0) = e^{t_0 X}$.

We next determine the values at dyadic subdivisions of t_0 . The homomorphism law gives

$$c(t_0/2)^2 = c(t_0) = e^{t_0 X}.$$

Both $c(t_0/2)$ and $e^{t_0 X/2}$ lie in the neighbourhood on which $\log$ is the inverse of $\exp$. Applying the local logarithm, or equivalently using uniqueness of the sufficiently small square root, gives

$$c(t_0/2) = e^{t_0 X/2}.$$

Repeating the argument,

$$c(t_0/2^k) = e^{t_0 X/2^k} \quad (k \in \mathbb{N}).$$

For every integer m with $0 \leq m \leq 2^k$,

$$c(mt_0/2^k) = c(t_0/2^k)^m = e^{mt_0 X/2^k}.$$

The dyadic points $\{mt_0/2^k\}$ are dense in $[0, t_0]$. Both c and $t \mapsto e^{tX}$ are continuous, hence

$$c(t) = e^{tX} \quad (0 \leq t \leq t_0).$$

The equality for $-t_0 \leq t \leq 0$ follows from

$$c(-t) = c(t)^{-1} = e^{-tX}.$$

Extension to every real t . Given $t \in \mathbb{R}$, choose $N \in \mathbb{N}$ with $|t/N| \leq t_0$. Then

$$c(t) = c(t/N)^N = (e^{(t/N)X})^N = e^{tX}.$$

This proves existence and completes the proof. ■

A useful conceptual consequence

For matrix groups, continuity of a homomorphism $\mathbb{R} \rightarrow \mathrm{GL}(n, \mathbb{K})$ is much stronger than it first appears: it forces analyticity. No separate differentiability assumption is needed.

15.4 The Lie–Trotter product formula**Theorem 15.5: Lie–Trotter product formula**

For arbitrary $X, Y \in \mathrm{Mat}_n(\mathbb{K})$, with no commutativity assumption,

$$\boxed{e^{X+Y} = \lim_{m \rightarrow \infty} \left(e^{X/m} e^{Y/m} \right)^m.} \quad (15.14)$$

The limit is in any matrix norm, and therefore entry by entry as well.

证明. For large m , expand each exponential to second order in m^{-1} :

$$e^{X/m} = I + \frac{X}{m} + O(m^{-2}), \quad e^{Y/m} = I + \frac{Y}{m} + O(m^{-2}).$$

The notation $O(m^{-2})$ means a matrix whose norm is at most C/m^2 for a constant C independent of sufficiently large m . Multiplying,

$$e^{X/m} e^{Y/m} = I + \frac{X+Y}{m} + O(m^{-2}). \quad (15.15)$$

This matrix approaches I , so the local logarithm is defined for all sufficiently large m . Since $\log(I + E) = E + O(\|E\|^2)$,

$$\log\left(e^{X/m} e^{Y/m}\right) = \frac{X+Y}{m} + O(m^{-2}).$$

Exponentiating after multiplying by m ,

$$\begin{aligned} \left(e^{X/m} e^{Y/m}\right)^m &= \exp\left[m \log\left(e^{X/m} e^{Y/m}\right)\right] \\ &= \exp(X + Y + O(m^{-1})). \end{aligned}$$

The matrix exponential is continuous, so the right-hand side tends to e^{X+Y} . ■

Why the formula is important

Equation (15.14) separates a sum of noncommuting generators into many alternating short evolutions. This is a basic tool in quantum Monte Carlo methods, imaginary-time path integrals, operator splitting, and the numerical solution of differential equations.

15.4.1 Application: free energy and imaginary-time response

The handwritten lecture applies the product formula to a standard problem from quantum statistical mechanics. Let

$$H(\lambda) = H_0 + \lambda A, \quad \beta := \frac{1}{k_B T},$$

where H_0 and A are Hermitian matrices. They need not commute. The partition function and free energy are

$$Z(\lambda) = \text{tr} e^{-\beta H(\lambda)}, \quad F(\lambda) = -k_B T \log Z(\lambda) = -\frac{1}{\beta} \log Z(\lambda). \quad (15.16)$$

For any observable B , its thermal expectation value at the chosen λ is

$$\langle B \rangle_\lambda := \frac{1}{Z(\lambda)} \text{tr} \left(B e^{-\beta H(\lambda)} \right).$$

The derivative of a noncommutative exponential cannot be found by simply moving $H'(\lambda)$ through the exponential. The correct finite-dimensional Duhamel formula is

$$\frac{d}{d\lambda} e^{M(\lambda)} = \int_0^1 e^{(1-s)M(\lambda)} M'(\lambda) e^{sM(\lambda)} ds. \quad (15.17)$$

It follows either by differentiating the power series and regrouping the possible insertion positions of M' , or by applying the Lie–Trotter formula to $e^{M(\lambda) + \delta M'(\lambda)}$ and taking $\delta \rightarrow 0$.

For $M = -\beta H$, formula (15.17) becomes

$$\frac{\partial}{\partial \lambda} e^{-\beta H} = - \int_0^\beta e^{-(\beta-\tau)H} A e^{-\tau H} d\tau. \quad (15.18)$$

Taking the trace and using cyclic invariance,

$$\text{tr}(BCD) = \text{tr}(CDB) = \text{tr}(DBC),$$

gives

$$\begin{aligned} Z'(\lambda) &= - \int_0^\beta \text{tr} \left(e^{-(\beta-\tau)H} A e^{-\tau H} \right) d\tau \\ &= - \int_0^\beta \text{tr} \left(A e^{-\beta H} \right) d\tau \\ &= -\beta \text{tr} \left(A e^{-\beta H} \right). \end{aligned}$$

Consequently,

$$\boxed{\frac{\partial F}{\partial \lambda} = -\frac{1}{\beta} \frac{Z'}{Z} = \langle A \rangle_\lambda.} \quad (15.19)$$

This conclusion is valid even when $[H_0, A] \neq 0$. The cyclic invariance of the trace is precisely what makes the first derivative look classical.

For the second derivative, define the imaginary-time Heisenberg operator

$$A(\tau) := e^{\tau H} A e^{-\tau H}, \quad 0 \leq \tau \leq \beta. \quad (15.20)$$

Differentiating $\langle A \rangle = Z^{-1} \text{tr}(Ae^{-\beta H})$, and using (15.18), yields

$$\begin{aligned} \frac{\partial^2 F}{\partial \lambda^2} &= -\frac{Z'}{Z^2} \text{tr}(Ae^{-\beta H}) + \frac{1}{Z} \text{tr}\left(A \frac{\partial}{\partial \lambda} e^{-\beta H}\right) \\ &= \beta \langle A \rangle^2 - \frac{1}{Z} \int_0^\beta \text{tr}\left(Ae^{-(\beta-\tau)H} Ae^{-\tau H}\right) d\tau \\ &= \boxed{\beta \langle A \rangle^2 - \int_0^\beta \langle A(\tau) A(0) \rangle d\tau.} \end{aligned} \quad (15.21)$$

The integral is the imaginary-time, or Kubo–Mori, correlation. This is the thermodynamic response quantity requested in the handwritten calculation.

If A commutes with H , then $A(\tau) = A$, and (15.21) reduces to

$$\frac{\partial^2 F}{\partial \lambda^2} = \beta \langle A \rangle^2 - \beta \langle A^2 \rangle = -\beta (\langle A^2 \rangle - \langle A \rangle^2). \quad (15.22)$$

Thus, for the convention $H = H_0 + \lambda A$, the curvature is minus β times the variance in the commuting case.

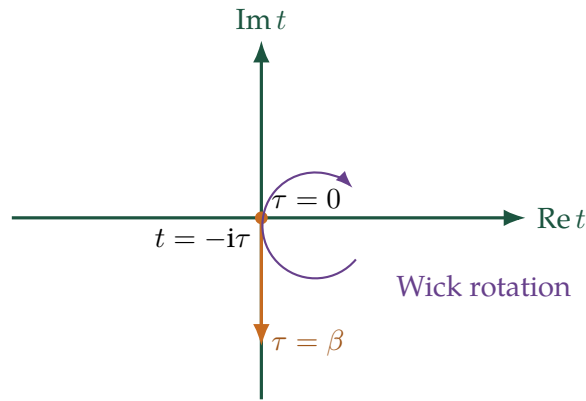


图 15.2: Real time lies on the horizontal axis. Thermal correlations evolve down the imaginary-time segment $t = -i\tau$, $0 \leq \tau \leq \beta$.

What cyclicity can and cannot do

Cyclicity is sufficient to reduce Z' to a single expectation value. For the second derivative, two copies of A occur at different imaginary times, and cyclicity cannot in general turn the integral into $\beta \langle A^2 \rangle$. That simplification is valid only when the relevant operators commute.

15.5 The Lie algebra of a closed matrix group

Definition 15.4: Linear Lie group and its Lie algebra

A linear Lie group over $\mathbb{K}$ is a subgroup

$$G \leq \mathrm{GL}(n, \mathbb{K})$$

that is closed in the matrix topology. Its Lie algebra is

$$\mathrm{Lie}(G) := \{X \in \mathrm{Mat}_n(\mathbb{K}) : e^{tX} \in G \text{ for every } t \in \mathbb{R}\}. \quad (15.23)$$

We often abbreviate $\mathfrak{g} = \mathrm{Lie}(G)$.

The definition says exactly that $X \in \mathfrak{g}$ when the one-parameter subgroup generated by X lies entirely inside G . Notice that the parameter t is real even when $\mathbb{K} = \mathbb{C}$.

Theorem 15.6: The set $\mathrm{Lie}(G)$ is a Lie algebra

Let $G \leq \mathrm{GL}(n, \mathbb{K})$ be closed. Then $\mathfrak{g} = \mathrm{Lie}(G)$ is a real vector subspace of $\mathrm{Mat}_n(\mathbb{K})$, is closed in the matrix topology, and is closed under the commutator bracket

$$[X, Y] = XY - YX.$$

Hence $\mathfrak{g}$ is a real matrix Lie algebra.

证明. Addition. Suppose $X, Y \in \mathfrak{g}$. For every $t \in \mathbb{R}$ and $m \in \mathbb{N}$,

$$e^{tX/m} \in G, \quad e^{tY/m} \in G,$$

so, because G is a subgroup,

$$\left(e^{tX/m} e^{tY/m}\right)^m \in G.$$

The Lie–Trotter formula gives

$$e^{t(X+Y)} = \lim_{m \rightarrow \infty} \left(e^{tX/m} e^{tY/m}\right)^m.$$

Closedness of G implies $e^{t(X+Y)} \in G$. Since this is true for every t , $X + Y \in \mathfrak{g}$.

Real scalar multiplication. If $\lambda \in \mathbb{R}$, then

$$e^{t(\lambda X)} = e^{(t\lambda)X} \in G \quad (t \in \mathbb{R}),$$

so $\lambda X \in \mathfrak{g}$.

Closedness of $\mathfrak{g}$. Let $X_j \in \mathfrak{g}$ and $X_j \rightarrow X$. For every fixed t , continuity of the exponential gives

$$e^{tX_j} \rightarrow e^{tX}.$$

Every e^{tX_j} lies in G , and G is closed; hence $e^{tX} \in G$. Therefore $X \in \mathfrak{g}$.

Closure under brackets. For $X, Y \in \mathfrak{g}$, define

$$C(t) := e^{tY} X e^{-tY}.$$

We first check that $C(t) \in \mathfrak{g}$. For every $s \in \mathbb{R}$, equivariance of the exponential gives

$$e^{sC(t)} = e^{tY} e^{sX} e^{-tY} \in G.$$

Thus $C(t) \in \mathfrak{g}$. Since $\mathfrak{g}$ is a vector space,

$$\frac{C(t) - C(0)}{t} \in \mathfrak{g} \quad (t \neq 0).$$

Since $\mathfrak{g}$ is closed, the limit belongs to $\mathfrak{g}$:

$$\begin{aligned} C'(0) &= \left. \frac{d}{dt} (e^{tY} X e^{-tY}) \right|_{t=0} \\ &= YX - XY \\ &= -[X, Y]. \end{aligned}$$

Finally, real scalar closure gives $[X, Y] \in \mathfrak{g}$. ■

Lemma 15.2: Invariance under group conjugation

For $A \in G$ and $X \in \mathfrak{g}$,

$$AXA^{-1} \in \mathfrak{g}. \quad (15.24)$$

证明. For every $t \in \mathbb{R}$,

$$e^{tAXA^{-1}} = A e^{tX} A^{-1}.$$

All three factors on the right lie in G , so the product lies in G . Definition (15.23) now gives $AXA^{-1} \in \mathfrak{g}$. ■

Analogy with a normal subgroup

Equation (15.24) resembles the defining condition for a normal subgroup: $\mathfrak{g}$ is invariant under conjugation by elements of G . Nevertheless $\mathfrak{g}$ is not a subgroup of G . Its operation as a vector space is addition, its zero is the zero matrix, and usually its elements are not group elements at all. By contrast, the identity of G is I , and the group operation is matrix multiplication.

Definition 15.5: Complex linear Lie group

Suppose $G \leq \mathrm{GL}(n, \mathbb{C})$. We call G a *complex linear Lie group* if $\mathrm{Lie}(G)$ is a complex vector subspace of $\mathrm{Mat}_n(\mathbb{C})$. Equivalently,

$$X \in \mathrm{Lie}(G) \implies iX \in \mathrm{Lie}(G).$$

Without this extra property, $\mathrm{Lie}(G)$ is regarded as a real Lie algebra even though its matrices have complex entries.

The groups $GL(n, \mathbb{C})$ and $SL(n, \mathbb{C})$ are complex linear Lie groups. The unitary groups are not: multiplying a skew-Hermitian matrix by i makes it Hermitian rather than skew-Hermitian.

15.5.1 Exponentials lie in the identity component

Let G° denote the connected component of I in G . If $X \in \mathfrak{g}$, the path

$$[0, 1] \longrightarrow G, \quad s \longmapsto e^{sX},$$

joins I to e^X . Therefore

$$\boxed{e^X \in G^\circ \text{ for every } X \in \mathfrak{g}.} \quad (15.25)$$

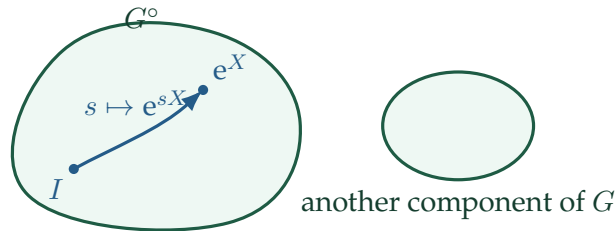


图 15.3: Every exponential generated by $\text{Lie}(G)$ lies in the identity component.

Proposition 15.4: Abelian groups and Abelian Lie algebras

If G is Abelian, then $\text{Lie}(G)$ is Abelian.

证明. For $X, Y \in \mathfrak{g}$, the group elements e^{tX} and e^{sY} commute for every s, t . Hence the group commutator

$$K(t, s) := e^{tX} e^{sY} e^{-tX} e^{-sY}$$

is identically I . Differentiating once in each variable at $(t, s) = (0, 0)$ gives

$$0 = \left. \frac{\partial^2 K}{\partial t \partial s} \right|_{(0,0)} = XY - YX = [X, Y].$$

Thus all brackets in $\mathfrak{g}$ vanish. ■

The converse needs connectedness. It will follow in Section 15.9: if G is connected and $\text{Lie}(G)$ is Abelian, then G is Abelian.

15.6 Classical groups and their Lie algebras

The definition (15.23), together with the identities already proved, quickly recovers the familiar classical matrix Lie algebras.

15.6.1 General and special linear groups

Example 15.2: $\mathrm{GL}(n, \mathbb{C})$ and $\mathrm{GL}(n, \mathbb{R})$

Every matrix exponential is invertible. Therefore

$$\mathrm{Lie}(\mathrm{GL}(n, \mathbb{C})) = \mathfrak{gl}(n, \mathbb{C}) = \mathrm{Mat}_n(\mathbb{C}),$$

which is a complex vector space, whereas

$$\mathrm{Lie}(\mathrm{GL}(n, \mathbb{R})) = \mathfrak{gl}(n, \mathbb{R}) = \mathrm{Mat}_n(\mathbb{R}),$$

which is a real vector space. If X is real, e^{tX} is real for real t ; there is no reason for e^{itX} to be real.

Example 15.3: $\mathrm{SL}(n, \mathbb{K})$

By (15.6),

$$\det(e^{tX}) = e^{t \operatorname{tr} X}.$$

If $\operatorname{tr} X = 0$, then $\det(e^{tX}) = 1$ for every t , so $X \in \mathrm{Lie}(\mathrm{SL}(n, \mathbb{K}))$.

Conversely, if $e^{tX} \in \mathrm{SL}(n, \mathbb{K})$ for every real t , then

$$1 = e^{t \operatorname{tr} X}.$$

Differentiating at $t = 0$ gives $\operatorname{tr} X = 0$. Hence

$$\mathrm{Lie}(\mathrm{SL}(n, \mathbb{K})) = \mathfrak{sl}(n, \mathbb{K}) = \{X \in \mathrm{Mat}_n(\mathbb{K}) : \operatorname{tr} X = 0\}.$$

For $\mathbb{K} = \mathbb{C}$, this is a complex linear Lie algebra; for $\mathbb{K} = \mathbb{R}$, it is real.

15.6.2 Unitary and orthogonal groups

Example 15.4: The unitary Lie algebra

For $X \in \text{Mat}_n(\mathbb{C})$,

$$\begin{aligned} e^{tX} \in U(n) \quad \text{for every } t &\iff (e^{tX})^\dagger e^{tX} = I \quad \text{for every } t \\ &\iff e^{tX^\dagger} e^{tX} = I \quad \text{for every } t. \end{aligned}$$

If $X^\dagger = -X$, the two exponents commute and the last product equals $e^{-tX} e^{tX} = I$. Conversely, differentiating the unitary condition at $t = 0$ gives

$$X^\dagger + X = 0.$$

Therefore

$$\mathfrak{u}(n) = \text{Lie}(U(n)) = \{X \in \text{Mat}_n(\mathbb{C}) : X^\dagger = -X\}. \quad (15.26)$$

Adding the determinant-one condition adds the infinitesimal trace-zero condition:

$$\mathfrak{su}(n) = \text{Lie}(SU(n)) = \{X : X^\dagger = -X, \text{tr } X = 0\}. \quad (15.27)$$

Neither $U(n)$ nor $SU(n)$ is a complex linear Lie group. Indeed, if $X^\dagger = -X$, then

$$(iX)^\dagger = -iX^\dagger = iX,$$

so iX is Hermitian, not skew-Hermitian, unless $X = 0$.

Example 15.5: The orthogonal Lie algebra

Exactly the same calculation with transpose in place of adjoint gives

$$\mathfrak{o}(n) = \text{Lie}(O(n)) = \{X \in \text{Mat}_n(\mathbb{R}) : X^T = -X\}. \quad (15.28)$$

A real skew-symmetric matrix has zero diagonal, hence zero trace. Therefore the additional condition $\det A = 1$ does not reduce the Lie algebra:

$$\text{Lie}(SO(n)) = \mathfrak{so}(n) = \mathfrak{o}(n). \quad (15.29)$$

Thus $O(n)$ and its identity component $SO(n)$ have the same infinitesimal directions, even though $O(n)$ has two connected components.

Group G	$\text{Lie}(G)$	scalar field of the Lie algebra
$\text{GL}(n, \mathbb{C})$	all $X \in \text{Mat}_n(\mathbb{C})$	complex
$\text{SL}(n, \mathbb{C})$	$\text{tr } X = 0$	complex
$\text{GL}(n, \mathbb{R})$	all $X \in \text{Mat}_n(\mathbb{R})$	real
$\text{SL}(n, \mathbb{R})$	$\text{tr } X = 0$	real
$U(n)$	$X^\dagger = -X$	real
$SU(n)$	$X^\dagger = -X, \text{tr } X = 0$	real
$O(n), SO(n)$	$X^T = -X, X \text{ real}$	real

表 15.1: Classical matrix groups and their Lie algebras.

15.6.3 The physics convention

Mathematicians usually put skew-Hermitian generators in $\mathfrak{u}(n)$, so a one-parameter unitary group is written e^{tX} . In physics, one normally prefers Hermitian generators and writes

$$U(t) = e^{itH}.$$

The two conventions differ by the factor i :

$$X = iH, \quad X^\dagger = -X \iff H^\dagger = H.$$

Accordingly, in the physics convention one may identify

$$\mathfrak{u}(n)_{\text{phys}} = \{H : H^\dagger = H\},$$

and

$$\mathfrak{su}(n)_{\text{phys}} = \{H : H^\dagger = H, \text{tr } H = 0\}.$$

One must then also transport the bracket. If $X = iH$ and $Y = iK$, the skew-Hermitian commutator corresponds to the Hermitian bracket

$$[H, K]_{\text{phys}} := i(HK - KH)$$

up to the sign convention chosen by the author. Keeping track of this factor prevents apparent disagreements between mathematics and physics texts.

15.7 The exponential map of a linear Lie group

Definition 15.6: Group exponential map

For a linear Lie group G with $\mathfrak{g} = \text{Lie}(G)$, the *exponential map* of G is the restriction

$$\exp : \mathfrak{g} \longrightarrow G, \quad X \longmapsto e^X. \quad (15.30)$$

The map need be neither injective nor surjective globally, although it is always locally bijective at 0.

Example 15.6: Failure of injectivity for $SU(2)$

Let

$$X = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \in \mathfrak{su}(2).$$

Then

$$e^{0X} = I, \quad e^{2\pi X} = \begin{pmatrix} e^{2\pi i} & 0 \\ 0 & e^{-2\pi i} \end{pmatrix} = I.$$

Since $0 \neq 2\pi X$, the map $\exp : \mathfrak{su}(2) \rightarrow SU(2)$ is not injective.

Example 15.7: Failure of surjectivity for $SL(2, \mathbb{C})$

Consider

$$A = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}.$$

Its determinant is 1, so $A \in SL(2, \mathbb{C})$. Its only eigenvalue is -1 , with algebraic multiplicity two, but

$$A + I = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

has a one-dimensional kernel. Hence A is not diagonalizable.

Suppose, for contradiction, that $A = e^X$ for some $X \in \mathfrak{sl}(2, \mathbb{C})$. If μ_1, μ_2 are the eigenvalues of X , spectral mapping gives $e^{\mu_1} = e^{\mu_2} = -1$. Thus

$$\mu_j = (2k_j + 1)\pi i \quad (k_j \in \mathbb{Z}).$$

Because $\text{tr } X = 0$, $\mu_2 = -\mu_1$; in particular the two eigenvalues are distinct, since neither can be zero. Therefore X is diagonalizable. In a basis diagonalizing X ,

$$e^X = \text{diag}(-1, -1) = -I,$$

contradicting $A \neq -I$. Consequently

$$\exp : \mathfrak{sl}(2, \mathbb{C}) \longrightarrow SL(2, \mathbb{C})$$

is not surjective.

15.7.1 Local bijectivity on a closed matrix group**Theorem 15.7: Local exponential theorem**

Let $G \leq GL(n, \mathbb{K})$ be a closed matrix group and $\mathfrak{g} = \text{Lie}(G)$. There are neighbourhoods $\mathcal{U}_G$ of 0 in $\mathfrak{g}$ and $\mathcal{V}_G$ of I in G such that

$$\exp : \mathcal{U}_G \longrightarrow \mathcal{V}_G \tag{15.31}$$

is a smooth bijection with smooth inverse $\log$. Equivalently, every element of G sufficiently close to I has a unique sufficiently small logarithm belonging to $\mathfrak{g}$.

证明. The ambient exponential is a smooth bijection from a neighbourhood $\mathcal{U} \subset \text{Mat}_n(\mathbb{K})$ of 0 onto a neighbourhood $\mathcal{V} \subset \text{GL}(n, \mathbb{K})$ of I , with inverse given by the convergent logarithm series. By the closed-subgroup theorem proved in Chapter 13, a closed matrix group is an embedded submanifold, and its tangent space at I is precisely the space of infinitesimal one-parameter subgroups. In the present notation that space is $\mathfrak{g}$. The differential of the exponential at 0 is

$$(\text{d exp})_0(X) = \left. \frac{d}{dt} e^{tX} \right|_{t=0} = X,$$

so its restriction from $\mathfrak{g}$ to $T_I G$ is the identity. The inverse-function theorem on these equal-dimensional manifolds therefore gives neighbourhoods $\mathcal{U}_G$ and $\mathcal{V}_G$ for which (15.31) is a diffeomorphism.

The direct tangent-space verification $T_I G = \mathfrak{g}$, which also removes any possible ambiguity in this use of the closed-subgroup theorem, is given below. ■

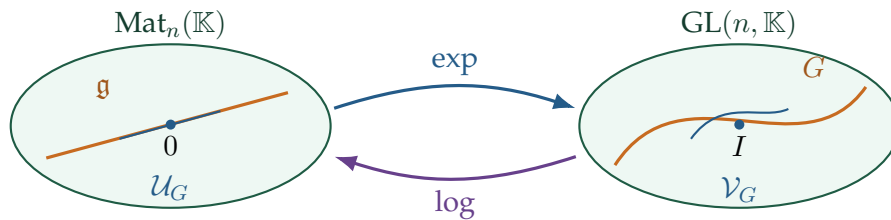


图 15.4: Restriction of the ambient local exponential chart to $\mathfrak{g}$ and G .

15.8 The tangent-space interpretation

Definition 15.7: Tangent space to a matrix group

Let $G \leq \text{GL}(n, \mathbb{K})$ be a linear Lie group and let $A \in G$. The tangent space at A is

$$T_A G := \{ \dot{c}(0) : c : (-\varepsilon, \varepsilon) \rightarrow G \text{ is smooth and } c(0) = A \}. \quad (15.32)$$

Thus a tangent vector is the velocity of a smooth curve in G .

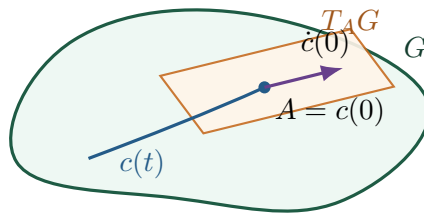


图 15.5: A smooth curve in G and its tangent vector at A .

Theorem 15.8: Lie algebra equals the tangent space at the identity

For every linear Lie group G ,

$$T_I G = \text{Lie}(G). \quad (15.33)$$

证明. $\text{Lie}(G) \subseteq T_I G$. If $X \in \text{Lie}(G)$, the curve

$$c_X(t) = e^{tX}$$

lies in G , satisfies $c_X(0) = I$, and has $\dot{c}_X(0) = X$. Hence $X \in T_I G$.

$T_I G \subseteq \text{Lie}(G)$. Let $X \in T_I G$. Then there is a smooth curve

$$c : (-\varepsilon, \varepsilon) \rightarrow G, \quad c(0) = I, \quad \dot{c}(0) = X.$$

For sufficiently small t , $c(t)$ lies in the neighbourhood $\mathcal{V}_G$ from the local exponential theorem above. Therefore $\log c(t) \in \mathfrak{g}$. Since $\mathfrak{g}$ is a vector space,

$$\frac{\log c(t)}{t} \in \mathfrak{g} \quad (t \neq 0).$$

Now expand the logarithm:

$$\log c(t) = (c(t) - I) - \frac{1}{2}(c(t) - I)^2 + \cdots.$$

Because $c(t) - I = tX + o(t)$, every term of degree at least two is $o(t)$. Thus

$$\lim_{t \rightarrow 0} \frac{\log c(t)}{t} = \lim_{t \rightarrow 0} \frac{c(t) - I}{t} = \dot{c}(0) = X.$$

The Lie algebra $\mathfrak{g}$ is closed, so $X \in \mathfrak{g}$. ■

Proposition 15.5: Tangent spaces at arbitrary points

For every $A \in G$,

$$T_A G = A T_I G = A \mathfrak{g}. \quad (15.34)$$

Equivalently, right translation gives $T_A G = \mathfrak{g}A$.

证明. If $c(0) = I$ and $\dot{c}(0) = X$, then $\bar{c}(t) := Ac(t)$ satisfies

$$\bar{c}(0) = A, \quad \dot{\bar{c}}(0) = AX.$$

This proves $A\mathfrak{g} \subseteq T_A G$. Conversely, if $\bar{c}(0) = A$, then $c(t) := A^{-1}\bar{c}(t)$ is a curve through I , and

$$\dot{c}(0) = A^{-1}\dot{\bar{c}}(0) \in \mathfrak{g}.$$

Thus $\dot{\bar{c}}(0) \in A\mathfrak{g}$, proving equality. ■

Definition 15.8: Dimension of a linear Lie group

The dimension of G is

$$\dim G := \dim_{\mathbb{R}} \text{Lie}(G). \quad (15.35)$$

Even when the matrices are complex, the Lie algebra is counted as a real vector space unless G has explicitly been declared a complex Lie group and a complex dimension is requested.

Since every tangent space is a translate of $T_I G$, all tangent spaces have the same dimension. The closed-subgroup theorem therefore realizes G as an embedded submanifold of $\text{Mat}_n(\mathbb{K})$ of

dimension $\dim_{\mathbb{R}} \mathfrak{g}$.

15.9 Connected groups are generated by exponentials

A Lie group is locally Euclidean and therefore locally path connected. Consequently, a connected Lie group is path connected. If G is connected and $A \in G$, we may choose a continuous path

$$c : [0, 1] \longrightarrow G, \quad c(0) = I, \quad c(1) = A.$$

Uniform continuity of c allows the path to be divided into sufficiently short pieces whose relative increments lie in the local exponential neighbourhood.

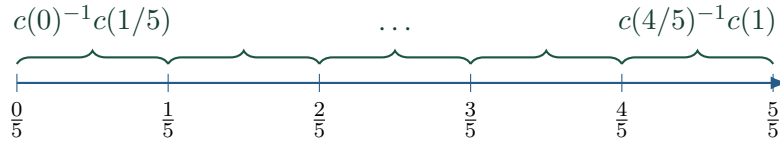


图 15.6: Divide a path into pieces whose relative changes are all close to the identity.

Theorem 15.9: Finite product of exponentials

If G is a connected linear Lie group, then every $A \in G$ can be written

$$A = e^{X_1} e^{X_2} \cdots e^{X_m}, \quad X_j \in \text{Lie}(G). \quad (15.36)$$

证明. Choose a path c from I to A . Let $\mathcal{V}_G$ be the neighbourhood of I on which the group logarithm exists. Because the map

$$(s, t) \longmapsto c(s)^{-1}c(t)$$

is continuous on the compact square $[0, 1]^2$, it is uniformly continuous. We may therefore choose m large enough that

$$c((j-1)/m)^{-1}c(j/m) \in \mathcal{V}_G \quad (j = 1, \dots, m).$$

By the local exponential theorem, for each j there is an $X_j \in \mathfrak{g}$ such that

$$c((j-1)/m)^{-1}c(j/m) = e^{X_j}.$$

Multiplying these equations in order makes all intermediate path values cancel:

$$\begin{aligned} A &= c(0)^{-1}c(1) \\ &= [c(0)^{-1}c(1/m)] [c(1/m)^{-1}c(2/m)] \cdots [c((m-1)/m)^{-1}c(1)] \\ &= e^{X_1} e^{X_2} \cdots e^{X_m}. \end{aligned}$$

■

One exponential versus a product

Connectedness guarantees a *finite product* of exponentials. It does not guarantee that $A = e^X$ for one X . The defective matrix in $SL(2, \mathbb{C})$ above shows why the distinction is essential.

Theorem 15.10: Connected group with Abelian Lie algebra

If G is connected and $\text{Lie}(G)$ is Abelian, then G is Abelian.

证明. If $X, Y \in \mathfrak{g}$, then $[X, Y] = 0$, so

$$e^X e^Y = e^{X+Y} = e^Y e^X.$$

Write arbitrary $A, B \in G$ as products of exponentials:

$$A = e^{X_1} \cdots e^{X_m}, \quad B = e^{Y_1} \cdots e^{Y_r}.$$

Every X_i commutes with every Y_j , so all the exponential factors commute. Reordering them gives $AB = BA$. ■

Combining the two implications just proved, we obtain

$$G \text{ Abelian} \implies \text{Lie}(G) \text{ Abelian}$$

without any connectedness assumption, and

$$\text{Lie}(G) \text{ Abelian} \implies G \text{ Abelian}$$

when G is connected.

15.10 When one exponential is enough

15.10.1 Surjectivity for $SU(n)$

Theorem 15.11: Every special unitary matrix is an exponential

For every $A \in SU(n)$, there exists $X \in \mathfrak{su}(n)$ such that

$$A = e^X.$$

Thus $\exp : \mathfrak{su}(n) \rightarrow SU(n)$ is surjective.

证明. A unitary matrix is normal, so the spectral theorem gives a unitary matrix S and phases $\theta_1, \dots, \theta_n \in \mathbb{R}$ such that

$$A = SDS^\dagger, \quad D = \text{diag}(e^{i\theta_1}, \dots, e^{i\theta_n}).$$

Because $\det A = 1$,

$$1 = \det D = e^{i(\theta_1 + \cdots + \theta_n)}.$$

Therefore

$$\theta_1 + \cdots + \theta_n = 2\pi k \quad \text{for some } k \in \mathbb{Z}.$$

Replace, for example, θ_n by

$$\theta'_n := \theta_n - 2\pi k.$$

This does not change $e^{i\theta_n}$, but now

$$\theta_1 + \cdots + \theta_{n-1} + \theta'_n = 0.$$

Set

$$\Theta := \text{diag}(\theta_1, \dots, \theta_{n-1}, \theta'_n), \quad X := iS\Theta S^\dagger.$$

Then

$$X^\dagger = -iS\Theta S^\dagger = -X, \quad \text{tr } X = i \text{tr } \Theta = 0,$$

so $X \in \mathfrak{su}(n)$. Finally,

$$e^X = Se^{i\Theta}S^\dagger = SDS^\dagger = A.$$

■

The phase correction that must not be skipped

Choosing all phases in $[0, 2\pi)$ only implies that their sum is an integer multiple of 2π , not literally zero. Subtracting that multiple from one phase leaves the unitary eigenvalue unchanged and makes the logarithm traceless. This small adjustment is essential for the generator to lie in $\mathfrak{su}(n)$.

15.10.2 Surjectivity for $SO(n)$

Theorem 15.12: Every special orthogonal matrix is an exponential

For every $A \in SO(n)$, there exists $X \in \mathfrak{so}(n)$ such that

$$A = e^X.$$

证明. The real spectral theorem for orthogonal matrices gives an orthogonal matrix Q for which $Q^T A Q$ is block diagonal with 2×2 rotation blocks

$$R(\theta_j) = \begin{pmatrix} \cos \theta_j & -\sin \theta_j \\ \sin \theta_j & \cos \theta_j \end{pmatrix}$$

and 1×1 blocks 1 or -1 . Since $\det A = 1$, the number of -1 blocks is even. Pair two -1 blocks and regard their direct sum as $R(\pi)$. We may therefore write

$$Q^T A Q = \text{diag}(R(\theta_1), \dots, R(\theta_r), I_{n-2r}).$$

Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Since $J^2 = -I_2$, separating the exponential series into even and odd powers gives

$$e^{\theta J} = I_2 \cos \theta + J \sin \theta = R(\theta).$$

Now define

$$X := Q \operatorname{diag}(\theta_1 J, \dots, \theta_r J, 0_{n-2r}) Q^T.$$

Every block $\theta_j J$ is real and skew-symmetric; hence $X^T = -X$, so $X \in \mathfrak{so}(n)$. Moreover,

$$e^X = Q \operatorname{diag}(R(\theta_1), \dots, R(\theta_r), I) Q^T = A.$$

■

Why a local argument is not enough here

It is tempting to embed $SO(n)$ in $SU(n)$, write $A = e^X$, and infer from $AA^T = I$ that $X + X^T = 0$. That inference is valid only in a neighbourhood where the logarithm is unique, or when the required commutativity has been established. The rotation-block construction above is global and directly produces a real skew-symmetric logarithm.

Theorem 15.13: Connected compact groups have surjective exponential

If G is a connected compact linear Lie group, then

$$G = \exp(\operatorname{Lie}(G)). \quad (15.37)$$

证明. The maximal-torus theorem says that every element A of a compact connected Lie group lies in some maximal torus $T \leq G$. A torus is isomorphic to

$$U(1)^r = \mathbb{R}^r / (2\pi\mathbb{Z})^r$$

for some r , and its exponential map

$$\operatorname{Lie}(T) \cong \mathbb{R}^r \longrightarrow T$$

is surjective, coordinate by coordinate:

$$(\theta_1, \dots, \theta_r) \longmapsto (e^{i\theta_1}, \dots, e^{i\theta_r}).$$

Thus $A = e^X$ for some $X \in \operatorname{Lie}(T) \subseteq \operatorname{Lie}(G)$.

■

Role of compactness

The theorem explains the surjectivity for $SU(n)$ and $SO(n)$, both of which are compact and connected. Compactness cannot be omitted: $\operatorname{SL}(2, \mathbb{C})$ is connected but the earlier defective matrix is not one exponential.

15.11 The identity component has the same Lie algebra

Theorem 15.14: The identity component of a linear Lie group

Let $G \leq \mathrm{GL}(n, \mathbb{K})$ be a linear Lie group. Then

1. G° is a closed normal subgroup of G , hence itself a linear Lie group;
- 2.

$$\mathrm{Lie}(G^\circ) = \mathrm{Lie}(G). \quad (15.38)$$

证明. The fact that the identity component is a normal subgroup was proved topologically in Chapter 13: multiplication, inversion, and conjugation send connected sets to connected sets.

It remains useful to see directly why G° is closed in the matrix topology. Let $A_m \in G^\circ$ and suppose $A_m \rightarrow A$. Since G is closed, $A \in G$. Furthermore,

$$A_m^{-1}A \rightarrow I.$$

For sufficiently large m , the local exponential theorem gives

$$A_m^{-1}A = e^X \quad \text{for some } X \in \mathfrak{g}.$$

There is a path in G from I to A_m , because $A_m \in G^\circ$. The curve

$$s \mapsto A_m e^{sX}, \quad 0 \leq s \leq 1,$$

then joins A_m to $A_m e^X = A$. Concatenating the two paths joins I to A , so $A \in G^\circ$. Thus G° is closed.

Because $G^\circ \subseteq G$,

$$\mathrm{Lie}(G^\circ) \subseteq \mathrm{Lie}(G).$$

Conversely, take $X \in \mathrm{Lie}(G)$. For each fixed $t \in \mathbb{R}$, the path

$$[0, 1] \rightarrow G, \quad s \mapsto e^{stX},$$

joins I to e^{tX} . Hence $e^{tX} \in G^\circ$ for every t , which says $X \in \mathrm{Lie}(G^\circ)$. This proves the reverse inclusion and (15.38). ■

Example 15.8: $O(n)$ and $SO(n)$

The determinant is continuous and takes only the values ± 1 on $O(n)$. Therefore no path in $O(n)$ can join a determinant-+1 matrix to a determinant-−1 matrix. The identity component is

$$O(n)^\circ = SO(n),$$

and the theorem gives

$$\mathrm{Lie}(O(n)) = \mathrm{Lie}(SO(n)) = \mathfrak{so}(n).$$

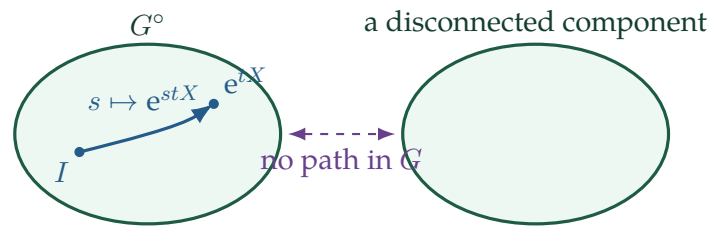


图 15.7: Every one-parameter subgroup through I stays in G° ; the other components introduce no new infinitesimal directions.

What the Lie algebra cannot see

The Lie algebra records the geometry of G arbitrarily close to the identity. It therefore determines the infinitesimal structure of G° , but it does not record how many disconnected components the whole group has. More precisely, the discrete component group G/G° is invisible to $\text{Lie}(G)$. This is why $O(n)$ and $SO(n)$ have the same Lie algebra even though the first group has an additional determinant- -1 component.

What to retain from this chapter

1. The matrix exponential is the absolutely convergent series

$$e^X = \sum_{k=0}^{\infty} \frac{X^k}{k!}, \quad \|e^X\|_F \leq e^{\|X\|_F}.$$

It is always invertible, with inverse e^{-X} .

2. If $XY = YX$, then $e^{X+Y} = e^X e^Y$. For arbitrary matrices,

$$e^{X+Y} = \lim_{m \rightarrow \infty} (e^{X/m} e^{Y/m})^m.$$

3. Similarity and determinant behave naturally:

$$e^{AXA^{-1}} = Ae^X A^{-1}, \quad \det(e^X) = e^{\text{tr } X}.$$

4. Near I ,

$$\log A = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} (A - I)^k$$

is the inverse of the exponential.

5. Every continuous one-parameter subgroup of $\text{GL}(n, \mathbb{K})$ is uniquely $c(t) = e^{tX}$, with $X = \dot{c}(0)$.

6. For a closed matrix group,

$$\text{Lie}(G) = \{X : e^{tX} \in G \text{ for all } t \in \mathbb{R}\}$$

is a closed real vector space, is closed under commutators, and is invariant under $X \mapsto AXA^{-1}$ for $A \in G$.

7. The same space is the tangent space at the identity:

$$T_I G = \text{Lie}(G), \quad T_A G = A \text{Lie}(G).$$

8. The exponential is a local diffeomorphism from $\text{Lie}(G)$ to G near 0 and I , but it need not be globally injective or surjective.

9. Every element of a connected linear Lie group is a finite product of exponentials. Every element is a *single* exponential when the group is connected and compact. In particular, the exponential maps onto $SU(n)$ and $SO(n)$.

10. The identity component is a closed normal linear Lie subgroup and

$$\text{Lie}(G^\circ) = \text{Lie}(G).$$

Derivations and the Adjoint Action

Road map

The adjoint construction lets elements of a Lie algebra act as linear operators on the Lie algebra itself. We begin with the physics hierarchy of states, operators, and superoperators. We then define derivations and prove in full detail that they form a Lie subalgebra of $\mathfrak{gl}(\mathfrak{g})$. The Jacobi identity produces a distinguished family of derivations, $\text{ad}_X(Y) = [X, Y]$, called inner derivations. We prove

$$[\text{ad}_X, \text{ad}_Y] = \text{ad}_{[X, Y]},$$

identify the kernel of ad with the centre of $\mathfrak{g}$, and deduce $\mathfrak{g}/Z(\mathfrak{g}) \cong \text{ider}(\mathfrak{g})$. Finally, we relate the infinitesimal map ad to the group adjoint action Ad .

16.1 From operators to superoperators

In quantum mechanics, a Hilbert space V is the space of state vectors. At this stage V is simply a vector space with an inner product; it is not itself a Lie algebra. A linear operator is a linear map

$$X : V \longrightarrow V.$$

The collection of all such operators is

$$\mathfrak{gl}(V) := \text{End}(V) = \{X : V \rightarrow V : X \text{ is linear}\}.$$

Composition turns $\text{End}(V)$ into an associative algebra. Forgetting the associative product but retaining its commutator gives the operator Lie algebra

$$[X_1, X_2] := X_1 \circ X_2 - X_2 \circ X_1. \quad (16.1)$$

We shall write

$$\mathfrak{g} := \mathfrak{gl}(V)$$

in this motivating example.

For a fixed $X \in \mathfrak{g}$, the rule

$$Y \longmapsto [X, Y]$$

is linear in Y . It is therefore a linear map from $\mathfrak{g}$ to itself. Such maps act on operators rather than directly on state vectors and are consequently called *superoperators* in physics.

Definition 16.1: The superoperator algebra

For any Lie algebra $\mathfrak{g}$,

$$\mathfrak{gl}(\mathfrak{g}) := \text{End}(\mathfrak{g}) = \{\mathcal{Y} : \mathfrak{g} \rightarrow \mathfrak{g} : \mathcal{Y} \text{ is linear}\}.$$

It is itself a Lie algebra under the commutator

$$[\mathcal{Y}_1, \mathcal{Y}_2] := \mathcal{Y}_1 \circ \mathcal{Y}_2 - \mathcal{Y}_2 \circ \mathcal{Y}_1.$$

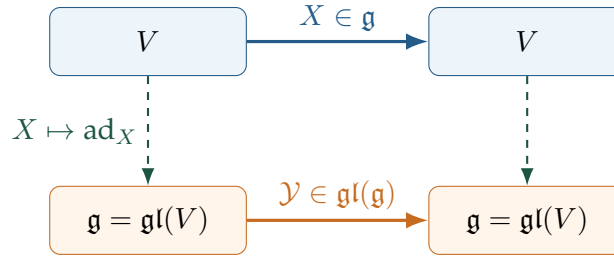


图 16.1: An operator acts on states, whereas a superoperator acts linearly on operators. The adjoint construction sends an operator X to the superoperator $Y \mapsto [X, Y]$.

Thus $\mathfrak{gl}(\mathfrak{g})$ constructs new operators linearly from old ones. The central example will be the operation

$$Y \mapsto [X, Y],$$

which turns an old operator Y into the commutator with a fixed X .

16.2 Derivations of a Lie algebra

Definition 16.2: Derivation

Let $\mathfrak{g}$ be a Lie algebra over a field $\mathbb{K}$. A linear map

$$D : \mathfrak{g} \longrightarrow \mathfrak{g}$$

is a *derivation* if it obeys the Leibniz rule for the Lie bracket:

$$D([X, Y]) = [D(X), Y] + [X, D(Y)] \quad (X, Y \in \mathfrak{g}). \quad (16.2)$$

The vector space of all derivations is denoted

$$\text{Der}(\mathfrak{g}) := \{D \in \mathfrak{gl}(\mathfrak{g}) : D \text{ satisfies (16.2)}\}.$$

Equation (16.2) is the product rule adapted to the Lie bracket. It should not be confused with the homomorphism condition

$$D([X, Y]) = [D(X), D(Y)].$$

A derivation differentiates a bracket; it does not usually preserve the bracket as a Lie-algebra homomorphism.

Theorem 16.1: Derivations form a Lie algebra

For every Lie algebra $\mathfrak{g}$,

$$\text{Der}(\mathfrak{g}) \leq \mathfrak{gl}(\mathfrak{g})$$

is a Lie subalgebra. Its bracket is the operator commutator

$$[D_1, D_2] = D_1 D_2 - D_2 D_1.$$

证明. We check vector-space closure and bracket closure separately.

Linear combinations. Let $D_1, D_2 \in \text{Der}(\mathfrak{g})$ and $\alpha_1, \alpha_2 \in \mathbb{K}$. Then

$$\begin{aligned} (\alpha_1 D_1 + \alpha_2 D_2)([X, Y]) &= \alpha_1 D_1([X, Y]) + \alpha_2 D_2([X, Y]) \\ &= \alpha_1 [D_1 X, Y] + \alpha_1 [X, D_1 Y] \\ &\quad + \alpha_2 [D_2 X, Y] + \alpha_2 [X, D_2 Y] \\ &= [(\alpha_1 D_1 + \alpha_2 D_2)X, Y] \\ &\quad + [X, (\alpha_1 D_1 + \alpha_2 D_2)Y]. \end{aligned}$$

Thus $\alpha_1 D_1 + \alpha_2 D_2$ is again a derivation.

Commutators. For $D_1, D_2 \in \text{Der}(\mathfrak{g})$, expand both compositions using the derivation rule:

$$\begin{aligned} [D_1, D_2]([X, Y]) &= D_1 D_2([X, Y]) - D_2 D_1([X, Y]) \\ &= D_1([D_2 X, Y] + [X, D_2 Y]) \\ &\quad - D_2([D_1 X, Y] + [X, D_1 Y]) \\ &= [D_1 D_2 X, Y] + [D_2 X, D_1 Y] \\ &\quad + [D_1 X, D_2 Y] + [X, D_1 D_2 Y] \\ &\quad - [D_2 D_1 X, Y] - [D_1 X, D_2 Y] \\ &\quad - [D_2 X, D_1 Y] - [X, D_2 D_1 Y]. \end{aligned}$$

The two mixed pairs cancel. What remains is

$$\begin{aligned} [D_1, D_2]([X, Y]) &= [(D_1 D_2 - D_2 D_1)X, Y] \\ &\quad + [X, (D_1 D_2 - D_2 D_1)Y] \\ &= [[D_1, D_2]X, Y] + [X, [D_1, D_2]Y]. \end{aligned}$$

Hence $[D_1, D_2]$ is a derivation. This proves that $\text{Der}(\mathfrak{g})$ is a Lie subalgebra. ■

16.3 Inner derivations

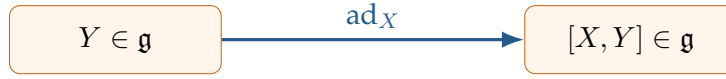
Every Lie algebra comes with a distinguished family of derivations obtained from its own elements.

Definition 16.3: Infinitesimal adjoint action

For $X \in \mathfrak{g}$, define

$$\mathrm{ad}_X : \mathfrak{g} \longrightarrow \mathfrak{g}, \quad \mathrm{ad}_X(Y) := [X, Y]. \quad (16.3)$$

This map is called the *adjoint action of X* or the *infinitesimal adjoint action*.



the fixed element X acts on all of $\mathfrak{g}$

图 16.2: The adjoint action of a fixed X sends Y to the bracket $[X, Y]$.

Proposition 16.1: Every adjoint action is a derivation

For every $X \in \mathfrak{g}$,

$$\mathrm{ad}_X \in \mathrm{Der}(\mathfrak{g}).$$

证明. Let $Y, Z \in \mathfrak{g}$. The Jacobi identity gives

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0.$$

Using antisymmetry twice,

$$[Y, [Z, X]] = -[Y, [X, Z]], \quad [Z, [X, Y]] = -[[X, Y], Z].$$

Therefore

$$\begin{aligned} \mathrm{ad}_X([Y, Z]) &= [X, [Y, Z]] \\ &= [Y, [X, Z]] + [[X, Y], Z] \\ &= [Y, \mathrm{ad}_X Z] + [\mathrm{ad}_X Y, Z] \\ &= [\mathrm{ad}_X Y, Z] + [Y, \mathrm{ad}_X Z]. \end{aligned}$$

This is exactly the derivation rule. ■

Definition 16.4: Inner and outer derivations

The set

$$\mathrm{id}(\mathfrak{g}) := \{\mathrm{ad}_X : X \in \mathfrak{g}\} \quad (16.4)$$

is the set of *inner derivations*. A derivation not belonging to this set is called *outer*. Once we prove that $\mathrm{id}(\mathfrak{g})$ is an ideal in $\mathrm{Der}(\mathfrak{g})$, the quotient

$$\mathrm{Out}(\mathfrak{g}) := \mathrm{Der}(\mathfrak{g}) / \mathrm{id}(\mathfrak{g})$$

will be the Lie algebra of outer-derivation classes.

Proposition 16.2: Bracket of inner derivations

For every $X, Y \in \mathfrak{g}$,

$$[\mathrm{ad}_X, \mathrm{ad}_Y] = \mathrm{ad}_{[X, Y]}. \quad (16.5)$$

Consequently,

$$\mathrm{id}(\mathfrak{g}) \leq \mathrm{Der}(\mathfrak{g})$$

is a Lie subalgebra.

证明. Apply the left-hand side to an arbitrary $Z \in \mathfrak{g}$:

$$\begin{aligned} [\mathrm{ad}_X, \mathrm{ad}_Y](Z) &= \mathrm{ad}_X(\mathrm{ad}_Y Z) - \mathrm{ad}_Y(\mathrm{ad}_X Z) \\ &= [X, [Y, Z]] - [Y, [X, Z]] \\ &= [X, [Y, Z]] + [Y, [Z, X]] \\ &= -[Z, [X, Y]] \quad \text{by the Jacobi identity} \\ &= [[X, Y], Z] \\ &= \mathrm{ad}_{[X, Y]}(Z). \end{aligned}$$

Since the two linear maps agree on every Z , they are equal. ■

Proposition 16.3: Inner derivations form an ideal

For $D \in \mathrm{Der}(\mathfrak{g})$ and $X \in \mathfrak{g}$,

$$[D, \mathrm{ad}_X] = \mathrm{ad}_{D(X)}. \quad (16.6)$$

Hence $\mathrm{id}(\mathfrak{g}) \triangleleft \mathrm{Der}(\mathfrak{g})$.

证明. For $Y \in \mathfrak{g}$,

$$\begin{aligned} [D, \mathrm{ad}_X](Y) &= D([X, Y]) - [X, D(Y)] \\ &= [D(X), Y] + [X, D(Y)] - [X, D(Y)] \\ &= [D(X), Y] \\ &= \mathrm{ad}_{D(X)}(Y). \end{aligned}$$
■

16.4 The adjoint representation and the centre

Theorem 16.2: The adjoint representation of a Lie algebra

The map

$$\mathrm{ad} : \mathfrak{g} \longrightarrow \mathfrak{gl}(\mathfrak{g}), \quad X \longmapsto \mathrm{ad}_X, \quad (16.7)$$

is a Lie-algebra homomorphism. Its image and kernel are

$$\mathrm{im}(\mathrm{ad}) = \mathrm{id}(\mathfrak{g}), \quad \mathrm{ker}(\mathrm{ad}) = Z(\mathfrak{g}). \quad (16.8)$$

证明. Linearity follows directly from bilinearity of the bracket:

$$\begin{aligned}\mathrm{ad}_{\alpha X + \beta Y}(Z) &= [\alpha X + \beta Y, Z] \\ &= \alpha[X, Z] + \beta[Y, Z] \\ &= (\alpha \mathrm{ad}_X + \beta \mathrm{ad}_Y)(Z).\end{aligned}$$

Identity (16.5) says precisely that brackets are preserved:

$$\mathrm{ad}_{[X, Y]} = [\mathrm{ad}_X, \mathrm{ad}_Y].$$

Thus ad is a Lie-algebra homomorphism.

Its image is the set of all maps ad_X , which is $\mathrm{id}\mathrm{er}(\mathfrak{g})$ by definition. Finally,

$$\begin{aligned}\ker(\mathrm{ad}) &= \{X \in \mathfrak{g} : \mathrm{ad}_X = 0\} \\ &= \{X \in \mathfrak{g} : [X, Y] = 0 \text{ for every } Y \in \mathfrak{g}\} \\ &= Z(\mathfrak{g}),\end{aligned}$$

the centre of $\mathfrak{g}$. ■

Theorem 16.3: Inner derivations as a quotient

The first isomorphism theorem applied to the adjoint representation gives

$$\boxed{\mathfrak{g}/Z(\mathfrak{g}) \cong \mathrm{id}\mathrm{er}(\mathfrak{g})}. \quad (16.9)$$

In particular, if $\mathfrak{g}$ is centreless, then

$$\boxed{\mathfrak{g} \cong \mathrm{id}\mathrm{er}(\mathfrak{g}), \quad X \longmapsto [X, \cdot]}. \quad (16.10)$$

证明. The homomorphism ad has kernel $Z(\mathfrak{g})$ and image $\mathrm{id}\mathrm{er}(\mathfrak{g})$. The first isomorphism theorem gives

$$\mathfrak{g}/\ker(\mathrm{ad}) \cong \mathrm{im}(\mathrm{ad}),$$

which is (16.9). If $Z(\mathfrak{g}) = \{0\}$, then the quotient on the left is simply $\mathfrak{g}$. ■

16.5 The group adjoint action

For a linear Lie group G with $\mathfrak{g} = \mathrm{Lie}(G)$, conjugation by a group element preserves $\mathfrak{g}$, as proved in the preceding chapter.

Definition 16.5: Group adjoint action

For $A \in G$, define

$$\mathrm{Ad}_A : \mathfrak{g} \longrightarrow \mathfrak{g}, \quad \mathrm{Ad}_A(X) := AXA^{-1}. \quad (16.11)$$

The map

$$\mathrm{Ad} : G \longrightarrow \mathrm{GL}(\mathfrak{g}), \quad A \longmapsto \mathrm{Ad}_A$$

is called the *adjoint representation of the group*.

Proposition 16.4: The group adjoint map is a representationFor $A, B \in G$,

$$\text{Ad}_{AB} = \text{Ad}_A \circ \text{Ad}_B. \quad (16.12)$$

Thus $\text{Ad} : G \rightarrow \text{GL}(\mathfrak{g})$ is a group homomorphism.证明. For $X \in \mathfrak{g}$,

$$\text{Ad}_{AB}(X) = (AB)X(AB)^{-1} = A(BXB^{-1})A^{-1} = \text{Ad}_A(\text{Ad}_B X).$$

■

Theorem 16.4: Relation between Ad and adFor $X \in \mathfrak{g}$,

$$\text{Ad}_{e^{tX}} = e^{t \text{ad}_X} \quad (t \in \mathbb{R}). \quad (16.13)$$

Equivalently, the differential of the group representation Ad at the identity is the Lie-algebra representation ad:

$$\left. \frac{d}{dt} \text{Ad}_{e^{tX}} \right|_{t=0} = \text{ad}_X. \quad (16.14)$$

证明. Fix $Y \in \mathfrak{g}$ and put

$$F(t) := \text{Ad}_{e^{tX}}(Y) = e^{tX} Y e^{-tX}.$$

Differentiate:

$$\begin{aligned} F'(t) &= X e^{tX} Y e^{-tX} - e^{tX} Y e^{-tX} X \\ &= [X, F(t)] \\ &= \text{ad}_X(F(t)). \end{aligned}$$

The initial condition is $F(0) = Y$. The unique solution of the constant-coefficient linear differential equation

$$F'(t) = \text{ad}_X F(t), \quad F(0) = Y,$$

is

$$F(t) = e^{t \text{ad}_X} Y.$$

Since this holds for every Y , identity (16.13) follows. Differentiating it at $t = 0$ gives (16.14). ■

16.6 Examples

Example 16.1: An Abelian Lie algebra

If $\mathfrak{g}$ is Abelian, then every bracket is zero. Consequently

$$\mathrm{ad}_X = 0 \quad \text{for every } X,$$

so

$$\mathrm{id}(\mathfrak{g}) = \{0\}, \quad Z(\mathfrak{g}) = \mathfrak{g}.$$

On the other hand, the derivation rule becomes $D(0) = 0$, which every linear map satisfies. Hence

$$\mathrm{Der}(\mathfrak{g}) = \mathfrak{gl}(\mathfrak{g}).$$

Thus a nonzero linear endomorphism of a nonzero Abelian Lie algebra is an outer derivation.

Example 16.2: The full matrix algebra

Let $\mathfrak{g} = \mathfrak{gl}(V)$ with commutator bracket. Then

$$\mathrm{ad}_X(Y) = XY - YX.$$

The centre consists of the scalar operators:

$$Z(\mathfrak{gl}(V)) = \{\lambda I : \lambda \in \mathbb{K}\}$$

when V is finite-dimensional and nonzero. Therefore

$$\mathfrak{gl}(V)/\mathbb{K}I \cong \mathrm{id}(\mathfrak{gl}(V)).$$

Two matrices X and $X + \lambda I$ produce the same inner derivation, because the scalar matrix commutes with every operator.

Example 16.3: $\mathfrak{so}(3)$ and the cross product

Under the isomorphism $\mathfrak{so}(3) \cong (\mathbb{R}^3, \times)$ from the preceding chapters,

$$\mathrm{ad}_x(y) = x \times y.$$

The only vector whose cross product with every y vanishes is $x = 0$. Thus $Z(\mathfrak{so}(3)) = \{0\}$, and

$$\mathfrak{so}(3) \cong \mathrm{id}(\mathfrak{so}(3)).$$

In coordinates,

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \longmapsto \mathrm{ad}_x = \begin{pmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{pmatrix},$$

because multiplication of this matrix by y produces $x \times y$.

What to retain from this chapter

1. $\mathfrak{gl}(\mathfrak{g})$ is the Lie algebra of linear superoperators $\mathfrak{g} \rightarrow \mathfrak{g}$, with commutator bracket.
2. A derivation satisfies the Lie-bracket product rule

$$D([X, Y]) = [DX, Y] + [X, DY].$$

The derivations form a Lie subalgebra

$$\text{Der}(\mathfrak{g}) \leq \mathfrak{gl}(\mathfrak{g}).$$

3. Every element $X \in \mathfrak{g}$ defines an inner derivation

$$\text{ad}_X(Y) = [X, Y].$$

The Jacobi identity is exactly what proves that this map is a derivation.

4. Inner derivations satisfy

$$[\text{ad}_X, \text{ad}_Y] = \text{ad}_{[X, Y]},$$

so $\text{ad} : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ is a Lie-algebra homomorphism.

- 5.

$$\text{im}(\text{ad}) = \text{ider}(\mathfrak{g}), \quad \ker(\text{ad}) = Z(\mathfrak{g}).$$

Hence

$$\mathfrak{g}/Z(\mathfrak{g}) \cong \text{ider}(\mathfrak{g}).$$

6. Inner derivations form an ideal in all derivations:

$$[D, \text{ad}_X] = \text{ad}_{D(X)}.$$

7. At group level,

$$\text{Ad}_A(X) = AXA^{-1}, \quad \text{Ad}_{AB} = \text{Ad}_A \text{Ad}_B.$$

The group and algebra adjoint actions are related by

$$\text{Ad}_{e^{tX}} = e^{t \text{ad}_X}.$$

Lie Group Homomorphisms and Their Differentials

Road map

A homomorphism of matrix Lie groups carries one-parameter subgroups to one-parameter subgroups. This simple observation produces a unique real linear map between the corresponding Lie algebras. We prove carefully that the induced map commutes with exponentials, respects conjugation and Lie brackets, and is computed by differentiating the group homomorphism along e^{tX} . We then revisit the adjoint representation, derive Hadamard's lemma and the Heisenberg evolution series, show that a homomorphism on a connected group is determined by its Lie-algebra map, and prove that every continuous matrix-group homomorphism is automatically smooth. The chapter ends by relating the centre of a connected Lie group to the centre of its Lie algebra.

17.1 Group and Lie-algebra homomorphisms

Definition 17.1: Lie group homomorphism

Let G and H be linear Lie groups. A map

$$\Phi : G \longrightarrow H$$

is a *Lie group homomorphism* if

1. it is a group homomorphism:

$$\Phi(AB) = \Phi(A)\Phi(B) \quad (A, B \in G);$$

2. it is continuous.

Continuity is enough in the definition: we shall prove later in this chapter that such a homomorphism is automatically smooth.

Definition 17.2: Lie-algebra homomorphism

Let $\mathfrak{g}$ and $\mathfrak{h}$ be Lie algebras. A map

$$\psi : \mathfrak{g} \longrightarrow \mathfrak{h}$$

is a *Lie-algebra homomorphism* if it is linear and

$$\psi([X, Y]) = [\psi(X), \psi(Y)] \quad (X, Y \in \mathfrak{g}). \quad (17.1)$$

The goal is to pass canonically from Φ to ψ .

17.2 Differentiating a Lie group homomorphism**Theorem 17.1: The homomorphism–exponential correspondence**

Let G, H be linear Lie groups, with

$$\mathfrak{g} = \text{Lie}(G), \quad \mathfrak{h} = \text{Lie}(H),$$

and let $\Phi : G \rightarrow H$ be a Lie group homomorphism. There is a unique real-linear map

$$\psi : \mathfrak{g} \longrightarrow \mathfrak{h}$$

such that

$$\Phi(e^X) = e^{\psi(X)} \quad (X \in \mathfrak{g}). \quad (17.2)$$

Equivalently, the following diagram commutes:

$$\begin{array}{ccc} \mathfrak{g} & \xrightarrow{\psi} & \mathfrak{h} \\ \exp \downarrow & & \downarrow \exp \\ G & \xrightarrow{\Phi} & H \end{array}$$

The map is computed by

$$\psi(X) = \left. \frac{d}{dt} \Phi(e^{tX}) \right|_{t=0}. \quad (17.3)$$

Moreover,

$$\psi(AXA^{-1}) = \Phi(A)\psi(X)\Phi(A)^{-1}, \quad A \in G, X \in \mathfrak{g}, \quad (17.4)$$

$$\psi([X, Y]) = [\psi(X), \psi(Y)], \quad X, Y \in \mathfrak{g}. \quad (17.5)$$

Thus ψ is in fact a Lie-algebra homomorphism.

证明. Step 1: construct $\psi(X)$ from a one-parameter subgroup. Fix $X \in \mathfrak{g}$. The map

$$\mathbb{R} \longrightarrow H, \quad t \longmapsto \Phi(e^{tX})$$

is continuous. It is a homomorphism because

$$\begin{aligned}\Phi(e^{(t+s)X}) &= \Phi(e^{tX}e^{sX}) \\ &= \Phi(e^{tX})\Phi(e^{sX}).\end{aligned}$$

It is therefore a one-parameter subgroup of H . By the classification of one-parameter matrix subgroups, there is a unique $Z \in \mathfrak{h}$ such that

$$\Phi(e^{tX}) = e^{tZ} \quad (t \in \mathbb{R}).$$

Define

$$\psi(X) := Z.$$

Setting $t = 1$ gives (17.2). Differentiating at $t = 0$ gives (17.3).

Step 2: homogeneity over $\mathbb{R}$. For $s, t \in \mathbb{R}$,

$$e^{t\psi(sX)} = \Phi(e^{tsX}) = e^{ts\psi(X)}.$$

Differentiating with respect to t at 0 gives

$$\psi(sX) = s\psi(X).$$

This proves real homogeneity. Even if the groups consist of complex matrices, the natural Lie algebras here are real unless extra complex linearity is separately established.

Step 3: additivity. Use the Lie–Trotter product formula:

$$e^{t(X+Y)} = \lim_{m \rightarrow \infty} \left(e^{tX/m} e^{tY/m} \right)^m.$$

Continuity of Φ and its homomorphism property give

$$\begin{aligned}e^{t\psi(X+Y)} &= \Phi(e^{t(X+Y)}) \\ &= \lim_{m \rightarrow \infty} \left(\Phi(e^{tX/m}) \Phi(e^{tY/m}) \right)^m \\ &= \lim_{m \rightarrow \infty} \left(e^{t\psi(X)/m} e^{t\psi(Y)/m} \right)^m \\ &= e^{t(\psi(X)+\psi(Y))}.\end{aligned}$$

Differentiating at $t = 0$ yields

$$\psi(X + Y) = \psi(X) + \psi(Y).$$

Together with Step 2, this proves that ψ is real linear.

Step 4: uniqueness. Suppose another real-linear map $\psi' : \mathfrak{g} \rightarrow \mathfrak{h}$ satisfies $\Phi(e^X) = e^{\psi'(X)}$. Replacing X by tX ,

$$e^{t\psi'(X)} = \Phi(e^{tX}) = e^{t\psi(X)}.$$

Differentiating at $t = 0$ gives $\psi'(X) = \psi(X)$ for every X .

Step 5: compatibility with conjugation. For $A \in G$, $X \in \mathfrak{g}$, and $t \in \mathbb{R}$,

$$\begin{aligned} e^{t\psi(AXA^{-1})} &= \Phi(e^{tAXA^{-1}}) \\ &= \Phi(Ae^{tX}A^{-1}) \\ &= \Phi(A)\Phi(e^{tX})\Phi(A)^{-1} \\ &= \Phi(A)e^{t\psi(X)}\Phi(A)^{-1}. \end{aligned}$$

Differentiation at $t = 0$ gives (17.4).

Step 6: preservation of brackets. For $X, Y \in \mathfrak{g}$,

$$[X, Y] = \left. \frac{d}{dt} (e^{tX}Y e^{-tX}) \right|_{t=0}.$$

Because ψ is linear, it commutes with differentiation of this finite-dimensional curve:

$$\begin{aligned} \psi([X, Y]) &= \left. \frac{d}{dt} \psi(e^{tX}Y e^{-tX}) \right|_{t=0} \\ &= \left. \frac{d}{dt} (\Phi(e^{tX})\psi(Y)\Phi(e^{tX})^{-1}) \right|_{t=0} \\ &= \left. \frac{d}{dt} (e^{t\psi(X)}\psi(Y)e^{-t\psi(X)}) \right|_{t=0} \\ &= [\psi(X), \psi(Y)]. \end{aligned}$$

This is (17.5). ■

What the theorem does and does not say

Every continuous Lie group homomorphism differentiates to a Lie-algebra homomorphism. The converse integration problem is subtler: a Lie-algebra homomorphism need not descend to a prescribed non-simply-connected group. Existence becomes clean when the source group is simply connected; the covering-group chapters will explain the missing global information.

Example 17.1: The determinant differentiates to the trace

Take

$$\Phi = \det : \mathrm{GL}(n, \mathbb{C}) \longrightarrow \mathbb{C}^\times.$$

The target has Abelian Lie algebra $\mathbb{C}$, written additively. Since

$$\det(e^{tX}) = e^{t \operatorname{tr} X},$$

the induced Lie-algebra homomorphism is

$$\psi(X) = \operatorname{tr} X.$$

The group-level kernel is $\mathrm{SL}(n, \mathbb{C})$, and the infinitesimal kernel is $\mathfrak{sl}(n, \mathbb{C})$.

17.3 The adjoint homomorphism revisited

Let $G \leq \mathrm{GL}(n, \mathbb{K})$ be a linear Lie group with $\mathfrak{g} = \mathrm{Lie}(G)$. For $A \in G$, define

$$\mathrm{Ad}_A : \mathfrak{g} \longrightarrow \mathfrak{g}, \quad \mathrm{Ad}_A(X) = AXA^{-1}.$$

This is well defined because conjugation by A preserves $\mathfrak{g}$.

Proposition 17.1: Each Ad_A is a Lie-algebra automorphism

For $A \in G$, the map Ad_A is linear, bijective, and preserves brackets:

$$\mathrm{Ad}_A([X, Y]) = [\mathrm{Ad}_A X, \mathrm{Ad}_A Y].$$

证明. Linearity follows from

$$\begin{aligned} \mathrm{Ad}_A(\lambda_1 X_1 + \lambda_2 X_2) &= A(\lambda_1 X_1 + \lambda_2 X_2)A^{-1} \\ &= \lambda_1 AX_1A^{-1} + \lambda_2 AX_2A^{-1}. \end{aligned}$$

Its inverse is $\mathrm{Ad}_{A^{-1}}$, so it is bijective. Finally,

$$\begin{aligned} \mathrm{Ad}_A([X, Y]) &= A(XY - YX)A^{-1} \\ &= (AXA^{-1})(AYA^{-1}) - (AYA^{-1})(AXA^{-1}) \\ &= [\mathrm{Ad}_A X, \mathrm{Ad}_A Y]. \end{aligned}$$

■

Let $\mathrm{GL}(\mathfrak{g})$ denote the group of invertible real-linear maps $\mathfrak{g} \rightarrow \mathfrak{g}$. If $d = \dim_{\mathbb{R}} \mathfrak{g} = \dim G$, then after choosing a basis

$$\mathrm{GL}(\mathfrak{g}) \cong \mathrm{GL}(d, \mathbb{R}),$$

so it is itself a matrix Lie group.

Theorem 17.2: The group adjoint representation

The map

$$\mathrm{Ad} : G \longrightarrow \mathrm{GL}(\mathfrak{g}), \quad A \longmapsto \mathrm{Ad}_A \tag{17.6}$$

is a Lie group homomorphism.

证明. For $A, B \in G$ and $X \in \mathfrak{g}$,

$$\begin{aligned} \mathrm{Ad}_{AB}(X) &= (AB)X(AB)^{-1} \\ &= A(BXB^{-1})A^{-1} \\ &= \mathrm{Ad}_A(\mathrm{Ad}_B X). \end{aligned}$$

Thus

$$\mathrm{Ad}_{AB} = \mathrm{Ad}_A \mathrm{Ad}_B.$$

For continuity, if $A_m \rightarrow A$ in G , then

$$A_m X A_m^{-1} \rightarrow A X A^{-1}$$

for every X , because matrix multiplication and inversion are continuous. In finite dimensions this is exactly continuity of $\text{Ad}_{A_m} \rightarrow \text{Ad}_A$ as linear operators. ■

The Lie algebra of $\text{GL}(\mathfrak{g})$ is $\mathfrak{gl}(\mathfrak{g})$. Applying the preceding theorem on differentiating homomorphisms gives an associated Lie-algebra homomorphism

$$\text{ad} : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g}).$$

For $X, Y \in \mathfrak{g}$,

$$\begin{aligned} \text{ad}_X(Y) &= \left. \frac{d}{dt} \text{Ad}_{e^{tX}}(Y) \right|_{t=0} \\ &= \left. \frac{d}{dt} (e^{tX} Y e^{-tX}) \right|_{t=0} \\ &= XY - YX \\ &= [X, Y]. \end{aligned}$$

This recovers exactly the infinitesimal adjoint action defined in the preceding chapter.

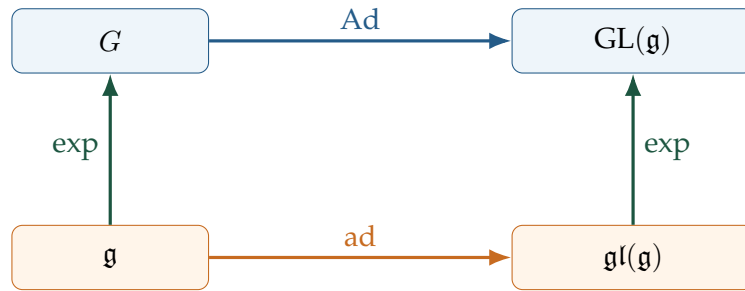


图 17.1: The adjoint representation commutes with exponential: $\text{Ad}_{e^X} = e^{\text{ad}_X}$.

17.4 Hadamard's lemma

Applying the exponential compatibility $\Phi(e^X) = e^{\psi(X)}$ to $\Phi = \text{Ad}$ and $\psi = \text{ad}$ gives

$$\text{Ad}_{e^X} = e^{\text{ad}_X}. \quad (17.7)$$

Acting on $Y \in \mathfrak{g}$, we obtain the following formula.

Theorem 17.3: Hadamard's lemma

For matrices X, Y , or more generally whenever the relevant exponential and adjoint action are defined,

$$e^X Y e^{-X} = Y + [X, Y] + \frac{1}{2!} [X, [X, Y]] + \frac{1}{3!} [X, [X, [X, Y]]] + \cdots \quad (17.8)$$

Equivalently,

$$e^X Y e^{-X} = \sum_{k=0}^{\infty} \frac{1}{k!} \operatorname{ad}_X^k(Y).$$

证明. Equation (17.7) gives

$$e^X Y e^{-X} = \operatorname{Ad}_{e^X}(Y) = e^{\operatorname{ad}_X}(Y).$$

Expand the exponential of the linear operator ad_X :

$$e^{\operatorname{ad}_X}(Y) = \sum_{k=0}^{\infty} \frac{1}{k!} \operatorname{ad}_X^k(Y).$$

The first iterates are

$$\operatorname{ad}_X^0(Y) = Y, \quad \operatorname{ad}_X^1(Y) = [X, Y], \quad \operatorname{ad}_X^2(Y) = [X, [X, Y]],$$

and so on, which is (17.8). ■

Example 17.2: Heisenberg evolution of an observable

Let $\mathcal{H}$ be a Hilbert space, H a Hamiltonian, and A an observable. In finite dimensions all operators are bounded, so every series below converges without domain subtleties. The Heisenberg-picture time dependence is

$$A(t) = e^{iHt} A e^{-iHt} = \operatorname{Ad}_{e^{iHt}} A.$$

Hadamard's lemma gives

$$\begin{aligned} A(t) &= e^{\operatorname{ad}_{iHt}} A \\ &= \sum_{k=0}^{\infty} \frac{(it)^k}{k!} \underbrace{[H, [H, \dots, [H, A] \dots]]}_{k \text{ brackets}}. \end{aligned} \tag{17.9}$$

Differentiating directly,

$$\frac{dA(t)}{dt} = i[H, A(t)],$$

the Heisenberg equation of motion in units $\hbar = 1$.

If A is an eigenoperator of ad_H ,

$$[H, A] = \omega A, \quad \omega \in \mathbb{C},$$

then induction gives

$$\operatorname{ad}_H^k(A) = \omega^k A.$$

Therefore

$$A(t) = \sum_{k=0}^{\infty} \frac{(it\omega)^k}{k!} A = e^{i\omega t} A.$$

17.5 Connected groups are determined infinitesimally

Theorem 17.4: Uniqueness on a connected source

Let G, H be linear Lie groups and assume that G is connected. Let

$$\Phi_1, \Phi_2 : G \longrightarrow H$$

be Lie group homomorphisms, with induced Lie-algebra homomorphisms

$$\psi_1, \psi_2 : \mathfrak{g} \longrightarrow \mathfrak{h}.$$

If $\psi_1 = \psi_2$, then

$$\Phi_1 = \Phi_2.$$

证明. Every $A \in G$ is a finite product of exponentials:

$$A = e^{X_1} \cdots e^{X_m}, \quad X_j \in \mathfrak{g}.$$

Using the homomorphism property and exponential compatibility,

$$\begin{aligned} \Phi_1(A) &= \Phi_1(e^{X_1}) \cdots \Phi_1(e^{X_m}) \\ &= e^{\psi_1(X_1)} \cdots e^{\psi_1(X_m)} \\ &= e^{\psi_2(X_1)} \cdots e^{\psi_2(X_m)} \\ &= \Phi_2(e^{X_1}) \cdots \Phi_2(e^{X_m}) \\ &= \Phi_2(A). \end{aligned}$$

■

Why connectedness appears

The differential sees only the identity component. On a disconnected group, two homomorphisms can agree on G° , hence have the same Lie-algebra map, but differ on other components. Connectedness eliminates this discrete ambiguity.

17.6 Continuity implies smoothness

Theorem 17.5: Automatic smoothness

Every continuous homomorphism between linear Lie groups is smooth (indeed, real analytic).

证明. Let $\Phi : G \rightarrow H$ be continuous, and let $\psi : \mathfrak{g} \rightarrow \mathfrak{h}$ be its induced real-linear map. Fix $A \in G$. If B is sufficiently close to A , then

$$A^{-1}B$$

is close to I and lies in the local exponential neighbourhood of G . Hence

$$A^{-1}B = e^X, \quad X = \log(A^{-1}B) \in \mathfrak{g}.$$

Therefore

$$\begin{aligned}
 \Phi(B) &= \Phi(Ae^X) \\
 &= \Phi(A)\Phi(e^X) \\
 &= \Phi(A)e^{\psi(X)} \\
 &= \Phi(A)\exp(\psi(\log(A^{-1}B))).
 \end{aligned} \tag{17.10}$$

Near A , this is a composition of smooth maps:

$$B \mapsto A^{-1}B \mapsto \log(A^{-1}B) \mapsto \psi(\log(A^{-1}B)) \mapsto \exp(\psi(\log(A^{-1}B))) \mapsto \Phi(A)\exp(\dots).$$

Thus Φ is smooth in a neighbourhood of every A , hence smooth on all of G . Since the matrix exponential and local logarithm are analytic power series, the same argument gives real analyticity. ■

17.7 Centres of connected groups and Lie algebras

Theorem 17.6: Centre correspondence

If G is a connected linear Lie group and $\mathfrak{g} = \text{Lie}(G)$, then

$$Z(\mathfrak{g}) = \text{Lie}(Z(G)). \tag{17.11}$$

证明. We prove a chain of equivalent conditions for $X \in \mathfrak{g}$.

First,

$$\begin{aligned}
 X \in Z(\mathfrak{g}) &\iff [Y, X] = 0 \quad \text{for every } Y \in \mathfrak{g} \\
 &\iff \text{ad}_Y(X) = 0 \quad \text{for every } Y \in \mathfrak{g}.
 \end{aligned}$$

This is equivalent to

$$e^{t \text{ad}_Y} X = X \quad \text{for every } Y \in \mathfrak{g}, t \in \mathbb{R}.$$

The forward implication is immediate from the exponential series. For the reverse implication, differentiate at $t = 0$. It is important that the equality holds for *all* t ; the isolated equation $e^T = I$ would not by itself imply $T = 0$.

Using $e^{t \text{ad}_Y} = \text{Ad}_{e^{tY}}$, we have

$$X \in Z(\mathfrak{g}) \iff \text{Ad}_{e^{tY}} X = X \quad (Y \in \mathfrak{g}, t \in \mathbb{R}).$$

Because G is connected, every $g \in G$ is a finite product

$$g = e^{Y_1} \dots e^{Y_m}.$$

The adjoint map is a group homomorphism, so invariance under each exponential factor is equivalent to invariance under every $g \in G$:

$$X \in Z(\mathfrak{g}) \iff \text{Ad}_g X = X \quad (g \in G).$$

In matrix notation this is

$$gXg^{-1} = X \iff gX = Xg \quad (g \in G).$$

If $gX = Xg$, then g commutes with every power X^k , and hence with the exponential series:

$$ge^{tX} = e^{tX}g \quad (g \in G, t \in \mathbb{R}).$$

Conversely, if this exponential commutation holds for every t , differentiation at $t = 0$ gives $gX = Xg$. Therefore

$$\begin{aligned} X \in Z(\mathfrak{g}) &\iff e^{tX} \in Z(G) \quad \text{for every } t \in \mathbb{R} \\ &\iff X \in \text{Lie}(Z(G)). \end{aligned}$$

This proves (17.11). ■

Connectedness is essential

Without connectedness, commuting with every infinitesimal direction only guarantees commutation with G° . A disconnected component can act nontrivially by conjugation while remaining invisible to $\text{Lie}(G)$.

What to retain from this chapter

1. A Lie group homomorphism is a continuous group homomorphism. It is automatically smooth.

2. Every $\Phi : G \rightarrow H$ induces a unique real-linear map

$$\psi : \text{Lie}(G) \rightarrow \text{Lie}(H)$$

satisfying

$$\Phi(e^X) = e^{\psi(X)}, \quad \psi(X) = \left. \frac{d}{dt} \Phi(e^{tX}) \right|_{t=0}.$$

3. The induced map respects conjugation and brackets:

$$\psi(AXA^{-1}) = \Phi(A)\psi(X)\Phi(A)^{-1}, \quad \psi([X, Y]) = [\psi(X), \psi(Y)].$$

Hence ψ is a Lie-algebra homomorphism.

4. The group adjoint representation

$$\text{Ad} : G \rightarrow \text{GL}(\mathfrak{g})$$

differentiates to

$$\text{ad} : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g}), \quad \text{ad}_X(Y) = [X, Y].$$

- 5.

$$\text{Ad}_{e^X} = e^{\text{ad}_X},$$

which yields Hadamard's lemma

$$e^X Y e^{-X} = Y + [X, Y] + \frac{1}{2!} [X, [X, Y]] + \cdots.$$

6. If the source group is connected, two Lie group homomorphisms with the same induced Lie-algebra homomorphism are equal.
7. For a connected linear Lie group,

$$Z(\text{Lie}(G)) = \text{Lie}(Z(G)).$$

The Double Cover $SU(2) \rightarrow SO(3)$

Road map

The purpose of this chapter is to construct, from first principles, a surjective Lie group homomorphism

$$R : SU(2) \longrightarrow SO(3)$$

whose kernel is $\{I_2, -I_2\}$. Thus every three-dimensional rotation has exactly two preimages in $SU(2)$, and

$$SO(3) \cong SU(2)/\{\pm I_2\}.$$

We shall:

1. identify $\mathfrak{su}(2)$ with a three-dimensional real inner product space using the Pauli matrices;
2. let $SU(2)$ act on this space by conjugation and write that action as a 3×3 real matrix $R(u)$;
3. verify directly that R is continuous, multiplicative, and orthogonal with determinant 1;
4. compute the differential $\mathfrak{su}(2) \rightarrow \mathfrak{so}(3)$, obtaining the familiar cross-product generators of rotations;
5. prove surjectivity and compute the kernel;
6. explain why $SU(2)$ is the three-sphere, while $SO(3)$ is a closed three-ball with antipodal boundary points identified.

18.1 Pauli matrices and the real space $\mathfrak{su}(2)$

Recall the Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

They are Hermitian and traceless. Their basic multiplication law is

$$\sigma_\alpha \sigma_\beta = \delta_{\alpha\beta} I_2 + i \sum_{\gamma=1}^3 \varepsilon_{\alpha\beta\gamma} \sigma_\gamma, \quad (18.1)$$

where $\varepsilon_{\alpha\beta\gamma}$ is the Levi–Civita symbol: it is +1 for even permutations of 123, −1 for odd permutations, and 0 when two indices coincide. Consequently,

$$[\sigma_\alpha, \sigma_\beta] = 2i \sum_{\gamma} \varepsilon_{\alpha\beta\gamma} \sigma_\gamma, \quad (18.2)$$

$$\text{tr}(\sigma_\alpha \sigma_\beta) = 2\delta_{\alpha\beta}. \quad (18.3)$$

The Lie algebra of $SU(2)$ is

$$\mathfrak{su}(2) = \left\{ X \in \mathfrak{gl}(2, \mathbb{C}) : X^\dagger = -X, \quad \text{tr } X = 0 \right\}.$$

Every $X \in \mathfrak{su}(2)$ has a unique expression

$$X = i \sum_{\alpha=1}^3 y_\alpha \sigma_\alpha = i \mathbf{y} \cdot \boldsymbol{\sigma}, \quad \mathbf{y} = (y_1, y_2, y_3) \in \mathbb{R}^3. \quad (18.4)$$

Indeed, a general anti-Hermitian traceless 2×2 matrix is

$$\begin{pmatrix} iy_3 & y_2 + iy_1 \\ -y_2 + iy_1 & -iy_3 \end{pmatrix} = iy_1 \sigma_1 + iy_2 \sigma_2 + iy_3 \sigma_3.$$

Thus

$$\dim_{\mathbb{R}} \mathfrak{su}(2) = 3, \quad \{i\sigma_1, i\sigma_2, i\sigma_3\}$$

is a real basis.

18.1.1 The normalized Frobenius inner product

On the real vector space $\mathfrak{su}(2)$, define

$$\langle X, Y \rangle = \frac{1}{2} \text{tr}(X^\dagger Y). \quad (18.5)$$

This is the suitably normalized Frobenius inner product. By (18.3),

$$\langle i\sigma_\alpha, i\sigma_\beta \rangle = \frac{1}{2} \text{tr}((i\sigma_\alpha)^\dagger i\sigma_\beta) = \frac{1}{2} \text{tr}(\sigma_\alpha \sigma_\beta) = \delta_{\alpha\beta}.$$

Hence the basis $\{i\sigma_\alpha\}$ is orthonormal. Therefore, if

$$Z = \sum_{\alpha=1}^3 z_\alpha i\sigma_\alpha,$$

then its coordinates are recovered by

$$z_\alpha = \langle i\sigma_\alpha, Z \rangle.$$

18.2 Constructing the map by the adjoint action

For $u \in SU(2)$, conjugation preserves anti-Hermiticity and trace:

$$(uXu^\dagger)^\dagger = uX^\dagger u^\dagger = -uXu^\dagger, \quad \text{tr}(uXu^\dagger) = \text{tr } X.$$

Thus

$$\text{Ad}_u : \mathfrak{su}(2) \longrightarrow \mathfrak{su}(2), \quad X \longmapsto uXu^{-1} = uXu^\dagger,$$

is a real-linear automorphism.

Definition 18.1: The rotation matrix associated with u

For $u \in SU(2)$, let $R(u)$ be the matrix of Ad_u in the ordered orthonormal basis $\{\mathbf{i}\sigma_1, \mathbf{i}\sigma_2, \mathbf{i}\sigma_3\}$. Thus

$$u(\mathbf{i}\sigma_\beta)u^\dagger = \sum_{\alpha=1}^3 R(u)_{\alpha\beta} \mathbf{i}\sigma_\alpha. \quad (18.6)$$

Taking the inner product with $\mathbf{i}\sigma_\alpha$ gives the explicit coordinate formula appearing in the handwritten derivation:

$$\begin{aligned} R(u)_{\alpha\beta} &= \langle \mathbf{i}\sigma_\alpha, u(\mathbf{i}\sigma_\beta)u^\dagger \rangle \\ &= \frac{1}{2} \text{tr}((\mathbf{i}\sigma_\alpha)^\dagger u(\mathbf{i}\sigma_\beta)u^\dagger) \\ &= \boxed{\frac{1}{2} \text{tr}(\sigma_\alpha u \sigma_\beta u^\dagger)}. \end{aligned} \quad (18.7)$$

Although complex matrices occur on the right, the result is real because it is the coordinate of an element of the real vector space $\mathfrak{su}(2)$ in a real basis.

18.2.1 The homomorphism property, index by index

Let $u_1, u_2 \in SU(2)$. Starting from the defining expansion,

$$\begin{aligned} (u_1 u_2)(\mathbf{i}\sigma_\beta)(u_1 u_2)^\dagger &= u_1 (u_2(\mathbf{i}\sigma_\beta)u_2^\dagger) u_1^\dagger \\ &= u_1 \left(\sum_{\mu} R(u_2)_{\mu\beta} \mathbf{i}\sigma_\mu \right) u_1^\dagger \\ &= \sum_{\mu} R(u_2)_{\mu\beta} \sum_{\alpha} R(u_1)_{\alpha\mu} \mathbf{i}\sigma_\alpha \\ &= \sum_{\alpha} \left(\sum_{\mu} R(u_1)_{\alpha\mu} R(u_2)_{\mu\beta} \right) \mathbf{i}\sigma_\alpha. \end{aligned}$$

Uniqueness of coordinates yields

$$R(u_1 u_2)_{\alpha\beta} = \sum_{\mu} R(u_1)_{\alpha\mu} R(u_2)_{\mu\beta}.$$

In matrix form,

$$\boxed{R(u_1 u_2) = R(u_1) R(u_2)}. \quad (18.8)$$

Also,

$$R(I_2) = I_3, \quad R(u^{-1}) = R(u)^{-1}.$$

18.2.2 Why $R(u)$ is orthogonal

Conjugation by a unitary matrix preserves the Frobenius inner product:

$$\begin{aligned} \langle uXu^\dagger, uYu^\dagger \rangle &= \frac{1}{2} \operatorname{tr}((uXu^\dagger)^\dagger uYu^\dagger) \\ &= \frac{1}{2} \operatorname{tr}(uX^\dagger u^\dagger uYu^\dagger) \\ &= \frac{1}{2} \operatorname{tr}(uX^\dagger Yu^\dagger) \\ &= \frac{1}{2} \operatorname{tr}(X^\dagger Y) = \langle X, Y \rangle. \end{aligned}$$

Applying this to the orthonormal basis vectors gives

$$\begin{aligned} \delta_{\alpha\beta} &= \langle u(i\sigma_\alpha)u^\dagger, u(i\sigma_\beta)u^\dagger \rangle \\ &= \sum_{\gamma, \mu} R(u)_{\gamma\alpha} R(u)_{\mu\beta} \langle i\sigma_\gamma, i\sigma_\mu \rangle \\ &= \sum_{\gamma} R(u)_{\gamma\alpha} R(u)_{\gamma\beta}. \end{aligned}$$

Therefore

$$R(u)^\top R(u) = I_3, \quad R(u) \in O(3). \quad (18.9)$$

In particular,

$$R(u^\dagger) = R(u)^{-1} = R(u)^\top,$$

which also follows directly from the trace formula:

$$R(u^\dagger)_{\alpha\beta} = \frac{1}{2} \operatorname{tr}(\sigma_\alpha u^\dagger \sigma_\beta u) = R(u)_{\beta\alpha}.$$

18.2.3 Continuity

Every entry

$$R(u)_{\alpha\beta} = \frac{1}{2} \operatorname{tr}(\sigma_\alpha u \sigma_\beta u^\dagger)$$

is a polynomial expression in the real and imaginary parts of the entries of u . Hence every matrix entry is continuous, so

$$R : SU(2) \longrightarrow O(3)$$

is continuous. Together with (18.8), this proves that R is a Lie group homomorphism once we establish that its image lies in the component $SO(3)$. We shall do so explicitly after computing its differential.

18.3 The induced Lie-algebra homomorphism

Let

$$r : \mathfrak{su}(2) \longrightarrow \mathfrak{so}(3)$$

be the differential of R at the identity. By definition,

$$r(X) = \left. \frac{d}{dt} R(e^{tX}) \right|_{t=0}. \quad (18.10)$$

Since R is the matrix representation of the adjoint action, the compatibility of homomorphisms with exponentials gives

$$R(e^X) = \text{the matrix of } \text{Ad}_{e^X} = \text{the matrix of } e^{\text{ad}_X}.$$

Therefore $r(X)$ is precisely the matrix of ad_X in the basis $\{\mathbf{i}\sigma_\alpha\}$.

For completeness, differentiate the entries directly:

$$\begin{aligned} r(X)_{\alpha\beta} &= \left. \frac{d}{dt} \frac{1}{2} \text{tr}(\sigma_\alpha e^{tX} \sigma_\beta e^{-tX}) \right|_{t=0} \\ &= \frac{1}{2} \text{tr}(\sigma_\alpha (X\sigma_\beta - \sigma_\beta X)) \\ &= \langle \mathbf{i}\sigma_\alpha, [X, \mathbf{i}\sigma_\beta] \rangle. \end{aligned} \quad (18.11)$$

18.3.1 Cross-product matrices and the sign convention

For $\mathbf{y} \in \mathbb{R}^3$, define $T(\mathbf{y}) \in \mathfrak{so}(3)$ by

$$T(\mathbf{y})\mathbf{v} = \mathbf{y} \times \mathbf{v}.$$

Writing

$$T(\mathbf{y}) = y_1 T_1 + y_2 T_2 + y_3 T_3,$$

the three generators are

$$T_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad T_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad T_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (18.12)$$

Thus

$$T(\mathbf{y}) = \begin{pmatrix} 0 & -y_3 & y_2 \\ y_3 & 0 & -y_1 \\ -y_2 & y_1 & 0 \end{pmatrix}.$$

Let $X = \mathbf{i} \mathbf{y} \cdot \boldsymbol{\sigma}$. From (18.2),

$$\begin{aligned} [X, \mathbf{i}\sigma_\beta] &= \sum_{\gamma} y_{\gamma} [\mathbf{i}\sigma_{\gamma}, \mathbf{i}\sigma_{\beta}] \\ &= -2 \sum_{\gamma, \alpha} y_{\gamma} \varepsilon_{\gamma\beta\alpha} \mathbf{i}\sigma_{\alpha}. \end{aligned}$$

Since

$$T(\mathbf{y})_{\alpha\beta} = \sum_{\gamma} \varepsilon_{\gamma\beta\alpha} y_{\gamma},$$

equation (18.11) gives

$$r(\mathbf{i} \mathbf{y} \cdot \boldsymbol{\sigma}) = -2T(\mathbf{y}). \quad (18.13)$$

The minus sign is not mysterious: it comes from choosing $\mathbf{i}\sigma_{\alpha}$, rather than $-\mathbf{i}\sigma_{\alpha}$, as the basis of $\mathfrak{su}(2)$, together with the convention $T(\mathbf{y})\mathbf{v} = \mathbf{y} \times \mathbf{v}$.

Proposition 18.1: The Lie algebras are isomorphic

The differential r is an isomorphism

$$\mathfrak{su}(2) \cong \mathfrak{so}(3).$$

With

$$e_{\alpha} = \mathbf{i}\sigma_{\alpha}, \quad f_{\alpha} = -2T_{\alpha},$$

we have $r(e_{\alpha}) = f_{\alpha}$, and both bases have the same structure constants:

$$[e_{\alpha}, e_{\beta}] = -2 \sum_{\gamma} \varepsilon_{\alpha\beta\gamma} e_{\gamma}, \quad [f_{\alpha}, f_{\beta}] = -2 \sum_{\gamma} \varepsilon_{\alpha\beta\gamma} f_{\gamma}.$$

证明. The first bracket formula follows from (18.2). The cross-product generators satisfy

$$[T_{\alpha}, T_{\beta}] = \sum_{\gamma} \varepsilon_{\alpha\beta\gamma} T_{\gamma}.$$

Hence

$$\begin{aligned} [-2T_{\alpha}, -2T_{\beta}] &= 4[T_{\alpha}, T_{\beta}] \\ &= 4 \sum_{\gamma} \varepsilon_{\alpha\beta\gamma} T_{\gamma} \\ &= -2 \sum_{\gamma} \varepsilon_{\alpha\beta\gamma} (-2T_{\gamma}). \end{aligned}$$

Thus $e_{\alpha} \mapsto f_{\alpha}$ preserves brackets. It maps a basis of the three-dimensional space $\mathfrak{su}(2)$ to a basis of the three-dimensional space $\mathfrak{so}(3)$, so it is an isomorphism. ■

18.4 The explicit rotation generated by u

18.4.1 Every element of $SU(2)$ is an exponential

A matrix $u \in SU(2)$ has the form

$$u = \begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix}, \quad |a|^2 + |b|^2 = 1. \quad (18.14)$$

To see this, write the first row as (a, b) . A unit vector orthogonal to it is $(-b^*, a^*)$; the determinant-one condition fixes the otherwise possible unit complex phase.

Write

$$a = a_0 + ia_3, \quad b = a_2 + ia_1, \quad a_j \in \mathbb{R}.$$

Then

$$u = a_0 I_2 + i \sum_{\alpha=1}^3 a_\alpha \sigma_\alpha, \quad a_0^2 + a_1^2 + a_2^2 + a_3^2 = 1. \quad (18.15)$$

Choose $s \in [0, \pi]$ such that $a_0 = \cos s$. When $\sin s \neq 0$, put

$$\mathbf{n} = \frac{(a_1, a_2, a_3)}{\sin s}, \quad \|\mathbf{n}\| = 1.$$

At $\sin s = 0$, any unit $\mathbf{n}$ may be used. Because

$$(\mathbf{n} \cdot \boldsymbol{\sigma})^2 = I_2,$$

its powers separate into even and odd terms, and therefore

$$\begin{aligned} \exp(is \mathbf{n} \cdot \boldsymbol{\sigma}) &= \sum_{k=0}^{\infty} \frac{(is \mathbf{n} \cdot \boldsymbol{\sigma})^k}{k!} \\ &= \cos s I_2 + i \sin s \mathbf{n} \cdot \boldsymbol{\sigma}. \end{aligned} \quad (18.16)$$

Thus every $u \in SU(2)$ can be written

$$u = e^{is \mathbf{n} \cdot \boldsymbol{\sigma}}.$$

18.4.2 Exponentiating the differential

Let

$$u = e^{iy \cdot \boldsymbol{\sigma}}, \quad \mathbf{y} = s\mathbf{n}, \quad \|\mathbf{n}\| = 1.$$

The exponential compatibility of a Lie group homomorphism, together with (18.13), gives

$$\boxed{R(u) = R(e^{iy \cdot \boldsymbol{\sigma}}) = e^{-2T(\mathbf{y})} = e^{-2sT(\mathbf{n})}.} \quad (18.17)$$

For a unit vector $\mathbf{n}$,

$$T(\mathbf{n})^2 = \mathbf{n}\mathbf{n}^T - I_3, \quad T(\mathbf{n})^3 = -T(\mathbf{n}).$$

The exponential series then gives Rodrigues' formula

$$\begin{aligned} e^{\theta T(\mathbf{n})} &= I_3 + \sin \theta T(\mathbf{n}) + (1 - \cos \theta) T(\mathbf{n})^2 \\ &= \cos \theta I_3 + (1 - \cos \theta) \mathbf{n}\mathbf{n}^T + \sin \theta T(\mathbf{n}). \end{aligned} \quad (18.18)$$

It fixes $\mathbf{n}$ and rotates the plane $\mathbf{n}^\perp$ through angle θ , with the right-hand-rule orientation. Therefore

$$R(e^{is \mathbf{n} \cdot \boldsymbol{\sigma}})$$

is the rotation about the axis $\mathbf{n}$ through angle $-2s$. Equivalently, the conventional rotation through angle θ about $\mathbf{n}$ is obtained from either of the two matrices

$$u_\pm = \pm \exp\left(-\frac{i\theta}{2} \mathbf{n} \cdot \boldsymbol{\sigma}\right). \quad (18.19)$$

Conjugation identity in vector form

Equation (18.6) is equivalent to

$$u \mathbf{i}(\mathbf{v} \cdot \boldsymbol{\sigma}) u^\dagger = \mathbf{i}((R(u)\mathbf{v}) \cdot \boldsymbol{\sigma}) \quad (\mathbf{v} \in \mathbb{R}^3).$$

Thus conjugating a traceless anti-Hermitian 2×2 matrix by u performs an ordinary three-dimensional rotation on its coefficient vector.

18.4.3 The determinant is 1

We already know that $R(u) \in O(3)$. Formula (18.17) gives the orientation directly:

$$\begin{aligned} \det R(u) &= \det(e^{-2T(\mathbf{y})}) \\ &= \exp(\operatorname{tr}(-2T(\mathbf{y}))) \\ &= e^0 = 1, \end{aligned}$$

because every skew-symmetric matrix has zero diagonal and hence zero trace. Consequently,

$$R(u) \in SO(3) \quad (u \in SU(2)). \quad (18.20)$$

This completes the verification that

$$R : SU(2) \longrightarrow SO(3)$$

is a continuous group homomorphism.

18.5 Surjectivity and the two-element kernel**18.5.1 Every rotation has a preimage****Proposition 18.2: Surjectivity**

The homomorphism

$$R : SU(2) \longrightarrow SO(3)$$

is onto.

证明. Every $Q \in SO(3)$ has a real eigenvector $\mathbf{n}$ with eigenvalue 1. Indeed, a real cubic characteristic polynomial has a real root, and the eigenvalues of an orthogonal matrix have absolute value 1. Nonreal eigenvalues occur in conjugate pairs whose product is 1; since $\det Q = 1$, the remaining real eigenvalue must be 1.

The orthogonal plane $\mathbf{n}^\perp$ is Q -invariant, and the restriction of Q to that oriented plane is a two-dimensional rotation through some angle θ . Thus, by Rodrigues' formula,

$$Q = e^{\theta T(\mathbf{n})}.$$

Set

$$u = \exp\left(-\frac{\mathbf{i}\theta}{2} \mathbf{n} \cdot \boldsymbol{\sigma}\right) \in SU(2).$$

Equation (18.17) gives $R(u) = Q$. ■

18.5.2 Computing the kernel from the exponential formula

Let

$$u = e^{is\mathbf{n} \cdot \boldsymbol{\sigma}}, \quad \|\mathbf{n}\| = 1.$$

The skew-symmetric matrix $T(\mathbf{n})$ has eigenvalues

$$0, \quad i, \quad -i.$$

The zero eigenvalue corresponds to the rotation axis $\mathbf{n}$; on $\mathbf{n}^\perp$, the matrix is a quarter-turn. Hence the eigenvalues of

$$R(u) = e^{-2sT(\mathbf{n})}$$

are

$$1, \quad e^{-2is}, \quad e^{2is}.$$

Therefore

$$R(u) = I_3 \iff e^{2is} = 1 \iff s = m\pi \quad (m \in \mathbb{Z}).$$

Using (18.16),

$$u = \cos(m\pi)I_2 + i \sin(m\pi)\mathbf{n} \cdot \boldsymbol{\sigma} = (-1)^m I_2.$$

Thus

$$\ker R = \{I_2, -I_2\}. \tag{18.21}$$

A second, basis-free kernel check

If $R(u) = I_3$, then

$$u\sigma_\alpha u^\dagger = \sigma_\alpha \quad (\alpha = 1, 2, 3),$$

so u commutes with all Pauli matrices. Commutation with σ_3 forces u to be diagonal, and commutation with σ_1 forces its two diagonal entries to be equal. Hence $u = \lambda I_2$. Unitarity gives $|\lambda| = 1$, while $\det u = \lambda^2 = 1$; therefore $\lambda = \pm 1$.

Theorem 18.1: $SU(2)$ double-covers $SO(3)$

The adjoint construction gives a surjective Lie group homomorphism

$$R : SU(2) \longrightarrow SO(3)$$

with kernel $\{\pm I_2\}$. It induces a Lie group isomorphism

$$SU(2)/\{\pm I_2\} \cong SO(3). \tag{18.22}$$

Moreover, R is a two-sheeted covering map.

证明. Surjectivity and the kernel have already been proved. The first isomorphism theorem gives the quotient isomorphism.

To see the covering property concretely, note that the differential

$$r : \mathfrak{su}(2) \longrightarrow \mathfrak{so}(3)$$

is an isomorphism. The inverse function theorem therefore provides a neighbourhood U of I_2 on which R is a diffeomorphism onto a neighbourhood V of I_3 . Shrink U , if necessary, so that $U \cap (-U) = \emptyset$.

If $g \in R^{-1}(V)$, there is a unique $u \in U$ with $R(u) = R(g)$. Then

$$R(gu^{-1}) = I_3,$$

so $gu^{-1} = \pm I_2$, or $g = \pm u$. Consequently,

$$R^{-1}(V) = U \sqcup (-U),$$

and R maps each of the two pieces diffeomorphically onto V . Left translation gives the same description over a neighbourhood of every point of $SO(3)$. Hence R is a two-sheeted covering. ■

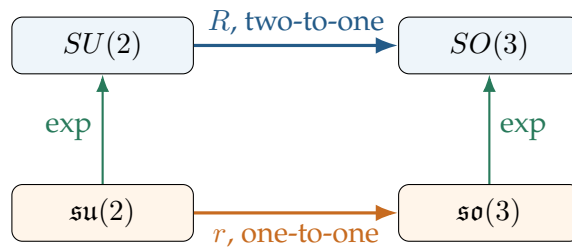


图 18.1: The group map is two-to-one, although its differential is an isomorphism. The square commutes: $R(e^X) = e^{r(X)}$.

18.6 Axis-angle coordinates and the topology of $SO(3)$

Every element of $SO(3)$ can be described by a unit axis $\mathbf{n} \in S^2$ and an angle $\theta \in [0, \pi]$. Associate to the rotation the vector

$$\mathbf{x} = \theta \mathbf{n} \in \mathbb{R}^3.$$

All vectors with $\|\mathbf{x}\| < \pi$ describe distinct rotations. On the boundary $\|\mathbf{x}\| = \pi$, however,

$$(\mathbf{n}, \pi) \quad \text{and} \quad (-\mathbf{n}, \pi)$$

describe the same rotation. A half-turn about $-\mathbf{n}$ is the same linear transformation as a half-turn about $\mathbf{n}$.

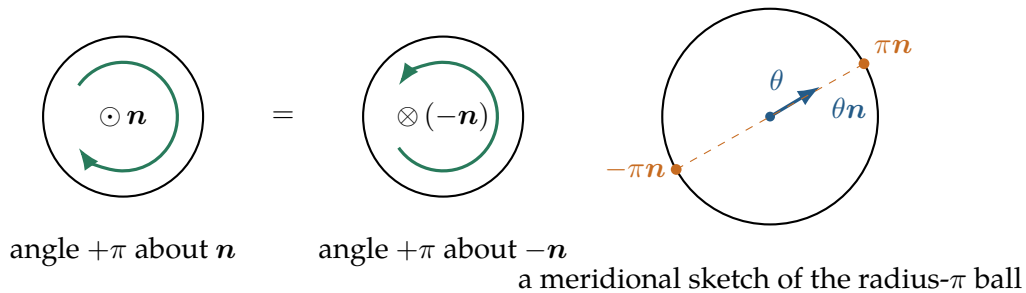


图 18.2: Axis-angle coordinates. Antipodal points on the boundary sphere represent the same rotation by π .

It follows that, topologically,

$$SO(3) \cong B_\pi^3 / (x \sim -x \text{ for } \|x\| = \pi) \cong \mathbb{RP}^3. \quad (18.23)$$

Here B_π^3 is the *three*-dimensional closed ball of radius π . Its two-dimensional drawings are only cross-sectional sketches; the rotation group itself is a three-dimensional manifold, which may be embedded in a higher-dimensional Euclidean space but is not an ordinary solid torus.

18.6.1 Closed paths through the identity

In the ball model, a small closed curve lying in the interior can be continuously shrunk to the origin, which represents the identity rotation. A path that begins at a boundary point πn and ends at its antipode $-\pi n$ becomes a closed path only *after* the boundary identification. It cannot be shrunk to a point in $SO(3)$. Traversing that path twice, however, produces a contractible loop.

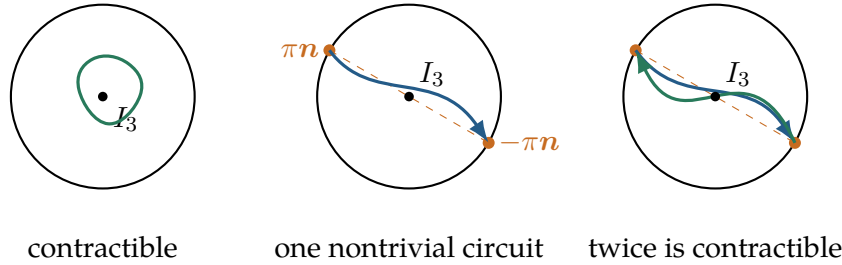


图 18.3: Schematic loops in the antipodally identified ball model of $SO(3)$.

The covering map makes this distinction precise. Consider the loop

$$\gamma(t) = \text{rotation through angle } 2\pi t \text{ about } n, \quad 0 \leq t \leq 1.$$

Its lift to $SU(2)$ beginning at I_2 is

$$\tilde{\gamma}(t) = \exp(-i\pi t n \cdot \sigma).$$

The endpoints are

$$\tilde{\gamma}(0) = I_2, \quad \tilde{\gamma}(1) = -I_2.$$

Thus the lifted path is not closed. A contractible loop has a closed lift when lifted from a chosen point, so γ is not contractible. If the rotation is carried through 4π , the lift ends at

$$\exp(-2\pi i n \cdot \sigma) = I_2;$$

that doubled loop is contractible.

18.7 $SU(2)$ is the three-sphere

The parametrization (18.15) gives a continuous bijection

$$\begin{aligned} S^3 &= \{(a_0, a_1, a_2, a_3) \in \mathbb{R}^4 : a_0^2 + a_1^2 + a_2^2 + a_3^2 = 1\} \\ &\longrightarrow SU(2), \\ (a_0, a_1, a_2, a_3) &\longmapsto a_0 I_2 + i \sum_{\alpha=1}^3 a_\alpha \sigma_\alpha. \end{aligned}$$

Its inverse reads off the four real coefficients and is continuous. Therefore it is a homeomorphism:

$$\boxed{SU(2) \cong S^3.} \quad (18.24)$$

The sphere S^3 is simply connected. One standard proof covers it by the complements of the north and south poles. Each complement is stereographically homeomorphic to $\mathbb{R}^3$, hence is simply connected, and their intersection is path-connected. The Seifert–van Kampen theorem then gives

$$\pi_1(S^3) = \{1\}.$$

Consequently $SU(2)$ is simply connected.

Because $SU(2)$ is simply connected and

$$R : SU(2) \longrightarrow SO(3)$$

is a two-sheeted covering, $SU(2)$ is the universal covering group of $SO(3)$. Its deck transformation group is

$$\{u \mapsto u, u \mapsto -u\} \cong \mathbb{Z}_2.$$

Equivalently,

$$\boxed{\pi_1(SO(3)) \cong \mathbb{Z}_2, \quad \pi_1(SU(2)) = \{1\}.} \quad (18.25)$$

Why identical Lie algebras do not imply identical groups

The differential r proves $\mathfrak{su}(2) \cong \mathfrak{so}(3)$, so the two groups have the same local infinitesimal structure. Globally they differ:

$$SU(2) \cong S^3 \text{ is simply connected,} \quad SO(3) \cong \mathbb{RP}^3 \text{ has fundamental group } \mathbb{Z}_2.$$

Lie algebras determine connected Lie groups only up to covering and discrete central quotients.

What to retain from this chapter

1. The real Lie algebra

$$\mathfrak{su}(2) = \text{span}_{\mathbb{R}}\{\mathbf{i}\sigma_1, \mathbf{i}\sigma_2, \mathbf{i}\sigma_3\}$$

is three-dimensional, and the displayed basis is orthonormal for $\langle X, Y \rangle = \frac{1}{2} \text{tr}(X^\dagger Y)$.

2. Conjugation defines

$$u(\mathbf{i}\sigma_\beta)u^\dagger = \sum_{\alpha} R(u)_{\alpha\beta} \mathbf{i}\sigma_\alpha, \quad R(u)_{\alpha\beta} = \frac{1}{2} \text{tr}(\sigma_\alpha u \sigma_\beta u^\dagger).$$

3. The map $R : SU(2) \rightarrow SO(3)$ is a continuous group homomorphism. Its differential is

$$r(\mathbf{i} \mathbf{y} \cdot \boldsymbol{\sigma}) = -2T(\mathbf{y}), \quad T(\mathbf{y})\mathbf{v} = \mathbf{y} \times \mathbf{v}.$$

Hence $\mathfrak{su}(2) \cong \mathfrak{so}(3)$.

4. If $u = e^{\mathbf{i} s \mathbf{n} \cdot \boldsymbol{\sigma}}$, then

$$R(u) = e^{-2sT(\mathbf{n})}.$$

Thus the angle in $SO(3)$ is twice the corresponding $SU(2)$ half-angle, with the sign fixed by our basis convention.

5. The map is surjective and

$$\ker R = \{\pm I_2\}, \quad SU(2)/\{\pm I_2\} \cong SO(3).$$

It is a two-sheeted covering map.

6. Topologically,

$$SU(2) \cong S^3, \quad SO(3) \cong B_\pi^3 / (\text{antipodal boundary points}) \cong \mathbb{RP}^3.$$

7. Therefore

$$\pi_1(SU(2)) = \{1\}, \quad \pi_1(SO(3)) \cong \mathbb{Z}_2.$$

A 2π rotation represents the nontrivial loop class, whereas a 4π rotation is contractible.

Covering Groups, Spin Groups, and Spinors

Road map

The preceding chapter exhibited a striking phenomenon:

$$\mathfrak{su}(2) \cong \mathfrak{so}(3), \quad SU(2)/\{\pm I_2\} \cong SO(3),$$

but $SU(2)$ and $SO(3)$ are not isomorphic as Lie groups. The correct global language is that $SU(2)$ is a *covering group* of $SO(3)$. This chapter develops that idea and then applies it to quantum spin:

1. covering homomorphisms and their discrete central kernels;
2. the universal covering group and the classification of connected covers by subgroups of the fundamental group;
3. the covers of $U(1)$;
4. Pauli spinors and the sign acquired by a spin- $\frac{1}{2}$ state under a 2π rotation;
5. the groups $\text{Spin}(n)$, their low-dimensional accidental isomorphisms, and their centres;
6. the universal and metaplectic covers of $\text{Sp}(2n, \mathbb{R})$;
7. the spin cover $SL(2, \mathbb{C}) \rightarrow SO^+(1, 3)$.

19.1 Covering homomorphisms

Suppose G and $\tilde{G}$ are Lie groups and

$$\Phi : \tilde{G} \longrightarrow G$$

is a Lie group homomorphism. Its induced Lie-algebra homomorphism is

$$\phi : \tilde{\mathfrak{g}} \longrightarrow \mathfrak{g}, \quad \phi(X) = \left. \frac{d}{dt} \Phi(e^{tX}) \right|_{t=0}.$$

The exponential compatibility diagram is

$$\begin{array}{ccc}
 \tilde{G} & \xrightarrow{\Phi} & G \\
 \exp \uparrow & & \uparrow \exp \\
 \tilde{\mathfrak{g}} & \xrightarrow{\phi} & \mathfrak{g}
 \end{array}
 \quad \Phi(e^X) = e^{\phi(X)}.$$

Definition 19.1: Covering group and covering homomorphism

The Lie group $\tilde{G}$ is a *covering group* of G , and $\Phi : \tilde{G} \rightarrow G$ is a *covering homomorphism*, if:

1. Φ is surjective;
2. the induced map $\phi : \tilde{\mathfrak{g}} \rightarrow \mathfrak{g}$ is an isomorphism;
3. $\ker \Phi$ is discrete in the ordinary manifold topology.

The three conditions imply the usual topological covering property. Indeed, because $\phi = d\Phi_e$ is invertible, the inverse function theorem gives an identity neighbourhood $U \subset \tilde{G}$ on which Φ is a diffeomorphism onto an identity neighbourhood $V \subset G$. Since $\ker \Phi$ is discrete, U may be chosen so that the translates

$$kU, \quad k \in \ker \Phi,$$

are pairwise disjoint. Every preimage of V differs from its unique representative in U by a kernel element, so

$$\Phi^{-1}(V) = \bigsqcup_{k \in \ker \Phi} kU.$$

Each kU maps diffeomorphically onto V . Translating this picture over every point of G proves that Φ is a covering map.

The quotient description

Every covering homomorphism satisfies

$$G \cong \tilde{G} / \ker \Phi.$$

The Lie algebra does not see the kernel because a discrete subgroup has zero-dimensional tangent space.

19.1.1 A discrete kernel is central when the cover is connected

Proposition 19.1: Centrality of the kernel

Let $\Phi : \tilde{G} \rightarrow G$ be a covering homomorphism. If $\tilde{G}$ is connected, then

$$\ker \Phi \subseteq Z(\tilde{G}). \quad (19.1)$$

In particular, $\ker \Phi$ is abelian.

证明. Take $k \in \ker \Phi$ and $g \in \tilde{G}$. Because $\tilde{G}$ is connected, it is path-connected, so there is a smooth path

$$g : [0, 1] \longrightarrow \tilde{G}, \quad g(0) = \tilde{e}, \quad g(1) = g.$$

Define

$$f(t) = g(t)kg(t)^{-1}.$$

Then

$$\begin{aligned} \Phi(f(t)) &= \Phi(g(t))\Phi(k)\Phi(g(t))^{-1} \\ &= \Phi(g(t))e\Phi(g(t))^{-1} = e. \end{aligned}$$

Thus $f(t) \in \ker \Phi$ for every t . The map f is continuous, whereas $\ker \Phi$ is discrete. Since $[0, 1]$ is connected, f must be constant. Therefore

$$f(0) = k = f(1) = gkg^{-1}.$$

As g was arbitrary, $k \in Z(\tilde{G})$. ■

This argument is often summarized by saying that a connected covering group “wraps around” the base group, and each deck displacement lies in its centre.

19.2 First examples

19.2.1 The double cover of $SO(3)$

The map constructed in the preceding chapter,

$$R : SU(2) \longrightarrow SO(3),$$

has all the required properties:

$$R \text{ is surjective,} \quad dR_e : \mathfrak{su}(2) \xrightarrow{\sim} \mathfrak{so}(3), \quad \ker R = \{\pm I_2\} \cong \mathbb{Z}_2.$$

Hence $SU(2)$ is a double cover of $SO(3)$.

19.2.2 The exponential cover of the circle

Let

$$\Phi : \mathbb{R} \longrightarrow U(1), \quad t \longmapsto e^{it}. \tag{19.2}$$

This is a continuous group homomorphism because

$$\Phi(t_1 + t_2) = e^{i(t_1+t_2)} = e^{it_1}e^{it_2} = \Phi(t_1)\Phi(t_2).$$

It is surjective. Its Lie-algebra map is

$$\mathbb{R} \longrightarrow i\mathbb{R}, \quad x \longmapsto ix,$$

which is an isomorphism of one-dimensional abelian real Lie algebras. Its kernel is

$$\ker \Phi = \{t \in \mathbb{R} : e^{it} = 1\} = 2\pi\mathbb{Z}. \quad (19.3)$$

Thus

$$\mathbb{R}/(2\pi\mathbb{Z}) \cong U(1).$$

The covering group $\mathbb{R}$ is simply connected, while the base circle is not.

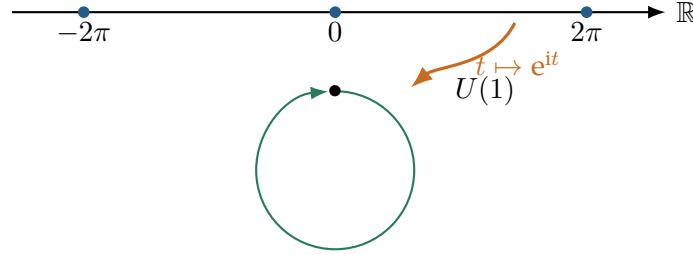


图 19.1: Each interval of length 2π wraps once around the circle.

There are also non-universal connected covers of the circle. For a fixed integer $n \geq 1$,

$$\Phi_n : U(1) \longrightarrow U(1), \quad z \longmapsto z^n, \quad (19.4)$$

is an n -sheeted covering homomorphism. Its kernel is the group of n -th roots of unity,

$$\ker \Phi_n = \{e^{2\pi i k/n} : k = 0, 1, \dots, n-1\} \cong \mathbb{Z}_n.$$

The covering group is again $U(1)$, and so it is not simply connected. This example shows that a cover of a Lie group is not unique.

19.3 The universal covering group

Definition 19.2: Universal cover

Let G be a connected Lie group with Lie algebra $\mathfrak{g}$. A *universal covering group* of G is a connected, simply connected Lie group $\tilde{G}_{\text{univ}}$ with

$$\text{Lie}(\tilde{G}_{\text{univ}}) \cong \mathfrak{g}$$

and a surjective covering homomorphism

$$p : \tilde{G}_{\text{univ}} \longrightarrow G.$$

Theorem 19.1: Existence and uniqueness of the universal group cover

Every connected Lie group has a universal covering group, unique up to a Lie group isomorphism commuting with the covering maps. For its covering homomorphism p ,

$$\ker p \cong \pi_1(G) \subseteq Z(\tilde{G}_{\text{univ}}), \quad G \cong \tilde{G}_{\text{univ}} / \ker p. \quad (19.5)$$

The universal cover “unwraps” every closed path that cannot be deformed to the identity

loop while keeping its base point fixed. If G is disconnected, one first applies the construction to the identity component G° ; the Lie algebra controls only that component.

19.3.1 Every connected cover comes from the universal one

Let

$$p : \tilde{G}_{\text{univ}} \longrightarrow G, \quad K = \ker p \cong \pi_1(G).$$

For every subgroup $H \leq K$, the quotient

$$\tilde{G}_{\text{univ}}/H$$

is a connected covering group of G . Conversely, every connected covering group of G is isomorphic to one of these quotients. The extreme cases are

$$H = \{e\} : \tilde{G}_{\text{univ}}, \quad H = K : G.$$

Thus the universal cover is maximal: all other connected covers are its quotients by subgroups of the discrete central kernel.

For the circle,

$$\widetilde{U(1)}_{\text{univ}} = \mathbb{R}, \quad K = 2\pi\mathbb{Z}.$$

Quotienting by $2\pi n\mathbb{Z}$ gives another circle, and the induced map to $U(1)$ is the n -sheeted cover $z \mapsto z^n$.

19.4 Pauli spinors in a magnetic field

Consider a spin- $\frac{1}{2}$ particle in a nonzero external magnetic field $\mathbf{B} \in \mathbb{R}^3$. In convenient units its Hamiltonian is

$$H(\mathbf{B}) = -\frac{1}{2}\mathbf{B} \cdot \boldsymbol{\sigma}. \quad (19.6)$$

The Hilbert space is $\mathbb{C}^2$, and the spin observable is

$$\mathbf{S} = \frac{1}{2}\boldsymbol{\sigma}.$$

Because

$$(\mathbf{B} \cdot \boldsymbol{\sigma})^2 = \|\mathbf{B}\|^2 I_2,$$

the eigenvalues of $\mathbf{B} \cdot \boldsymbol{\sigma}$ are $\pm\|\mathbf{B}\|$. Therefore the two energy levels are

$$E_0(\mathbf{B}) = -\frac{\|\mathbf{B}\|}{2}, \quad E_1(\mathbf{B}) = +\frac{\|\mathbf{B}\|}{2}.$$

This is the Zeeman splitting.

19.4.1 The direction-dependent ground state

Write

$$\mathbf{B} = \|\mathbf{B}\| \begin{pmatrix} \cos \varphi \sin \vartheta \\ \sin \varphi \sin \vartheta \\ \cos \vartheta \end{pmatrix} = \|\mathbf{B}\| \mathbf{n}, \quad \|\mathbf{n}\| = 1. \quad (19.7)$$

A normalized ground-state spinor is

$$|\psi_0(\vartheta, \varphi)\rangle = \begin{pmatrix} \cos(\vartheta/2) \\ e^{i\varphi} \sin(\vartheta/2) \end{pmatrix}. \quad (19.8)$$

It depends only on the direction of $\mathbf{B}$, not on its magnitude.

To check the eigenvalue equation, note that

$$\mathbf{n} \cdot \boldsymbol{\sigma} = \begin{pmatrix} \cos \vartheta & e^{-i\varphi} \sin \vartheta \\ e^{i\varphi} \sin \vartheta & -\cos \vartheta \end{pmatrix}.$$

Multiplication gives

$$\begin{aligned} (\mathbf{n} \cdot \boldsymbol{\sigma})|\psi_0\rangle &= \begin{pmatrix} \cos \vartheta \cos(\vartheta/2) + \sin \vartheta \sin(\vartheta/2) \\ e^{i\varphi} (\sin \vartheta \cos(\vartheta/2) - \cos \vartheta \sin(\vartheta/2)) \end{pmatrix} \\ &= \begin{pmatrix} \cos(\vartheta/2) \\ e^{i\varphi} \sin(\vartheta/2) \end{pmatrix} = |\psi_0\rangle. \end{aligned}$$

Hence

$$H(\mathbf{B})|\psi_0\rangle = -\frac{\|\mathbf{B}\|}{2}|\psi_0\rangle.$$

19.4.2 The spin expectation aligns with the field

Put

$$c = \cos(\vartheta/2), \quad s = \sin(\vartheta/2).$$

Then the three component calculations are

$$\begin{aligned} \langle \psi_0 | \sigma_1 | \psi_0 \rangle &= (c, e^{-i\varphi} s) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} c \\ e^{i\varphi} s \end{pmatrix} \\ &= cs(e^{i\varphi} + e^{-i\varphi}) = 2cs \cos \varphi = \sin \vartheta \cos \varphi, \\ \langle \psi_0 | \sigma_2 | \psi_0 \rangle &= (c, e^{-i\varphi} s) \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} c \\ e^{i\varphi} s \end{pmatrix} \\ &= -ics e^{i\varphi} + ics e^{-i\varphi} = \sin \vartheta \sin \varphi, \\ \langle \psi_0 | \sigma_3 | \psi_0 \rangle &= (c, e^{-i\varphi} s) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} c \\ e^{i\varphi} s \end{pmatrix} \\ &= c^2 - s^2 = \cos \vartheta. \end{aligned}$$

The half-angle identities

$$2 \cos(\vartheta/2) \sin(\vartheta/2) = \sin \vartheta, \quad \cos^2(\vartheta/2) - \sin^2(\vartheta/2) = \cos \vartheta$$

were used in the last steps. Therefore

$$\boxed{\langle \psi_0 | \mathbf{S} | \psi_0 \rangle = \frac{1}{2} \mathbf{n} = \frac{1}{2} \frac{\mathbf{B}}{\|\mathbf{B}\|}.} \quad (19.9)$$

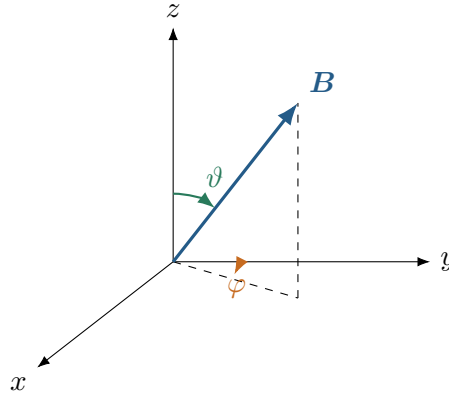


图 19.2: The ground-state spin expectation points along the magnetic-field direction.

19.5 Rotations of a spinor

We use the active viewpoint: the physical state is rotated,

$$|\psi\rangle \mapsto |\psi'\rangle = U(R)|\psi\rangle,$$

while the laboratory measurement apparatus and the operators $\mathbf{S} = \frac{1}{2}\boldsymbol{\sigma}$ are kept fixed.

Under an ordinary rotation $R \in SO(3)$, a classical vector transforms as $\mathbf{v} \mapsto R\mathbf{v}$. Therefore the quantum expectation value must obey

$$\langle \psi' | \mathbf{S} | \psi' \rangle = R \langle \psi | \mathbf{S} | \psi \rangle.$$

Since this should hold for every state, the corresponding operator identity is

$$\boxed{U(R)^\dagger \mathbf{S} U(R) = R\mathbf{S}.} \quad (19.10)$$

In components,

$$U(R)^\dagger S_\alpha U(R) = \sum_{\beta=1}^3 R_{\alpha\beta} S_\beta.$$

Wigner's theorem says that a symmetry preserving transition probabilities acts on rays by a unitary or antiunitary operator. Rotations vary continuously from the identity, so the relevant implementation is unitary. Its phase can be chosen locally so that, for an infinitesimal rotation, $U(R)$ is close to I_2 ; in particular $U(R) \in SU(2)$.

For a rotation

$$R = \mathbf{e}^{\theta T(\mathbf{n})} \in SO(3),$$

the double-cover calculation of the preceding chapter gives two possible lifts:

$$U_{\pm}(R) = \pm \exp\left(-\frac{i\theta}{2} \mathbf{n} \cdot \boldsymbol{\sigma}\right). \quad (19.11)$$

The lift continuous from the identity is

$$U(R) = \exp\left(-\frac{i\theta}{2} \mathbf{n} \cdot \boldsymbol{\sigma}\right).$$

Equation (19.10) is exactly the adjoint identity

$$U^{\dagger} \sigma_{\alpha} U = \sum_{\beta} R_{\alpha\beta} \sigma_{\beta}.$$

19.5.1 A 2π rotation changes the spinor's sign

Because $(\mathbf{n} \cdot \boldsymbol{\sigma})^2 = I_2$,

$$\begin{aligned} U(2\pi, \mathbf{n}) &= \exp(-i\pi \mathbf{n} \cdot \boldsymbol{\sigma}) \\ &= \cos \pi I_2 - i \sin \pi \mathbf{n} \cdot \boldsymbol{\sigma} = -I_2. \end{aligned} \quad (19.12)$$

Thus a 360° rotation acts by

$$|\psi\rangle \mapsto -|\psi\rangle.$$

A second full turn gives $U(4\pi, \mathbf{n}) = I_2$.

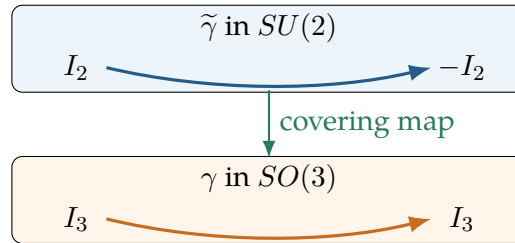


图 19.3: The closed 2π -rotation loop in $SO(3)$ lifts to a path from I_2 to $-I_2$ in $SU(2)$.

The sign is a quantum-mechanical phase. A single isolated ray does not distinguish $|\psi\rangle$ from $-|\psi\rangle$, but the relative sign is observable in interference. It is the representation-theoretic shadow of the nontrivial loop in $\pi_1(SO(3)) \cong \mathbb{Z}_2$, and it may be described as a topological or Berry phase of π .

Definition 19.3: Pauli spinor

A two-component complex object transforming according to the defining representation of $SU(2)$,

$$\psi \mapsto U\psi, \quad U \in SU(2),$$

is called a *Pauli spinor*. Its representation space is $\mathbb{C}^2$.

19.6 Spin groups

The group $SO(n)$ is connected for every $n \geq 2$. Its fundamental group is

$$\pi_1(SO(2)) \cong \mathbb{Z}, \quad \pi_1(SO(n)) \cong \mathbb{Z}_2 \quad (n \geq 3). \quad (19.13)$$

For $n \geq 4$, the fibration

$$SO(n-1) \longrightarrow SO(n) \longrightarrow S^{n-1}, \quad A \longmapsto Ae_n,$$

and its long exact homotopy sequence reduce $\pi_1(SO(n))$ inductively to $\pi_1(SO(3))$. The base sphere S^{n-1} is simply connected for $n \geq 3$, which is why the answer stabilizes at $\mathbb{Z}_2$.

Definition 19.4: Spin group

For $n \geq 3$, the *spin group* is the universal covering group

$$\text{Spin}(n) := \widetilde{SO(n)}_{\text{univ}}.$$

It has Lie algebra

$$\text{Lie}(\text{Spin}(n)) \cong \mathfrak{so}(n).$$

There is a canonical double covering homomorphism

$$\rho_n : \text{Spin}(n) \longrightarrow SO(n), \quad \ker \rho_n = \{\pm 1\}, \quad (19.14)$$

and hence

$$SO(n) \cong \text{Spin}(n)/\{\pm 1\} \quad (n \geq 3). \quad (19.15)$$

The exceptional notation in dimension two

The universal cover of $SO(2) \cong U(1)$ is

$$\mathbb{R} \longrightarrow SO(2), \quad t \longmapsto \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix},$$

with kernel $2\pi\mathbb{Z}$. Standard Clifford-algebra notation instead uses $\text{Spin}(2) \cong U(1)$ for the *double* cover $z \mapsto z^2$ of $SO(2)$. Thus the conventional compact $\text{Spin}(2)$ is not the universal cover. Keeping these two objects distinct resolves the apparent conflict in the low-dimensional table.

19.6.1 Low-dimensional spin groups and their centres

Several low-dimensional spin groups are isomorphic to more familiar classical groups. These are sometimes called accidental isomorphisms:

$$\text{Spin}(3) \cong SU(2), \quad \text{Spin}(4) \cong SU(2) \times SU(2),$$

$$\text{Spin}(5) \cong Sp(2), \quad \text{Spin}(6) \cong SU(4).$$

Here $Sp(2)$ means the compact symplectic group of 2×2 quaternionic unitary matrices, equivalently $USp(4)$; it should not be confused with the real group $Sp(4, \mathbb{R})$.

n	$\dim SO(n)$	$\pi_1(SO(n))$	$\widetilde{SO(n)}_{\text{univ}}$	kernel	centre of the cover	$Z(SO(n))$
2	1	$\mathbb{Z}$	$\mathbb{R}$	$2\pi\mathbb{Z}$	$\mathbb{R}$	$SO(2)$
3	3	$\mathbb{Z}_2$	$SU(2)$	$\{\pm 1\}$	$\mathbb{Z}_2$	$\{1\}$
4	6	$\mathbb{Z}_2$	$SU(2) \times SU(2)$	$\{\pm(1, 1)\}$	$\mathbb{Z}_2 \times \mathbb{Z}_2$	$\mathbb{Z}_2$
5	10	$\mathbb{Z}_2$	$Sp(2)$	$\{\pm 1\}$	$\mathbb{Z}_2$	$\{1\}$
6	15	$\mathbb{Z}_2$	$SU(4)$	$\{\pm 1\}$	$\mathbb{Z}_4$	$\mathbb{Z}_2$
7	21	$\mathbb{Z}_2$	$\text{Spin}(7)$	$\{\pm 1\}$	$\mathbb{Z}_2$	$\{1\}$
8	28	$\mathbb{Z}_2$	$\text{Spin}(8)$	$\{\pm 1\}$	$\mathbb{Z}_2 \times \mathbb{Z}_2$	$\mathbb{Z}_2$

表 19.1: The low-dimensional covering groups, kernels, and centres.

The dimension formula in the table is

$$\dim SO(n) = \dim \mathfrak{so}(n) = \frac{n(n-1)}{2}.$$

For $n \geq 3$, the centre of $\text{Spin}(n)$ follows the pattern

$$Z(\text{Spin}(n)) \cong \begin{cases} \mathbb{Z}_2, & n \text{ odd}, \\ \mathbb{Z}_4, & n \equiv 2 \pmod{4}, \\ \mathbb{Z}_2 \times \mathbb{Z}_2, & n \equiv 0 \pmod{4}, \end{cases} \quad (19.16)$$

whereas

$$Z(SO(n)) = \begin{cases} \{I_n\}, & n \text{ odd}, \\ \{\pm I_n\} \cong \mathbb{Z}_2, & n \text{ even} \end{cases} \quad (n \geq 3). \quad (19.17)$$

The covering kernel $\{\pm 1\}$ is always central, but it need not be the entire centre of $\text{Spin}(n)$.

19.7 Two further families of covers

19.7.1 The special unitary groups

For every $n \geq 2$, $SU(n)$ is simply connected. One way to see the inductive structure is through the fibration

$$SU(n-1) \longrightarrow SU(n) \longrightarrow S^{2n-1}, \quad U \longmapsto Ue_n.$$

The base S^{2n-1} is simply connected, and the induction starts with

$$SU(2) \cong S^3.$$

Also $SU(1) = \{1\}$. Therefore there is nothing further to unwrap:

$$\widetilde{SU(n)}_{\text{univ}} = SU(n).$$

19.7.2 The real symplectic group and the metaplectic group

The real symplectic group

$$\mathrm{Sp}(2n, \mathbb{R}) = \left\{ M \in GL(2n, \mathbb{R}) : M^T J M = J \right\}, \quad J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix},$$

is connected but not simply connected. Its fundamental group and centre are

$$\pi_1(\mathrm{Sp}(2n, \mathbb{R})) \cong \mathbb{Z}, \quad Z(\mathrm{Sp}(2n, \mathbb{R})) = \{\pm I_{2n}\}. \quad (19.18)$$

Consequently its universal covering homomorphism

$$p : \widetilde{\mathrm{Sp}}(2n, \mathbb{R}) \longrightarrow \mathrm{Sp}(2n, \mathbb{R})$$

has infinitely many sheets and

$$\ker p \cong \mathbb{Z}.$$

The universal covering group $\widetilde{\mathrm{Sp}}(2n, \mathbb{R})$ is a genuine Lie group, but it is not a linear matrix Lie group: no faithful finite-dimensional matrix representation realizes it as a closed subgroup of some $GL(N, \mathbb{C})$.

The subgroup $2\mathbb{Z}$ of the universal kernel gives an important intermediate quotient:

$$\mathrm{Mp}(2n, \mathbb{R}) := \widetilde{\mathrm{Sp}}(2n, \mathbb{R}) / (2\mathbb{Z}). \quad (19.19)$$

There is a double covering homomorphism

$$\mathrm{Mp}(2n, \mathbb{R}) \longrightarrow \mathrm{Sp}(2n, \mathbb{R}), \quad \ker = \{\pm 1\}. \quad (19.20)$$

The group $\mathrm{Mp}(2n, \mathbb{R})$ is the *metaplectic group*. The quotient relations are

$$\mathrm{Mp}(2n, \mathbb{R}) \cong \widetilde{\mathrm{Sp}}(2n, \mathbb{R}) / (2\mathbb{Z}), \quad \mathrm{Sp}(2n, \mathbb{R}) \cong \widetilde{\mathrm{Sp}}(2n, \mathbb{R}) / \mathbb{Z},$$

and hence

$$\mathrm{Sp}(2n, \mathbb{R}) \cong \mathrm{Mp}(2n, \mathbb{R}) / \mathbb{Z}_2.$$

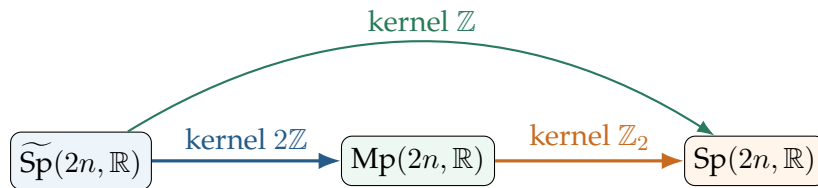


图 19.4: The universal and metaplectic covers of the real symplectic group.

This resembles $SO(2)$: both $SO(2)$ and $\mathrm{Sp}(2n, \mathbb{R})$ have fundamental group $\mathbb{Z}$, so both possess an infinite universal cover and nontrivial finite intermediate covers. The essential difference is that $SO(2)$ is abelian and compact, whereas $\mathrm{Sp}(2n, \mathbb{R})$ is noncompact and semisimple.

19.8 The proper orthochronous Lorentz group

The full Lorentz group $SO(1, 3)$ is not connected. The component containing the identity is the proper orthochronous Lorentz group

$$SO^+(1, 3).$$

Its Lie algebra is

$$\mathfrak{so}(1, 3) = \left\{ X \in \mathfrak{gl}(4, \mathbb{R}) : X^T \eta + \eta X = 0 \right\}, \quad \eta = \text{diag}(1, -1, -1, -1). \quad (19.21)$$

As real Lie algebras,

$$\mathfrak{so}(1, 3) \cong \mathfrak{sl}(2, \mathbb{C}).$$

Furthermore,

$$\pi_1(SO^+(1, 3)) \cong \mathbb{Z}_2,$$

and its universal covering group is

$$\text{Spin}^+(1, 3) \cong SL(2, \mathbb{C}).$$

The covering map can be seen explicitly. Associate

$$x = (x^0, x^1, x^2, x^3) \in \mathbb{R}^{1,3}$$

with the Hermitian matrix

$$X = x^0 I_2 + \sum_{j=1}^3 x^j \sigma_j.$$

Its determinant is the Minkowski quadratic form:

$$\det X = (x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2.$$

For $A \in SL(2, \mathbb{C})$, define

$$X \mapsto AXA^\dagger.$$

This preserves Hermiticity and determinant, so it determines a Lorentz transformation $\Lambda(A) \in SO^+(1, 3)$. The assignment

$$\Lambda : SL(2, \mathbb{C}) \longrightarrow SO^+(1, 3)$$

is a surjective Lie group homomorphism with

$$\ker \Lambda = \{\pm I_2\}.$$

Therefore

$$\boxed{SO^+(1, 3) \cong SL(2, \mathbb{C})/\{\pm I_2\}.} \quad (19.22)$$

This is the Lorentzian analogue of

$$SO(3) \cong SU(2)/\{\pm I_2\}.$$

What to retain from this chapter

1. A covering homomorphism $\Phi : \tilde{G} \rightarrow G$ is surjective, induces an isomorphism of Lie algebras, and has discrete kernel. It realizes

$$G \cong \tilde{G} / \ker \Phi.$$

2. If $\tilde{G}$ is connected, then

$$\ker \Phi \subseteq Z(\tilde{G}).$$

The proof conjugates a kernel element along a path and uses discreteness.

3. The universal cover of a connected Lie group is connected and simply connected, and

$$\ker p \cong \pi_1(G).$$

Every connected cover is obtained by quotienting the universal cover by a subgroup of this kernel.

4. For the circle,

$$\mathbb{R} \rightarrow U(1), \quad t \mapsto e^{it}, \quad \ker = 2\pi\mathbb{Z},$$

while $z \mapsto z^n$ gives non-universal n -sheeted covers.

5. A Pauli spinor transforms by

$$U(R) = \exp\left(-\frac{i\theta}{2} \mathbf{n} \cdot \boldsymbol{\sigma}\right),$$

and

$$U(2\pi, \mathbf{n}) = -I_2, \quad U(4\pi, \mathbf{n}) = I_2.$$

6. For $n \geq 3$,

$$\text{Spin}(n) = \widetilde{SO(n)}_{\text{univ}}, \quad SO(n) \cong \text{Spin}(n)/\{\pm 1\}.$$

In particular, $\text{Spin}(3) \cong SU(2)$.

7. The real symplectic group has

$$\pi_1(\text{Sp}(2n, \mathbb{R})) \cong \mathbb{Z}.$$

Its universal cover has infinitely many sheets, while $\text{Mp}(2n, \mathbb{R})$ is its distinguished double cover.

8. The identity component of the Lorentz group satisfies

$$SO^+(1, 3) \cong SL(2, \mathbb{C})/\{\pm I_2\}.$$

Linear Representations of Groups

Road map

Representation theory replaces an abstract algebraic object by linear maps on a vector space. Once this is done, one can use matrices, eigenvalues, inner products, tensor products, and invariant subspaces. This is especially important in quantum mechanics, where the state space is already a Hilbert space and symmetries act linearly or unitarily.

In this chapter we shall:

1. define linear and matrix representations;
2. prove carefully how the matrix multiplication law follows from the abstract homomorphism law;
3. introduce faithful, trivial, and unitary representations;
4. work out a two-dimensional representation of the quaternion group;
5. construct scalar, vector, and tensor representations of rotation groups;
6. derive Neumann's principle for material tensors and reproduce the full crystal-system table of independent tensor components;
7. construct infinite-dimensional representations on solution spaces of the Helmholtz and Schrödinger equations.

20.1 From an abstract group to linear maps

Let V be a vector space over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Its general linear group is

$$GL(V) = \{A : V \rightarrow V : A \text{ is linear and bijective}\}.$$

If $V = \mathbb{K}^n$, then

$$GL(V) = GL(n, \mathbb{K}) = \{A \in \text{Mat}(n, \mathbb{K}) : \det A \neq 0\}.$$

Definition 20.1: Linear representation

Let G be a group and V a vector space over $\mathbb{K}$. A *linear representation* of G on V is a group homomorphism

$$D : G \longrightarrow GL(V), \quad a \longmapsto D(a).$$

Thus

$$D(a_1 a_2) = D(a_1) D(a_2) \quad (a_1, a_2 \in G). \quad (20.1)$$

The vector space V is the *representation space*. Its dimension is the dimension of the representation:

$$\dim D := \dim_{\mathbb{K}} V.$$

The homomorphism law immediately implies

$$D(e) = I_V, \quad D(a^{-1}) = D(a)^{-1}.$$

Indeed,

$$D(e) = D(e)^2,$$

and $D(e)$ is invertible, so $D(e) = I_V$. Then

$$D(a) D(a^{-1}) = D(e) = I_V.$$

20.1.1 Linear versus matrix representations

Choose an ordered basis

$$\mathcal{B} = (e_1, \dots, e_n)$$

of an n -dimensional representation space V . For $a \in G$, write

$$D(a)e_j = \sum_{i=1}^n D_{\mathcal{B}}(a)_{ij} e_i. \quad (20.2)$$

The coefficients form an invertible matrix

$$D_{\mathcal{B}}(a) = (D_{\mathcal{B}}(a)_{ij}) \in GL(n, \mathbb{K}).$$

The map

$$D_{\mathcal{B}} : G \longrightarrow GL(n, \mathbb{K})$$

is called the *matrix representation* in the basis $\mathcal{B}$.

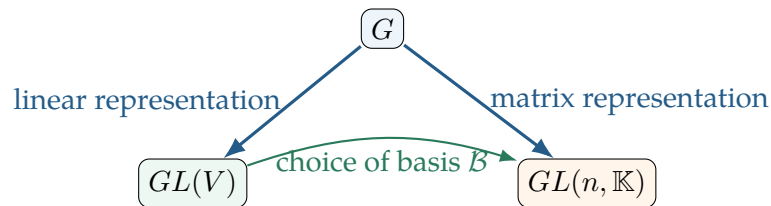


图 20.1: A matrix representation is the coordinate form of a linear representation after choosing a basis.

20.1.2 The multiplication law in indices

Starting from (20.1),

$$D(a)D(b) = D(ab).$$

Apply both sides to a basis vector e_j :

$$\begin{aligned} D(ab)e_j &= \sum_i D_{\mathcal{B}}(ab)_{ij}e_i, \\ D(a)D(b)e_j &= D(a) \left(\sum_k D_{\mathcal{B}}(b)_{kj}e_k \right) \\ &= \sum_k D_{\mathcal{B}}(b)_{kj}D(a)e_k \\ &= \sum_k D_{\mathcal{B}}(b)_{kj} \sum_i D_{\mathcal{B}}(a)_{ik}e_i \\ &= \sum_i \left(\sum_k D_{\mathcal{B}}(a)_{ik}D_{\mathcal{B}}(b)_{kj} \right) e_i. \end{aligned}$$

Uniqueness of the coordinate expansion gives

$$D_{\mathcal{B}}(ab)_{ij} = \sum_k D_{\mathcal{B}}(a)_{ik}D_{\mathcal{B}}(b)_{kj}.$$

This is exactly the matrix multiplication rule

$$D_{\mathcal{B}}(ab) = D_{\mathcal{B}}(a)D_{\mathcal{B}}(b).$$

20.2 A first finite-group example: the quaternion group

The quaternion group is

$$Q_8 = \{\pm 1, \pm \mathbf{i}, \pm \mathbf{j}, \pm \mathbf{k}\},$$

with defining relations

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1. \quad (20.3)$$

Equivalently,

$$\mathbf{ij} = \mathbf{k}, \quad \mathbf{ji} = -\mathbf{k},$$

and their cyclic analogues.

Define $D : Q_8 \rightarrow GL(2, \mathbb{C})$ on the generators by

$$\begin{aligned} D(1) &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, & D(-1) &= \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \\ D(\mathbf{i}) &= \begin{pmatrix} \mathbf{i} & 0 \\ 0 & -\mathbf{i} \end{pmatrix} = \mathbf{i}\sigma_3, & D(\mathbf{j}) &= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \mathbf{i}\sigma_2, \\ D(\mathbf{k}) &= \begin{pmatrix} 0 & \mathbf{i} \\ \mathbf{i} & 0 \end{pmatrix} = \mathbf{i}\sigma_1. \end{aligned} \quad (20.4)$$

The negative elements are represented by

$$D(-q) = -D(q).$$

Let us check the defining relations rather than merely assert them:

$$\begin{aligned} D(\mathbf{i})^2 &= D(\mathbf{j})^2 = D(\mathbf{k})^2 = -I_2 = D(-1), \\ D(\mathbf{i})D(\mathbf{j}) &= \begin{pmatrix} \mathbf{i} & 0 \\ 0 & -\mathbf{i} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & \mathbf{i} \\ \mathbf{i} & 0 \end{pmatrix} = D(\mathbf{k}), \\ D(\mathbf{j})D(\mathbf{i}) &= -D(\mathbf{k}). \end{aligned}$$

Thus every relation in Q_8 is respected, so D is a two-dimensional complex representation.

Defining representations

Matrix groups such as $GL(n, \mathbb{K})$, $O(n)$, $SO(n)$, and $SU(n)$ come with a natural *defining representation* on $\mathbb{K}^n$:

$$D(g)v = gv.$$

An abstract group, by contrast, may possess many inequivalent representations of different dimensions.

20.3 Important adjectives for representations

Definition 20.2: Faithful representation

A representation $D : G \rightarrow GL(V)$ is *faithful* if it is one-to-one, or equivalently if

$$\ker D = \{e\}.$$

A representation may preserve fewer distinctions than the original group. Every element in $\ker D$ acts as the identity on V , so those group elements become invisible in the representation.

Theorem 20.1: A representation is faithful on the quotient by its kernel

For any representation $D : G \rightarrow GL(V)$, the induced map

$$\overline{D} : G / \ker D \longrightarrow GL(V), \quad a \ker D \longmapsto D(a),$$

is a faithful representation of $G / \ker D$.

证明. The kernel of a group homomorphism is normal, so the quotient is defined. If $a \ker D = b \ker D$, then $b^{-1}a \in \ker D$, hence

$$D(a) = D(b)D(b^{-1}a) = D(b),$$

so $\overline{D}$ is well-defined. It is a homomorphism because

$$\overline{D}((a \ker D)(b \ker D)) = D(ab) = D(a)D(b).$$

Finally,

$$\overline{D}(a \ker D) = I_V \implies a \in \ker D \implies a \ker D = \ker D.$$

Thus $\ker \overline{D}$ is trivial. ■

Definition 20.3: Trivial representation

The *trivial* or *unit representation* is

$$D : G \longrightarrow GL(V), \quad D(a) = I_V \quad (a \in G).$$

It deliberately discards all information about the group action.

Definition 20.4: Unitary representation

Let V be a complex inner-product space. A representation

$$D : G \longrightarrow GL(V)$$

is *unitary* if

$$\langle D(a)v_1, D(a)v_2 \rangle = \langle v_1, v_2 \rangle \quad (a \in G, v_1, v_2 \in V). \quad (20.5)$$

Equivalently,

$$D(a)^\dagger = D(a)^{-1} = D(a^{-1}).$$

20.4 Tensor representations of the rotation group

Let $G = SO(3)$, or let G be a subgroup of $SO(3)$ describing the proper rotational symmetries of a material.

20.4.1 Scalars and vectors

A scalar s transforms under the one-dimensional trivial representation:

$$s' = D(R)s = s, \quad D(R) = 1.$$

A vector $\mathbf{v} \in \mathbb{R}^3$ transforms under the defining representation:

$$\mathbf{v}' = D(R)\mathbf{v} = R\mathbf{v},$$

or, in components,

$$v'_i = \sum_{j=1}^3 R_{ij}v_j. \quad (20.6)$$

20.4.2 Rank-two tensors

Let $C = (C_{ij})$ be a rank-two Cartesian tensor. Under a rotation,

$$C'_{ij} = \sum_{k,\ell=1}^3 R_{ik} R_{j\ell} C_{k\ell}. \quad (20.7)$$

In matrix notation,

$$C' = R C R^T. \quad (20.8)$$

The nine-dimensional space of rank-two tensors is

$$\mathbb{R}^3 \otimes \mathbb{R}^3.$$

If the vector representation is $D^{(1)}(R) = R$, then the tensor-product representation is

$$D^{(1)} \otimes D^{(1)} : R \longmapsto R \otimes R.$$

With the double index (i, j) , its matrix entries are

$$D^{(2)}(R)_{(i,j),(k,\ell)} = R_{ik} R_{j\ell}.$$

Equation (20.7) is simply

$$C' = D^{(2)}(R)C$$

after the two indices are flattened into one nine-valued index.

20.5 Linear response and Neumann's principle

Consider Ohm's law in an anisotropic material:

$$j_i = \sum_j \sigma_{ij} E_j, \quad \text{or} \quad \mathbf{j} = \sigma \mathbf{E}. \quad (20.9)$$

Here $\mathbf{j}$ is the current density, $\mathbf{E}$ is the electric field, and σ is the rank-two conductivity tensor.

Rotate the coordinate frame by R . The vector components become

$$\mathbf{j}' = R\mathbf{j}, \quad \mathbf{E}' = R\mathbf{E}.$$

Since $\mathbf{E} = R^T \mathbf{E}'$,

$$\mathbf{j}' = R\sigma\mathbf{E} = R\sigma R^T \mathbf{E}'.$$

Thus the conductivity components in the rotated frame are

$$\boxed{\sigma' = R\sigma R^T, \quad \sigma'_{ij} = \sum_{k,\ell} R_{ik} R_{j\ell} \sigma_{k\ell}.} \quad (20.10)$$

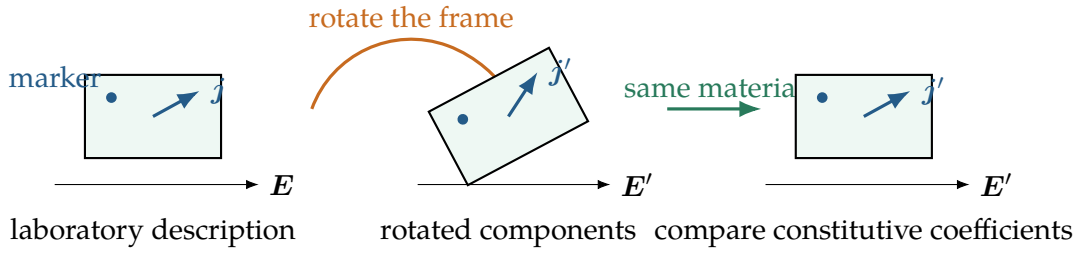


图 20.2: A constitutive law is independent of the coordinate description; the tensor components transform with the rotated basis.

Theorem 20.2: Neumann's principle

Every tensor describing a physical property of a crystal is invariant under every symmetry transformation of that crystal.

For the conductivity tensor and a material symmetry group $G \leq SO(3)$, Neumann's principle says

$$\sigma = R\sigma R^T \quad (R \in G). \quad (20.11)$$

Since $R^{-1} = R^T$, this is equivalent to

$$[R, \sigma] = 0 \quad (R \in G). \quad (20.12)$$

Thus the allowed material tensors form the invariant subspace

$$(\mathbb{R}^3 \otimes \mathbb{R}^3)^G = \{C : D^{(2)}(R)C = C \text{ for every } R \in G\}.$$

For the proper rotational symmetry group $\mathcal{O}$ of a cube, the defining three-dimensional representation is irreducible and its commutant consists only of scalar matrices. Schur's lemma therefore gives

$$\sigma = \sigma_0 I_3. \quad (20.13)$$

The conductivity of a cubic crystal is isotropic even though the crystal itself has only a finite rotation group.

20.5.1 Hooke's law and fourth-rank tensors

In linear elasticity,

$$\tau_{ij} = \sum_{k,\ell} C_{ijkl} \varepsilon_{kl}, \quad (20.14)$$

where τ is the stress tensor, ε is the strain tensor, and C is the rank-four stiffness tensor. Before imposing symmetries, a general rank-four tensor in three dimensions has $3^4 = 81$ components. The physical minor and major symmetries

$$C_{ijkl} = C_{jikl} = C_{ijlk} = C_{klij}$$

reduce this to 21. Crystal symmetries reduce it further. A cubic crystal has only three independent elastic constants, conventionally

$$C_{11}, \quad C_{12}, \quad C_{44}.$$

An isotropic solid has only two, the Lamé parameters.

Bravais system	Point-group symbol		Number of independent components			
	International	Schoenflies	first	second	third	fourth
triclinic	1	C_1	3	6	18	21
	$\bar{1}$	S_2	0	6	0	21
monoclinic	2	C_2	1	4	8	13
	m	C_{1h}	2	4	10	13
	$2/m$	C_{2h}	0	4	0	13
orthorhombic	222	D_2	0	3	3	9
	$mm2$	C_{2v}	1	3	5	9
	mmm	D_{2h}	0	3	0	9
tetragonal	4	C_4	1	2	4	7
	$\bar{4}$	S_4	0	2	4	7
	$4/m$	C_{4h}	0	2	0	7
	422	D_4	0	2	1	6
	$4mm$	C_{4v}	1	2	3	6
	$\bar{4}2m$	D_{2d}	0	2	2	6
	$4/mmm$	D_{4h}	0	2	0	6
rhombohedral	3	C_3	1	2	6	7
	$\bar{3}$	S_6	0	2	0	7
	32	D_3	0	2	2	6
	$3m$	C_{3v}	1	2	4	6
	$\bar{3}m$	D_{3d}	0	2	0	6
hexagonal	6	C_6	1	2	4	5
	$\bar{6}$	C_{3h}	0	2	2	5
	$6/m$	C_{6h}	0	2	0	5
	622	D_6	0	2	1	5
	$6mm$	C_{6v}	1	2	3	5
	$\bar{6}m2$	D_{3h}	0	2	1	5
	$6/mmm$	D_{6h}	0	2	0	5
cubic	23	T	0	1	1	3
	$m\bar{3}$	T_h	0	1	0	3
	432	O	0	1	0	3
	$\bar{4}3m$	T_d	0	1	1	3
	$m\bar{3}m$	O_h	0	1	0	3
isotropic	—	—	0	1	0	2

表 20.1: Number of independent elements of physical tensor properties of orders one through four for the 32 crystallographic point groups. The second-, third-, and fourth-order columns incorporate the conventional internal index symmetries of the corresponding material-property tensors. The data reproduce the table cited in the handwritten notes from J. W. Morris, *Notes on the Thermodynamics of Solids* (2008).

20.6 Infinite-dimensional representations on function spaces

Representations need not be finite-dimensional. Symmetry groups often act naturally on spaces of functions, differential-equation solutions, or quantum wavefunctions.

20.6.1 The Helmholtz equation

Fix $k > 0$, and let

$$\mathcal{S}_k = \{f : \mathbb{R}^3 \rightarrow \mathbb{R} : \Delta f + k^2 f = 0\}. \quad (20.15)$$

Because the Helmholtz equation is linear, $\mathcal{S}_k$ is a vector space, generally infinite-dimensional.

Let $E(3) = \mathbb{R}^3 \rtimes O(3)$ be the Euclidean isometry group. Write an isometry as $g = (\mathbf{a}, R)$, acting by

$$g\mathbf{x} = \mathbf{a} + R\mathbf{x}, \quad g^{-1}\mathbf{x} = R^{-1}(\mathbf{x} - \mathbf{a}).$$

Define the pullback action

$$[D(g)f](\mathbf{x}) := f(g^{-1}\mathbf{x}) = f(R^{-1}(\mathbf{x} - \mathbf{a})). \quad (20.16)$$

The map $D(g) : \mathcal{S}_k \rightarrow \mathcal{S}_k$ is linear:

$$D(g)(z_1 f_1 + z_2 f_2) = z_1 D(g)f_1 + z_2 D(g)f_2.$$

It is a homomorphism because

$$\begin{aligned} [D(g_1)D(g_2)f](\mathbf{x}) &= [D(g_2)f](g_1^{-1}\mathbf{x}) \\ &= f(g_2^{-1}g_1^{-1}\mathbf{x}) \\ &= f((g_1g_2)^{-1}\mathbf{x}) \\ &= [D(g_1g_2)f](\mathbf{x}). \end{aligned}$$

Finally, the Laplacian is invariant under translations and orthogonal transformations:

$$\Delta(f \circ g^{-1}) = (\Delta f) \circ g^{-1}.$$

Therefore

$$(\Delta + k^2)D(g)f = D(g)(\Delta + k^2)f = 0.$$

Hence

$$D : E(3) \longrightarrow GL(\mathcal{S}_k)$$

is an infinite-dimensional representation of the Euclidean group.

20.6.2 Rotationally invariant Schrödinger operators

Let

$$H = -\frac{\hbar^2}{2m}\Delta + V(\mathbf{x}) \quad (20.17)$$

act on $L^2(\mathbb{R}^3)$, and fix an energy E . Its eigenspace is

$$\mathcal{E}_E = \{\psi \in L^2(\mathbb{R}^3) : H\psi = E\psi\}.$$

Assume a subgroup $G \leq SO(3)$ preserves the potential:

$$V(R\mathbf{x}) = V(\mathbf{x}) \quad (R \in G). \quad (20.18)$$

Define

$$[D(R)\psi](\mathbf{x}) = \psi(R^{-1}\mathbf{x}). \quad (20.19)$$

The Laplacian is rotation-invariant and

$$V(\mathbf{x})\psi(R^{-1}\mathbf{x}) = V(R^{-1}\mathbf{x})\psi(R^{-1}\mathbf{x}).$$

Thus

$$\begin{aligned} H[D(R)\psi](\mathbf{x}) &= [H\psi](R^{-1}\mathbf{x}) \\ &= E\psi(R^{-1}\mathbf{x}) = E[D(R)\psi](\mathbf{x}). \end{aligned}$$

Therefore each energy eigenspace is G -invariant, and

$$D : G \longrightarrow GL(\mathcal{E}_E)$$

is a representation.

Its dimension is the degeneracy of the energy E :

$$\dim D = \dim \mathcal{E}_E.$$

The representation is unitary. Indeed, with the usual L^2 inner product and the change of variables $\mathbf{y} = R^{-1}\mathbf{x}$,

$$\begin{aligned} \langle D(R)\psi, D(R)\phi \rangle &= \int_{\mathbb{R}^3} \psi(R^{-1}\mathbf{x})^* \phi(R^{-1}\mathbf{x}) \, d^3x \\ &= \int_{\mathbb{R}^3} \psi(\mathbf{y})^* \phi(\mathbf{y}) \, d^3y \\ &= \langle \psi, \phi \rangle, \end{aligned}$$

because $|\det R| = 1$.

Symmetry and degeneracy

Since H commutes with every $D(R)$,

$$HD(R) = D(R)H,$$

symmetry maps an energy eigenstate to another state of the same energy. Nontrivial representation multiplets therefore produce systematic degeneracies. The precise relation between irreducibility and degeneracy will be developed in the next chapters.

What to retain from this chapter

1. A representation of G on V is a homomorphism

$$D : G \rightarrow GL(V), \quad D(ab) = D(a)D(b).$$

Its dimension is $\dim V$.

2. Choosing a basis converts a linear representation into a matrix representation. A different basis changes all matrices by a simultaneous similarity transformation.
3. A representation is faithful exactly when $\ker D = \{e\}$. In general it induces a faithful representation of $G/\ker D$.

4. A unitary representation preserves inner products and satisfies

$$D(a)^\dagger = D(a)^{-1} = D(a^{-1}).$$

5. Under rotations, scalars transform trivially, vectors by R , and rank-two tensors by

$$C' = RCR^\top, \quad C'_{ij} = R_{ik}R_{j\ell}C_{k\ell}.$$

6. Neumann's principle requires every material-property tensor to be invariant under the material's symmetry group. For conductivity,

$$R\sigma R^\top = \sigma.$$

Symmetry can drastically reduce the number of independent tensor components.

7. Isometries act on functions by pullback,

$$[D(g)f](\mathbf{x}) = f(g^{-1}\mathbf{x}),$$

producing infinite-dimensional representations on invariant solution spaces.

8. If a Schrödinger Hamiltonian is invariant under G , each energy eigenspace is a unitary representation space of G , and its dimension is the degeneracy of that energy.

Equivalent Representations, Sums, and Products

Road map

The same abstract representation may be written on different vector spaces or in different bases. We first make this statement precise through *equivalence of representations* and simultaneous similarity of all representing matrices. We then prove, by averaging an inner product over a finite group, that every finite-group representation on a finite-dimensional complex inner-product space is equivalent to a unitary representation. The last two parts construct new representations from old ones: the direct sum acts independently on two summands, whereas the tensor product acts simultaneously on two factors.

21.1 Equivalent representations

Let G be a group, let V, V' be vector spaces over the same field $\mathbb{K}$, and let

$$D : G \longrightarrow GL(V), \quad D' : G \longrightarrow GL(V')$$

be representations.

Definition 21.1: Equivalent representations

The representations D and D' are *equivalent*, written $D \sim D'$, if there exists a vector-space isomorphism $S : V \rightarrow V'$ such that, for every $a \in G$,

$$D(a) = S^{-1}D'(a)S. \tag{21.1}$$

Equivalently,

$$D'(a)S = SD(a) \quad (a \in G).$$

The isomorphism S is called an *intertwining isomorphism*, or simply an *intertwiner*.

The last equation says exactly that the following diagram commutes:

$$\begin{array}{ccc}
 V & \xrightarrow{D(a)} & V \\
 S \downarrow & & \downarrow S \\
 V' & \xrightarrow{D'(a)} & V'
 \end{array}$$

图 21.1: Equivalence means that applying $D(a)$ and then changing vector space gives the same answer as changing vector space first and applying $D'(a)$.

Indeed, starting with $v \in V$, the upper-right route produces

$$S(D(a)v),$$

whereas the lower-left route produces

$$D'(a)(Sv).$$

Equality for every v is precisely $SD(a) = D'(a)S$. Thus equivalent representations describe the same group action after the vectors have merely been relabelled by S .

21.1.1 Why this is an equivalence relation

Equivalence of representations satisfies the three axioms of an equivalence relation.

1. **Reflexivity.** Taking $S = \text{id}_V$ gives $D(a) = S^{-1}D(a)S$, so $D \sim D$.
2. **Symmetry.** If $D(a) = S^{-1}D'(a)S$, then multiplication by S on the left and S^{-1} on the right gives

$$D'(a) = (S^{-1})^{-1}D(a)S^{-1}.$$

Hence $D' \sim D$, with intertwiner S^{-1} .

3. **Transitivity.** Suppose

$$D(a) = S^{-1}D'(a)S, \quad D'(a) = T^{-1}D''(a)T.$$

Substitution yields

$$D(a) = S^{-1}T^{-1}D''(a)TS = (TS)^{-1}D''(a)(TS).$$

Thus $D \sim D''$, with intertwiner TS .

It is therefore natural to classify representations only up to equivalence.

If $V = V'$, Equation (21.1) says that the operators $D(a)$ and $D'(a)$ are similar. One must not overlook an important strengthening: there must be *one fixed* invertible map S that works simultaneously for every $a \in G$. Allowing a different similarity transformation for each group element would not define equivalence of representations.

21.2 Matrix representations and changes of basis

Assume that V is n -dimensional. Choose an ordered basis

$$\mathcal{B} = (e_1, \dots, e_n)$$

and let

$$T_{\mathcal{B}} : V \longrightarrow \mathbb{K}^n, \quad v = \sum_{i=1}^n v_i e_i \longmapsto [v]_{\mathcal{B}} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix},$$

be the coordinate isomorphism. The matrix representation determined by $\mathcal{B}$ is

$$D_{\mathcal{B}}(a) = T_{\mathcal{B}} D(a) T_{\mathcal{B}}^{-1}. \quad (21.2)$$

Choose a second ordered basis

$$\mathcal{B}' = (e'_1, \dots, e'_n)$$

with coordinate isomorphism $T_{\mathcal{B}'}$, and define the coordinate change

$$S := T_{\mathcal{B}'} T_{\mathcal{B}}^{-1} : \mathbb{K}^n \longrightarrow \mathbb{K}^n. \quad (21.3)$$

Thus

$$[v]_{\mathcal{B}'} = S[v]_{\mathcal{B}}.$$

Using Equation (21.2) twice,

$$\begin{aligned} D_{\mathcal{B}'}(a) &= T_{\mathcal{B}'} D(a) T_{\mathcal{B}'}^{-1} \\ &= (T_{\mathcal{B}'} T_{\mathcal{B}}^{-1}) (T_{\mathcal{B}} D(a) T_{\mathcal{B}}^{-1}) (T_{\mathcal{B}} T_{\mathcal{B}'}^{-1}) \\ &= S D_{\mathcal{B}}(a) S^{-1}. \end{aligned}$$

Equivalently,

$$D_{\mathcal{B}}(a) = S^{-1} D_{\mathcal{B}'}(a) S. \quad (21.4)$$

This is exactly the equivalence relation from the previous section.

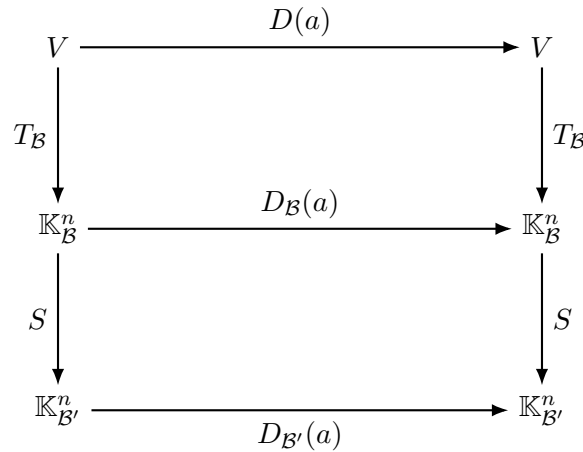


图 21.2: The same abstract operator $D(a)$, expressed in two coordinate systems. Every rectangle commutes.

Convention check

Some books define the change-of-basis matrix in the opposite direction, $P = T_{\mathcal{B}}T_{\mathcal{B}'}^{-1} = S^{-1}$. They consequently write $D_{\mathcal{B}'}(a) = P^{-1}D_{\mathcal{B}}(a)P$. The formulas look reversed, but the mathematical statement is identical once the direction of the coordinate map has been fixed.

21.3 Unitarising a finite-group representation

Let V be a finite-dimensional complex inner-product space with original inner product $\langle v, w \rangle$. Recall that the inner product is conjugate linear in its first argument and linear in its second argument.

Theorem 21.1: Every finite-group representation is unitarisable

Let G be a finite group and let

$$D : G \longrightarrow GL(V)$$

be a representation on a finite-dimensional complex inner-product space $(V, \langle \cdot, \cdot \rangle)$. Then D is equivalent to a unitary representation.

证明. The given operators $D(a)$ need not preserve the original inner product. We repair this by averaging all of its translates over the group. Define

$$\{v, w\} := \frac{1}{|G|} \sum_{a \in G} \langle D(a)v, D(a)w \rangle. \quad (21.5)$$

The factor $1/|G|$ is a convenient normalization: it does not affect positivity or invariance, but makes the operation a literal group average.

We verify carefully that $\{\cdot, \cdot\}$ is an inner product.

1. **Sesquilinearity.** Each summand is conjugate linear in v and linear in w , because both $D(a)$ and the original inner product have these properties. A finite sum and a positive scalar multiple preserve them.
2. **Conjugate symmetry.**

$$\{w, v\} = \frac{1}{|G|} \sum_{a \in G} \langle D(a)w, D(a)v \rangle = \overline{\{v, w\}}.$$

3. **Positive definiteness.** For every v ,

$$\{v, v\} = \frac{1}{|G|} \sum_{a \in G} \|D(a)v\|^2 \geq 0.$$

If $v \neq 0$, then the term corresponding to the identity e is $\|D(e)v\|^2 = \|v\|^2 > 0$. Therefore $\{v, v\} > 0$.

The crucial property is invariance under D . Fix $b \in G$. Then

$$\begin{aligned}\{D(b)v, D(b)w\} &= \frac{1}{|G|} \sum_{a \in G} \langle D(a)D(b)v, D(a)D(b)w \rangle \\ &= \frac{1}{|G|} \sum_{a \in G} \langle D(ab)v, D(ab)w \rangle.\end{aligned}$$

Right multiplication by b ,

$$G \longrightarrow G, \quad a \longmapsto ab,$$

is a bijection, with inverse $c \mapsto cb^{-1}$. Consequently the elements ab , as a runs through G , run through every element of G exactly once. Relabelling $c = ab$ therefore gives

$$\{D(b)v, D(b)w\} = \frac{1}{|G|} \sum_{c \in G} \langle D(c)v, D(c)w \rangle = \{v, w\}.$$

Thus D is unitary with respect to the averaged inner product.

It remains to express this as an equivalent representation that is unitary with respect to the *original* inner product. Choose an orthonormal basis $(e_1, \dots, e_n)$ for $\langle \cdot, \cdot \rangle$, and choose an orthonormal basis $(q_1, \dots, q_n)$ for $\{ \cdot, \cdot \}$. Define the linear isomorphism $S : V \rightarrow V$ by

$$Se_i = q_i \quad (1 \leq i \leq n).$$

For

$$v = \sum_i v_i e_i, \quad w = \sum_j w_j e_j,$$

we compute

$$\begin{aligned}\{Sv, Sw\} &= \left\{ \sum_i v_i q_i, \sum_j w_j q_j \right\} \\ &= \sum_{i,j} v_i^* w_j \{q_i, q_j\} \\ &= \sum_{i,j} v_i^* w_j \delta_{ij} = \sum_i v_i^* w_i = \langle v, w \rangle.\end{aligned}$$

Equivalently,

$$\{x, y\} = \langle S^{-1}x, S^{-1}y \rangle \quad (x, y \in V). \quad (21.6)$$

Now define

$$D'(a) := S^{-1}D(a)S.$$

By construction $D' \sim D$. For $v, w \in V$, use Equation (21.6) and the invariance just proved:

$$\begin{aligned}\langle D'(a)v, D'(a)w \rangle &= \langle S^{-1}D(a)Sv, S^{-1}D(a)Sw \rangle \\ &= \{D(a)Sv, D(a)Sw\} \\ &= \{Sv, Sw\} \\ &= \langle v, w \rangle.\end{aligned}$$

Thus D' is unitary in the original inner product, as required. ■

Infinite dimension and compact groups

The argument also applies when $\dim V = \infty$, provided the analytic conditions making the averaged inner product and the resulting isomorphism meaningful are satisfied. Its essential ingredient is an invariant, normalized group average.

For a finite group the average is

$$M(f) := \frac{1}{|G|} \sum_{a \in G} f(a)$$

and, for a compact Lie group such as $SO(n)$, it is replaced by integration against normalized Haar measure μ :

$$M(f) := \int_G f(a) \, d\mu(a), \quad \mu(G) = 1.$$

The Haar-invariance identity

$$\int_G f(ab) \, d\mu(a) = \int_G f(a) \, d\mu(a),$$

replaces the relabelling step in the finite-group proof. Under the usual continuity and finite-dimensionality assumptions, every representation of a compact group can therefore be made unitary.

21.4 Direct sums of vector spaces and linear maps

21.4.1 Sums of subspaces

A subset $U \subseteq V$ is a *linear subspace* if

$$u_1, u_2 \in U, \quad \alpha_1, \alpha_2 \in \mathbb{K} \quad \implies \quad \alpha_1 u_1 + \alpha_2 u_2 \in U.$$

For two subspaces $U_1, U_2 \leq V$, their sum is

$$U_1 + U_2 := \{u_1 + u_2 : u_1 \in U_1, u_2 \in U_2\}. \quad (21.7)$$

This is again a subspace. In finite dimension,

$$\dim(U_1 + U_2) = \dim U_1 + \dim U_2 - \dim(U_1 \cap U_2). \quad (21.8)$$

The subtraction is necessary because the directions lying in the intersection were counted once in $\dim U_1$ and once again in $\dim U_2$.

Definition 21.2: Internal direct sum

The sum $U_1 + U_2$ is a *direct sum*, denoted $U_1 \oplus U_2$, if $U_1 \cap U_2 = \{0\}$. Equivalently, in finite dimension,

$$\dim(U_1 + U_2) = \dim U_1 + \dim U_2.$$

Proposition 21.1: Direct sum and unique decomposition

The sum $U_1 + U_2$ is direct if and only if every $v \in U_1 + U_2$ has a unique expression

$$v = u_1 + u_2, \quad u_1 \in U_1, \quad u_2 \in U_2.$$

证明. First suppose $U_1 \cap U_2 = \{0\}$, and suppose

$$v = u_1 + u_2 = u'_1 + u'_2.$$

Rearranging gives

$$u_1 - u'_1 = u'_2 - u_2.$$

The left-hand side lies in U_1 , while the right-hand side lies in U_2 . Their common value is therefore in $U_1 \cap U_2 = \{0\}$. Consequently $u_1 = u'_1$ and $u_2 = u'_2$, proving uniqueness.

Conversely, suppose every decomposition is unique. If $x \in U_1 \cap U_2$, then

$$x = x + 0 = 0 + x$$

are two decompositions with components in U_1 and U_2 . Uniqueness forces $x = 0$. Hence $U_1 \cap U_2 = \{0\}$. ■

21.4.2 Direct sums of maps

Let $f_1 : U_1 \rightarrow U_1$ and $f_2 : U_2 \rightarrow U_2$ be linear. On $U_1 \oplus U_2$, define

$$(f_1 \oplus f_2)(u_1 + u_2) := f_1(u_1) + f_2(u_2). \quad (21.9)$$

The unique-decomposition property makes this definition unambiguous. Linearity follows component by component. For example,

$$\begin{aligned} (f_1 \oplus f_2)(\alpha(u_1 + u_2) + \beta(u'_1 + u'_2)) &= f_1(\alpha u_1 + \beta u'_1) + f_2(\alpha u_2 + \beta u'_2) \\ &= \alpha(f_1 \oplus f_2)(u_1 + u_2) + \beta(f_1 \oplus f_2)(u'_1 + u'_2). \end{aligned}$$

Choose a basis

$$\mathcal{B}_1 = (u_1^{(1)}, \dots, u_{n_1}^{(1)}) \quad \text{of } U_1$$

and a basis

$$\mathcal{B}_2 = (u_1^{(2)}, \dots, u_{n_2}^{(2)}) \quad \text{of } U_2.$$

Their concatenation $\mathcal{B}_1 \parallel \mathcal{B}_2$ is a basis of $U_1 \oplus U_2$. If $A_1 = [f_1]_{\mathcal{B}_1}$ and $A_2 = [f_2]_{\mathcal{B}_2}$, then

$$[f_1 \oplus f_2]_{\mathcal{B}_1 \parallel \mathcal{B}_2} = \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}. \quad (21.10)$$

The zero off-diagonal blocks record that $f_1 \oplus f_2$ never mixes a vector from one summand into the other.

21.5 Direct sums of representations

Definition 21.3: Direct-sum representation

Let

$$D_1 : G \longrightarrow GL(V_1), \quad D_2 : G \longrightarrow GL(V_2)$$

be representations. Their *direct sum* is the map

$$D_1 \oplus D_2 : G \longrightarrow GL(V_1 \oplus V_2)$$

defined by

$$(D_1 \oplus D_2)(a)(v_1 + v_2) = D_1(a)v_1 + D_2(a)v_2. \quad (21.11)$$

In operator notation,

$$(D_1 \oplus D_2)(a) = D_1(a) \oplus D_2(a).$$

Let us verify that this is genuinely a representation. For $a, b \in G$ and $v = v_1 + v_2$,

$$\begin{aligned} (D_1 \oplus D_2)(a)(D_1 \oplus D_2)(b)v &= (D_1 \oplus D_2)(a)(D_1(b)v_1 + D_2(b)v_2) \\ &= D_1(a)D_1(b)v_1 + D_2(a)D_2(b)v_2 \\ &= D_1(ab)v_1 + D_2(ab)v_2 \\ &= (D_1 \oplus D_2)(ab)v. \end{aligned}$$

Moreover,

$$(D_1 \oplus D_2)(e) = I_{V_1} \oplus I_{V_2} = I_{V_1 \oplus V_2}.$$

In bases adapted to $V_1 \oplus V_2$, the representing matrices are

$$[D_1 \oplus D_2](a) = \begin{pmatrix} D_1(a) & 0 \\ 0 & D_2(a) \end{pmatrix}, \quad \dim(D_1 \oplus D_2) = \dim D_1 + \dim D_2. \quad (21.12)$$

$D_1(a)$ $\dim V_1$	0
0	$D_2(a)$ $\dim V_2$

图 21.3: The block-diagonal matrix of a direct-sum representation in a basis adapted to the two summands.

The block-diagonal appearance is basis-dependent. An arbitrary change of basis on $V_1 \oplus V_2$ can mix the two displayed blocks and make the matrix look dense. The invariant statement is that the representation space contains two complementary invariant subspaces. In a later chapter this observation will lead to reducible and irreducible representations.

21.6 Tensor products of vector spaces and maps

21.6.1 The tensor-product space

Let V_1, V_2 be vector spaces over $\mathbb{K}$. Their tensor product is the vector space

$$V_1 \otimes V_2 = \text{span}\{v_1 \otimes v_2 : v_1 \in V_1, v_2 \in V_2\},$$

subject to the bilinearity relations

$$(v_1 + v'_1) \otimes v_2 = v_1 \otimes v_2 + v'_1 \otimes v_2, \quad (21.13)$$

$$v_1 \otimes (v_2 + v'_2) = v_1 \otimes v_2 + v_1 \otimes v'_2, \quad (21.14)$$

$$\lambda(v_1 \otimes v_2) = (\lambda v_1) \otimes v_2 = v_1 \otimes (\lambda v_2), \quad \lambda \in \mathbb{K}. \quad (21.15)$$

In particular,

$$0 \otimes v_2 = (0 + 0) \otimes v_2 = 0 \otimes v_2 + 0 \otimes v_2,$$

so subtracting $0 \otimes v_2$ from both sides gives

$$0 \otimes v_2 = 0.$$

Similarly $v_1 \otimes 0 = 0$.

A vector of the form $v_1 \otimes v_2$ is called a *simple* or *pure* tensor. A general tensor is a finite sum of pure tensors and need not itself be pure.

Suppose

$$(v_1^{(1)}, \dots, v_{n_1}^{(1)}) \quad \text{and} \quad (v_1^{(2)}, \dots, v_{n_2}^{(2)})$$

are bases of V_1 and V_2 . Then

$$\{v_i^{(1)} \otimes v_j^{(2)} : 1 \leq i \leq n_1, 1 \leq j \leq n_2\} \quad (21.16)$$

is a basis of $V_1 \otimes V_2$. There are $n_1 n_2$ basis elements, and therefore

$$\dim(V_1 \otimes V_2) = (\dim V_1)(\dim V_2). \quad (21.17)$$

To see explicitly why the spanning property holds, write

$$v_1 = \sum_i x_i v_i^{(1)}, \quad v_2 = \sum_j y_j v_j^{(2)}.$$

Bilinearity gives

$$v_1 \otimes v_2 = \sum_{i,j} x_i y_j (v_i^{(1)} \otimes v_j^{(2)}).$$

Since arbitrary tensors are sums of pure tensors, the displayed tensors span the whole tensor product.

21.6.2 Tensor products of linear maps

For linear maps $f_1 : V_1 \rightarrow V_1$ and $f_2 : V_2 \rightarrow V_2$, define

$$f_1 \otimes f_2 : V_1 \otimes V_2 \longrightarrow V_1 \otimes V_2$$

first on pure tensors by

$$(f_1 \otimes f_2)(v_1 \otimes v_2) := f_1(v_1) \otimes f_2(v_2), \quad (21.18)$$

and then extend linearly to all tensors. The bilinearity relations ensure that the definition is well-defined.

Tensor products of maps obey the useful composition rule

$$(f_1 \otimes f_2)(g_1 \otimes g_2) = (f_1 g_1) \otimes (f_2 g_2). \quad (21.19)$$

It is enough to check this on a pure tensor:

$$\begin{aligned} (f_1 \otimes f_2)(g_1 \otimes g_2)(v_1 \otimes v_2) &= (f_1 \otimes f_2)(g_1(v_1) \otimes g_2(v_2)) \\ &= f_1(g_1(v_1)) \otimes f_2(g_2(v_2)) \\ &= ((f_1 g_1) \otimes (f_2 g_2))(v_1 \otimes v_2). \end{aligned}$$

21.7 Tensor products of representations

Definition 21.4: Tensor-product representation

Let

$$D_1 : G \longrightarrow GL(V_1), \quad D_2 : G \longrightarrow GL(V_2)$$

be representations. Their *tensor product*, also called the *product representation*, is

$$D_1 \otimes D_2 : G \longrightarrow GL(V_1 \otimes V_2)$$

defined by

$$(D_1 \otimes D_2)(a) := D_1(a) \otimes D_2(a). \quad (21.20)$$

Thus, on a pure tensor,

$$(D_1 \otimes D_2)(a)(v_1 \otimes v_2) = D_1(a)v_1 \otimes D_2(a)v_2.$$

The representation law follows directly from Equation (21.19):

$$\begin{aligned} (D_1 \otimes D_2)(a)(D_1 \otimes D_2)(b) &= (D_1(a)D_1(b)) \otimes (D_2(a)D_2(b)) \\ &= D_1(ab) \otimes D_2(ab) \\ &= (D_1 \otimes D_2)(ab). \end{aligned}$$

Also,

$$(D_1 \otimes D_2)(e) = I_{V_1} \otimes I_{V_2} = I_{V_1 \otimes V_2}.$$

Therefore the definition indeed produces a representation, of dimension

$$\dim(D_1 \otimes D_2) = (\dim D_1)(\dim D_2). \quad (21.21)$$

In the product basis from Equation (21.16), the matrix of $(D_1 \otimes D_2)(a)$ is the Kronecker product of the two matrices. If

$$D_1(a) = (A_{ki}), \quad D_2(a) = (B_{\ell j}),$$

then

$$\begin{aligned} (D_1(a) \otimes D_2(a))(v_i^{(1)} \otimes v_j^{(2)}) &= \left(\sum_k A_{ki} v_k^{(1)} \right) \otimes \left(\sum_\ell B_{\ell j} v_\ell^{(2)} \right) \\ &= \sum_{k,\ell} A_{ki} B_{\ell j} (v_k^{(1)} \otimes v_\ell^{(2)}). \end{aligned}$$

Hence the matrix entry with row pair (k, ℓ) and column pair (i, j) is

$$(D_1(a) \otimes D_2(a))_{(k,\ell),(i,j)} = A_{ki} B_{\ell j}.$$

Direct sum versus tensor product

The two constructions combine dimensions in different ways:

$$\dim(D_1 \oplus D_2) = \dim D_1 + \dim D_2, \quad \dim(D_1 \otimes D_2) = (\dim D_1)(\dim D_2).$$

In a direct sum, a group element acts on either component separately. In a tensor product, the same group element acts on both factors at once.

What to retain from this chapter

1. Representations D and D' are equivalent if a single isomorphism S intertwines every group action:

$$D(a) = S^{-1}D'(a)S \quad (a \in G).$$

2. Choosing a different basis gives an equivalent matrix representation. All representing matrices undergo the same similarity transformation.
3. For a finite group, averaging any inner product,

$$\{v, w\} = \frac{1}{|G|} \sum_{a \in G} \langle D(a)v, D(a)w \rangle,$$

produces a positive-definite G -invariant inner product. It follows that every finite-dimensional representation of a finite group is equivalent to a unitary representation.

4. The condition $U_1 \cap U_2 = \{0\}$ is equivalent to the uniqueness of every decomposition $u_1 + u_2$ in $U_1 \oplus U_2$.
5. Direct-sum representations act componentwise and have block-diagonal matrices in an adapted basis:

$$(D_1 \oplus D_2)(a) = \begin{pmatrix} D_1(a) & 0 \\ 0 & D_2(a) \end{pmatrix}.$$

6. Tensor products are bilinear, have the product basis $v_i^{(1)} \otimes v_j^{(2)}$, and satisfy

$$\dim(V_1 \otimes V_2) = \dim V_1 \dim V_2.$$

7. The product representation acts by

$$(D_1 \otimes D_2)(a)(v_1 \otimes v_2) = D_1(a)v_1 \otimes D_2(a)v_2.$$

Invariant Subspaces and Irreducibility

Road map

Diagonalisation is one of the central problems of ordinary matrix theory. For a family of matrices representing a group, the analogous problem is to find one basis, valid for every group element, in which all matrices have the smallest possible blocks. The correct language is that of invariant subspaces, reducible and irreducible representations, and complete reducibility. After developing this language, we prove the orthogonal complement theorem, Maschke's theorem, Schur's lemma and its principal consequences, the one-dimensionality of complex irreducible representations of Abelian groups, and uniqueness of irreducible multiplicities.

22.1 Invariant subspaces and restricted representations

Let

$$D : G \longrightarrow GL(V)$$

be a representation of a group G on a vector space V .

Definition 22.1: G -invariant subspace

A linear subspace $U \leq V$ is *invariant under G* , or *D -invariant*, if

$$D(a)U \subseteq U \quad \text{for every } a \in G. \quad (22.1)$$

Equivalently, $u \in U$ implies $D(a)u \in U$ for every $a \in G$.

Because $D(a)$ is invertible and $D(a)^{-1} = D(a^{-1})$, the inclusion in Equation (22.1) is actually an equality:

$$D(a)U = U.$$

Indeed, $D(a)U \subseteq U$, while applying invariance to a^{-1} gives $D(a^{-1})U \subseteq U$; multiplication by $D(a)$ then gives $U \subseteq D(a)U$.

The two *trivial invariant subspaces* are always

$$\{0\} \quad \text{and} \quad V.$$

Any invariant subspace different from both is called nontrivial.

If U is invariant, every $D(a)$ restricts to an invertible linear map on U . We may therefore define

$$D_U : G \longrightarrow GL(U), \quad D_U(a) := D(a)|_U. \quad (22.2)$$

For $u \in U$,

$$D_U(a)D_U(b)u = D(a)D(b)u = D(ab)u = D_U(ab)u,$$

so D_U is a representation. If U is nontrivial and $\dim V < \infty$, then

$$0 < \dim D_U = \dim U < \dim V = \dim D.$$

This is the first way in which an invariant subspace reduces the effective dimension of a representation.

Proposition 22.1: Restriction of a unitary representation

If V is an inner-product space, D is unitary and U is invariant, then D_U is unitary with respect to the inner product restricted to U .

证明. For $u_1, u_2 \in U$ and $a \in G$,

$$\langle D_U(a)u_1, D_U(a)u_2 \rangle = \langle D(a)u_1, D(a)u_2 \rangle = \langle u_1, u_2 \rangle.$$

Thus each $D_U(a)$ preserves the restricted inner product. ■

22.2 Reducibility and block-triangular form

Definition 22.2: Reducible and irreducible representations

The representation $D : G \rightarrow GL(V)$ is *reducible* if V contains a nontrivial invariant subspace:

$$\exists U \leq V, \quad U \neq \{0\}, \quad U \neq V, \quad D(a)U \subseteq U \quad (a \in G).$$

If no such subspace exists, D is *irreducible*.

22.2.1 An adapted basis

Suppose D is reducible and U is an invariant subspace with

$$\dim U = k, \quad \dim V = n, \quad 0 < k < n.$$

Choose a basis $(e_1, \dots, e_k)$ of U and extend it to a basis

$$\mathcal{B} = (e_1, \dots, e_k, e_{k+1}, \dots, e_n)$$

of V . Define

$$W := \text{span}\{e_{k+1}, \dots, e_n\}.$$

As vector spaces,

$$V = U \oplus W,$$

but W has not been assumed invariant.

Write

$$D(a)e_j = \sum_{i=1}^n D_{ij}(a)e_i.$$

For $1 \leq j \leq k$, invariance gives $D(a)e_j \in U$, so the coefficients of $e_{k+1}, \dots, e_n$ vanish:

$$D_{ij}(a) = 0 \quad (1 \leq j \leq k, k+1 \leq i \leq n).$$

Consequently every representing matrix has the same block-upper-triangular form

$$[D(a)]_{\mathcal{B}} = \begin{pmatrix} D_U(a) & N(a) \\ 0 & D''(a) \end{pmatrix}. \quad (22.3)$$

Here the lower-left zero block expresses invariance of U . The block $N(a)$ measures how vectors chosen in W may acquire components in U .

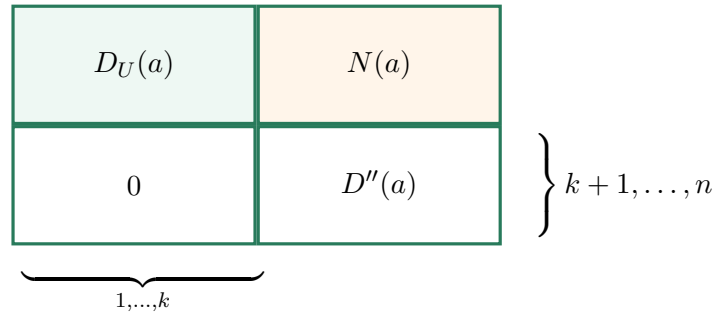


图 22.1: A reducible representation in a basis adapted to one invariant subspace. Only the lower-left block is forced to vanish.

If $u \in U$, its adapted coordinate column has the form

$$[u]_{\mathcal{B}} = \begin{pmatrix} x \\ 0 \end{pmatrix},$$

and Equation (22.3) gives

$$[D(a)u]_{\mathcal{B}} = \begin{pmatrix} D_U(a)x \\ 0 \end{pmatrix} \in U.$$

In contrast, for $w \in W$,

$$[w]_{\mathcal{B}} = \begin{pmatrix} 0 \\ y \end{pmatrix} \implies [D(a)w]_{\mathcal{B}} = \begin{pmatrix} N(a)y \\ D''(a)y \end{pmatrix}.$$

The first component need not vanish; thus $D(a)w \notin W$ in general.

22.2.2 Example: rotations about one axis

Let

$$G = \{R_z(\theta) : \theta \in \mathbb{R}\} \leq SO(3)$$

act on $V = \mathbb{R}^3$ by the defining representation $D(R) = R$. The xy -plane

$$U = \text{span}\{e_x, e_y\}$$

and the z -axis

$$\bar{U} = \text{span}\{e_z\}$$

are both invariant. If W is a slanted line complementary to U , however, rotating its nonzero horizontal component takes it out of W .

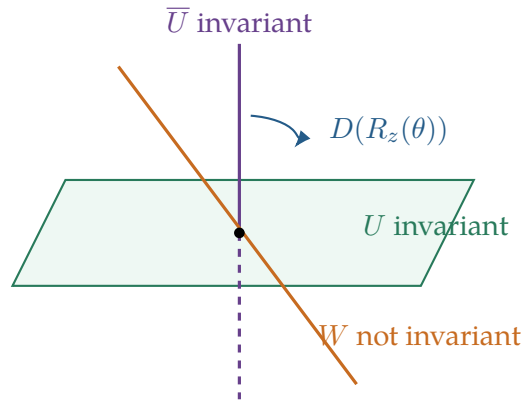


图 22.2: Rotations about the z -axis preserve the horizontal plane and the z -axis, but not a generic complementary line.

Thus the three-dimensional representation D is reducible. Its restriction to U is the usual real two-dimensional rotation representation, and its restriction to $\bar{U}$ is the one-dimensional trivial representation. The restriction D_W cannot be defined because W is not invariant. Over $\mathbb{R}$, the action of the full rotation circle on U is irreducible: a real invariant line would have to be preserved by every rotation, which is impossible. Over $\mathbb{C}$, its complexification splits into two one-dimensional weight spaces; thus the ground field matters.

22.3 Completely reducible representations

If both U and a complementary subspace W are invariant, then $N(a) = 0$ in Equation (22.3), and

$$V = U \oplus W, \quad D = D_U \oplus D_W.$$

In an adapted basis,

$$[D(a)]_{\mathcal{B}} = \begin{pmatrix} D_U(a) & 0 \\ 0 & D_W(a) \end{pmatrix} \quad (a \in G). \quad (22.4)$$

One-dimensional representations

Every one-dimensional representation is irreducible, because a one-dimensional vector space has no subspaces other than $\{0\}$ and itself.

Definition 22.3: Completely reducible representation

A representation $D : G \rightarrow GL(V)$ is *completely reducible* if its representation space is a direct sum of invariant irreducible subspaces,

$$V = \bigoplus_{\alpha} U_{\alpha},$$

and hence

$$D = \bigoplus_{\alpha} D_{\alpha}, \quad D_{\alpha} := D|_{U_{\alpha}}, \quad (22.5)$$

with every D_{α} irreducible.

For finite-dimensional V , complete reducibility is equivalent to the existence of one basis in which every $D(a)$ has the same block-diagonal pattern,

$$[D(a)] = \begin{pmatrix} D_1(a) & & 0 \\ & \ddots & \\ 0 & & D_r(a) \end{pmatrix},$$

where every block representation D_{α} is irreducible. The matrices inside the blocks vary with a , but the block sizes and positions do not.

22.3.1 A reducible representation that is not completely reducible

Consider the additive group $G = (\mathbb{R}, +)$ and

$$D(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}. \quad (22.6)$$

This is a real two-dimensional representation because

$$D(x)D(x') = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & x' \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & x+x' \\ 0 & 1 \end{pmatrix} = D(x+x'),$$

and $D(0) = I$. Its determinant is 1, so every $D(x)$ is invertible.

The line

$$U = \left\{ \begin{pmatrix} u \\ 0 \end{pmatrix} : u \in \mathbb{R} \right\} = \text{span}\{e_1\}$$

is invariant, since

$$D(x) \begin{pmatrix} u \\ 0 \end{pmatrix} = \begin{pmatrix} u \\ 0 \end{pmatrix}.$$

Thus D is reducible.

Suppose it were completely reducible. Then there would be an invariant one-dimensional complement U' to U . Write

$$U' = \text{span} \left\{ \begin{pmatrix} u'_1 \\ u'_2 \end{pmatrix} \right\}.$$

The condition $U' \neq U$ forces $u'_2 \neq 0$. But

$$D(x) \begin{pmatrix} u'_1 \\ u'_2 \end{pmatrix} = \begin{pmatrix} u'_1 + xu'_2 \\ u'_2 \end{pmatrix}.$$

If this vector were in U' , there would be a scalar λ_x such that

$$\begin{pmatrix} u'_1 + xu'_2 \\ u'_2 \end{pmatrix} = \lambda_x \begin{pmatrix} u'_1 \\ u'_2 \end{pmatrix}.$$

The second coordinate and $u'_2 \neq 0$ imply $\lambda_x = 1$; the first coordinate then implies $xu'_2 = 0$, which fails whenever $x \neq 0$. Therefore no invariant complement exists, and D is not completely reducible.

The exception and the rule

Representations need not be completely reducible, as the unipotent example shows. Nevertheless complete reducibility is the rule in several central settings: finite-dimensional unitary representations, finite-group representations over characteristic zero, and continuous finite-dimensional representations of compact groups.

22.4 Unitary representations and invariant complements

Theorem 22.1: Orthogonal complements are invariant

Let V be a finite-dimensional inner-product space and let

$$D : G \longrightarrow U(V)$$

be a unitary representation. If $U \leq V$ is invariant, then its orthogonal complement

$$U^\perp = \{w \in V : \langle u, w \rangle = 0 \text{ for every } u \in U\}$$

is invariant, and

$$V = U \oplus U^\perp, \quad D = D_U \oplus D_{U^\perp}.$$

证明. The set $U^\perp$ is a subspace. For instance, if $w_1, w_2 \in U^\perp$ and α, β are scalars, then for every $u \in U$,

$$\langle u, \alpha w_1 + \beta w_2 \rangle = \alpha \langle u, w_1 \rangle + \beta \langle u, w_2 \rangle = 0.$$

Choose an orthonormal basis

$$(e_1, \dots, e_k)$$

of U , and extend it to the orthonormal basis

$$(e_1, \dots, e_n)$$

of V . Then

$$U^\perp = \text{span}\{e_{k+1}, \dots, e_n\},$$

so every $v \in V$ has a unique orthogonal decomposition

$$v = u + w, \quad u \in U, \quad w \in U^\perp.$$

Hence $V = U \oplus U^\perp$.

It remains to prove invariance. Let $w \in U^\perp$, $u \in U$, and $a \in G$. Because $D(a)$ is unitary,

$$D(a)^\dagger = D(a)^{-1} = D(a^{-1}).$$

Therefore

$$\begin{aligned} \langle u, D(a)w \rangle &= \langle D(a)^\dagger u, w \rangle \\ &= \langle D(a^{-1})u, w \rangle. \end{aligned}$$

Since U is invariant, $D(a^{-1})u \in U$; since $w \in U^\perp$, the last inner product is zero. Thus

$$\langle u, D(a)w \rangle = 0 \quad \text{for every } u \in U,$$

which says $D(a)w \in U^\perp$. Hence $U^\perp$ is invariant. ■

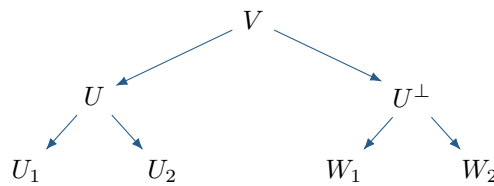
Theorem 22.2: Finite-dimensional unitary representations are completely reducible

Every finite-dimensional unitary representation is completely reducible.

证明. If D is irreducible, there is nothing to prove. Otherwise choose a nontrivial invariant subspace U . By the orthogonal-complement theorem above,

$$V = U \oplus U^\perp, \quad D = D_U \oplus D_{U^\perp}.$$

If either restricted representation is reducible, split that invariant subspace again into a nontrivial invariant subspace and its orthogonal complement. Continue this procedure:



At every genuine split, each child has strictly smaller positive dimension than its parent. Since $\dim V < \infty$, the process must terminate after finitely many steps. The terminal invariant subspaces are irreducible and their orthogonal direct sum is V . Hence D is completely reducible. ■

22.5 Maschke's theorem

Chapter 21 showed that a finite-group representation over $\mathbb{C}$ is equivalent to a unitary representation. We can now combine that result with the complete-reducibility theorem for unitary representations.

Theorem 22.3: Maschke's theorem over $\mathbb{R}$ or $\mathbb{C}$

Let G be finite and let $D : G \rightarrow GL(V)$ be a finite-dimensional real or complex representation. Then D is completely reducible.

Proof by unitarisation and pullback. Average an inner product over G as in Equation (21.5). With respect to this averaged inner product, D is unitary. Hence

$$V = U'_1 \oplus \cdots \oplus U'_r, \quad D' = D'_1 \oplus \cdots \oplus D'_r,$$

for irreducible invariant subspaces U'_α , after identifying D with an equivalent unitary representation D' . Let

$$D(a) = S^{-1} D'(a) S$$

and define

$$U_\alpha := S^{-1} U'_\alpha.$$

First, $V = \bigoplus_\alpha U_\alpha$. For any $v \in V$, write

$$Sv = u'_1 + \cdots + u'_r, \quad u'_\alpha \in U'_\alpha.$$

Then

$$v = S^{-1} u'_1 + \cdots + S^{-1} u'_r,$$

so the U_α span V . If

$$u_1 + \cdots + u_r = \bar{u}_1 + \cdots + \bar{u}_r, \quad u_\alpha, \bar{u}_\alpha \in U_\alpha,$$

applying S gives two decompositions in $\bigoplus_\alpha U'_\alpha$, hence $Su_\alpha = S\bar{u}_\alpha$ and, because S is injective, $u_\alpha = \bar{u}_\alpha$.

Second, each U_α is invariant:

$$D(a)U_\alpha = S^{-1} D'(a) S S^{-1} U'_\alpha = S^{-1} D'(a) U'_\alpha \subseteq S^{-1} U'_\alpha = U_\alpha.$$

Finally, $D|_{U_\alpha}$ is irreducible. If $\bar{U}_\alpha$ were a nontrivial invariant subspace of U_α , then $S\bar{U}_\alpha$ would be a nontrivial invariant subspace of U'_α , contradicting irreducibility of D'_α . Therefore the pulled-back decomposition is a complete reduction of D . ■

22.5.1 The averaged-projection proof

There is a more algebraic proof that displays directly how to manufacture an invariant complement. Let $U \leq V$ be invariant. Choose any vector-space complement W , so $V = U \oplus W$, and let

$$P : V \longrightarrow U$$

be the projection along W : $P(u + w) = u$. The map P need not commute with D . Average all its conjugates:

$$\bar{P} := \frac{1}{|G|} \sum_{a \in G} D(a) P D(a)^{-1}. \quad (22.7)$$

For $b \in G$,

$$\begin{aligned} D(b)\overline{P}D(b)^{-1} &= \frac{1}{|G|} \sum_{a \in G} D(ba)PD(ba)^{-1} \\ &= \overline{P}, \end{aligned}$$

because $a \mapsto ba$ permutes G . Hence

$$D(b)\overline{P} = \overline{P}D(b).$$

Moreover, for $u \in U$, invariance gives $D(a)^{-1}u \in U$, so

$$D(a)PD(a)^{-1}u = D(a)D(a)^{-1}u = u.$$

Thus $\overline{P}|_U = I_U$, while every summand in Equation (22.7) has image in U ; consequently $\text{im } \overline{P} = U$ and $\overline{P}^2 = \overline{P}$. Its kernel is an invariant complement:

$$V = U \oplus \ker \overline{P}.$$

Indeed, commutation gives

$$\overline{P}(D(b)v) = D(b)\overline{P}(v) = 0 \quad (v \in \ker \overline{P}).$$

Repeatedly applying this construction proves complete reducibility.

General field and compact-group forms

The projection proof works over a field $\mathbb{K}$ whenever $|G|$ is invertible in $\mathbb{K}$, that is, $\text{char } \mathbb{K} \nmid |G|$. This is the usual algebraic form of Maschke's theorem. If the characteristic divides $|G|$, complete reducibility can fail.

For a compact group, the finite average is replaced by normalized Haar integration:

$$\overline{P} = \int_G D(a)PD(a)^{-1} d\mu(a).$$

Thus every continuous finite-dimensional representation of a compact group over $\mathbb{R}$ or $\mathbb{C}$ is completely reducible.

22.5.2 A comparison of the hypotheses

The preceding results are easy to confuse, so it is useful to display their logical roles explicitly.

Hypothesis or ingredient	Invariant orthogonal complement	Unitary complete reduction	Maschke / compact averaging
$ G < \infty$	—	—	sufficient
Normalized group average exists	—	—	essential
$\dim V < \infty$	used for $V = U \oplus U^\perp$ here	ensures termination	standard assumption
V has an inner product	required	required	constructed if needed
D is unitary	required	required	obtained by averaging

表 22.1: How the hypotheses enter the complete-reducibility theorems.

22.6 Schur's lemma

A central theorem of representation theory is Schur's lemma—so central that it is often simply called *the lemma of representation theory*. Let

$$D : G \rightarrow GL(V), \quad D' : G \rightarrow GL(V')$$

be irreducible representations, and let $A : V \rightarrow V'$ be an intertwining map:

$$D'(a)A = AD(a) \quad (a \in G). \quad (22.8)$$

The equation is represented by the commutative square

$$\begin{array}{ccc}
 V & \xrightarrow{A} & V' \\
 D(a) \downarrow & & \downarrow D'(a) \\
 V & \xrightarrow{A} & V'
 \end{array}$$

Theorem 22.4: Schur's lemma, intertwiner form

For irreducible representations D on V and D' on V' , an intertwiner $A : V \rightarrow V'$ satisfies exactly one of the following:

1. $A = 0$;
2. A is an isomorphism, so $D \sim D'$ and $\dim V = \dim V'$.

No finite-dimensional hypothesis is needed for this algebraic statement.

证明. We first show that $\ker A \leq V$ is D -invariant. If $v \in \ker A$, then $Av = 0$, and

$$A(D(a)v) = D'(a)Av = 0.$$

Thus $D(a)v \in \ker A$. Irreducibility of D implies

$$\ker A = \{0\} \quad \text{or} \quad \ker A = V.$$

Next, $\text{im } A \leq V'$ is D' -invariant. If $v' \in \text{im } A$, then $v' = Av$ for some $v \in V$, and

$$D'(a)v' = D'(a)Av = AD(a)v \in \text{im } A.$$

Irreducibility of D' implies

$$\text{im } A = \{0\} \quad \text{or} \quad \text{im } A = V'.$$

If $\ker A = V$, then $A = 0$. If $\text{im } A = \{0\}$, again $A = 0$. These are the first alternative. Otherwise $\ker A = \{0\}$ and $\text{im } A = V'$, so A is both injective and surjective and hence is an isomorphism. Rearranging the intertwining relation gives

$$D'(a) = AD(a)A^{-1},$$

so $D' \sim D$. This is the second alternative. ■

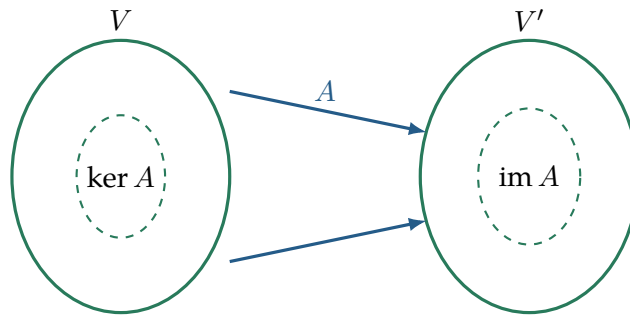


图 22.3: The kernel and image of an intertwiner are invariant. For irreducible source and target they can only be trivial or the whole space.

Proposition 22.2: Scalar commutant over $\mathbb{C}$

Let $D : G \rightarrow GL(V)$ be an irreducible representation on a nonzero finite-dimensional complex vector space. If $A : V \rightarrow V$ commutes with the representation,

$$[D(a), A] = D(a)A - AD(a) = 0 \quad (a \in G),$$

then

$$A = \lambda I_V \quad \text{for some } \lambda \in \mathbb{C}.$$

证明. Because $V \neq \{0\}$ is finite-dimensional over $\mathbb{C}$, the characteristic polynomial of A has a root. Let λ be an eigenvalue and choose $v \neq 0$ with $Av = \lambda v$. Define

$$\bar{A} := A - \lambda I_V.$$

The map $\bar{A}$ also commutes with every $D(a)$, so it is an intertwiner from D to itself. Moreover,

$$\bar{A}v = (A - \lambda I_V)v = 0,$$

and hence $\ker \bar{A} \neq \{0\}$. By Schur's intertwiner lemma above, a nonzero intertwiner of an irreducible representation with itself must be an isomorphism. Since $\bar{A}$ has nonzero kernel it is not an isomorphism, so the only possibility is $\bar{A} = 0$. Therefore $A = \lambda I_V$. ■

Why the complex field matters

The eigenvalue step is automatic over $\mathbb{C}$ but not over $\mathbb{R}$. A real irreducible representation can have a larger division algebra of commuting operators; real, complex and quaternionic types all occur. The scalar conclusion above should therefore not be transferred to real representations without additional hypotheses.

22.7 The commutant and inequivalent irreducible summands

Suppose a complex representation decomposes as

$$D = D_1 \oplus \cdots \oplus D_r$$

on

$$V = V_1 \oplus \cdots \oplus V_r,$$

where the D_α are irreducible and pairwise inequivalent. Let $A : V \rightarrow V$ commute with D .

For clarity, first take $r = 2$. In the corresponding block basis write

$$D(a) = \begin{pmatrix} D_1(a) & 0 \\ 0 & D_2(a) \end{pmatrix}, \quad A = \begin{pmatrix} A_1 & Q_1 \\ Q_2 & A_2 \end{pmatrix}.$$

The equation $D(a)A = AD(a)$ becomes

$$\begin{pmatrix} D_1(a)A_1 & D_1(a)Q_1 \\ D_2(a)Q_2 & D_2(a)A_2 \end{pmatrix} = \begin{pmatrix} A_1D_1(a) & Q_1D_2(a) \\ Q_2D_1(a) & A_2D_2(a) \end{pmatrix}.$$

Thus

$$\begin{aligned} D_1(a)A_1 &= A_1D_1(a), & D_2(a)A_2 &= A_2D_2(a), \\ D_1(a)Q_1 &= Q_1D_2(a), & D_2(a)Q_2 &= Q_2D_1(a). \end{aligned}$$

The scalar-commutant form of Schur's lemma gives

$$A_1 = \lambda_1 I_{V_1}, \quad A_2 = \lambda_2 I_{V_2}.$$

The off-diagonal block $Q_1 : V_2 \rightarrow V_1$ is an intertwiner from D_2 to D_1 . Since these irreducibles are inequivalent, Schur's intertwiner lemma forces $Q_1 = 0$. Similarly $Q_2 = 0$. Therefore

$$A = \begin{pmatrix} \lambda_1 I_{V_1} & 0 \\ 0 & \lambda_2 I_{V_2} \end{pmatrix}.$$

The same block-by-block argument for arbitrary r gives

$$A = \lambda_1 I_{V_1} \oplus \cdots \oplus \lambda_r I_{V_r}. \quad (22.9)$$

Thus each irreducible invariant subspace V_α is a subspace on which every commuting operator acts by a scalar. If the numbers λ_α are pairwise distinct, the V_α are exactly the eigenspaces of A . If some scalars coincide, the corresponding eigenspace is the direct sum of all V_α carrying that common scalar. This qualification makes precise the eigenspace statement in the handwritten argument.

22.8 Irreducible representations of Abelian groups

Theorem 22.5: Complex irreducibles of Abelian groups

Every nonzero finite-dimensional irreducible complex representation of an Abelian group is one-dimensional.

证明. Let $D : G \rightarrow GL(V)$ be irreducible and let $a \in G$. Since G is Abelian, for every $b \in G$,

$$D(a)D(b) = D(ab) = D(ba) = D(b)D(a).$$

Thus $D(a)$ commutes with the whole representation. By the scalar-commutant form of Schur's lemma,

$$D(a) = \lambda(a)I_V$$

for some nonzero $\lambda(a) \in \mathbb{C}$. Choose any nonzero $e \in V$. The line $L = \text{span}\{e\}$ is invariant because

$$D(a)e = \lambda(a)e \in L \quad (a \in G).$$

Irreducibility and $L \neq \{0\}$ force $L = V$. Hence $\dim V = 1$. ■

Example 22.1: Simultaneous diagonalisation of an Abelian unitary action

Let $D : G \rightarrow U(V)$ be a finite-dimensional unitary representation of an Abelian group. Every $D(a)$ is normal, and all the $D(a)$ commute:

$$D(a)D(b) = D(ab) = D(ba) = D(b)D(a).$$

A commuting family of normal operators on a finite-dimensional complex inner-product space is simultaneously unitarily diagonalizable. Hence there is an orthonormal basis $(e_1, \dots, e_n)$ such that

$$D(a)e_\alpha = \lambda_\alpha(a)e_\alpha \quad (a \in G).$$

Each line

$$U_\alpha = \text{span}\{e_\alpha\}$$

is invariant and carries the one-dimensional representation

$$D_\alpha(a) = \lambda_\alpha(a).$$

Therefore

$$V = U_1 \oplus \dots \oplus U_n, \quad D = D_1 \oplus \dots \oplus D_n.$$

This is the concrete matrix form of the theorem: every finite-dimensional unitary representation of an Abelian group decomposes into one-dimensional irreducibles.

22.9 Uniqueness of irreducible multiplicities

Theorem 22.6: Uniqueness of a complete reduction

Let D be a finite-dimensional completely reducible complex representation. Grouping equivalent summands, write

$$D \sim n_1 D^{(1)} \oplus n_2 D^{(2)} \oplus \cdots \oplus n_r D^{(r)}, \quad (22.10)$$

where the $D^{(\alpha)}$ are pairwise inequivalent irreducible representations and

$$n_\alpha D^{(\alpha)} := \underbrace{D^{(\alpha)} \oplus \cdots \oplus D^{(\alpha)}}_{n_\alpha \text{ copies}}.$$

Then the set of irreducible equivalence classes $\{D^{(1)}, \dots, D^{(r)}\}$ and every multiplicity n_α are uniquely determined by D , up to reordering.

证明. For representations E on W and D on V , let

$$\text{Hom}_G(E, D) = \{A : W \rightarrow V : AE(a) = D(a)A \text{ for every } a \in G\}$$

be the vector space of intertwiners.

Fix an irreducible representation E . If $D = \bigoplus_{\alpha,j} D^{(\alpha)}$, then an intertwiner $A : E \rightarrow D$ is equivalent to a collection of component intertwiners

$$A_{\alpha,j} : E \longrightarrow D^{(\alpha)}.$$

Hence

$$\text{Hom}_G(E, D) \cong \bigoplus_{\alpha=1}^r \bigoplus_{j=1}^{n_\alpha} \text{Hom}_G(E, D^{(\alpha)}).$$

Schur's lemma says

$$\dim_{\mathbb{C}} \text{Hom}_G(E, D^{(\alpha)}) = \begin{cases} 1, & E \sim D^{(\alpha)}, \\ 0, & E \not\sim D^{(\alpha)}. \end{cases}$$

Indeed, in the equivalent case every intertwiner becomes a scalar multiple of a fixed equivalence isomorphism, while in the inequivalent case every intertwiner is zero.

Taking $E = D^{(\beta)}$ therefore gives

$$n_\beta = \dim_{\mathbb{C}} \text{Hom}_G(D^{(\beta)}, D). \quad (22.11)$$

The right-hand side depends only on the original representation D and the equivalence class of $D^{(\beta)}$, not on a chosen decomposition. Thus every multiplicity is intrinsic. Moreover, an irreducible class occurs in D exactly when this Hom space is nonzero, so the list of occurring classes is intrinsic as well. This proves uniqueness up to equivalence and reordering. ■

22.10 The defining representation of $GL(n, \mathbb{C})$

Example 22.2: Irreducibility of the defining representation

Let $G = GL(n, \mathbb{C})$ act on $V = \mathbb{C}^n$ by its defining representation

$$D : G \longrightarrow GL(\mathbb{C}^n), \quad D(A) = A.$$

We show that D is irreducible.

Suppose $U \leq \mathbb{C}^n$ is a nonzero proper invariant subspace. Let $\dim U = k$, where $0 < k < n$, choose a basis $(e_1, \dots, e_k)$ of U , and extend it to a basis $(e_1, \dots, e_n)$ of $\mathbb{C}^n$. Define an invertible linear map A by cyclically permuting this basis:

$$Ae_i = e_{i+1} \quad (1 \leq i < n), \quad Ae_n = e_1.$$

This A lies in $GL(n, \mathbb{C})$. But

$$e_k \in U, \quad Ae_k = e_{k+1} \notin U,$$

contradicting invariance under every element of $GL(n, \mathbb{C})$. Therefore the only invariant subspaces are $\{0\}$ and $\mathbb{C}^n$, so the defining representation is irreducible.

Example 22.3: The centre of $GL(n, \mathbb{C})$

Let $A \in GL(n, \mathbb{C})$ commute with every $B \in GL(n, \mathbb{C})$:

$$AB = BA \quad (B \in GL(n, \mathbb{C})).$$

Regard $B \mapsto B$ as the defining irreducible representation. The commutation relation says that A is an intertwiner from this representation to itself:

$$\begin{array}{ccc} \mathbb{C}^n & \xrightarrow{A} & \mathbb{C}^n \\ \downarrow B & & \downarrow B \\ \mathbb{C}^n & \xrightarrow{A} & \mathbb{C}^n \end{array}$$

By the scalar-commutant corollary to Schur's lemma,

$$A = \lambda I_n$$

for some $\lambda \in \mathbb{C}$. Since A is invertible, $\lambda \neq 0$. Conversely every nonzero scalar matrix commutes with every matrix. Hence

$$Z(GL(n, \mathbb{C})) = \{\lambda I_n : \lambda \in \mathbb{C}^\times\}.$$

What to retain from this chapter

1. $U \leq V$ is invariant if $D(a)U \subseteq U$ for every a . Restricting D to U gives a smaller representation D_U .
2. A representation is reducible exactly when it has a nontrivial invariant subspace. In an adapted basis all matrices have the simultaneous block form

$$D(a) = \begin{pmatrix} D_U(a) & N(a) \\ 0 & D''(a) \end{pmatrix}.$$

3. Complete reducibility requires invariant complementary subspaces and produces simultaneous block-diagonal form with irreducible blocks. Reducibility alone does not guarantee this.
4. If D is unitary and U is invariant, then $U^\perp$ is invariant. Repeated orthogonal splitting proves that every finite-dimensional unitary representation is completely reducible.
5. Maschke's theorem follows by averaging an inner product or a projection. A finite-group representation is completely reducible when the field characteristic does not divide $|G|$. The same is true for continuous finite-dimensional representations of compact groups.
6. Schur's lemma says that an intertwiner between irreducibles is either zero or an isomorphism. Over $\mathbb{C}$, every commuting endomorphism of a finite-dimensional irreducible representation is λI .
7. Finite-dimensional complex irreducible representations of Abelian groups are one-dimensional.
8. In a completely reducible complex representation, the irreducible equivalence classes and their multiplicities are unique. The multiplicity of E is

$$\dim_{\mathbb{C}} \operatorname{Hom}_G(E, D).$$

Symmetries and Degeneracies

This chapter applies the representation theory developed above to quantum mechanics. Its central question is very concrete:

If a Hamiltonian is invariant under a group of transformations, what can one know about its eigenvalues and eigenvectors before solving the full eigenvalue problem?

The answer separates two pieces of information. The dimensions of the irreducible representations determine the degeneracies forced by symmetry, whereas their multiplicities determine the smaller matrices that still have to be diagonalised. We first derive this statement in complete generality and then work it out for rotations, spin systems, open chains, and rings.

23.1 Quantum transformations and Wigner's theorem

Let V be a complex Hilbert space. A nonzero vector $\psi \in V$ describes a pure state, but its phase has no physical significance: ψ and $e^{i\theta}\psi$ represent the same ray. For normalised states, the transition probability is

$$P(\psi_1, \psi_2) = |\langle \psi_1, \psi_2 \rangle|^2.$$

Thus a transformation of pure states must be a bijection of rays that preserves this quantity.

Theorem 23.1: Wigner's theorem

Let $\mathcal{D}$ be a bijective transformation of the rays of V such that

$$|\langle \mathcal{D}\psi_1, \mathcal{D}\psi_2 \rangle|^2 = |\langle \psi_1, \psi_2 \rangle|^2 \quad (\psi_1, \psi_2 \in V).$$

After choosing one phase on each transformed ray, $\mathcal{D}$ can be represented either by a unitary operator or by an antiunitary operator.

Here an antiunitary operator is conjugate-linear:

$$\mathcal{D}(\alpha_1\psi_1 + \alpha_2\psi_2) = \alpha_1^*\mathcal{D}\psi_1 + \alpha_2^*\mathcal{D}\psi_2, \quad \langle \mathcal{D}\psi_1, \mathcal{D}\psi_2 \rangle = \langle \psi_1, \psi_2 \rangle^*.$$

In particular, $\mathcal{D}^\dagger \mathcal{D} = I$. A useful mnemonic for conjugate-linearity is

$$c\mathcal{D} = \mathcal{D}c^* \quad (c \in \mathbb{C}),$$

where the equality means equality of the corresponding maps.

Detailed proof idea. Choose an orthonormal basis (e_j) . Preservation of transition probabilities sends the mutually orthogonal rays $[e_j]$ to mutually orthogonal rays. Choose unit vectors f_j on those image rays. A ray with a representative

$$\psi = \sum_j c_j e_j$$

has transition probability $|c_j|^2$ to $[e_j]$, so its image has a representative

$$\psi' = \sum_j c'_j f_j, \quad |c'_j| = |c_j|.$$

Use the irrelevant overall phase to arrange $c'_1 = c_1$ whenever $c_1 \neq 0$. Comparing transition probabilities with the rays of $(e_1 + e_j)/\sqrt{2}$ determines the real parts of c'_j , while comparison with $(e_1 + ie_j)/\sqrt{2}$ determines the imaginary parts. The consistency of these comparisons leaves exactly two possibilities, the same one for every j :

$$c'_j = c_j \quad \text{for all } j, \quad \text{or} \quad c'_j = c_j^* \quad \text{for all } j.$$

The map obtained by $e_j \mapsto f_j$ is therefore respectively linear or conjugate-linear, and it preserves the inner product in the unitary or antiunitary sense. This is the essential argument; the zero coefficients are handled by continuity or by omitting them from the comparisons. ■

For a transformation group G , the operators assigned to group elements need not multiply exactly after phases have been chosen. In general,

$$\mathcal{D}(a)\mathcal{D}(b) = \omega(a, b)\mathcal{D}(ab), \quad |\omega(a, b)| = 1.$$

This is a *projective* unitary or antiunitary representation. Passing to a suitable covering group often converts a projective representation into an ordinary representation; spin representations of $SO(3)$ and ordinary representations of $SU(2)$ are the standard example.

23.1.1 Two different uses of a unitary operator

It is important not to confuse a change of description with a physical transformation of the system.

1. In a passive change of quantum description, both states and observables are transformed:

$$|\psi\rangle \mapsto \mathcal{D}|\psi\rangle, \quad A \mapsto \mathcal{D}A\mathcal{D}^\dagger.$$

Nothing measurable changes, because

$$\begin{aligned} \langle \mathcal{D}\psi | \mathcal{D}A\mathcal{D}^\dagger | \mathcal{D}\psi \rangle &= \langle \psi | \mathcal{D}^\dagger \mathcal{D}A\mathcal{D}^\dagger \mathcal{D} | \psi \rangle \\ &= \langle \psi | A | \psi \rangle. \end{aligned}$$

This viewpoint is especially useful for a time-dependent change of frame.

2. In an active transformation, the physical system is changed while the measuring apparatus is held fixed. One may transform the state alone,

$$|\psi\rangle \mapsto \mathcal{D}(a)|\psi\rangle, \quad A \text{ fixed},$$

or, equivalently, hold the state fixed and transform the observable.

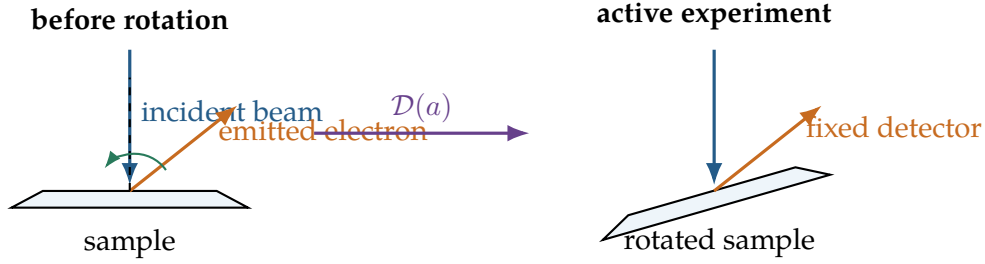


图 23.1: In photoemission, rotating the sample while leaving the incoming beam and detector fixed is a physical transformation, not merely a change of coordinates.

23.2 Hamiltonian symmetry and the first examples

Let $H : V \rightarrow V$ be a linear Hamiltonian and let

$$D : G \longrightarrow U(V)$$

be a concrete unitary representation of a group G on the physical Hilbert space. The Hamiltonian has the symmetry G when

$$D(a)HD(a)^{-1} = H \quad (a \in G),$$

or, equivalently,

$$[H, D(a)] = 0 \quad (a \in G).$$

Specifying the physical representation is essential. The trivial representation $D(a) = I$ commutes with every Hamiltonian, but it contains no information about the system.

If D is irreducible on a finite-dimensional complex space, the scalar commutant result proved in the preceding chapter immediately gives

$$H = EI.$$

Every state then has the same energy. Physical representations are more often reducible, and the purpose of this chapter is to understand the resulting block structure.

Example 23.1: Rotational symmetry of a central potential

For a particle in three dimensions with

$$H = \frac{\mathbf{p}^2}{2m} + V(r), \quad r = \|\mathbf{x}\|,$$

the rotations are represented by

$$D(\hat{\mathbf{n}}, \phi) = \exp\left(-\frac{i}{\hbar}\phi \hat{\mathbf{n}} \cdot \mathbf{J}\right).$$

Both $\mathbf{p}^2$ and r are rotational scalars, so $[H, D(\hat{\mathbf{n}}, \phi)] = 0$ for every axis and angle.

For the hydrogen atom, the familiar notation

$$E_{n\ell m} = E_n$$

contains two logically different statements. Rotational $SO(3)$ symmetry forces the $2\ell + 1$ values $m = -\ell, \dots, \ell$ at fixed (n, ℓ) to have the same energy. Equality of the energies for different values of ℓ is not forced by $SO(3)$; it is an “accidental” degeneracy from the viewpoint of that group and is explained by the larger dynamical symmetry generated by angular momentum and the Runge–Lenz vector.

Example 23.2: Parity does not force degeneracy

For the one-dimensional harmonic oscillator,

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2,$$

define parity by

$$(\Pi\psi)(x) = \psi(-x).$$

Then $\Pi^{-1} = \Pi^\dagger = \Pi$, $\Pi^2 = I$, and $[H, \Pi] = 0$. Hence

$$D(e) = I, \quad D(s) = \Pi$$

is an infinite-dimensional unitary representation of $\mathbb{Z}_2 = \{e, s\}$. Nevertheless, the oscillator energies

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right)$$

are nondegenerate. The reason is that the irreducible complex representations of the Abelian group $\mathbb{Z}_2$ are one-dimensional. The energy eigenstates alternate between even and odd parity, but symmetry does not require two states to share an energy.

Antiunitary symmetries

Time reversal is normally antiunitary. Its invariance condition is still $\Theta H \Theta^{-1} = H$, interpreted as an equality of maps, but the commutant calculation below uses complex-linear intertwiners and therefore assumes a unitary representation. Antiunitary symmetry can lead to extra phenomena, notably Kramers degeneracy when $\Theta^2 = -I$.

23.3 The symmetry-adapted decomposition

Assume from now on that V is finite-dimensional, D is unitary, and H is linear with $[H, D(a)] = 0$ for every $a \in G$. Unitarity makes D completely reducible, so

$$D \cong \bigoplus_{\alpha=1}^k \bigoplus_{i=1}^{n_\alpha} D_i^{(\alpha)} = \bigoplus_{\alpha=1}^k n_\alpha D^{(\alpha)}.$$

Correspondingly,

$$V = \bigoplus_{\alpha=1}^k \bigoplus_{i=1}^{n_\alpha} V_i^{(\alpha)}.$$

The label α distinguishes inequivalent irreducible types. For a fixed α , the label $i = 1, \dots, n_\alpha$ distinguishes equivalent copies. Write

$$d_\alpha = \dim V_i^{(\alpha)};$$

this dimension is independent of i .

23.3.1 Constructing a common basis in the equivalent copies

Choose a basis

$$\{q_{im}^{(\alpha)} : m = 1, \dots, d_\alpha\}$$

of each $V_i^{(\alpha)}$. The union of these bases spans V . This is the representation-theoretic part of the problem and does not require knowing H . For a Lie group one often obtains these vectors by solving the simultaneous structure imposed by the Hermitian generators in

$$D(a) = \exp\left(-\frac{i}{\hbar} \eta_a^r J_r\right).$$

Equivalent copies initially need not have equal matrices. If $\tilde{D}_i^{(\alpha)}$ and $\tilde{D}_j^{(\alpha)}$ are equivalent, there is an invertible intertwiner $S_{ij}^{(\alpha)}$ such that

$$\tilde{D}_i^{(\alpha)}(a) = (S_{ij}^{(\alpha)})^{-1} \tilde{D}_j^{(\alpha)}(a) S_{ij}^{(\alpha)}.$$

Fix one reference copy j . Changing the basis in copy i by the intertwiner $S_{ij}^{(\alpha)}$ makes the matrices equal:

$$D_i^{(\alpha)}(a) = D_j^{(\alpha)}(a) = D^{(\alpha)}(a) \quad \text{for every } i, j \text{ and } a.$$

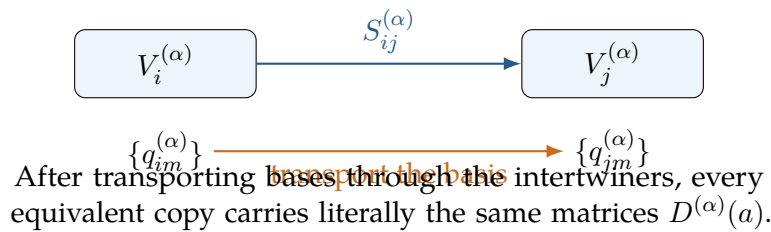


图 23.2: Equalising the matrix representatives on equivalent irreducible summands.

In the resulting basis $e_{im}^{(\alpha)}$, the matrix entries of the representation are

$$D_{ij;mn}^{(\alpha\beta)}(a) = \delta_{\alpha\beta} \delta_{ij} D_{mn}^{(\alpha)}(a).$$

The first pair (α, i) selects an irreducible block, while m selects a component inside that block.

23.3.2 The block-by-block commutator calculation

Write the most general Hamiltonian block from $V_j^{(\beta)}$ to $V_i^{(\alpha)}$ as $H_{ij}^{(\alpha\beta)}$. The equation $HD = DH$ gives

$$\sum_{\eta,k} H_{ik}^{(\alpha\eta)} D_{kj}^{(\eta\beta)}(a) = \sum_{\eta,k} D_{ik}^{(\alpha\eta)}(a) H_{kj}^{(\eta\beta)}.$$

Substituting the Kronecker deltas from (23.3.1) collapses both sums:

$$H_{ij}^{(\alpha\beta)} D^{(\beta)}(a) = D^{(\alpha)}(a) H_{ij}^{(\alpha\beta)} \quad (a \in G).$$

Thus every Hamiltonian block is an intertwiner.

If $\alpha \neq \beta$, the irreducible representations are inequivalent, so Schur's lemma gives

$$H_{ij}^{(\alpha\beta)} = 0.$$

If $\alpha = \beta$, the two matrices on the sides of (23.3.2) are the same irreducible representation. The complex scalar-commutant form of Schur's lemma gives

$$H_{ij}^{(\alpha\alpha)} = C_{ij}^{(\alpha)} I_{d_\alpha}$$

for a scalar $C_{ij}^{(\alpha)}$.

Theorem 23.2: Hamiltonians in a symmetry-adapted basis

Suppose

$$[H, D(a)] = 0 \quad (a \in G), \quad D \cong \bigoplus_{\alpha} n_{\alpha} D^{(\alpha)},$$

where the $D^{(\alpha)}$ are pairwise inequivalent finite-dimensional irreducible unitary complex representations. There is a basis, determined by D and independent of H , in which

$$H_{ij;mn}^{(\alpha\beta)} = \delta_{\alpha\beta} C_{ij}^{(\alpha)} \delta_{mn}.$$

Equivalently,

$$V \cong \bigoplus_{\alpha} (\mathbb{C}^{n_{\alpha}} \otimes W_{\alpha}), \quad D(a) = \bigoplus_{\alpha} (I_{n_{\alpha}} \otimes D^{(\alpha)}(a)),$$

and

$$H = \bigoplus_{\alpha} (C^{(\alpha)} \otimes I_{d_{\alpha}}).$$

When $H = H^{\dagger}$, every multiplicity-space matrix $C^{(\alpha)}$ is Hermitian. Indeed,

$$H^{\dagger} = \bigoplus_{\alpha} (C^{(\alpha)\dagger} \otimes I_{d_{\alpha}}),$$

so equality with H implies $C^{(\alpha)\dagger} = C^{(\alpha)}$. Diagonalise it by a unitary matrix U_{α} :

$$U_{\alpha}^{\dagger} C^{(\alpha)} U_{\alpha} = \text{diag}(E_{\alpha 1}, \dots, E_{\alpha n_{\alpha}}).$$

The change of basis $U_{\alpha} \otimes I_{d_{\alpha}}$ mixes only equivalent copies and therefore leaves $D(a)$ in the same

repeated-block form. Each eigenvalue $E_{\alpha r}$ occurs at least d_α times:

$$\underbrace{E_{\alpha r}, \dots, E_{\alpha r}}_{d_\alpha \text{ copies}}$$

This is the *symmetry-induced degeneracy*.

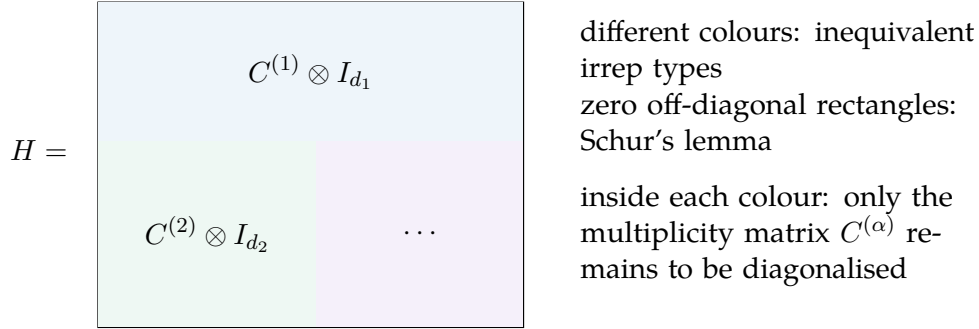


图 23.3: The universal block form of a symmetry-invariant Hamiltonian.

23.3.3 What symmetry does and does not determine

The total dimension is

$$\dim V = \sum_{\alpha} n_{\alpha} d_{\alpha}.$$

Without symmetry one diagonalises one matrix of that size. With symmetry, one diagonalises the smaller matrices $C^{(\alpha)}$, whose sizes are $n_{\alpha} \times n_{\alpha}$.

Representation data	Consequence for H	Physical meaning
$d_{\alpha} = \dim D^{(\alpha)}$	every eigenvalue from $C^{(\alpha)}$ is repeated at least d_{α} times	symmetry-enforced degeneracy
n_{α} , the number of equivalent copies	size of the matrix $C^{(\alpha)}$	remaining diagonalisation problem
all $d_{\alpha} = 1$	no degeneracy is forced	typical for finite-dimensional complex irreps of Abelian groups
all $n_{\alpha} = 1$	every $C^{(\alpha)}$ is a scalar	the symmetry basis already diagonalises H

Two eigenvalues may coincide even when symmetry does not require them to. For example, if $H = H(\lambda)$, levels from distinct irreducible sectors can cross as λ varies:

$$E_{\alpha r}(\lambda_0) = E_{\beta s}(\lambda_0).$$

Such a coincidence is called an *accidental degeneracy relative to G* .

Sometimes a larger group $\overline{G} \supset G$ explains the coincidence. The apparently distinct G -irreducible spaces may combine into a single larger-dimensional irreducible representation of $\overline{G}$. Relative to the larger group there are fewer sectors, larger d_{α} , and often smaller multiplicities n_{α} .

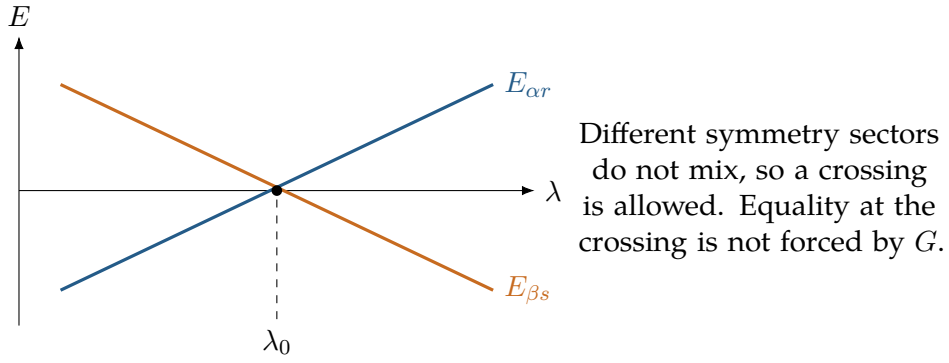


图 23.4: An accidental degeneracy at a parameter-dependent level crossing.

The hydrogenic equality across different ℓ values is the standard example. Thus “accidental” means unexplained by the symmetry currently being used, not necessarily devoid of a deeper cause.

A basis-independent statement

The decomposition

$$V \cong \bigoplus_{\alpha} \mathbb{C}^{n_{\alpha}} \otimes W_{\alpha}$$

is called the decomposition into *isotypic components*. The commutant of $D(G)$ is

$$\text{End}_G(V) \cong \bigoplus_{\alpha} M_{n_{\alpha}}(\mathbb{C}) \otimes I_{W_{\alpha}}.$$

The Hamiltonian theorem is precisely the statement that every G -invariant Hamiltonian belongs to this commutant.

23.4 The spin- $\frac{1}{2}$ representation

Let $V = \mathbb{C}^2$ with orthonormal basis $\{|+\rangle, |-\rangle\}$, where

$$|+\rangle = |\tfrac{1}{2}\rangle, \quad |-\rangle = |-\tfrac{1}{2}\rangle.$$

In units with $\hbar = 1$, define

$$S_{\alpha} = \frac{1}{2} \sum_{m, m' = \pm 1/2} |m\rangle (\sigma_{\alpha})_{mm'} \langle m'|, \quad \alpha = x, y, z,$$

where

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Direct multiplication gives

$$\sigma_{\alpha} \sigma_{\beta} = \delta_{\alpha\beta} I + i \sum_{\gamma} \varepsilon_{\alpha\beta\gamma} \sigma_{\gamma}.$$

Indeed, each Pauli matrix squares to I , while, for example,

$$\sigma_x \sigma_y = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = i\sigma_z, \quad \sigma_y \sigma_x = -i\sigma_z.$$

Taking the difference of the two orders yields

$$[S_\alpha, S_\beta] = i \sum_\gamma \varepsilon_{\alpha\beta\gamma} S_\gamma.$$

The Casimir operator is

$$\begin{aligned} \mathbf{S}^2 &= S_x^2 + S_y^2 + S_z^2 \\ &= \frac{1}{4} (\sigma_x^2 + \sigma_y^2 + \sigma_z^2) \\ &= \frac{3}{4} I = s(s+1)I, \end{aligned}$$

so $s = \frac{1}{2}$. Also

$$S_z |\pm\rangle = \pm \frac{1}{2} |\pm\rangle.$$

For $\mathbf{a} = \phi \hat{\mathbf{n}} \in \mathbb{R}^3$, define

$$D^{(1/2)}(\mathbf{a}) = \exp(-i\mathbf{a} \cdot \mathbf{S}) = \exp\left(-\frac{i}{2} \mathbf{a} \cdot \boldsymbol{\sigma}\right).$$

Because

$$(\hat{\mathbf{n}} \cdot \boldsymbol{\sigma})^2 = I,$$

separating the exponential series into its even and odd powers gives the explicit formula

$$D^{(1/2)}(\hat{\mathbf{n}}, \phi) = \cos \frac{\phi}{2} I - i \sin \frac{\phi}{2} \hat{\mathbf{n}} \cdot \boldsymbol{\sigma}.$$

It is unitary because its exponent is anti-Hermitian. It also has determinant one:

$$\det D^{(1/2)}(\mathbf{a}) = \det \exp\left(-\frac{i}{2} \mathbf{a} \cdot \boldsymbol{\sigma}\right) = \exp\left(\operatorname{tr}\left[-\frac{i}{2} \mathbf{a} \cdot \boldsymbol{\sigma}\right]\right) = 1,$$

since every Pauli matrix is traceless. Thus these are matrices in $SU(2)$.

Proposition 23.1: Irreducibility of spin one-half

The defining two-dimensional representation $D^{(1/2)}$ of $SU(2)$ is irreducible.

证明. A proper nonzero invariant subspace of $\mathbb{C}^2$ would be one-dimensional, say $W = \mathbb{C}v$. Invariance under every $\exp(-itS_\alpha)$ implies, after differentiating at $t = 0$, that $S_\alpha v \in W$ for all α . Thus v would be a common eigenvector of σ_x and σ_z .

The eigenvectors of σ_z are multiples of $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$, but

$$\sigma_x \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \sigma_x \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Neither σ_z -eigenline is invariant under σ_x . Hence no such W exists. ■

The spin operators transform as a three-dimensional vector:

$$D^{(1/2)}(\mathbf{a})S_\beta D^{(1/2)}(\mathbf{a})^\dagger = \sum_{\alpha=x,y,z} R_{\alpha\beta}(\mathbf{a})S_\alpha,$$

where $R(\mathbf{a}) \in SO(3)$ is the ordinary axis-angle rotation. To see the infinitesimal mechanism, put $X = -i\phi \hat{\mathbf{n}} \cdot \mathbf{S}$. Hadamard's lemma gives

$$e^X S_\beta e^{-X} = S_\beta + [X, S_\beta] + \frac{1}{2!}[X, [X, S_\beta]] + \cdots.$$

Using the spin commutation relations,

$$[X, S_\beta] = \phi \sum_{\alpha,\gamma} \hat{n}_\gamma \varepsilon_{\gamma\beta\alpha} S_\alpha.$$

This is exactly the infinitesimal generator of a real rotation acting on a three-vector. Exponentiating it gives (23.4).

23.5 Two spin- $\frac{1}{2}$ particles

Let

$$V = V_1 \otimes V_2, \quad \dim V = 4,$$

and let $\mathbf{S}_1 = \mathbf{S} \otimes I$ and $\mathbf{S}_2 = I \otimes \mathbf{S}$. Consider the isotropic exchange Hamiltonian

$$H = J \mathbf{S}_1 \cdot \mathbf{S}_2 = J \sum_{\alpha=x,y,z} S_{1\alpha} S_{2\alpha}.$$

The rotation representation on the product space is

$$D(\mathbf{a}) = D^{(1/2)}(\mathbf{a}) \otimes D^{(1/2)}(\mathbf{a}).$$

Because operators acting on distinct tensor factors commute,

$$\begin{aligned} D(\mathbf{a}) &= e^{-i\mathbf{a} \cdot \mathbf{S}_1} e^{-i\mathbf{a} \cdot \mathbf{S}_2} \\ &= e^{-i\mathbf{a} \cdot (\mathbf{S}_1 + \mathbf{S}_2)} = e^{-i\mathbf{a} \cdot \mathbf{S}_{\text{tot}}}, \end{aligned}$$

where $\mathbf{S}_{\text{tot}} = \mathbf{S}_1 + \mathbf{S}_2$.

The Hamiltonian is invariant. Using (23.4) on each factor,

$$\begin{aligned} D(\mathbf{a}) H D(\mathbf{a})^\dagger &= J \sum_{\alpha} \left(\sum_{\beta} R_{\beta\alpha} S_{1\beta} \right) \left(\sum_{\gamma} R_{\gamma\alpha} S_{2\gamma} \right) \\ &= J \sum_{\beta,\gamma} \left(\sum_{\alpha} R_{\beta\alpha} R_{\gamma\alpha} \right) S_{1\beta} S_{2\gamma} \\ &= J \sum_{\beta,\gamma} \delta_{\beta\gamma} S_{1\beta} S_{2\gamma} = H, \end{aligned}$$

where $RR^\top = I$ was used in the third line.

23.5.1 Product and coupled bases

The uncoupled orthonormal basis is

$$|m_1 m_2\rangle = |m_1\rangle \otimes |m_2\rangle, \quad m_1, m_2 = \pm \frac{1}{2}.$$

The coupled basis consists of simultaneous eigenvectors of S_{tot}^2 and $S_{\text{tot},z}$:

$$|S, M\rangle, \quad S = 0, 1, \quad M = -S, -S + 1, \dots, S.$$

The change of basis is expressed by Clebsch–Gordan coefficients:

$$|S, M\rangle = \sum_{m_1, m_2} |m_1 m_2\rangle \langle m_1 m_2 | S, M\rangle,$$

$$|m_1 m_2\rangle = \sum_{S, M} |S, M\rangle \langle S, M | m_1 m_2\rangle.$$

The coefficient vanishes unless $M = m_1 + m_2$.

Starting with the unique highest-weight vector and lowering it gives the coefficients explicitly:

$$|1, 1\rangle = |++\rangle,$$

$$|1, 0\rangle = \frac{1}{\sqrt{2}}(|+-\rangle + |-+\rangle),$$

$$|1, -1\rangle = |--\rangle,$$

$$|0, 0\rangle = \frac{1}{\sqrt{2}}(|+-\rangle - |-+\rangle).$$

The first three symmetric states form the triplet V_1 ; the last antisymmetric state forms the singlet V_0 . Therefore

$$D^{(1/2)} \otimes D^{(1/2)} = D^{(0)} \oplus D^{(1)}, \quad \frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1.$$

There are no matrix elements between the $S = 0$ and $S = 1$ spaces, because they are inequivalent irreducible representations.

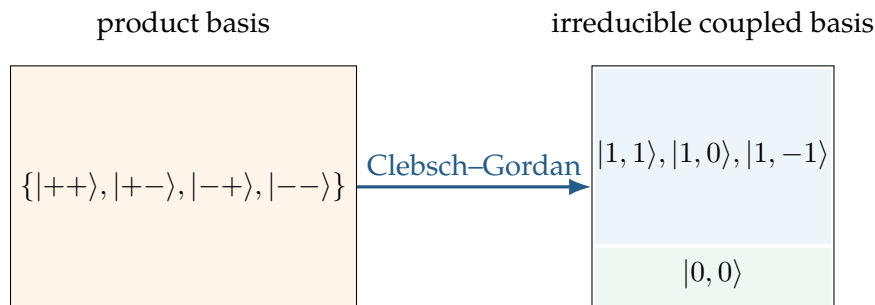


图 23.5: The product representation becomes a spin-one block and a spin-zero block in the coupled basis.

23.5.2 The spectrum without a 4×4 determinant

Since

$$S_{\text{tot}}^2 = S_1^2 + S_2^2 + 2S_1 \cdot S_2,$$

we have

$$\mathbf{S}_1 \cdot \mathbf{S}_2 = \frac{1}{2} (\mathbf{S}_{\text{tot}}^2 - \mathbf{S}_1^2 - \mathbf{S}_2^2).$$

On the subspace of total spin S ,

$$\begin{aligned} E_S &= \frac{J}{2} \left[S(S+1) - \frac{1}{2} \left(\frac{1}{2} + 1 \right) - \frac{1}{2} \left(\frac{1}{2} + 1 \right) \right] \\ &= \frac{J}{2} \left[S(S+1) - \frac{3}{2} \right]. \end{aligned}$$

Consequently,

$$E_0 = -\frac{3J}{4} \quad (\text{one singlet state}), \quad E_1 = \frac{J}{4} \quad (\text{three triplet states}).$$

For $J > 0$ the singlet lies below the triplet. For $J < 0$ the order is reversed: the threefold-degenerate triplet lies below the singlet.

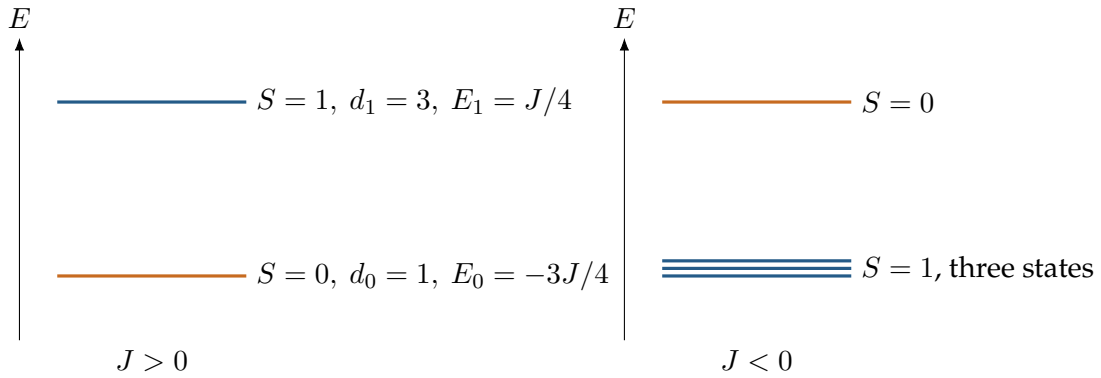


图 23.6: Singlet and triplet levels for the two signs of the exchange coupling. The closely drawn lines at right represent three states at the same energy, not three distinct energies.

23.5.3 General spin- j matrices

For any $j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$, the Wigner matrices are

$$D_{m'm}^{(j)}(\mathbf{a}) = \langle j, m' | \exp(-i\mathbf{a} \cdot \mathbf{J}) | j, m \rangle, \quad m, m' = -j, \dots, j.$$

The Casimir $\mathbf{J}^2$ commutes with all J_α , so rotations cannot mix different j -subspaces. Within a fixed j -subspace, the raising and lowering operators

$$J_\pm = J_x \pm iJ_y$$

connect every weight state:

$$J_\pm |j, m\rangle = \sqrt{(j \mp m)(j \pm m + 1)} |j, m \pm 1\rangle.$$

Any nonzero invariant subspace containing one weight vector therefore contains all $2j + 1$ weight vectors. Hence $D^{(j)}$ is irreducible.

23.6 Three and more spin- $\frac{1}{2}$ particles

The general angular-momentum addition rule is

$$s_1 \otimes s_2 = \bigoplus_{S=|s_1-s_2|}^{s_1+s_2} S,$$

where S increases in integer steps. This is the representation-theoretic form of the triangle inequality

$$|s_1 - s_2| \leq S \leq s_1 + s_2.$$

23.6.1 Three spins and a nontrivial multiplicity matrix

For three spin- $\frac{1}{2}$ spaces,

$$\begin{aligned} \left(\frac{1}{2}\right)^{\otimes 3} &= (0 \oplus 1) \otimes \frac{1}{2} \\ &= \frac{1}{2} \oplus \left(\frac{1}{2} \oplus \frac{3}{2}\right) \\ &= 2 \left(\frac{1}{2}\right) \oplus \frac{3}{2}. \end{aligned}$$

Thus

$$n_{1/2} = 2, \quad d_{1/2} = 2, \quad n_{3/2} = 1, \quad d_{3/2} = 4.$$

Every rotationally invariant Hamiltonian has the structure

$$H = \left(C^{(1/2)} \otimes I_2\right) \oplus \left(E_{3/2} I_4\right),$$

where $C^{(1/2)}$ is a 2×2 Hermitian matrix. The quartet energy is fourfold degenerate, and each eigenvalue of $C^{(1/2)}$ is a doublet.

Consider the open three-site exchange Hamiltonian from the handwritten notes,

$$H = J(\mathbf{S}_1 \cdot \mathbf{S}_2 + \mathbf{S}_2 \cdot \mathbf{S}_3).$$

It commutes with the total-spin representation because each dot product is a rotational scalar. The abstract block form already determines all degeneracies. To exhibit the remaining 2×2 problem explicitly, couple spins 1 and 2 first. In the $M = +\frac{1}{2}$ sector choose

$$\begin{aligned} |a\rangle &= |s_{12} = 0; S = \frac{1}{2}, M = \frac{1}{2}\rangle = \frac{1}{\sqrt{2}}(|+-+\rangle - |-++\rangle), \\ |b\rangle &= |s_{12} = 1; S = \frac{1}{2}, M = \frac{1}{2}\rangle \\ &= \sqrt{\frac{2}{3}}|++-\rangle - \frac{1}{\sqrt{6}}(|+-+\rangle + |-++\rangle). \end{aligned}$$

For any pair of spin- $\frac{1}{2}$ particles,

$$\begin{aligned} S_i \cdot S_j |++\rangle &= \frac{1}{4} |++\rangle, \\ S_i \cdot S_j |+-\rangle &= -\frac{1}{4} |+-\rangle + \frac{1}{2} |-+\rangle, \\ S_i \cdot S_j |-+\rangle &= -\frac{1}{4} |-+\rangle + \frac{1}{2} |+-\rangle, \\ S_i \cdot S_j |--\rangle &= \frac{1}{4} |--\rangle. \end{aligned}$$

Applying these four elementary rules to the states above gives

$$C^{(1/2)} = J \begin{pmatrix} -\frac{3}{4} & \frac{\sqrt{3}}{4} \\ \frac{\sqrt{3}}{4} & -\frac{1}{4} \end{pmatrix}.$$

Its characteristic polynomial is

$$\begin{aligned}\det(C^{(1/2)} - EI_2) &= \left(-\frac{3J}{4} - E\right) \left(-\frac{J}{4} - E\right) - \frac{3J^2}{16} \\ &= E(E + J).\end{aligned}$$

Hence the two doublet energies are

$$E_{1/2}^{(1)} = -J, \quad E_{1/2}^{(2)} = 0.$$

On the totally symmetric $S = \frac{3}{2}$ space, each neighbouring pair is a triplet and contributes $J/4$, so

$$E_{3/2} = \frac{J}{2}.$$

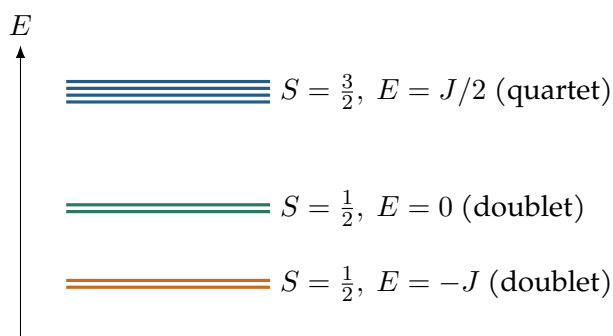


图 23.7: Spectrum of the open three-spin Hamiltonian for $J > 0$. The multiplicities 2, 2, 4 are dictated by rotation symmetry.

23.6.2 Four spins and the coupling tree

Adding one more spin gives

$$\begin{aligned}\left(\frac{1}{2}\right)^{\otimes 4} &= \left[2\left(\frac{1}{2}\right) \oplus \frac{3}{2}\right] \otimes \frac{1}{2} \\ &= 2(0 \oplus 1) \oplus (1 \oplus 2) \\ &= 2(0) \oplus 3(1) \oplus 1(2).\end{aligned}$$

The dimension check is

$$2 \cdot 1 + 3 \cdot 3 + 1 \cdot 5 = 2 + 9 + 5 = 16 = 2^4.$$

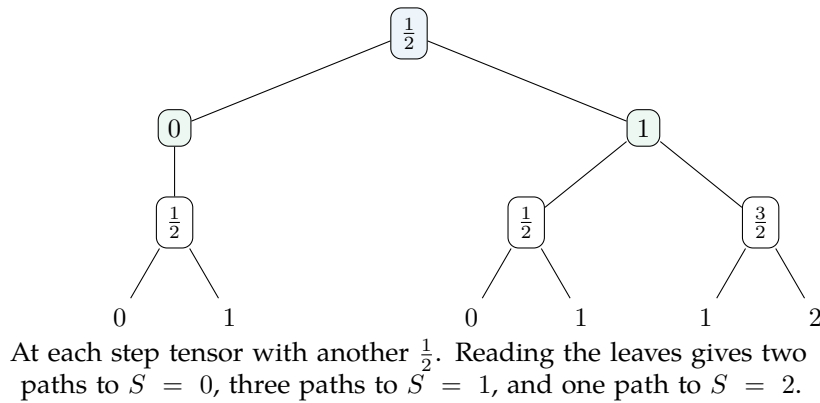


图 23.8: A coupling tree for four spin- $\frac{1}{2}$ particles. Different paths ending at the same S are the equivalent copies counted by n_S .

23.6.3 The half-Pascal triangle and its closed formula

Let $n_S(L)$ be the multiplicity of total spin S in $(\frac{1}{2})^{\otimes L}$. Adding the last spin and using the triangle rule shows that a final spin S can arise from $S - \frac{1}{2}$ or $S + \frac{1}{2}$ at the previous stage:

$$n_S(L+1) = n_{S-1/2}(L) + n_{S+1/2}(L),$$

with nonexistent or negative-spin multiplicities interpreted as zero. The first seven rows are

$L \backslash S$	0	$\frac{1}{2}$	1	$\frac{3}{2}$	2	$\frac{5}{2}$	3	$\frac{7}{2}$
1		1						
2	1		1					
3		2		1				
4	2		3		1			
5		5		4		1		
6	5		9		5		1	
7		14		14		6		1

This is the “half-Pascal triangle” drawn in the notes.

There is also a closed formula:

$$n_S(L) = \binom{L}{L/2 - S} - \binom{L}{L/2 - S - 1}.$$

Here a binomial coefficient is zero if its lower argument is not an integer between 0 and L .

证明. Fix a total magnetic quantum number M . A product state has

$$M = \frac{1}{2}(N_+ - N_-) = \frac{L}{2} - N_-.$$

Therefore the number of product-basis states with that M is

$$N_M = \binom{L}{L/2 - M}.$$

Every copy of a spin- S' irreducible representation contains exactly one state of magnetic quantum number M whenever $S' \geq |M|$. For $M \geq 0$,

$$N_M = \sum_{S' \geq M} n_{S'}(L).$$

Subtract the equation at $M = S + 1$ from the equation at $M = S$:

$$\begin{aligned} n_S(L) &= N_S - N_{S+1} \\ &= \binom{L}{L/2 - S} - \binom{L}{L/2 - S - 1}. \end{aligned}$$

This proves the formula. ■

As a final consistency check, the irreducible dimensions must recover the full tensor-product dimension:

$$\sum_S n_S(L)(2S + 1) = 2^L.$$

The left side counts each spin- S copy with its $2S + 1$ magnetic components, while the right side counts all choices of $+$ and $-$ on the L sites.

23.7 Tight-binding model on an open chain

Let

$$V = \text{span}\{e_1, \dots, e_L\}, \quad \langle e_i, e_j \rangle = \delta_{ij},$$

where $e_i = |i\rangle$ is an orbital localised at site i . A general tight-binding Hamiltonian is

$$H = \sum_{i,j=1}^L T_{ij} |i\rangle \langle j| = \sum_{i,j=1}^L T_{ij} e_i e_j^\dagger.$$

Hermiticity requires $T_{ji} = T_{ij}^*$. For a nearest-neighbour open chain with real uniform hopping T ,

$$H_{\text{open}} = T \sum_{i=1}^{L-1} (|i\rangle\langle i+1| + |i+1\rangle\langle i|).$$

Its matrix is tridiagonal:

$$H_{\text{open}} = T \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 1 & 0 & 1 & \ddots & \vdots \\ 0 & 1 & 0 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & 1 \\ 0 & \cdots & 0 & 1 & 0 \end{pmatrix}.$$

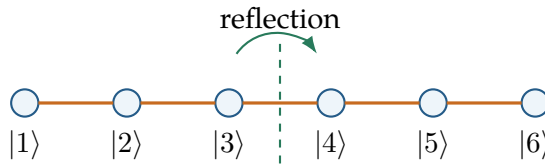


图 23.9: The open six-site chain and its only nontrivial spatial symmetry.

Let s reflect the chain:

$$D_{\text{chain}}(s)e_i = e_{L+1-i}, \quad D_{\text{chain}}(e) = I.$$

This is a unitary representation of $\mathbb{Z}_2$. In the site basis its nontrivial matrix is the anti-diagonal permutation matrix

$$D_{\text{chain}}(s) = \begin{pmatrix} 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & \cdots & 1 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 1 & \cdots & 0 & 0 \\ 1 & 0 & \cdots & 0 & 0 \end{pmatrix}.$$

Reflection maps every nearest-neighbour bond $(i, i+1)$ to the bond $(L-i, L+1-i)$, so

$$D_{\text{chain}}(s)H_{\text{open}}D_{\text{chain}}(s)^{-1} = H_{\text{open}}.$$

For even L , define

$$f_j^+ = \frac{e_j + e_{L+1-j}}{\sqrt{2}}, \quad f_j^- = \frac{e_j - e_{L+1-j}}{\sqrt{2}}, \quad j = 1, \dots, \frac{L}{2}.$$

Then

$$D_{\text{chain}}(s)f_j^+ = f_j^+, \quad D_{\text{chain}}(s)f_j^- = -f_j^-.$$

For every j , the line $\mathbb{C}f_j^+$ carries the even irrep. The line $\mathbb{C}f_j^-$ carries the odd irrep. Thus

$$D_{\text{chain}} = \frac{L}{2} D^{(+)} \oplus \frac{L}{2} D^{(-)} \quad (L \text{ even}).$$

For odd L , the fixed middle site is even, so the multiplicities are $(L+1)/2$ and $(L-1)/2$, respectively.

For the $L = 6$ example drawn in the notes, the invariant lines are

even	odd
$\text{span}\{e_1 + e_6\}$	$\text{span}\{e_1 - e_6\}$
$\text{span}\{e_2 + e_5\}$	$\text{span}\{e_2 - e_5\}$
$\text{span}\{e_3 + e_4\}$	$\text{span}\{e_3 - e_4\}$

and the symmetry-adapted Hamiltonian is

$$H_{\text{open}} \cong H_+ \oplus H_-,$$

with

$$H_+ = T \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \quad H_- = T \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix}.$$

For example, the $+1$ in the last diagonal entry of H_+ and the -1 in that of H_- come from the central bond $|3\rangle\langle 4| + |4\rangle\langle 3|$.

All complex irreducible representations of $\mathbb{Z}_2$ are one-dimensional. Hence $d_+ = d_- = 1$: parity creates two smaller diagonalisation problems but forces no degeneracy. This can be checked by solving the chain completely. Put

$$\psi_n = \sqrt{\frac{2}{L+1}} \sum_{j=1}^L \sin\left(\frac{n\pi j}{L+1}\right) e_j, \quad n = 1, \dots, L.$$

Using

$$\sin((j-1)q) + \sin((j+1)q) = 2 \cos q \sin(jq)$$

and the boundary values $\sin 0 = \sin(n\pi) = 0$, one obtains

$$H_{\text{open}} \psi_n = 2T \cos\left(\frac{n\pi}{L+1}\right) \psi_n.$$

The cosine is strictly decreasing on $(0, \pi)$, so these L energies are distinct. Reflection gives

$$\begin{aligned} \sin\left(\frac{n\pi(L+1-j)}{L+1}\right) &= \sin\left(n\pi - \frac{n\pi j}{L+1}\right) \\ &= (-1)^{n+1} \sin\left(\frac{n\pi j}{L+1}\right), \end{aligned}$$

and therefore

$$D_{\text{chain}}(s) \psi_n = (-1)^{n+1} \psi_n.$$

The exact eigenvectors alternate between the two parity sectors.

23.8 Tight-binding model on a ring

Close the chain by identifying $|L+1\rangle = |1\rangle$:

$$H_{\text{ring}} = T \sum_{i=1}^L (|i\rangle\langle i+1| + |i+1\rangle\langle i|).$$

The spatial symmetry is now the dihedral group

$$D_L = \langle r, s \mid r^L = e, s^2 = e, srs = r^{-1} \rangle,$$

of order $2L$. On the site basis, choose the generators

$$D_{\text{ring}}(r)e_i = e_{i-1}, \quad D_{\text{ring}}(s)e_i = e_{2-i},$$

where indices are read modulo L , with $e_0 = e_L$. The first generator is one step around the ring; the second is reflection about the axis through site 1. Both merely permute the bonds in (23.8), so

$$[H_{\text{ring}}, D_{\text{ring}}(a)] = 0 \quad (a \in D_L).$$

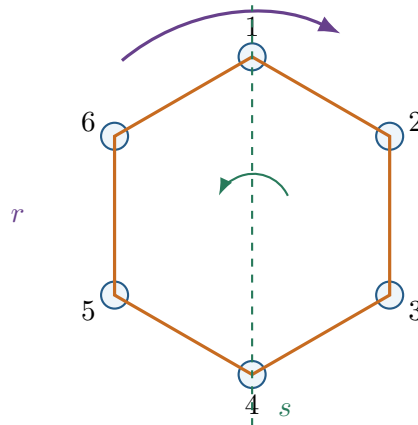


图 23.10: The six-site ring. Rotations and reflections generate the non-Abelian group D_6 .

23.8.1 Real Fourier subspaces

Put

$$\theta_k = \frac{2\pi k}{L}.$$

The constant vector

$$f_0 = \frac{1}{\sqrt{L}} \sum_{i=1}^L e_i$$

spans the trivial representation. When L is even, the alternating vector

$$f_{L/2} = \frac{1}{\sqrt{L}} \sum_{i=1}^L (-1)^i e_i$$

spans another one-dimensional representation:

$$D(r)f_{L/2} = -f_{L/2}, \quad D(s)f_{L/2} = f_{L/2}.$$

For

$$k = 1, \dots, \left\lfloor \frac{L-1}{2} \right\rfloor,$$

define

$$c_k = \sqrt{\frac{2}{L}} \sum_{i=1}^L \cos(\theta_k i) e_i,$$

$$s_k = \sqrt{\frac{2}{L}} \sum_{i=1}^L \sin(\theta_k i) e_i.$$

The elementary discrete orthogonality identities for sines and cosines show that all these vectors form an orthonormal basis.

We now compute the group action rather than merely quote it. Because $D(r)e_i = e_{i-1}$, relabelling $j = i - 1$ gives

$$\begin{aligned} D(r)c_k &= \sqrt{\frac{2}{L}} \sum_i \cos(\theta_k i) e_{i-1} \\ &= \sqrt{\frac{2}{L}} \sum_j \cos(\theta_k (j+1)) e_j \\ &= \cos \theta_k c_k - \sin \theta_k s_k. \end{aligned}$$

The same calculation with sine gives

$$D(r)s_k = \sin \theta_k c_k + \cos \theta_k s_k.$$

Thus, with column-vector coordinates in the ordered basis (c_k, s_k) ,

$$D^{(k)}(r) = \begin{pmatrix} \cos \theta_k & \sin \theta_k \\ -\sin \theta_k & \cos \theta_k \end{pmatrix}.$$

If the opposite convention $D(r)e_i = e_{i+1}$ is chosen, this matrix is transposed; the representation is equivalent and all conclusions are unchanged.

For the reflection, relabel $j = 2 - i$:

$$\begin{aligned} D(s)c_k &= \sqrt{\frac{2}{L}} \sum_j \cos(\theta_k (2-j)) e_j \\ &= \cos(2\theta_k) c_k + \sin(2\theta_k) s_k, \\ D(s)s_k &= \sin(2\theta_k) c_k - \cos(2\theta_k) s_k. \end{aligned}$$

Therefore

$$D^{(k)}(s) = \begin{pmatrix} \cos(2\theta_k) & \sin(2\theta_k) \\ \sin(2\theta_k) & -\cos(2\theta_k) \end{pmatrix}.$$

This matrix squares to I_2 , while direct multiplication verifies

$$D^{(k)}(s)D^{(k)}(r)D^{(k)}(s) = D^{(k)}(r)^{-1}.$$

23.8.2 The complete $L = 6$ calculation

For the six-site example in the handwritten pages, use the ordered subspaces

subspace	orthonormal basis	dimension
V_1	$f_1 = \frac{1}{\sqrt{6}} \sum_{i=1}^6 e_i$	1
V_2	$f_2 = \frac{1}{\sqrt{6}} \sum_{i=1}^6 (-1)^i e_i$	1
V_3	$f_3 = \frac{1}{\sqrt{3}} \sum_i \cos\left(\frac{2\pi i}{3}\right) e_i, \quad f_4 = \frac{1}{\sqrt{3}} \sum_i \sin\left(\frac{2\pi i}{3}\right) e_i$	2
V_4	$f_5 = \frac{1}{\sqrt{3}} \sum_i \cos\left(\frac{\pi i}{3}\right) e_i, \quad f_6 = \frac{1}{\sqrt{3}} \sum_i \sin\left(\frac{\pi i}{3}\right) e_i$	2

The one-dimensional actions are

$$\begin{array}{c|cc} & r & s \\ \hline D^{(1)} & 1 & 1 \\ D^{(2)} & -1 & 1 \end{array}.$$

For V_3 , where $\theta = 2\pi/3$,

$$D^{(3)}(r) = \begin{pmatrix} \cos \frac{2\pi}{3} & \sin \frac{2\pi}{3} \\ -\sin \frac{2\pi}{3} & \cos \frac{2\pi}{3} \end{pmatrix},$$

$$D^{(3)}(s) = \begin{pmatrix} \cos \frac{2\pi}{3} & -\sin \frac{2\pi}{3} \\ -\sin \frac{2\pi}{3} & -\cos \frac{2\pi}{3} \end{pmatrix}.$$

This second matrix follows from the general reflection formula, with $2\theta = 4\pi/3$. In writing it we used $\cos(4\pi/3) = \cos(2\pi/3)$ and $\sin(4\pi/3) = -\sin(2\pi/3)$.

For V_4 , where $\theta = \pi/3$,

$$D^{(4)}(r) = \begin{pmatrix} \cos \frac{\pi}{3} & \sin \frac{\pi}{3} \\ -\sin \frac{\pi}{3} & \cos \frac{\pi}{3} \end{pmatrix},$$

$$D^{(4)}(s) = \begin{pmatrix} -\cos \frac{\pi}{3} & \sin \frac{\pi}{3} \\ \sin \frac{\pi}{3} & \cos \frac{\pi}{3} \end{pmatrix}.$$

The two-dimensional representations are not equivalent. Equivalent matrices have equal traces, but

$$\text{tr } D^{(3)}(r) = 2 \cos \frac{2\pi}{3} = -1, \quad \text{tr } D^{(4)}(r) = 2 \cos \frac{\pi}{3} = 1.$$

This is the trace test carried out in the notes. Consequently,

$$D_{\text{ring}} = D^{(1)} \oplus D^{(2)} \oplus D^{(3)} \oplus D^{(4)} \quad (L = 6),$$

and all four summands are mutually inequivalent.

For completeness, the handwritten final-page calculation of the V_3 rotation is reproduced in

one uninterrupted chain:

$$\begin{aligned}
 D^{(3)}(r)f_3 &= \frac{1}{\sqrt{3}} \sum_i \cos\left(\frac{2\pi i}{3}\right) D(r)e_i \\
 &= \frac{1}{\sqrt{3}} \sum_j \cos\left(\frac{2\pi(j+1)}{3}\right) e_j \\
 &= \frac{1}{\sqrt{3}} \sum_j \left[\cos\left(\frac{2\pi j}{3}\right) \cos \frac{2\pi}{3} - \sin\left(\frac{2\pi j}{3}\right) \sin \frac{2\pi}{3} \right] e_j \\
 &= \cos \frac{2\pi}{3} f_3 - \sin \frac{2\pi}{3} f_4.
 \end{aligned}$$

Analogously,

$$D^{(3)}(r)f_4 = \sin \frac{2\pi}{3} f_3 + \cos \frac{2\pi}{3} f_4,$$

which gives exactly the matrix stated above.

23.8.3 Irreducible decomposition and energy degeneracies

For even L , the vertex representation decomposes as

$$D_{\text{ring}} = D^{(0)} \oplus D^{(L/2)} \oplus \bigoplus_{k=1}^{L/2-1} D^{(k)},$$

The first two summands are one-dimensional. Every $D^{(k)}$ in the sum is two-dimensional. For odd L , only $D^{(0)}$ is one-dimensional and

$$D_{\text{ring}} = D^{(0)} \oplus \bigoplus_{k=1}^{(L-1)/2} D^{(k)}.$$

The two-dimensional spaces are irreducible: over $\mathbb{C}$, the rotation has eigenlines with eigenvalues $e^{\pm i\theta_k}$, but reflection interchanges those lines, so neither is invariant under the full dihedral group. The trace

$$\text{tr } D^{(k)}(r) = 2 \cos \theta_k$$

also shows that the listed summands are pairwise inequivalent.

Every summand appears once. Therefore the general symmetry theorem says that H_{ring} is already scalar on each listed subspace. The scalar can be found directly:

$$\begin{aligned}
 H_{\text{ring}}c_k &= T \sqrt{\frac{2}{L}} \sum_i \cos(\theta_k i) (e_{i-1} + e_{i+1}) \\
 &= 2T \cos \theta_k c_k,
 \end{aligned}$$

and similarly

$$H_{\text{ring}}s_k = 2T \cos \theta_k s_k.$$

Thus

$$E_k = 2T \cos \left(\frac{2\pi k}{L} \right).$$

The modes k and $L - k$, combined in the real basis (c_k, s_k) , have the same energy. Their twofold degeneracy is forced by the two-dimensional irreducible representation of D_L .

For $L = 6$, the complete result is

subspace	k	E_k	degeneracy
V_1	0	$2T$	1
V_4	1	T	2
V_3	2	$-T$	2
V_2	3	$-2T$	1

and the Hamiltonian in the symmetry basis has the schematic form

$$H_{\text{ring}} \cong (2T) \oplus (-2T) \oplus (-TI_2) \oplus (TI_2).$$

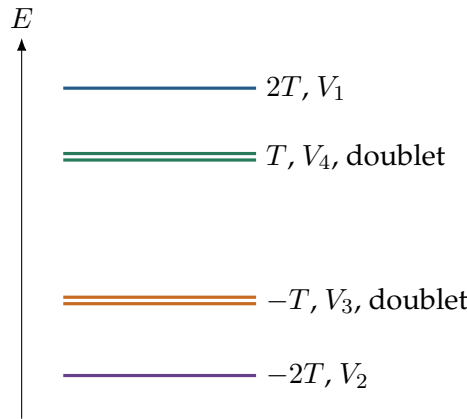


图 23.11: The six-site ring spectrum for $T > 0$. The two middle levels are doubly degenerate because V_3 and V_4 are two-dimensional irreducible D_6 -spaces.

The case $L = 4q + 2$, such as $L = 6$, has no zero-energy mode because no integer k satisfies $2\pi k/L = \pi/2$. If $L = 4q$, the allowed mode $k = L/4$ instead gives

$$E_{L/4} = 0$$

and belongs to a two-dimensional dihedral irrep, so the zero energy is twofold degenerate. This is the distinction between the $L = 4q + 2$ and $L = 4q$ rings noted in the lecture.

Why reflections matter

If one uses only the cyclic translation group C_L , all complex irreducible representations are one-dimensional. The two Fourier modes $e^{i\theta_k i}$ and $e^{-i\theta_k i}$ are then distinct one-dimensional sectors. Enlarging the symmetry to the full dihedral group adds a reflection that exchanges them; together they form one two-dimensional irreducible space, and the equality of their energies is then visibly symmetry-enforced.

What to retain from this chapter

1. Wigner's theorem represents transition-probability-preserving transformations by unitary or antiunitary operators, up to phase.
2. A unitary symmetry D of a Hamiltonian satisfies $D(a)HD(a)^{-1} = H$. The concrete physical action of D , not merely the abstract group, is essential.

3. If

$$D \cong \bigoplus_{\alpha} n_{\alpha} D^{(\alpha)},$$

then a symmetry-adapted basis gives

$$H = \bigoplus_{\alpha} \left(C^{(\alpha)} \otimes I_{d_{\alpha}} \right).$$

This follows by applying Schur's lemma to every Hamiltonian block.

4. The irrep dimension d_{α} is the minimum symmetry-induced degeneracy. The multiplicity n_{α} is the size of the matrix $C^{(\alpha)}$ still to be diagonalised.
5. A degeneracy unexplained by the chosen group is accidental relative to that group; a larger symmetry can turn it into an ordinary irrep degeneracy.
6. The spin- $\frac{1}{2}$ representation is generated by $S_{\alpha} = \sigma_{\alpha}/2$, and

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1.$$

The two-spin Heisenberg Hamiltonian therefore has a singlet and a triplet whose three states have equal energy.

7. For three spins,

$$\left(\frac{1}{2} \right)^{\otimes 3} = 2 \left(\frac{1}{2} \right) \oplus \frac{3}{2},$$

so an invariant Hamiltonian contains a 2×2 multiplicity matrix for the doublets and one scalar for the quartet.

8. The multiplicity of total spin S among L spin- $\frac{1}{2}$ particles is

$$n_S(L) = \binom{L}{L/2 - S} - \binom{L}{L/2 - S - 1}.$$

9. Reflection symmetry splits an open chain into even and odd blocks but forces no degeneracy, because the irreps of $\mathbb{Z}_2$ are one-dimensional.
10. The dihedral vertex representation of a ring contains irreducible Fourier subspaces of dimension two. On each such subspace,

$$E_k = 2T \cos \left(\frac{2\pi k}{L} \right)$$

is therefore twofold degenerate.